Fall 2026 Lecture Notes

STAT 8561

Linear Statistical Analysis I

Chi-Kuang Yeh

Assistant Professor in Statistics

Department of Mathematics and Statistics

Georgia State University

Personal website

chikuang.github.io

Course website

chikuang.github.io/teaching/2026F_stat8561/

Updated 2026-10-27

Table of contents

Preface **8**
Office Hours . . . 8
Grade Distribution . . . 8
Assignment . . . 9
Exam . . . 9
Project . . . 9
Topics and Corresponding Lectures . . . 9
Recommended Textbooks . . . 10
Side Readings . . . 10

Introduction **11**

1 Matrix Representation and Notation **14**
1.1 Why Linear Statistical Analysis? . . . 14
1.2 A unifying point of view . . . 15
1.3 Basic Notation . . . 15
1.3.1 Inner product and norm . . . 17
1.3.2 Expectation and Covariance Operators . . . 18
1.4 Statistical Models . . . 19
1.4.1 General idea . . . 19
1.4.2 Deterministic part and random part . . . 19
1.5 The Linear Regression Model . . . 20
1.5.1 Geometry Preview . . . 20
1.6 Why projection matters . . . 21
1.7 Discussion . . . 23
1.8 Practice Problems . . . 23
1.9 Take home Practice . . . 24
Summary / Recap . . . 24

2 Least Squares Estimation **25**
2.1 The Least Squares Criterion . . . 27
2.2 Derivation of the Least Squares Estimator in a matrix form . . . 30
2.3 Geometric Interpretation of Least Squares . . . 34
2.3.1 Least Squares as Projection . . . 34
2.3.2 Consequence when an Intercept Is Included . . . 40
2.4 The Hat Matrix . . . 41
2.5 The common name to refer to what we have done – Ordinary Least Squares (OLS) . . . 44
2.6 Sum of Squares Decomposition . . . 45
2.6.1 Uncentered Geometric Decomposition . . . 45
2.6.2 Centered Sum of Squares Decomposition . . . 45
2.6.3 Example: Simple Linear Regression . . . 46
2.7 What Happens Without Full Column Rank? . . . 53
2.8 Interpretation of the Geometry . . . 53
2.9 Looking Ahead: Statistical Properties of OLS . . . 55
2.10 Practice at Home . . . 55
Summary / Recap . . . 56

Side Fact: Matrix Calculus Facts Used in This Chapter . . . 56

3 Distribution Theory of OLS and Inference 58
3.1 Mean and Variance of the OLS Estimator . . . 60
3.1.1 Why Is OLS Unbiased? . . . 61
3.1.2 Variance of OLS . . . 61
3.2 The Normal Linear Model . . . 62
3.3 Distribution of the OLS Estimator . . . 64
3.4 Residual Sum of Squares and Estimation of σ^2 . . . 65
3.4.1 Residual Sum of Squares . . . 66
3.4.2 Distribution of SSE . . . 67
3.4.3 Why Are the Degrees of Freedom $n-p$? . . . 67
3.4.4 Estimating σ^2 . . . 69
3.5 Independence of OLS and Residual Variation . . . 70
3.5.1 Why Does Independence Occur? . . . 70
3.6 Inference for a Single Regression Coefficient . . . 72
3.6.1 Standard Error . . . 72
3.6.2 Why Does the t Distribution Appear? . . . 73
3.6.3 Confidence Interval for β_j . . . 74
3.6.4 Hypothesis Test for β_j . . . 74
3.7 Inference for Linear Combinations . . . 75
3.8 General Linear Hypotheses . . . 76
3.9 Inference for the Mean Response . . . 77
3.10 Prediction for a New Observation . . . 78
3.11 Worked Example . . . 79
3.12 R Demonstration . . . 82
3.13 Reading Standard Regression Output . . . 86
3.14 Practice at Home . . . 87
Summary / Recap . . . 88
3.15 Side Fact . . . 89

4 Model Comparison, Partial F Tests, and ANOVA 90
4.1 Total, Explained, and Unexplained Variation . . . 92
4.2 The Overall F Test . . . 95
4.3 Nested Models . . . 96
4.4 R Demonstration . . . 102
4.5 Practice Problems . . . 111
Summary / Recap . . . 112
Appendix: From the SSE Difference to the Overall F Test . . . 113
A geometric picture of the extra sum of squares . . . 113
Why subtract the two residual sums of squares? . . . 114
Recovering Theorem 4.2 from Theorem 4.3 . . . 114
Why does the ratio have an F distribution? . . . 115

5 Multiple Regression and Categorical Predictors 116
5.1 Multiple regression and adjusted interpretation . . . 117
5.2 Categorical predictors are columns of a design matrix . . . 120
5.3 A continuous predictor and a group indicator . . . 125
5.4 Interactions and conditional comparisons . . . 129
5.5 Match the hypothesis to the model comparison . . . 133
5.6 Summary . . . 135
5.7 Practice Problems . . . 135

6 Outliers, Leverage, and Influence **137**
6.1 See how one observation changes the fit . . . 137
6.2 High leverage: unusual predictor values . . . 139
6.3 Outliers: unusual responses given the predictors . . . 141
6.4 Influence: changes in estimates and Cook's distance . . . 144
6.5 Decide what is unusual, influential, and worth investigating . . . 146
6.6 Summary . . . 148
6.7 Practice Problems . . . 148

7 Transformations and Variable Selection **150**
7.1 A curved relationship can still be a linear model . . . 150
7.2 Choose a transformation, fit, and check again . . . 151
7.3 From candidate terms to variable selection . . . 156
7.4 Shrinkage: ridge, lasso, and elastic net . . . 159
7.5 Evaluate the complete procedure . . . 161
7.6 Summary . . . 163
7.7 Practice Problems . . . 163

I Optional topic **165**

Quantile Regression **166**
When the mean is not the whole question . . . 166
A mean model can miss changing spread . . . 168
Uncertainty and practical checks . . . 170
Application and self-study . . . 171

Mixed-Effects Models **173**
Why group membership changes the analysis . . . 173
Partial pooling and shrinkage . . . 174
Application: repeated measurements and random slopes . . . 177
Inference, model checks, and further reading . . . 180

Contrasts and Estimable Comparisons **183**
Comparisons of group means . . . 183
Conditional comparisons in the ToothGrowth data . . . 184
One comparison and a family of comparisons . . . 186
Coding and estimability . . . 188
Summary / Recap . . . 188
Further practice: a three-group comparison . . . 188
Practice . . . 190

References **192**

II Appendix **193**

Matrix Algebra Review **194**

Basic Mathematical Statistics **195**
Chi-square: a sum of squared standard normals . . . 195
Why normal quadratic forms are chi-square . . . 196
Student's t: estimating an unknown scale . . . 197
F: comparing two independent sums of squares . . . 198
Proof of the general linear hypothesis F test . . . 199

A compact regression example and reading guide 200

III Assignment 202

Assignment 1 203
Question 1: Matrix Representation of the Linear Model [8] 204
(a) [2] 204
(b) [2] 205
(c) [2] 205
(d) [2] 206
Question 2: Least Squares and Projection [12] 206
(a) [4] 206
(b) [3] 207
(c) [2] 208
(d) [3] 208
Question 3: Consequences and Geometry of Least Squares [12] 209
(a) Fitted Values and Residuals [3] 209
(b) Consequences of Including an Intercept [3] 210
(c) Residual-Maker Matrix [3] 210
(d) Rank and Uniqueness [3] 211
Question 4: Linear Regression in R [8] 212
(a) [4] 212
(b) [4] 213

Assignment 2 214
Question 1: Least Squares: Assumptions and Consequences [6] 215
(a) Changing the distributional assumption [2] 215
(b) What makes the minimizer unique? [2] 215
(c) Shifting every response [2] 216
Question 2: Projection Geometry and Rank [8] 216
(a) Projecting a second time [2] 216
(b) Adding a redundant predictor [2] 217
(c) Geometric consequences [4] 217
Question 3: Sampling Properties and the Role of Normality [8] 218
(a) Match the result to its assumptions [4] 218
(b) Why does independence follow? [2] 219
(c) Increasing the error variance [2] 219
Question 4: Interpreting Confidence and Prediction Intervals [6] 220
(a) Mean response versus one future response [2] 220
(b) Connecting a confidence interval and a test [2] 220
(c) What an interval means [2] 221
Question 5: Distribution and Inference [12] 221
(a) [3] 222
(b) [2] 222
(c) [4] 223
(d) [3] 223

Assignment 3 224
Question 1: What Model Comparisons Establish [10] 224
(a) Adding a predictor [2] 225
(b) The overall regression sum of squares [2] 225
(c) An overall rejection [2] 225
(d) Training fit [2] 225

(e) A missing-data trap [2] . . . 225
Question 2: ANOVA and Extra Sums of Squares [10] . . . 225
(a)-(h) Numerical answers [8] . . . 226
(i) Decision [2] . . . 226
Question 3: Does Horsepower Add Information Beyond Weight? [8] . . . 226
(a) The null hypothesis [2] . . . 227
(b) Reference distribution [2] . . . 227
(c) Connect the tests [2] . . . 227
(d) What does the result establish? [2] . . . 227
Question 4: Guided Introduction to Chapter 5 [12] . . . 228
(a) Group B's population mean [2] . . . 228
(b) Design rows [2] . . . 228
(c) Fitted means [2] . . . 228
(d) Adjusted group difference [2] . . . 228
(e) Interpret the slope [2] . . . 228
(f) Connect to Chapter 4 [2] . . . 229

Assignment 4 **230**

IV Exam 231

Exam 1 **232**
Question 1: Multiple Choice [50] . . . 233
1a) . . . 233
1b) . . . 233
1c) . . . 233
1d) . . . 234
1e) . . . 234
1f) . . . 234
1g) . . . 235
1h) . . . 235
1i) . . . 235
1j) . . . 236
Question 2: True or False [20] . . . 237
2a) . . . 237
2b) . . . 237
2c) . . . 237
2d) . . . 237
2e) . . . 238
Question 3: Conceptual [30] . . . 239
3a) Matrix representation and least squares [20] . . . 239
3b) Interpreting uncertainty [10] . . . 239
Answer Key and Grading Rubric . . . 241
Question 1: Multiple Choice [50] . . . 241
Question 2: True or False [20] . . . 241
Question 3: Conceptual [30] . . . 241

V Project 242

Instruction **243**
Group Members . . . 244
Project Overview . . . 245

Report writing model . . . 245
Assessment criteria . . . 246
How to get started . . . 246
A few possible questions . . . 247

Report **248**
Research question and data . . . 248
Linear regression analysis . . . 248
Machine learning comparison . . . 249
A fair evaluation . . . 249
Explain the findings . . . 251
Report format and reproducibility . . . 252
Writing model: The American Statistician . . . 252
Suggested report structure . . . 253
Preview and download the template . . . 253
Assessment criteria . . . 255
Submission . . . 256

Preface

The topic of this course includes statistical inference, Multivariate normal distribution, distribution of quadratic forms, linear models, regression models and experimental design models.

The course moves from Gaussian linear-model theory to regression, experimental comparisons, and model checking. Chapter 6 distinguishes outliers, leverage, and influence. Chapter 7 develops transformations and variable selection, including ridge and lasso. Later units develop experimental design, broader model checks, prediction assessment, and logistic regression. Poisson regression and fuller GLM theory remain supplementary reading. Advanced extensions are collected under **Optional topic**.

Prerequisites

Math 4751/6751 Mathematical Statistics I and Math 4752/6752 Mathematics and Statistics II.

Instructor

Chi-Kuang Yeh, Assistant Professor in the Department of Mathematics and Statistics, Georgia State University.

- Office: Suite 1407, 25 Park Place.
- Email: cyeh@gsu.edu.

Office Hours

Wednesdays, 9:00 a.m. to 12:00 p.m., or by appointment.

Grade Distribution

- Homework – 40%
- Exam 1 – 20%
- Exam 2 – 20%
- Final Project – 20%
- Attendance – *7%

Assignment

- ☒ Assignment 1: September 11, 2026
- ☒ Assignment 2: September 30, 2026
- ☐ Assignment 3: October 30, 2026
- ☐ Assignment 4: TBA

Exam

- ☒ Exam 1: October 14, 2026
- ☐ Exam 2, November 11, 2026

Project

Final assessment decision: The final assessment will be a **Final Project**, selected by a class vote of **20 to 5**.

See Instruction for the project description, requirements, and grading criteria.

- ☐ Report: Due on December 13, 2026 (20%)

Topics and Corresponding Lectures

Those chapters are based on the lecture notes. This part will be updated frequently.

Status	Chapter	Topic	Lecture
Done		Welcome and Overview	1
Done	Ch. 1	Math Notation	2–3
Done	Ch. 2	Least Squares Estimation	3–6
Done	Ch. 3	Inference on Least Square Estimation	7–10
Done	Ch. 4	Model Comparison, Partial F Tests, and ANOVA	11–17
Done		Exam 1	15
Next	Ch. 5	Multiple Regression and Categorical Predictors	18–
Planned	Ch. 6	Outliers, Leverage, and Influence	TBA
Planned	Ch. 7	Transformations and Variable Selection	TBA
Exam		Exam 2	23

Optional topic

Students interested in extensions can read Quantile Regression, Mixed-Effects Models, and Contrasts and Estimable Comparisons. These readings include explanations, graphical examples, R code, and further reading; they are supplementary to the scheduled chapters.

Recommended Textbooks

- Ding, Peng. *Linear Model and Extensions*. CRC Press, 2027. arXiv link
- Montgomery, Douglas C., Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 2021. (Montgomery et al. 2021)
- Seber, George AF, and Alan J. Lee. *Linear Regression Analysis*. John Wiley & Sons, 2003. (Seber and Lee 2003)

Side Readings

- Montgomery, Douglas C. (2017). *Design and Analysis of Experiments*. John Wiley & Sons. (Montgomery 2017)

Introduction

This course develops a common language for statistical modelling: specify the research question, construct a design matrix, estimate a mean structure, quantify uncertainty, and check whether the assumptions are credible.

The core sequence covers matrix notation and least squares; normal distributions and quadratic forms; inference and model comparison; categorical predictors and interactions; transformations, variable selection, and regularization; selected design, diagnostic, and validation applications; and an introduction to logistic regression.

The logistic lesson retains the linear predictor and changes the response distribution and mean link. Chapter 7 introduces ridge/lasso through prediction validation, with later practice devoted to tuning and interpretation. Poisson regression and fuller GLM theory remain supplementary reading. The supplementary appendices cover rank-deficient models, additional selection criteria, correlated errors/GLS, and GLM likelihood and computation. The **Optional topic** group contains contrasts, quantile regression, and mixed-effects models as self-study readings for students who want to explore distributional features or grouped observations.

Lectures, homework, two in-class exams, and a final assessment develop both mathematical reasoning and reproducible analysis in R. Core tasks include interpreting estimates and intervals, identifying the experimental unit, and explaining the limits of an analysis.

To illustrate the concepts we will be covering in this course, let's consider a simple example. Suppose we have a data set of heights and weights of individuals. We want to understand the relationship between height and weight, and we can use linear regression to model this relationship.

Example 1: First Example

```
# Load necessary libraries
library(ggplot2)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
# Create a sample data set

set.seed(8561)

theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

n <- 100
heights <- rnorm(n, mean = 170, sd = 10)
weights <- 0.5 * heights + rnorm(n, mean = 0, sd = 5)
df <- data.frame(heights, weights)
head(df, 5)
```

```
   heights  weights
1 179.2812 88.01599
2 168.0474 76.10017
3 178.3029 83.46016
4 178.7006 82.01763
5 172.6678 92.84592
```

```r
# Fit a linear regression model
model <- lm(weights ~ heights, data = df)
# Summarize the model
summary(model)
```

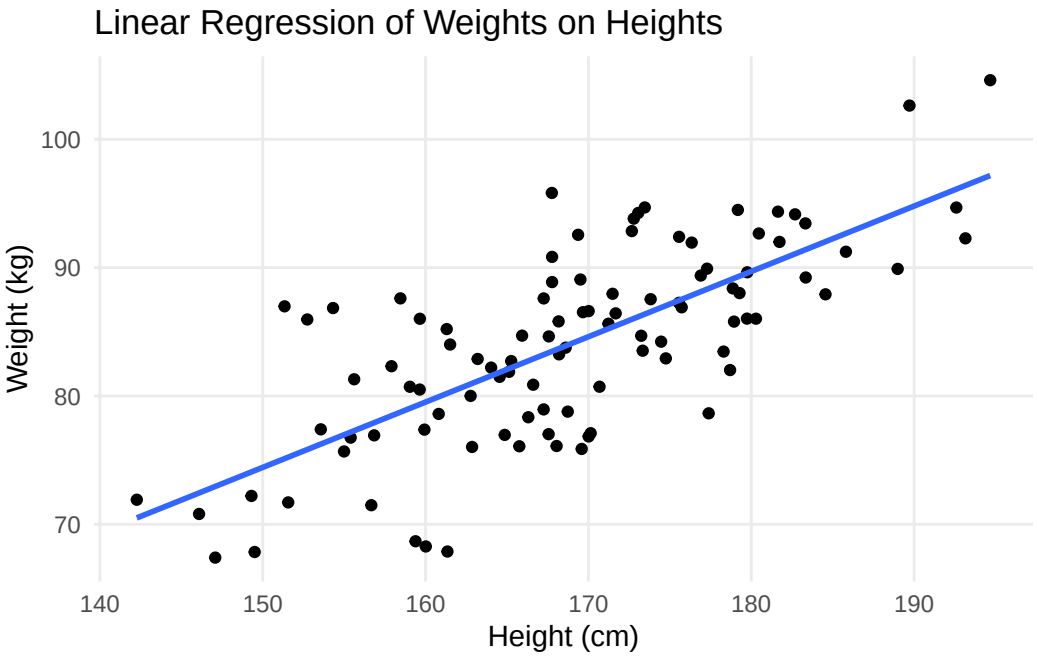
```
Call:
lm(formula = weights ~ heights, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-12.3303  -3.8998  -0.2191   3.2380  12.3478

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.92852    8.22226  -0.235    0.815
heights      0.50907    0.04857  10.481   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.251 on 98 degrees of freedom
Multiple R-squared:  0.5285,    Adjusted R-squared:  0.5237
F-statistic: 109.9 on 1 and 98 DF,  p-value: < 2.2e-16
```

```r
# Plot the data and the fitted line
ggplot(df, aes(x = heights, y = weights)) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  labs(title = "Linear Regression of Weights on Heights",
       x = "Height (cm)",
       y = "Weight (kg)")
```



In this example, we generated a data set of heights and weights, fitted a linear regression model to the data, and visualized the relationship between height and weight. This is just a simple example, but it illustrates the types of analyses we will be doing in this course. We will be covering much more complex models and data sets as we progress through the semester.

Throughout the course, we will be using **R** programming software.

Section 1

Matrix Representation and Notation

In this chapter, we introduce the mathematical language and statistical framework that will be used throughout the course. Our focus is on the notation of vectors and matrices, vectors of random variables, expectation and covariance operators, and the basic form of the linear regression model.

Learning Objectives

By the end of this chapter, students should be able to:

- use standard matrix notation for linear statistical models;
- distinguish between scalars, vectors, matrices, random variables, and random vectors;
- compute expectations and covariance matrices for random vectors;
- interpret the linear regression model in matrix form;
- understand why projection and least squares will play a central role in this course.

Reading

Recommended reading for this chapter:

- Seber and Lee (2003), Chapter 1:
  - 1.1 Notation
  - 1.2 Statistical Models
  - 1.3 Linear Regression Models
  - 1.4 Expectation and Covariance Operators
- Optional preview:
  - Chapter 2: Multivariate Normal Distribution

Section 1.1 Why Linear Statistical Analysis?

Linear statistical analysis is one of the central foundations of graduate statistics. Many methods that at first look different are built on the same underlying structure:

- regression
- analysis of variance
- analysis of covariance
- prediction
- model comparison

- and parts of generalized linear modelling

A major goal of this course is to see these topics under a unified framework.

Section 1.2 A unifying point of view

A large part of the course can be summarized by the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

- $\mathbf{Y}$ is a response vector,
- $\mathbf{X}$ is a design matrix,
- $\boldsymbol{\beta}$ is an unknown parameter vector,
- $\boldsymbol{\varepsilon}$ is a random error vector.

This compact expression contains a great deal of statistical structure. Over the semester, we will study how to estimate $\boldsymbol{\beta}$, quantify uncertainty, test hypotheses, diagnose model failures, and make predictions.

Section 1.3 Basic Notation

Scalars, vectors, and matrices

We use the following conventions throughout the course:

- scalars are written in ordinary (nonbold) type, such as a, b, n;
- vectors are written in boldface, such as $\mathbf{x}$, $\mathbf{y}$, $\boldsymbol{\beta}$, and the random response vector $\mathbf{Y}$;
- matrices are written in bold uppercase letters, such as $\mathbf{X}$, $\mathbf{A}$;
- random variables are often written in uppercase letters, such as Y;
- realizations of random variables are written in lowercase letters, such as y.

Individual entries remain scalars: Y_i is an entry of $\mathbf{Y}$, x_{ij} is an entry of $\mathbf{X}$, and β_j is an entry of $\boldsymbol{\beta}$. The scalar mean $\bar{Y}$ and variance σ^2 are not bold. We write the zero vector or matrix as $\mathbf{0}$ and the all-ones vector as $\mathbf{1}$; a scalar zero or one remains 0 or 1.

Univariate case

A random variable is **univariate** if it takes values in one dimension. We write

$$X \in \mathbb{R}.$$

A realization, or observed value, of X is denoted by x.

Some common quantities associated with a univariate random variable X include:

- **Mean**
$$\mu = \mathbb{E}(X).$$
- **Variance**
$$\sigma^2 = \mathrm{Var}(X) = \mathbb{E}\left[(X-\mu)^2\right].$$
- **kth moment**
$$\mathbb{E}(X^k).$$
- **Cumulative distribution function (CDF)**
$$F(x) = P(X \le x).$$
- **Probability density function (PDF)**, for a continuous random variable,
$$f(x).$$
- **Probability mass function (PMF)**, for a discrete random variable,
$$p(x) = P(X = x).$$
- **Quantile**
$$q_\alpha = F^{-1}(\alpha),$$
where $0 < \alpha < 1$.

For a random sample of size n, we write

$$X_1, X_2, \dots, X_n,$$

with observed values

$$x_1, x_2, \dots, x_n.$$

The sample mean is

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

and the sample variance is

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Multivariate case

A column vector in $\mathbb{R}^n$ is written as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = (x_1, x_2, \ldots, x_n).$$

An $n \times p$ matrix is written as

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}.$$

Transpose, inverse, and rank

If $\mathbf{A}$ is a matrix, then:

- $\mathbf{A}^\top$ denotes its transpose;
- $\mathbf{A}^{-1}$ denotes its inverse, when it exists;
- $\text{rank}(\mathbf{A})$ denotes its rank;
- $\mathbf{I}_n$ denotes the $n \times n$ identity matrix.

1.3.1 Inner product and norm

Definition 1: inner product in Eucludean space

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\mathbf{x}^\top \mathbf{y} = \sum_{i=1}^{n} x_i y_i.$$

Definition 2: Euclidean norm

For vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the inner product is

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

These ideas are fundamental because least squares estimation is based on minimizing squared Euclidean distance.

Definition 3: Random Vectors

A random vector is a vector whose entries are random variables. For example,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}$$

is an n-dimensional random vector.

In linear models, the response is naturally treated as a random vector.

Definition 4: Mean vector

The mean vector of $\mathbf{Y}$ is

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} \mathbb{E}[Y_1] \\ \mathbb{E}[Y_2] \\ \vdots \\ \mathbb{E}[Y_n] \end{pmatrix}.$$

We often write $\mathbb{E}[\mathbf{Y}] = \boldsymbol{\mu}$.

Definition 5: Covariance matrix

The covariance matrix of $\mathbf{Y}$ is

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}\left[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top\right].$$

Equivalently,

$$\mathrm{Var}(\mathbf{Y}) = \mathbb{E}[\mathbf{Y}\mathbf{Y}^\top] - \boldsymbol{\mu}\boldsymbol{\mu}^\top.$$

Example 1

Suppose $\mathbf{Y} = (Y_1, Y_2, Y_3)$.
Then the corresponding covariance matrix is

$$\mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} \mathrm{Var}(Y_1) & \mathrm{Cov}(Y_1, Y_2) & \mathrm{Cov}(Y_1, Y_3) \\ \mathrm{Cov}(Y_2, Y_1) & \mathrm{Var}(Y_2) & \mathrm{Cov}(Y_2, Y_3) \\ \mathrm{Cov}(Y_3, Y_1) & \mathrm{Cov}(Y_3, Y_2) & \mathrm{Var}(Y_3) \end{pmatrix}.$$

1.3.2 Expectation and Covariance Operators

Property 1: Linearity of expectation

If $\mathbf{A}$ is a constant matrix and $\mathbf{b}$ is a constant vector, then

$$\mathbb{E}[\mathbf{A}\mathbf{Y} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{Y}] + \mathbf{b}.$$

This is one of the most useful identities in the course.

Property 2: Covariance of linear transformations

If $\mathbf{A}$ is a constant matrix, then

$$\mathrm{Var}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\,\mathrm{Var}(\mathbf{Y})\,\mathbf{A}^\top.$$

More generally, if $\mathbf{Y}$ and $\mathbf{Z}$ are random vectors, then

$$\mathrm{Cov}(\mathbf{AY}, \mathbf{BZ}) = \mathbf{A}\,\mathrm{Cov}(\mathbf{Y}, \mathbf{Z})\,\mathbf{B}^\top.$$

These formulas are essential for deriving the variance of estimators later.

Example 2: independent components

If $Y_1, \ldots, Y_n$ are independent and each has variance σ^2, then

$$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

This is the most common starting assumption in classical linear regression.

Section 1.4 Statistical Models

A statistical model is a set of probability distributions that may plausibly describe the data-generating mechanism.

1.4.1 General idea

Suppose we observe data y from a random quantity Y. A model introduces assumptions about the distribution of Y, often indexed by an unknown parameter θ.

For example:

$$Y \sim N(\mu, \sigma^2)$$

with unknown parameters μ and σ^2.

In regression, the model is not only about the marginal distribution of the response but also about how the mean changes with explanatory variables.

1.4.2 Deterministic part and random part

A useful way to think about a statistical model is:

$$\text{data} = \text{systematic part} + \text{random part}.$$

For linear regression, this becomes

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

Here,

- $\beta_0 + \beta_1 x_i$ is the systematic part;
- ε_i is the random part.

Section 1.5 The Linear Regression Model

Perhaps the most simple model is the *simple linear regression* where there are only **one** independent variable and **one** response variable.

Definition 6: Simple linear regression

A simple linear regression model is defined as

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

with the typical assumption as

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2, \qquad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j.$$

Matrix form

We can write the model compactly as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}, \qquad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}.$$

This notation will allow us to treat simple regression, multiple regression, ANOVA, and ANCOVA in one common language.

Under the same assumption,

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0} \qquad \text{and} \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

we have the following.

$$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}, \quad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

These are immediate consequences of the expectation and covariance rules above.

1.5.1 Geometry Preview

A central idea in linear regression is projection.

The fitted values $\hat{\mathbf{Y}}$ will later be obtained by projecting $\mathbf{Y}$ onto the column space of $\mathbf{X}$.

Definition 7: Column space

The column space of $\mathbf{X}$ is

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\}.$$

This is the set of all mean vectors that the model can represent.

Section 1.6 Why projection matters

The least squares estimator chooses $\hat{\boldsymbol{\beta}}$ so that

$$\|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2$$

is minimized.

So regression is not only algebra. It is also geometry.

Example 3

Suppose we observe the following data:

x_i	0	1	2	3
y_i	1	3	3	5

Then

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

We will learn in Chapter 2 — Least Squares Estimation that the least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For now, the main point is to understand how the model is written and how the data are represented in matrix form.

R code

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)
mod <- lm(y ~ x)
summary(mod)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

```r
# Inspect the design matrix, fitted values, and residuals.
model.matrix(mod)
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```r
fitted(mod)
```

```
  1   2   3   4
1.2 2.4 3.6 4.8
```

```r
residuals(mod)
```

```
   1    2    3    4
-0.2  0.6 -0.6  0.2
```

```r
ggplot(data.frame(x, y), aes(x, y)) +
  geom_point(size = 2.5) +
  geom_abline(intercept = coef(mod)[1], slope = coef(mod)[2],
              colour = "steelblue4", linewidth = 0.8) +
  labs(x = "x", y = "y")
```

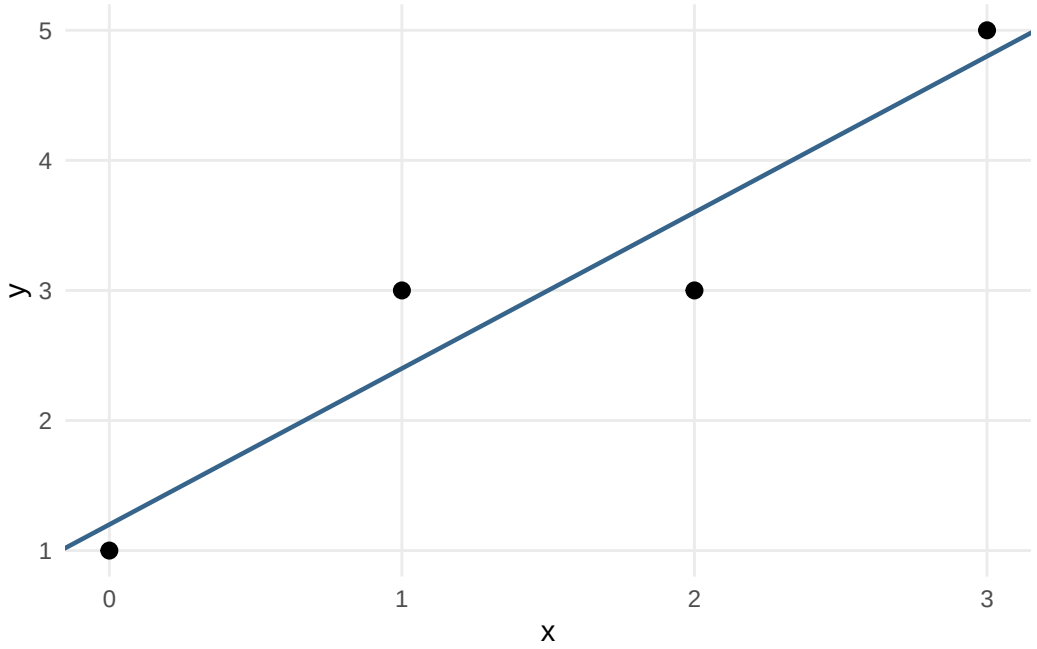


Figure 1.1: Four observations and their least-squares fitted line.

Section 1.7 Discussion

1. Why is it helpful to write regression models in matrix form rather than only scalar notation?
2. What does the covariance matrix tell us that separate variances do not?
3. What is the interpretation of the column space of $\mathbf{X}$?
4. In what sense is regression a projection problem?

Section 1.8 Practice Problems

Conceptual

1. Explain the difference between a random variable and a random vector.
2. Explain why $\mathrm{Var}(\mathbf{Y})$ must be a symmetric matrix.
3. Give an example of a statistical model outside regression.

Computational

Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix}$$

with

$$\mathbb{E}[\mathbf{Y}] = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \qquad \mathrm{Var}(\mathbf{Y}) = \begin{pmatrix} 4 & 1 \\ 1 & 9 \end{pmatrix}.$$

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$$

Compute:

1. $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$
2. $\mathrm{Var}(\mathbf{AY})$

Regression setup

For the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

write out the matrices $\mathbf{Y}$, $\mathbf{X}$, $\boldsymbol{\beta}$, and $\boldsymbol{\varepsilon}$ for $n = 5$ observations.

Section 1.9 Take home Practice

Complete the following:

- review matrix multiplication and transpose rules;
- derive $\mathbb{E}[\mathbf{AY} + \mathbf{b}]$ from first principles;
- derive $\mathrm{Var}(\mathbf{AY})$ using the definition of covariance;
- write the simple linear regression model in matrix form for a dataset of your choice;
- fit a simple regression in R and report:
- the estimated coefficients,
- fitted values,
- residuals,
- and a scatterplot with the fitted line.

Summary / Recap

- Write the model as $Y = X\beta + \varepsilon$: Y has n entries, X is $n \times p$, and β has p entries. Count columns, including the intercept.
- For a fixed design, $E(Y) = X\beta$ specifies the mean; $\mathrm{Var}(Y)$ specifies the uncertainty around that mean.
- Linear transformations obey $E(AY + b) = AE(Y) + b$ and $\mathrm{Var}(AY + b) = A\,\mathrm{Var}(Y)A^\top$.
- The column space of X contains the possible fitted mean vectors. Full column rank makes the coefficient representation unique.

Section 2

Least Squares Estimation

In this chapter, we study one of the fundamental ideas in linear models: **least squares estimation**.

Recall that the linear model is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

The unknown parameter vector $\boldsymbol{\beta}$ determines the mean structure of the response. Given observed data, our first goal is to estimate $\boldsymbol{\beta}$.

The central idea of least squares is simple:

> Choose the value of $\boldsymbol{\beta}$ for which the fitted response $\mathbf{X}\boldsymbol{\beta}$ is as close as possible to the observed response $\mathbf{Y}$.

This optimization problem leads to the **normal equations**, the **least squares estimator**, and a useful geometric interpretation in terms of **orthogonal projection**.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - formulate least squares estimation as an optimization problem;
> - define the residual sum of squares;
> - derive the normal equations;
> - obtain the least squares estimator when $\mathbf{X}$ has full column rank;
> - explain the role of the column space $\mathcal{C}(\mathbf{X})$;
> - interpret fitted values as an orthogonal projection;
> - explain why residuals are orthogonal to the model space;
> - define and interpret the hat matrix and residual-maker matrix;
> - verify symmetry and idempotence of projection matrices;
> - explain the decomposition $\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$;
> - distinguish the uniqueness of $\hat{\boldsymbol{\beta}}$ from the uniqueness of the fitted values.

Reading

Recommended reading for this week:

- Seber and Lee:
  - Chapter 3: sections on least squares estimation

- Montgomery, Peck, and Vining:
  - introductory sections on estimation in linear regression

Recap of the Linear Model

Recall from Week 1, the linear model can be written as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where

- $\mathbf{Y}$ is an $n \times 1$ response vector;
- $\mathbf{X}$ is an $n \times p$ design matrix;
- $\boldsymbol{\beta}$ is a $p \times 1$ parameter vector;
- $\boldsymbol{\varepsilon}$ is an $n \times 1$ error vector.

The design matrix may be written as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix},$$

where $\mathbf{x}_i \in \mathbb{R}^p$ contains the predictors associated with observation i.
Thus,

$$Y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i, \qquad i = 1, \ldots, n.$$

Under the classical linear model, we often assume

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

Consequently,

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \qquad \mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n.$$

Least Squares Does Not Require a Distributional Assumption

The least squares estimator is defined through the optimization problem

$$\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .$$

No distributional assumption on $\boldsymbol{\varepsilon}$ is required to define or compute the least squares estimator.
Assumptions such as

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$$

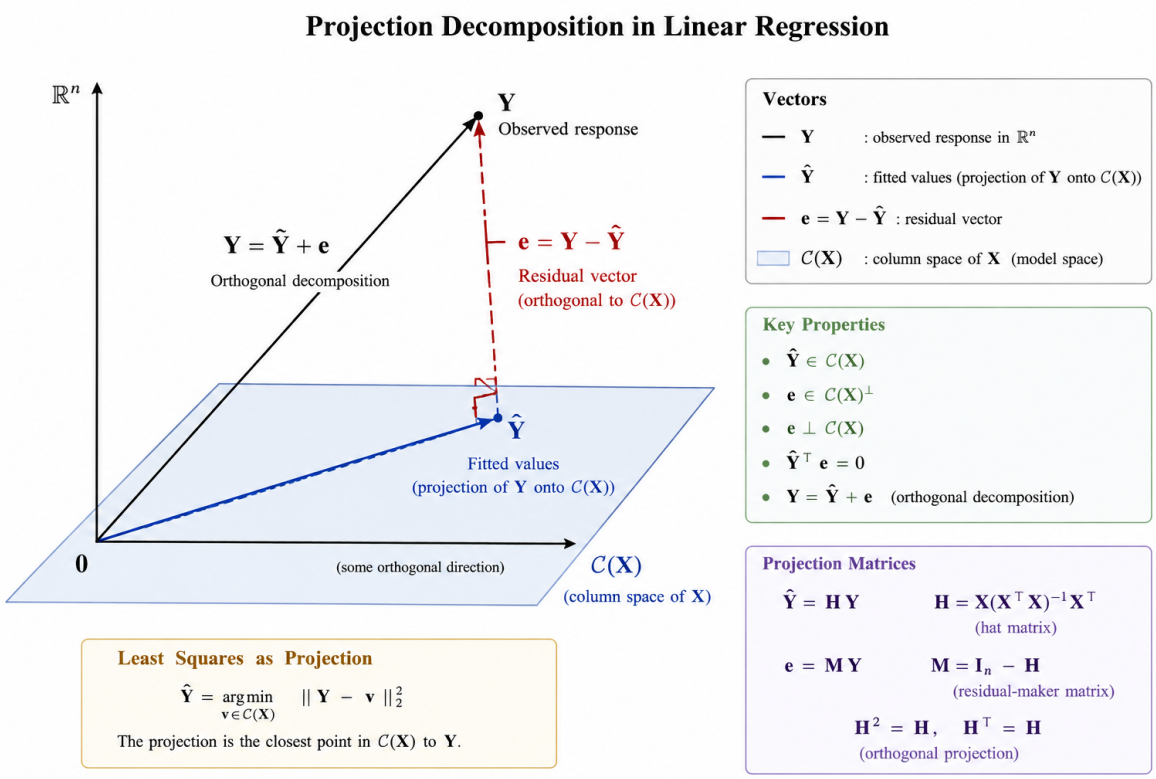


Figure 2.1: Geometric interpretation of least squares as an orthogonal projection.

are needed when we study the **statistical properties** of the estimator.
Normality will be introduced later when we study exact finite-sample inference.

Section 2.1 The Least Squares Criterion

Motivation

For any candidate value $\boldsymbol{\beta}$, the corresponding fitted vector is

$$\mathbf{X}\boldsymbol{\beta}.$$

The discrepancy between the observed response $\mathbf{Y}$ and this candidate fit is

$$\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

A natural idea is to choose $\boldsymbol{\beta}$ so that this discrepancy is as small as possible.

To measure its size, we use the squared Euclidean norm.

Residual Sum of Squares

The **residual sum of squares** associated with a candidate $\boldsymbol{\beta}$ is

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 .$$

Equivalently,

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}).$$

In scalar notation,

$$S(\boldsymbol{\beta}) = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i^\top \boldsymbol{\beta}\right)^2 .$$

This last expression explains the name **least squares**: we choose $\boldsymbol{\beta}$ to make the sum of the squared discrepancies as small as possible.

We therefore define

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta} \in \mathbb{R}^p} S(\boldsymbol{\beta}).$$

Definition 2.1 (Least Squares Estimator)
A **least squares estimator** is any value $\hat{\boldsymbol{\beta}}$ satisfying

$$S(\hat{\boldsymbol{\beta}}) \le S(\boldsymbol{\beta})$$

for every $\boldsymbol{\beta} \in \mathbb{R}^p$.

Example 2.1 (LSE for Simple Linear Regression)
Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

where

$$\mathbb{E}[\varepsilon_i] = 0, \qquad \mathrm{Var}(\varepsilon_i) = \sigma^2.$$

The fitted value at x_i is

$$\hat{Y}_i = \beta_0 + \beta_1 x_i.$$

The corresponding residual is

$$e_i = Y_i - \beta_0 - \beta_1 x_i.$$

The least squares method chooses β_0 and β_1 to minimize the **sum of squared residuals**

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1 x_i\right)^2 .$$

Derivation of the Normal Equations

To minimize $S(\beta_0, \beta_1)$, differentiate with respect to β_0 and β_1.
For β_0,

$$\frac{\partial S}{\partial \beta_0} = -2\sum_{i=1}^{n}\left(Y_i - \beta_0 - \beta_1 x_i\right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n}\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0.$$

Therefore,

$$\sum_{i=1}^{n} Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i.$$

Dividing by n,

$$\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x},$$

so

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

For β_1,

$$\frac{\partial S}{\partial \beta_1} = -2\sum_{i=1}^{n} x_i\left(Y_i - \beta_0 - \beta_1 x_i\right).$$

Setting this equal to zero gives

$$\sum_{i=1}^{n} x_i\left(Y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right) = 0.$$

Hence,

$$\sum_{i=1}^{n} x_i Y_i = \hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Substituting

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x},$$

we obtain

$$\sum_{i=1}^{n} x_i Y_i = \left(\bar{Y} - \hat{\beta}_1 \bar{x}\right)\sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2.$$

Since

$$\sum_{i=1}^{n} x_i = n\bar{x},$$

we have

$$\sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y} = \hat{\beta}_1 \left(\sum_{i=1}^{n} x_i^2 - n\bar{x}^2 \right).$$

Using

$$\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}) = \sum_{i=1}^{n} x_i Y_i - n\bar{x}\bar{Y},$$

and

$$\sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2,$$

we obtain

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}.$$

Therefore,

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

Alternative Notation
It is common to define

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2$$

and

$$S_{xy} = \sum_{i=1}^{n} (x_i - \bar{x})(Y_i - \bar{Y}).$$

Then

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

and

$$\hat{\beta}_0 = \bar{Y} - \frac{S_{xy}}{S_{xx}} \bar{x}.$$

Thus, the fitted regression line is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i.$$

Section 2.2 Derivation of the Least Squares Estimator in a matrix form

Expansion of the Criterion Starting from

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

we expand:

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - \mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} - \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Because

$$\mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta}$$

is a scalar,

$$\mathbf{Y}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y}.$$

Therefore,

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Derivation of the Normal Equations

Differentiate $S(\boldsymbol{\beta})$ with respect to $\boldsymbol{\beta}$:

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

At a minimum, the gradient is zero, so

$$-2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{0}.$$

Thus,

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

These are called the **normal equations**.

> **Why Are They Called the Normal Equations?**
>
> The normal equations can be written as
>
> $$\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}.$$
>
> Later we will define
>
> $$\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Therefore,

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

This means that $\mathbf{e}$ is perpendicular, or **normal**, to the column space of $\mathbf{X}$.
Thus, the word *normal* is being used in its geometric sense.

Why is the Stationary Point a Minimum?

The Hessian of the least squares criterion is

$$\nabla^2 S(\boldsymbol{\beta}) = 2\mathbf{X}^\top \mathbf{X}.$$

For any $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} = (\mathbf{X}\mathbf{a})^\top (\mathbf{X}\mathbf{a}) = \|\mathbf{X}\mathbf{a}\|_2^2 \geq 0.$$

Therefore, $\mathbf{X}^\top \mathbf{X}$ is positive semidefinite, and $S(\boldsymbol{\beta})$ is a convex function.

If $\mathbf{X}$ has full column rank, then for every $\mathbf{a} \neq \mathbf{0}$,

$$\mathbf{X}\mathbf{a} \neq \mathbf{0},$$

and hence

$$\mathbf{a}^\top \mathbf{X}^\top \mathbf{X}\mathbf{a} > 0.$$

Thus, $\mathbf{X}^\top \mathbf{X}$ is positive definite and the least squares minimizer is unique.

Theorem 1: Least Squares Estimator

A vector $\hat{\boldsymbol{\beta}}$ minimizes

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

if and only if it satisfies the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

If $\mathbf{X}$ has full column rank,

$$\operatorname{rank}(\mathbf{X}) = p,$$

then $\mathbf{X}^\top \mathbf{X}$ is invertible and the least squares estimator is unique:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

This is commonly called the **ordinary least squares estimator**, or **OLS estimator**.

When Is the Closed-Form Formula Valid?

The expression

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

exists if and only if $\mathbf{X}$ has full column rank:

$$\text{rank}(\mathbf{X}) = p.$$

This means that the columns of $\mathbf{X}$ are linearly independent.
If they are linearly dependent, then $\mathbf{X}^\top \mathbf{X}$ is singular and the inverse does not exist.
We will return to this issue later.

Example 2.2 (LSE for Simple Linear Regression continue)
As shown in Example 2.1, we know the expression of the LSE $\hat{\boldsymbol{\beta}}$. Here, we want to see its connection to the matrix Form.
For simple linear regression,

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

The general least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

For simple linear regression,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix},$$

and

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum_{i=1}^n Y_i \\ \sum_{i=1}^n x_i Y_i \end{pmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

which gives exactly

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(Y_i - \bar{Y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

and

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}.$$

Hence,

> The familiar simple linear regression formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ are special cases of the general matrix LSE
>
> $$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Section 2.3 Geometric Interpretation of Least Squares

The algebraic derivation gives us the estimator. However, the geometry explains **why** least squares has its important properties.

> **Recap: Column Space of the Design Matrix**
>
> The column space of $\mathbf{X}$ is
>
> $$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\boldsymbol{\beta} : \boldsymbol{\beta} \in \mathbb{R}^p\}.$$
>
> Thus, $\mathcal{C}(\mathbf{X})$ is the set of all response vectors that can be represented exactly by the linear model.
> If $\mathbf{X}$ has full column rank p, then
>
> $$\dim\{\mathcal{C}(\mathbf{X})\} = p.$$
>
> Because $\mathbf{X}$ has n rows,
>
> $$\mathcal{C}(\mathbf{X}) \subseteq \mathbb{R}^n.$$

2.3.1 Least Squares as Projection

The fitted value vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Because

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}),$$

least squares chooses the vector in $\mathcal{C}(\mathbf{X})$ that is closest to the observed vector $\mathbf{Y}$.

That is,

$$\hat{\mathbf{Y}} = \arg\min_{\mathbf{v} \in \mathcal{C}(\mathbf{X})} \|\mathbf{Y} - \mathbf{v}\|_2^2.$$

Therefore, $\hat{\mathbf{Y}}$ is the **orthogonal projection** of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$.

So we can say that the fitted vector $\hat{\mathbf{Y}}$ is constrained to live in the model space generated by the predictors. Least squares finds the point in that space closest to the observed data vector $\mathbf{Y}$.

Definition 2.2 (Orthogonality)
Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are said to be **orthogonal** if their inner product is zero:

$$\mathbf{u}^\top \mathbf{v} = 0.$$

We write

$$\mathbf{u} \perp \mathbf{v}.$$

Geometrically, two nonzero vectors are orthogonal if the angle between them is 90°.

Definition 2.3 (Residual Vector)
After obtaining the LSE, define the **residual vector**

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The individual residuals are

$$e_i = Y_i - \hat{Y}_i, \qquad i = 1, \dots, n.$$

Geometrically, $\mathbf{e}$ is the component of $\mathbf{Y}$ that is orthogonal to the model space $\mathcal{C}(\mathbf{X})$.

Proposition 2.1 (Orthogonality of the Residuals)
Recall that the LSE satisfies the normal equations $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$. If we rearrange, we have $\mathbf{X}^\top \left(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\right) = \mathbf{0}$.
Since the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}},$$

we obtain the fundamental least squares property

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

To understand what this means, write the design matrix as

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}^{(1)} & \mathbf{x}^{(2)} & \cdots & \mathbf{x}^{(p)} \end{pmatrix},$$

where $\mathbf{x}^{(j)}$ is the jth column of $\mathbf{X}$.
Then

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix} (\mathbf{x}^{(1)})^\top \mathbf{e} \\ (\mathbf{x}^{(2)})^\top \mathbf{e} \\ \vdots \\ (\mathbf{x}^{(p)})^\top \mathbf{e} \end{pmatrix} = \mathbf{0}.$$

Therefore,

$$(\mathbf{x}^{(j)})^\top \mathbf{e} = 0, \qquad j = 1, \dots, p.$$

Thus, the residual vector is orthogonal to every column of $\mathbf{X}$.
Because the column space $\mathcal{C}(\mathbf{X})$ consists of all linear combinations of the columns of $\mathbf{X}$, it follows that

$$\mathbf{e} \perp \mathcal{C}(\mathbf{X}).$$

Intuition. The columns of $\mathbf{X}$ represent all directions that the linear model is allowed to use. After least squares has chosen the best fitted value, the remaining error $\mathbf{e}$ has no component left in any of these directions. Otherwise, we could move the fitted value within $\mathcal{C}(\mathbf{X})$ and reduce the residual sum of squares further.

Proposition 2.2 (Fundamental Orthogonal Decomposition)
The fitted response vector is

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Therefore,

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X}).$$

In other words, $\hat{\mathbf{Y}}$ must lie in the space generated by the columns of $\mathbf{X}$.
Since

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

we can write

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.$$

The two components satisfy

$$\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$$

and

$$\mathbf{e} \in \mathcal{C}(\mathbf{X})^\perp.$$

Hence,

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0.$$

Geometrically, least squares projects $\mathbf{Y}$ onto the column space of $\mathbf{X}$:

$$\mathbf{Y} = \underbrace{\hat{\mathbf{Y}}}_{\text{part explained by the model}} + \underbrace{\mathbf{e}}_{\text{part orthogonal to the model space}}.$$

Because the two components are orthogonal, the Pythagorean theorem gives

$$\|\mathbf{Y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{e}\|^2.$$

This geometric decomposition is the foundation of least squares regression and will later lead naturally to the decomposition of sums of squares in ANOVA.

Exercise 2.1 (Computational Problems)
Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

1. Compute $\mathbf{X}^\top \mathbf{X}$.
2. Compute $\mathbf{X}^\top \mathbf{Y}$.
3. Find $\hat{\boldsymbol{\beta}}$.
4. Compute $\hat{\mathbf{Y}}$.
5. Compute the residual vector $\mathbf{e}$.
6. Verify that $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$.

Solution 2.1 (Solution for Exercise 2.1)
Hand calculation
We have

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 3 & 3 \\ 3 & 5 \end{pmatrix}.$$

Also,

$$\mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}.$$

Therefore,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 7/6 \\ 1/2 \end{pmatrix}.$$

The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}.$$

Thus, the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

R calculation

```r
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

# Define X and Y
X <- matrix(
  c(
    1, 0,
    1, 1,
    1, 2
  ),
  nrow = 3,
  byrow = TRUE
)

Y <- matrix(
  c(1, 2, 2),
  ncol = 1
)

# 1. Compute X'X
XtX <- t(X) %*% X
XtX
```

```
     [,1] [,2]
[1,]    3    3
[2,]    3    5
```

```r
# 2. Compute X'Y
XtY <- t(X) %*% Y
XtY
```

```
     [,1]
[1,]    5
[2,]    6
```

```r
# 3. Compute beta_hat
beta_hat <- solve(XtX) %*% XtY
beta_hat
```

```
         [,1]
[1,] 1.166667
[2,] 0.500000
```

```r
# 4. Compute fitted values
Y_hat <- X %*% beta_hat
Y_hat
```

```
         [,1]
[1,] 1.166667
[2,] 1.666667
[3,] 2.166667
```

```r
# 5. Compute residuals
e <- Y - Y_hat
e
```

```
           [,1]
[1,] -0.1666667
[2,]  0.3333333
[3,] -0.1666667
```

```r
# Optional checks
t(X) %*% e
```

```
              [,1]
[1,] -8.881784e-16
[2,] -8.881784e-16
```

```r
t(Y_hat) %*% e
```

```
              [,1]
[1,] -1.474129e-15
```

Exercise 2.2
Explain in words what the column space

$$\mathcal{C}(\mathbf{X})$$

represents in a linear regression model.

Solution 2.2
The column space of $\mathbf{X}$ is the set of all response patterns that can be represented by the linear model.
More intuitively,
$\mathcal{C}(\mathbf{X})$ describes everything the model is capable of fitting. Anything outside this space cannot be explained by the predictors in $\mathbf{X}$.

2.3.2 Consequence when an Intercept Is Included

If the model includes an intercept, then one column of $\mathbf{X}$ is

$$\mathbf{1}_n = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

Since the residual vector is orthogonal to every column of $\mathbf{X}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0.$$

Therefore,

$$\sum_{i=1}^{n} e_i = 0.$$

Since

$$e_i = Y_i - \hat{Y}_i,$$

we also have

$$\sum_{i=1}^{n} Y_i = \sum_{i=1}^{n} \hat{Y}_i.$$

Thus,

$$\bar{\hat{Y}} = \bar{Y}.$$

> **Intuition for the Mean of the Fitted Values**
>
> The intercept allows the fitted model to shift up or down. At the least squares solution, there can be no overall tendency for the fitted values to be systematically too high or too low. Therefore, the residuals sum to zero and the fitted values have the same mean as the observed responses.

Section 2.4 The Hat Matrix

In the full-column-rank case,

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

Substituting

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y},$$

we obtain

$$\hat{\mathbf{Y}} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

Definition 1: The Hat Matrix

Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top.$$

Then

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

The matrix $\mathbf{H}$ is called the **hat matrix** because it puts the "hat" on $\mathbf{Y}$:

$$\mathbf{Y} \quad \longrightarrow \quad \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Proposition 2.3 (Properties of the Hat Matrix)

When $\mathbf{X}$ has full column rank, the hat matrix satisfies:

i) Symmetry

$$\mathbf{H}^\top = \mathbf{H}.$$

ii) Idempotence

$$\mathbf{H}^2 = \mathbf{H}.$$

iii) Projection of the design matrix

$$\mathbf{H}\mathbf{X} = \mathbf{X}.$$

iv) Rank

$$\operatorname{rank}(\mathbf{H}) = p.$$

v) Trace

$$\operatorname{tr}(\mathbf{H}) = p.$$

A symmetric and idempotent matrix represents an orthogonal projection. Therefore, $\mathbf{H}$ is the orthogonal projection matrix onto

$$\mathcal{C}(\mathbf{X}).$$

Why it is symmetric

We have

$$\begin{aligned}\mathbf{H}^\top &= \left[\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\right]^\top \\ &= \mathbf{X}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\right]^\top\mathbf{X}^\top.\end{aligned}$$

Since $\mathbf{X}^\top\mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{H}^\top = \mathbf{H}.$$

Why it is idempotent

We have

$$\begin{aligned}\mathbf{H}^2 &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top \\ &= \mathbf{H}.\end{aligned}$$

Thus,

$$\mathbf{H}^2 = \mathbf{H}.$$

What Does Idempotence Mean Geometrically?

If we project $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ once, we obtain

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}.$$

Projecting the fitted vector again does nothing:

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \mathbf{H}\mathbf{Y} = \hat{\mathbf{Y}}.$$

Once a vector is already inside the model space, projecting it onto the same space leaves it unchanged.

Notice that, $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ and $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$, we have

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}.$$

Definition 2.4 (Residual-Maker Matrix)
Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Then

$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

The matrix $\mathbf{M}$ is called the **residual-maker matrix**.
It removes the component of $\mathbf{Y}$ lying in $\mathcal{C}(\mathbf{X})$.

Proposition 2.4 (Properties of the Residual-Maker Matrix)
Since

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H},$$

we obtain

$$\mathbf{M}^\top = \mathbf{M}$$

and

$$\mathbf{M}^2 = \mathbf{M}.$$

Thus, $\mathbf{M}$ is also an orthogonal projection matrix.
While $\mathbf{H}$ projects onto

$$\mathcal{C}(\mathbf{X}),$$

$\mathbf{M}$ projects onto

$$\mathcal{C}(\mathbf{X})^\perp.$$

In addition,

$$\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}.$$

Under full column rank,

$$\text{rank}(\mathbf{M}) = n - p$$

and

$$\text{tr}(\mathbf{M}) = n - p.$$

Two Complementary Projection Matrices

The matrices $\mathbf{H}$ and $\mathbf{M}$ split $\mathbb{R}^n$ into two orthogonal components:

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

For the observed response,

$$\mathbf{Y} = \underbrace{\mathbf{H}\mathbf{Y}}_{\hat{\mathbf{Y}}} + \underbrace{\mathbf{M}\mathbf{Y}}_{\mathbf{e}}.$$

Thus,

$$\mathbf{H} + \mathbf{M} = \mathbf{I}_n.$$

Section 2.5 The common name to refer to what we have done – Ordinary Least Squares (OLS)

Now that we have seen both the algebraic and geometric views of least squares, we can explain the name **ordinary least squares**, or **OLS**.

- **Squares:** we measure the discrepancy using squared residuals,

$$\sum_{i=1}^{n} e_i^2 = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2.$$

- **Least:** we choose $\boldsymbol{\beta}$ to make this quantity as small as possible.
- **Ordinary:** we use the usual unweighted sum of squared residuals, so each observation enters the criterion in the same way.

Thus,

$$\text{OLS} = \text{minimize the ordinary sum of squared residuals.}$$

Geometrically, OLS finds the point in $\mathcal{C}(\mathbf{X})$ that is closest to $\mathbf{Y}$ using ordinary Euclidean distance.

Later, we will consider extensions such as weighted least squares and generalized least squares.

Exercise 2.3 (Quick Check)
What do **ordinary**, **least**, and **squares** each mean in OLS?

Section 2.6 Sum of Squares Decomposition

2.6.1 Uncentered Geometric Decomposition

Since

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

and

$$\hat{\mathbf{Y}}^\top \mathbf{e} = 0,$$

we have

$$\begin{aligned}\mathbf{Y}^\top \mathbf{Y} &= (\hat{\mathbf{Y}} + \mathbf{e})^\top (\hat{\mathbf{Y}} + \mathbf{e}) \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + 2\hat{\mathbf{Y}}^\top \mathbf{e} + \mathbf{e}^\top \mathbf{e} \\ &= \hat{\mathbf{Y}}^\top \hat{\mathbf{Y}} + \mathbf{e}^\top \mathbf{e}.\end{aligned}$$

Therefore,

$$\|\mathbf{Y}\|_2^2 = \|\hat{\mathbf{Y}}\|_2^2 + \|\mathbf{e}\|_2^2.$$

This is a direct application of the Pythagorean theorem.

2.6.2 Centered Sum of Squares Decomposition

When the regression model contains an intercept,

$$\bar{\hat{Y}} = \bar{Y}.$$

Therefore,

$$Y_i - \bar{Y} = (\hat{Y}_i - \bar{Y}) + (Y_i - \hat{Y}_i).$$

The two components are orthogonal, giving

$$\sum_{i=1}^n (Y_i - \bar{Y})^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 + \sum_{i=1}^n (Y_i - \hat{Y}_i)^2.$$

These quantities will later be called

- **total sum of squares**
$$\text{SST} = \sum_{i=1}^{n}(Y_i - \bar{Y})^2;$$
- **regression sum of squares**
$$\text{SSR} = \sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2;$$
- **error sum of squares**
$$\text{SSE} = \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2.$$

Thus,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

! Centered vs. Uncentered Decomposition

The geometric identity

$$\mathbf{Y}^\top\mathbf{Y} = \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}$$

is an **uncentered** decomposition.
The familiar regression decomposition

$$\text{SST} = \text{SSR} + \text{SSE}$$

is a **centered** decomposition and relies on the model containing an intercept.
These two decompositions should not be confused.

2.6.3 Example: Simple Linear Regression

Example 1: Hand Calculation

Consider the simple regression dataset

x_i	0	1	2	3
y_i	1	3	3	5

.

We fit the model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i.$$

The response vector is

$$\mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix},$$

and the design matrix is

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

Step 1: Compute $\mathbf{X}^\top\mathbf{X}$
We have

$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}.$$

Step 2: Compute $\mathbf{X}^\top\mathbf{Y}$
We obtain

$$\mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}.$$

Step 3: Compute $\hat{\boldsymbol{\beta}}$
Since

$$(\mathbf{X}^\top\mathbf{X})^{-1} = \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix},$$

the least squares estimator is

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \\ &= \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}\begin{pmatrix} 12 \\ 24 \end{pmatrix} \\ &= \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix}. \end{aligned}$$

Thus,

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2,$$

and the fitted regression line is

$$\hat{Y} = 1.2 + 1.2x.$$

Step 4: Compute the Fitted Values
The fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Step 5: Compute the Residuals

The residual vector is

$$\begin{aligned}\mathbf{e} &= \mathbf{Y} - \hat{\mathbf{Y}} \\ &= \begin{pmatrix}1\\3\\3\\5\end{pmatrix} - \begin{pmatrix}1.2\\2.4\\3.6\\4.8\end{pmatrix} \\ &= \begin{pmatrix}-0.2\\0.6\\-0.6\\0.2\end{pmatrix}.\end{aligned}$$

Notice that

$$\sum_{i=1}^{4} e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0.$$

This occurs because the model contains an intercept.

Step 6: Verify Orthogonality

We compute

$$\mathbf{X}^\top \mathbf{e} = \begin{pmatrix}1 & 1 & 1 & 1\\0 & 1 & 2 & 3\end{pmatrix} \begin{pmatrix}-0.2\\0.6\\-0.6\\0.2\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}.$$

Thus,

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

We can also verify that
$\hat{\mathbf{Y}}^\top \mathbf{e} = 0$.

Step 7: Compute the Hat Matrix

The hat matrix is

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top.$$

For this example,

$$\mathbf{H} = \begin{pmatrix}0.7 & 0.4 & 0.1 & -0.2\\0.4 & 0.3 & 0.2 & 0.1\\0.1 & 0.2 & 0.3 & 0.4\\-0.2 & 0.1 & 0.4 & 0.7\end{pmatrix}.$$

Multiplying by $\mathbf{Y}$ gives

$$\mathbf{HY} = \begin{pmatrix}1.2\\2.4\\3.6\\4.8\end{pmatrix} = \hat{\mathbf{Y}}.$$

Thus, the hat matrix projects the observed response onto the model space.

Example 2: R Demonstration

We can verify the previous calculations in R.

```
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

df <- data.frame(
  x = x,
  y = y
)

fit <- lm(y ~ x, data = df)

coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

The estimated coefficients are

$$\hat{\beta}_0 = 1.2, \qquad \hat{\beta}_1 = 1.2.$$

We can also compute the estimator directly using matrix operations.

```
X <- model.matrix(fit)
X
```

```
  (Intercept) x
1           1 0
2           1 1
3           1 2
4           1 3
attr(,"assign")
[1] 0 1
```

```
Y <- matrix(y, ncol = 1)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

Compute the fitted values and residuals:

```
y_hat <- X %*% beta_hat
e <- Y - y_hat

y_hat
```

```
  [,1]
1  1.2
2  2.4
3  3.6
4  4.8
```

```
e
```

```
  [,1]
1 -0.2
2  0.6
3 -0.6
4  0.2
```

Check that the residuals sum to zero:

```
sum(e)
```

```
[1] -3.774758e-15
```

Compute the hat matrix:

```
H <-
  X %*%
  solve(t(X) %*% X) %*%
  t(X)

round(H, 4)
```

```
     1   2   3    4
1  0.7 0.4 0.1 -0.2
2  0.4 0.3 0.2  0.1
3  0.1 0.2 0.3  0.4
4 -0.2 0.1 0.4  0.7
```

Verify symmetry:

```
round(H - t(H), 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify idempotence:

```
round(H %*% H - H, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify the normal equations geometrically:

```
round(t(X) %*% e, 10)
```

```
            [,1]
(Intercept)    0
x              0
```

Verify that fitted values and residuals are orthogonal:

```
round(t(y_hat) %*% e, 10)
```

```
     [,1]
[1,]    0
```

Construct the residual-maker matrix:

```
M <- diag(nrow(X)) - H

round(M, 4)
```

```
     1    2    3    4
1  0.3 -0.4 -0.1  0.2
2 -0.4  0.7 -0.2 -0.1
3 -0.1 -0.2  0.7 -0.4
4  0.2 -0.1 -0.4  0.3
```

```
round(M %*% M - M, 10)
```

```
  1 2 3 4
1 0 0 0 0
2 0 0 0 0
3 0 0 0 0
4 0 0 0 0
```

Verify that

$$\mathbf{e} = \mathbf{MY}.$$

```
round(M %*% Y - e, 10)
```

```
  [,1]
1    0
2    0
3    0
4    0
```

Finally, plot the fitted regression line:

```
ggplot(df, aes(x, y)) +
  geom_point(size = 2.5) +
  geom_abline(intercept = coef(fit)[1], slope = coef(fit)[2],
              colour = "steelblue4", linewidth = 0.8) +
  labs(x = "x", y = "y")
```

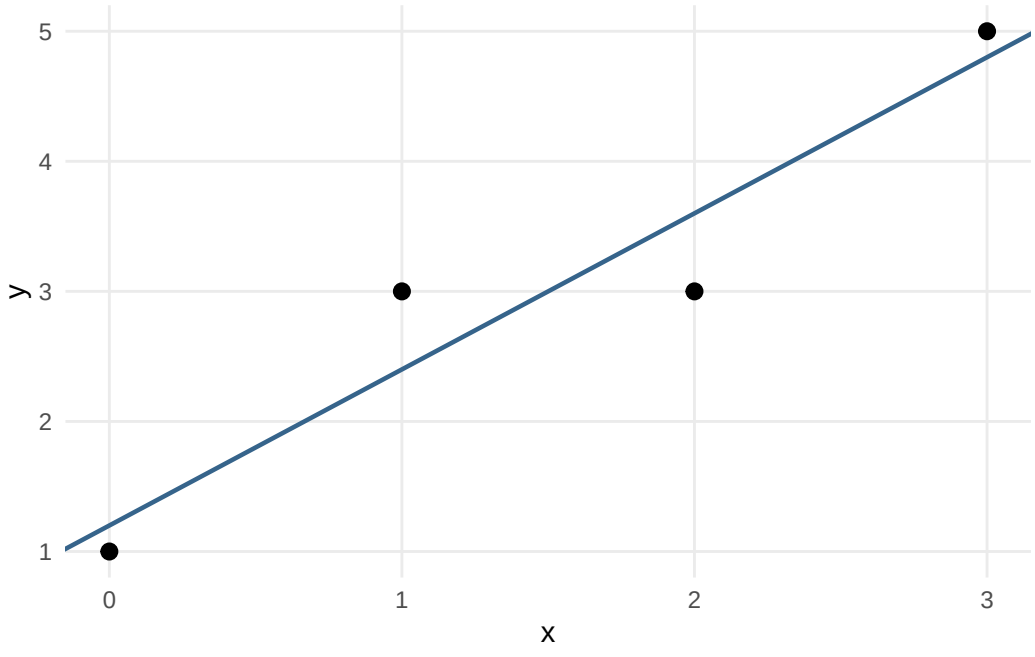


Figure 2.2: The least-squares line for the four-observation example.

Section 2.7 What Happens Without Full Column Rank?

Suppose

$$\text{rank}(\mathbf{X}) < p.$$

Then the columns of $\mathbf{X}$ are linearly dependent, and

$$\mathbf{X}^\top \mathbf{X}$$

is singular.

Therefore,

$$(\mathbf{X}^\top \mathbf{X})^{-1}$$

does not exist.

! A Subtle but Important Point

When $\mathbf{X}$ does not have full column rank,

- a least squares solution still exists;
- $\hat{\boldsymbol{\beta}}$ may not be unique;
- the fitted vector $\hat{\mathbf{Y}}$ is still unique;
- the residual vector $\mathbf{e}$ is still unique.

Thus,

$$\hat{\boldsymbol{\beta}} \text{ may not be unique, but } \hat{\mathbf{Y}} \text{ is unique.}$$

The reason is geometric: even if several coefficient vectors represent the same point in $\mathcal{C}(\mathbf{X})$, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.
We will discuss rank deficiency, generalized inverses, and estimability in more detail later.

Section 2.8 Interpretation of the Geometry

The observed vector

$$\mathbf{Y} \in \mathbb{R}^n.$$

When $\mathbf{X}$ has full column rank p, the model space

$$\mathcal{C}(\mathbf{X})$$

is a p-dimensional subspace of $\mathbb{R}^n$.

Least squares finds the point in this subspace that is closest to $\mathbf{Y}$.

The central relationships are

$$\min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2$$

$$\Downarrow$$

$$\mathbf{X}^\top(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$$

$$\Downarrow$$

$$\mathbf{e} \perp \mathcal{C}(\mathbf{X})$$

$$\Downarrow$$

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

$$\Downarrow$$

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}.$$

Thus,

- fitted values are orthogonal projections;
- residuals are orthogonal to the model space;
- the hat matrix is a projection matrix;
- the residual-maker matrix projects onto the orthogonal complement;
- sum-of-squares decompositions are consequences of the Pythagorean theorem.

Section 2.9 Looking Ahead: Statistical Properties of OLS

> **Looking Ahead**
>
> Everything we have done so far is primarily **algebraic and geometric**.
> We defined $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ by solving an optimization problem.
> However, under the statistical model, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, the response $\mathbf{Y}$ is random. Therefore, $\hat{\boldsymbol{\beta}}$ is also a random vector.
> In Chapter 3 — Distribution Theory of OLS and Inference, we will study quantities such as
>
> $$\mathbb{E}(\hat{\boldsymbol{\beta}}), \quad \text{and} \quad \mathrm{Var}(\hat{\boldsymbol{\beta}}),$$
>
> followed by estimation of σ^2, sampling distributions, confidence intervals, and hypothesis testing.

Section 2.10 Practice at Home

1. Explain why

   $$\mathbf{X}^\top\mathbf{e} = \mathbf{0}$$

   means that the residual vector is orthogonal to the column space $\mathcal{C}(\mathbf{X})$.

2. Explain why

   $$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

   is an orthogonal projection, and why

   $$\mathbf{H}^2 = \mathbf{H}.$$

3. If the model contains an intercept, explain why

   $$\sum_{i=1}^{n} e_i = 0$$

   and hence

   $$\bar{\hat{Y}} = \bar{Y}.$$

4. Let

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

   Compute

$$\hat{\boldsymbol{\beta}}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e},$$

   and verify that

$$\mathbf{X}^\top \mathbf{e} = \mathbf{0}.$$

5. Show that the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$$

   is symmetric and idempotent.

Summary / Recap

- Least squares minimizes $\|Y - X\beta\|^2$ and satisfies the normal equations $X^\top X\hat{\beta} = X^\top Y$.
- With full column rank, $\hat{\beta} = (X^\top X)^{-1}X^\top Y$. Without it, coefficients may be nonunique, but the fitted vector is unique.
- The fitted vector $\hat{Y} = HY$ is the orthogonal projection onto $\mathcal{C}(X)$; the residual $e = (I - H)Y$ satisfies $X^\top e = 0$.
- H and $I - H$ are symmetric, idempotent projection matrices. With an intercept, residuals sum to zero and fitted responses have the observed response mean.
- These are algebraic properties of the fit. Statistical claims about bias, variance and tests require assumptions about the errors.

Next, Chapter 3 supplies those assumptions and the sampling distributions used for inference.

Side Fact: Matrix Calculus Facts Used in This Chapter

> **Theorem 2: Matrix Calculus Facts**
>
> Let $\boldsymbol{\beta}$ be a column vector.
> For a constant vector $\mathbf{a}$,
>
> $$\frac{\partial}{\partial \boldsymbol{\beta}} (\mathbf{a}^\top \boldsymbol{\beta}) = \mathbf{a}.$$
>
> For a constant matrix $\mathbf{A}$,

$$\frac{\partial}{\partial \boldsymbol{\beta}} (\boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta}) = (\mathbf{A} + \mathbf{A}^\top)\boldsymbol{\beta}.$$

If $\mathbf{A}$ is symmetric,

$$\mathbf{A}^\top = \mathbf{A},$$

and therefore

$$\frac{\partial}{\partial \boldsymbol{\beta}} (\boldsymbol{\beta}^\top \mathbf{A} \boldsymbol{\beta}) = 2\mathbf{A}\boldsymbol{\beta}.$$

These identities justify the derivative used in the least squares criterion.

Section 3

Distribution Theory of OLS and Inference

In Chapter 2 — Least Squares Estimation, we studied least squares primarily from an **algebraic and geometric** point of view.

We obtained

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and showed that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$$

and

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

We now move from **estimation** to **statistical inference**.

Since $\mathbf{Y}$ is random, quantities such as

$$\hat{\boldsymbol{\beta}}, \qquad \hat{\mathbf{Y}}, \qquad \mathbf{e}, \qquad \text{SSE}$$

are also random.

Our goal is to understand their sampling distributions and use them to quantify uncertainty.

> Learning Objectives
>
> By the end of this chapter, students should be able to:
>
> - derive the expectation and variance of the OLS estimator;
> - state the normal linear model;
> - derive the distribution of the OLS estimator under normality;
> - explain the distribution of the residual sum of squares;
> - explain why the residual degrees of freedom are $n - p$;

- estimate σ^2 using the mean squared error;
- explain why $\hat{\boldsymbol{\beta}}$ and SSE are independent;
- construct confidence intervals for regression coefficients;
- perform hypothesis tests for regression coefficients;
- perform inference for linear combinations of regression coefficients;
- distinguish inference for a mean response from prediction of a future observation;
- interpret standard regression output from R.

Reading

Recommended reading for this chapter:

- Seber and Lee:
  - sections on distribution theory of least squares estimators;
  - estimation of the error variance;
  - inference for regression coefficients.
- Montgomery, Peck, and Vining:
  - sections on confidence intervals;
  - hypothesis testing;
  - estimation and prediction in linear regression.

Recap from Least Squares

From Chapter 2 — Least Squares Estimation, when $\mathbf{X}$ has full column rank,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

We also have

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \qquad \mathbf{e} = \mathbf{M}\mathbf{Y},$$

with

$$\mathbf{H} + \mathbf{M} = \mathbf{I}_n$$

and

$$\mathbf{H}\mathbf{M} = \mathbf{M}\mathbf{H} = \mathbf{0}.$$

Geometrically,

$$\mathbf{Y} = \hat{\mathbf{Y}} + \mathbf{e}$$

is an orthogonal decomposition.
In this chapter, we will see that this geometric decomposition also provides the foundation for regression inference.

> **!** Throughout This Chapter
>
> We treat the design matrix $\mathbf{X}$ as fixed.
> If the predictors are random, the results can instead be interpreted conditionally on the observed design matrix $\mathbf{X}$.

Section 3.1 Mean and Variance of the OLS Estimator

Before introducing normality, we can already study the expectation and variance of the OLS estimator.

Assume

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n.$$

Recall that

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}.$$

Substituting this into the OLS estimator gives

$$\begin{aligned}
\hat{\boldsymbol{\beta}} &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\
&= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\
&= \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.
\end{aligned}$$

Thus,

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}.$$

This representation will be useful throughout the chapter.

> **Proposition 3.1** (Mean and Variance of OLS)
> Under
>
> $$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$
>
> and
>
> $$\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

the OLS estimator satisfies

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

3.1.1 Why Is OLS Unbiased?

Using

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon},$$

we obtain

$$\begin{aligned}\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) &= \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbb{E}\left(\boldsymbol{\varepsilon}\right)\\ &= \boldsymbol{\beta}.\end{aligned}$$

Thus,

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}.$$

OLS is therefore an **unbiased estimator** of $\boldsymbol{\beta}$.

Intuitively, if we repeated the experiment many times using the same design matrix, the OLS estimates would fluctuate around the true parameter vector.

3.1.2 Variance of OLS

Similarly,

$$\begin{aligned}\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) &= \mathrm{Var}\left[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}\right]\\ &= (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\,\mathrm{Var}(\boldsymbol{\varepsilon})\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\\ &= \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.\end{aligned}$$

! Normality Is Not Needed Yet

The results

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\text{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}$$

do **not** require normally distributed errors.
They follow from the first two moment assumptions

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n.$$

Normality becomes important when we want exact finite sample distributions and inference.

Exercise 3.1 (Which Assumptions Are Needed?)
Suppose

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$$

and

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n.$$

Do we need to assume that the errors are normally distributed to conclude that

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}?$$

Solution 3.1
No.
Normality is **not** required for unbiasedness.
The result follows from

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\boldsymbol{\varepsilon}$$

and

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}.$$

Normality will be needed later for exact finite sample inference.

Section 3.2 The Normal Linear Model

To obtain exact sampling distributions, we now strengthen the assumptions.

Definition 3.1 (Normal Linear Model)
The **normal linear model** assumes

$$\mathbf{Y} \sim N_n\left(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n\right).$$

Equivalently,

$$\boldsymbol{\varepsilon} \sim N_n\left(\mathbf{0}, \sigma^2\mathbf{I}_n\right).$$

Under this assumption:

- each response has a normal distribution;
- the errors have common variance σ^2;
- the errors are independent;
- exact finite sample distributions can be obtained for many regression statistics.

i Why Does Normality Matter?

Least squares itself does not require normality.
Normality is introduced because it gives us exact distributions for

$$\hat{\boldsymbol{\beta}}, \qquad \text{SSE}, \qquad T, \qquad F.$$

These distributions allow us to construct exact finite sample confidence intervals and hypothesis tests.

Exercise 3.2 (What Changes When We Add Normality?)
Suppose the errors satisfy

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \qquad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n,$$

but are not normally distributed.
Which of the following can still hold?

1. OLS can still be computed.
2. OLS can still be unbiased.
3. We can still have

$$\text{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

4. Exact finite sample t and F distributions automatically hold.

Solution 3.2
Statements 1, 2, and 3 can still hold.
Statement 4 does not generally hold.

Normality is not required to define OLS or derive its mean and variance. Its main role here is to provide exact finite sample distributions for inference.

Section 3.3 Distribution of the OLS Estimator

Recall that

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}.$$

Thus, $\hat{\boldsymbol{\beta}}$ is a **linear transformation** of $\mathbf{Y}$.

A fundamental property of the multivariate normal distribution is that a linear transformation of a multivariate normal vector is again multivariate normal.

Theorem 3.1 (Distribution of the OLS Estimator)
Under the normal linear model,

$$\hat{\boldsymbol{\beta}} \sim N_p\left(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}\right).$$

Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

Then

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2 \mathbf{C}.$$

If c_{jj} denotes the jth diagonal element of $\mathbf{C}$, then

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Therefore,

$$\frac{\hat{\beta}_j - \beta_j}{\sigma \sqrt{c_{jj}}} \sim N(0,1).$$

Exercise 3.3 (Reading the Covariance Matrix)
Let

$$\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}.$$

If c_{jj} is the jth diagonal element of $\mathbf{C}$:

1. What is $\mathrm{Var}(\hat{\beta}_j)$?

2. What happens to the uncertainty of $\hat{\beta}_j$ if c_{jj} becomes larger?

Solution 3.3
The variance is

$$\text{Var}(\hat{\beta}_j) = \sigma^2 c_{jj}.$$

Therefore, a larger c_{jj} means greater sampling variability and less precise estimation of β_j. Thus, the precision of OLS depends not only on σ^2, but also on the design matrix $\mathbf{X}$.

There is still one difficulty.

The quantity

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}}$$

depends on the unknown parameter σ.

Before performing inference, we need to estimate σ^2.

Exercise 3.4 (Why Can We Not Use the Normal Statistic Directly?)
We have

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

Why can we usually not use this statistic directly for inference?

Solution 3.4
Because the error standard deviation σ is unknown.
We therefore need to estimate σ from the data.
Replacing σ with an estimator will lead naturally to the t distribution.

Section 3.4 Residual Sum of Squares and Estimation of σ^2

Recall from Chapter 2 — Least Squares Estimation that

$$\mathbf{e} = \mathbf{M}\mathbf{Y},$$

where

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

Since

$$\mathbf{MX} = \mathbf{0},$$

we have

$$\begin{aligned}\mathbf{e} &= \mathbf{MY} \\ &= \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) \\ &= \mathbf{M}\boldsymbol{\varepsilon}.\end{aligned}$$

Thus,

$$\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}.$$

The residual vector therefore contains the part of the random error that lies outside the model space.

3.4.1 Residual Sum of Squares

The residual sum of squares is

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}.$$

Since

$$\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon},$$

and $\mathbf{M}$ is symmetric and idempotent,

$$\begin{aligned}\text{SSE} &= \boldsymbol{\varepsilon}^\top \mathbf{M}^\top \mathbf{M}\boldsymbol{\varepsilon} \\ &= \boldsymbol{\varepsilon}^\top \mathbf{M}\boldsymbol{\varepsilon}.\end{aligned}$$

Therefore,

$$\text{SSE} = \boldsymbol{\varepsilon}^\top \mathbf{M}\boldsymbol{\varepsilon}.$$

3.4.2 Distribution of SSE

Under the normal linear model,

$$\frac{\boldsymbol{\varepsilon}}{\sigma} \sim N_n\left(\mathbf{0}, \mathbf{I}_n\right).$$

Let

$$\mathbf{Z} = \frac{\boldsymbol{\varepsilon}}{\sigma}.$$

Then

$$\frac{\text{SSE}}{\sigma^2} = \mathbf{Z}^\top \mathbf{M} \mathbf{Z}.$$

Recall that $\mathbf{M}$ is symmetric and idempotent and

$$\text{rank}(\mathbf{M}) = n - p.$$

Therefore,

Theorem 3.2 (Distribution of the Residual Sum of Squares)
Under the normal linear model,

$$\frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p}.$$

3.4.3 Why Are the Degrees of Freedom $n - p$?

Recall that

$$\text{rank}(\mathbf{H}) = p$$

and

$$\text{rank}(\mathbf{M}) = n - p.$$

The fitted vector lies in

$$\mathcal{C}(\mathbf{X}),$$

which has dimension p.

The residual vector lies in

$$\mathcal{C}(\mathbf{X})^{\perp},$$

which has dimension

$$n - p.$$

Therefore, the residual variation has

$$n - p$$

degrees of freedom.

Geometric Interpretation of Degrees of Freedom

The response space can be decomposed as

$$\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^{\perp}.$$

The fitted component occupies p dimensions.
The remaining

$$n - p$$

dimensions are available for residual variation.
This provides a geometric explanation for why estimating p regression parameters costs p degrees of freedom.

Exercise 3.5 (Degrees of Freedom)
Suppose a regression model has

$$n = 20$$

observations and

$$p = 3$$

regression parameters.

1. What is the residual degrees of freedom?
2. Why is it not 20?

Solution 3.5
The residual degrees of freedom are

$$n - p = 20 - 3 = 17.$$

It is not 20 because fitting the regression model uses $p = 3$ dimensions of the response space. The residual vector therefore lies in a space of dimension

$$20 - 3 = 17.$$

3.4.4 Estimating σ^2

Since a chi-square random variable with ν degrees of freedom has expectation ν,

$$\mathbb{E}\left(\frac{\text{SSE}}{\sigma^2}\right) = n - p.$$

Therefore,

$$\mathbb{E}(\text{SSE}) = (n - p)\sigma^2.$$

This gives us an unbiased estimator of the error variance.

Definition 3.2 (Mean Squared Error)
The **mean squared error** is defined as

$$\text{MSE} = \hat{\sigma}^2 = \frac{\text{SSE}}{n - p}.$$

It satisfies

$$\mathbb{E}\left(\hat{\sigma}^2\right) = \sigma^2.$$

The estimated error standard deviation is

$$\hat{\sigma} = \sqrt{\text{MSE}}.$$

Exercise 3.6 (Estimate the Error Variance)
Suppose

$$n = 20, \qquad p = 3, \qquad \text{SSE} = 34.$$

Compute

$$\hat{\sigma}^2$$

and

$$\hat{\sigma}.$$

Solution 3.6
The residual degrees of freedom are

$$n - p = 17.$$

Therefore,

$$\hat{\sigma}^2 = \frac{34}{17} = 2.$$

Hence,

$$\hat{\sigma} = \sqrt{2} \approx 1.414.$$

Section 3.5 Independence of OLS and Residual Variation

A crucial result under the normal linear model is

$$\hat{\boldsymbol{\beta}} \quad \text{and} \quad \text{SSE} \quad \text{are independent.}$$

This result is closely connected to the projection geometry from Chapter 2 — Least Squares Estimation.

3.5.1 Why Does Independence Occur?

Recall that

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$$

and

$$\mathbf{e} = \mathbf{M}\mathbf{Y}.$$

We also know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Thus, the fitted component and the residual component come from orthogonal parts of the response vector.

Geometry Becomes Probability

In Chapter 2 — Least Squares Estimation, we learned that

$$\hat{\mathbf{Y}} \perp \mathbf{e}.$$

This is a geometric statement.
Under the normal linear model, these components are jointly normal.
For jointly normal random vectors, zero covariance implies independence.
Thus, under normality, geometric orthogonality leads to probabilistic independence.

More formally,

$$\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$$

and

$$\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}.$$

Therefore,

$$\begin{aligned} \operatorname{Cov}\left(\hat{\boldsymbol{\beta}}, \mathbf{e}\right) &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M} \\ &= \mathbf{0}, \end{aligned}$$

because

$$\mathbf{X}^\top \mathbf{M} = \mathbf{0}.$$

Since the two random vectors are jointly normal,

$$\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \mathbf{e}.$$

Since

$$\text{SSE} = \mathbf{e}^\top \mathbf{e}$$

is a function of $\mathbf{e}$,

$$\hat{\boldsymbol{\beta}} \perp\!\!\!\perp \text{SSE}.$$

Exercise 3.7 (Does Orthogonal Always Mean Independent?)
We know that

$$\mathbf{H}\mathbf{M} = \mathbf{0}.$$

Does the orthogonality between the fitted and residual components automatically imply that they are independent?
Why is normality important?

Solution 3.7
No.
In general, zero covariance or orthogonality does **not** imply independence.
Under the normal linear model, the fitted and residual components are jointly normal.
For jointly normal random vectors, zero covariance implies independence.
Thus, normality is what allows us to move from orthogonality to independence.

Section 3.6 Inference for a Single Regression Coefficient

For the jth regression coefficient,

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

If σ were known, we could use

$$\frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

In practice, σ is unknown.

We replace it with

$$\hat{\sigma} = \sqrt{\frac{\text{SSE}}{n-p}}.$$

3.6.1 Standard Error

The estimated standard deviation of $\hat{\beta}_j$ is

Definition 3.3 (Standard Error of an OLS Coefficient)
The standard error of $\hat{\beta}_j$ is

$$\text{SE}\left(\hat{\beta}_j\right) = \hat{\sigma}\sqrt{c_{jj}}.$$

3.6.2 Why Does the t Distribution Appear?

We know that

$$Z = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{c_{jj}}} \sim N(0,1).$$

We also know that

$$U = \frac{\text{SSE}}{\sigma^2} \sim \chi^2_{n-p}.$$

In addition,

$$Z \quad \text{and} \quad U$$

are independent.

Therefore,

$$\frac{Z}{\sqrt{U/(n-p)}} \sim t_{n-p}.$$

Substituting the regression quantities gives

Theorem 3.3 (t Distribution for an OLS Coefficient)
Under the normal linear model,

$$T = \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{c_{jj}}} \sim t_{n-p}.$$

i Normal vs. t Distribution

If σ were known, we would use a normal distribution.
Because σ is unknown and must be estimated from the data, we introduce additional uncertainty.
The t distribution accounts for this additional uncertainty.
As the degrees of freedom increase,

$$t_\nu$$

approaches

$$N(0,1).$$

Exercise 3.8 (Why Do We Use a t Distribution?)
Suppose

$$\hat{\beta}_j \sim N\left(\beta_j, \sigma^2 c_{jj}\right).$$

Why do we use a t distribution rather than a standard normal distribution when performing inference for β_j?

Solution 3.8
Because σ is unknown.
We replace σ with the estimate $\hat{\sigma}$ obtained from the residual sum of squares.
This additional estimation uncertainty changes the standardized statistic from a normal distribution to a t distribution.

3.6.3 Confidence Interval for β_j

A $100(1-\alpha)\%$ confidence interval for β_j is

$$\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p}\, \mathrm{SE}\left(\hat{\beta}_j\right).$$

Equivalently,

$$\hat{\beta}_j \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{c_{jj}}.$$

3.6.4 Hypothesis Test for β_j

Suppose we want to test

$$H_0 : \beta_j = \beta_{j,0}$$

against

$$H_1 : \beta_j \neq \beta_{j,0}.$$

The test statistic is

$$T = \frac{\hat{\beta}_j - \beta_{j,0}}{\mathrm{SE}(\hat{\beta}_j)}.$$

Under H_0,

$$T \sim t_{n-p}.$$

For a two sided test at significance level α, reject H_0 if

$$|T| > t_{1-\alpha/2,\, n-p}.$$

Exercise 3.9 (Confidence Interval and Hypothesis Test)
Suppose we test

$$H_0 : \beta_j = 0$$

against

$$H_1 : \beta_j \neq 0$$

at significance level

$$\alpha = 0.05.$$

What should happen if the corresponding 95% confidence interval for β_j does not contain zero?

Solution 3.9
We reject

$$H_0 : \beta_j = 0$$

at the 5% significance level.
A two sided hypothesis test at level α and a $100(1-\alpha)\%$ confidence interval give equivalent conclusions.

Section 3.7 Inference for Linear Combinations

Sometimes the quantity of interest is not a single coefficient. Suppose we are interested in $\mathbf{a}^\top\boldsymbol{\beta}$, where $\mathbf{a}$ is a fixed $p \times 1$ vector. Since

$$\hat{\boldsymbol{\beta}} \sim N_p\left(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\right),$$

we have

$$\mathbf{a}^\top\hat{\boldsymbol{\beta}} \sim N\left(\mathbf{a}^\top\boldsymbol{\beta}, \sigma^2\mathbf{a}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{a}\right).$$

Therefore,

$$\frac{\mathbf{a}^\top\hat{\boldsymbol{\beta}} - \mathbf{a}^\top\boldsymbol{\beta}}{\hat{\sigma}\sqrt{\mathbf{a}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{a}}} \sim t_{n-p}.$$

A $100(1-\alpha)\%$ confidence interval for $\mathbf{a}^\top\boldsymbol{\beta}$ is

$$\mathbf{a}^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}.$$

i Why Is This Useful?

Many regression questions can be expressed as linear combinations of the regression coefficients. Examples include:

- a single coefficient;
- the difference between two coefficients;
- the mean response at a particular predictor value.

Thus, inference for a single regression coefficient is a special case of inference for a linear combination.

Exercise 3.10 (A Single Coefficient as a Linear Combination)
Suppose

$$\boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

What vector $\mathbf{a}$ would you choose so that $\mathbf{a}^\top \boldsymbol{\beta} = \beta_1$?

Solution 3.10
Choose

$$\mathbf{a} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Then $\mathbf{a}^\top \boldsymbol{\beta} = \beta_1$. Thus, inference for an individual coefficient is a special case of inference for a linear combination.

Section 3.8 General Linear Hypotheses

Sometimes we want to test several linear restrictions simultaneously. Suppose $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, where $\mathbf{C}$ is an $r \times p$ matrix of rank r and $\mathbf{d}$ is an $r \times 1$ vector.

Definition 3.4 (General Linear Hypothesis)
A hypothesis of the form

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$$

is called a **general linear hypothesis**. It allows several linear restrictions on the regression coefficients to be tested simultaneously.

The corresponding test statistic is

$$F = \frac{(\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})^\top \left[\mathbf{C}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{C}^\top\right]^{-1} (\mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d})}{r\hat{\sigma}^2}.$$

Under H_0,

$$F \sim F_{r,\, n-p}.$$

For the distribution background, see Section 3.15; the full derivations are in the Basic Mathematical Statistics appendix.

Relationship Between t and F

When only one restriction is being tested, $r = 1$. In this case,

$$F = T^2.$$

Thus, the familiar t test for one coefficient is a special case of the general F test. We will study F tests more thoroughly in Chapter 4 — Model Comparison, Partial F Tests, and ANOVA when we develop the ANOVA decomposition and compare nested regression models.

Exercise 3.11 (t Test or F Test?)
Suppose we want to test only $H_0 : \beta_2 = 0$. Could we use either a t test or an F test? How are the two test statistics related?

Solution 3.11
Yes. Because there is only one linear restriction, either test can be used. The corresponding statistics satisfy

$$F = T^2.$$

Therefore, the two tests produce the same conclusion.

Section 3.9 Inference for the Mean Response

Suppose we want to estimate the mean response at a predictor vector $\mathbf{x}_0$. The true mean response is

$$\mu(\mathbf{x}_0) = \mathbf{x}_0^\top \boldsymbol{\beta}.$$

Its estimator is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

Since this is a linear combination of $\hat{\boldsymbol{\beta}}$,

$$\text{Var}\left(\hat{\mu}(\mathbf{x}_0)\right) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0.$$

Thus,

$$\frac{\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} - \mathbf{x}_0^\top \boldsymbol{\beta}}{\hat{\sigma}\sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t_{n-p}.$$

Therefore, a $100(1-\alpha)\%$ confidence interval for the mean response is

$$\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

Section 3.10 Prediction for a New Observation

Now suppose we want to predict one new observation at the same predictor value $\mathbf{x}_0$:

$$Y_{\text{new}} = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_{\text{new}},$$

where

$$\varepsilon_{\text{new}} \sim N(0, \sigma^2)$$

and is independent of the original data. The predictor is

$$\hat{Y}_{\text{new}} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

The prediction error is

$$Y_{\text{new}} - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}.$$

Its variance is

$$\text{Var}\left(Y_{\text{new}} - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}\right) = \sigma^2 \left[1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0\right].$$

Therefore, a $100(1-\alpha)\%$ prediction interval is

$$\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{1-\alpha/2,\, n-p} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

! Mean Response vs. New Observation

The two intervals answer different questions.

Goal	Source of uncertainty
Estimate a mean response	Uncertainty in estimating the regression mean
Predict a new observation	Uncertainty in the regression mean plus new random variation

The additional 1 in $1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ represents the variability of the new observation itself. Therefore, at the same $\mathbf{x}_0$ and confidence level, the prediction interval is wider than the confidence interval for the mean response.

Exercise 3.12 (Mean Response or Prediction?)
Consider the following questions.

1. What is the **average stopping distance** of all cars travelling at 50 mph?
2. What stopping distance should we expect for **one particular future car** travelling at 50 mph?

Which question requires a confidence interval for the mean response, and which requires a prediction interval?

Solution 3.12
The first question concerns the **mean response**, so we use a confidence interval for the regression mean. The second question concerns **one future observation**, so we use a prediction interval. The prediction interval is wider because it also includes the random variation associated with the future observation.

Section 3.11 Worked Example

Consider again the simple regression dataset from Chapter 2 — Least Squares Estimation:

x_i	0	1	2	3
Y_i	1	3	3	5

The design matrix and response vector are

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad \mathbf{Y} = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 5 \end{pmatrix}.$$

From Chapter 2 — Least Squares Estimation,

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} 1.2 \\ 1.2 \end{pmatrix},$$

and

$$\hat{\mathbf{Y}} = \begin{pmatrix} 1.2 \\ 2.4 \\ 3.6 \\ 4.8 \end{pmatrix}.$$

Therefore,

$$\mathbf{e} = \begin{pmatrix} -0.2 \\ 0.6 \\ -0.6 \\ 0.2 \end{pmatrix}.$$

Estimate the Error Variance

The residual sum of squares is

$$\begin{aligned} \text{SSE} &= \mathbf{e}^\top \mathbf{e} \\ &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 \\ &= 0.8. \end{aligned}$$

Since $n = 4$ and $p = 2$, the residual degrees of freedom are $n - p = 2$. Thus,

$$\hat{\sigma}^2 = \frac{0.8}{2} = 0.4.$$

Therefore,

$$\hat{\sigma} = \sqrt{0.4} \approx 0.632.$$

Standard Error of the Slope

We have

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{20} \begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}.$$

For the slope,

$$c_{22} = \frac{4}{20} = 0.2.$$

Therefore,

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{0.4(0.2)} = \sqrt{0.08} \approx 0.283.$$

Test the Slope

Consider $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$. The test statistic is

$$T = \frac{1.2}{0.283} \approx 4.243.$$

The residual degrees of freedom are 2. For a 5% two sided test, $t_{0.975,2} \approx 4.303$. Since $|4.243| < 4.303$, we fail to reject H_0 at the 5% significance level.

i What Does This Example Show?

The estimated slope $\hat{\beta}_1 = 1.2$ is relatively large. However, there are only four observations, leaving only two residual degrees of freedom. Therefore, there is substantial uncertainty in the estimate. A large estimated effect does not automatically imply strong evidence against the null hypothesis.

Confidence Interval for the Slope

A 95% confidence interval is $1.2 \pm 4.303(0.283)$, which gives approximately

$$(-0.017,\ 2.417).$$

The interval contains zero, which agrees with the hypothesis test.

Exercise 3.13 (Check the Connection)
Without performing another hypothesis test, what conclusion can you draw from the fact that the 95% confidence interval $(-0.017,\ 2.417)$ contains zero?

Solution 3.13
We fail to reject $H_0 : \beta_1 = 0$ at the 5% significance level. This agrees with the previous t test.

Mean Response at $x_0 = 2$

For simple linear regression with an intercept,

$$\mathbf{x}_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The estimated mean response is

$$\hat{\mu}(\mathbf{x}_0) = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} = 3.6.$$

Also,

$$\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 = 0.3.$$

Therefore,

$$\mathrm{SE}\left(\hat{\mu}(\mathbf{x}_0)\right) = \sqrt{0.4(0.3)} \approx 0.346.$$

The 95% confidence interval for the mean response is approximately

$$(2.110,\ 5.090).$$

Prediction at $x_0 = 2$

For one new observation,

$$\mathrm{SE}_{\mathrm{pred}} = \sqrt{0.4(1 + 0.3)} \approx 0.721.$$

Therefore, the 95% prediction interval is approximately

$$(0.497,\ 6.703).$$

Notice that the prediction interval is substantially wider than the confidence interval for the mean response.

Section 3.12 R Demonstration

We can reproduce the previous calculations using R.

```r
x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)

summary(fit)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

Estimated Coefficients

```
coef(fit)
```

```
(Intercept)           x
        1.2         1.2
```

Residual Degrees of Freedom and Error Variance

```
df.residual(fit)
```

```
[1] 2
```

```
sigma(fit)
```

```
[1] 0.6324555
```

```
sigma(fit)^2
```

```
[1] 0.4
```

Covariance Matrix and Standard Errors

```r
vcov(fit)
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```r
sqrt(diag(vcov(fit)))
```

```
(Intercept)           x
  0.5291503   0.2828427
```

Verify the Formulas Using Matrix Operations

```r
X <- model.matrix(fit)
Y <- matrix(y, ncol = 1)

n <- nrow(X)
p <- ncol(X)

beta_hat <-
  solve(t(X) %*% X) %*%
  t(X) %*%
  Y

e <- Y - X %*% beta_hat

SSE <- as.numeric(t(e) %*% e)

sigma2_hat <- SSE / (n - p)

V_beta_hat <-
  sigma2_hat *
  solve(t(X) %*% X)

SE_beta_hat <-
  sqrt(diag(V_beta_hat))

beta_hat
```

```
            [,1]
(Intercept)  1.2
x            1.2
```

```
SSE
```

```
[1] 0.8
```

```
sigma2_hat
```

```
[1] 0.4
```

```
V_beta_hat
```

```
            (Intercept)     x
(Intercept)        0.28 -0.12
x                 -0.12  0.08
```

```
SE_beta_hat
```

```
(Intercept)           x
  0.5291503   0.2828427
```

Hypothesis Tests and Confidence Intervals

R automatically reports coefficient level t tests:

```
summary(fit)$coefficients
```

```
            Estimate Std. Error  t value  Pr(>|t|)
(Intercept)      1.2  0.5291503 2.267787 0.1514719
x                1.2  0.2828427 4.242641 0.0513167
```

Confidence intervals for the regression coefficients can be obtained using

```
confint(fit)
```

```
                 2.5 %   97.5 %
(Intercept) -1.07674982 3.476750
x           -0.01697397 2.416974
```

Mean Response and Prediction

At $x_0 = 2$, use

```r
newdat <- data.frame(x = 2)

predict(
  fit,
  newdata = newdat,
  interval = "confidence"
)
```

```
  fit      lwr      upr
1 3.6 2.109517 5.090483
```

```r
predict(
  fit,
  newdata = newdat,
  interval = "prediction"
)
```

```
  fit      lwr      upr
1 3.6 0.497313 6.702687
```

The first interval estimates the **mean response**. The second interval predicts **one future observation**.

Section 3.13 Reading Standard Regression Output

For

```r
summary(fit)
```

the coefficient table contains four important columns:

Column	Interpretation
`Estimate`	Estimated regression coefficient
`Std. Error`	Estimated standard deviation of the coefficient estimator
`t value`	t statistic for testing whether the coefficient equals zero
`Pr(>|t|)`	Two sided p value for the test

The output also includes:

- the residual standard error;
- the residual degrees of freedom;

- R^2;
- adjusted R^2;
- an overall F test.

We will study R^2, the ANOVA decomposition, and the overall F test more carefully in Chapter 4 — Model Comparison, Partial F Tests, and ANOVA.

Exercise 3.14 (Reading an R Regression Table)
Suppose R reports

	Estimate	Std. Error	t value	Pr(>
x_1	2.40	0.60	4.00	0.001

Answer the following.

1. What is $\hat{\beta}_1$?
2. What is $\mathrm{SE}(\hat{\beta}_1)$?
3. What null hypothesis is being tested by default?

Solution 3.14
The estimate is $\hat{\beta}_1 = 2.40$. The standard error is $\mathrm{SE}(\hat{\beta}_1) = 0.60$. The default coefficient level test is $H_0 : \beta_1 = 0$. The reported t statistic is

$$\frac{2.40}{0.60} = 4.$$

Section 3.14 Practice at Home

1. Starting from

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon},$$

derive

$$\mathbb{E}\left(\hat{\boldsymbol{\beta}}\right) = \boldsymbol{\beta}$$

and

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

2. Explain why normality is not required for OLS to be unbiased but is useful for exact finite sample inference.

3. Explain geometrically why the residual degrees of freedom are $n - p$.
4. Explain why $\hat{\boldsymbol{\beta}}$ and SSE are independent under the normal linear model.
5. Derive the t statistic for testing $H_0 : \beta_j = \beta_{j,0}$.
6. Explain the connection between a two sided hypothesis test and the corresponding confidence interval.
7. Explain the difference between a confidence interval for $\mathbf{x}_0^\top \boldsymbol{\beta}$ and a prediction interval for a future observation at $\mathbf{x}_0$.
8. Fit a simple regression model in R and identify:
    - the coefficient estimates;
    - their standard errors;
    - the t statistics;
    - the p values;
    - the residual standard error;
    - the residual degrees of freedom.

Summary / Recap

- For fixed full-rank X, $E(\varepsilon) = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2 I$ give $E(\hat{\beta}) = \beta$ and $\mathrm{Var}(\hat{\beta}) = \sigma^2 (X^\top X)^{-1}$.
- Adding normal errors gives $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-p}$, independently of $\hat{\beta}$. Estimate σ^2 by $\mathrm{MSE} = \mathrm{SSE}/(n-p)$.
- A fixed linear comparison uses its estimate and full covariance matrix to obtain an SE; normal-model standardization gives an exact t test or interval.
- A general linear hypothesis tests several linearly independent restrictions jointly with an F statistic.
- A confidence interval for a mean response and a prediction interval for a new response have different targets. The prediction variance includes an additional σ^2.

Next, Chapter 4 expresses joint tests through nested models and extra sums of squares.

Section 3.15 Side Fact

These three reference distributions are built from normal variables:

- **Chi-square** measures squared variation: if $Z_1, \dots, Z_r$ are independent $N(0,1)$ variables, then $U = \sum_{j=1}^{r} Z_j^2 \sim \chi_r^2$.
- **Student's** t standardizes a normal variable by an estimated standard deviation: if $Z \sim N(0,1)$ and $V \sim \chi_\nu^2$ are independent, then $T = Z/\sqrt{V/\nu} \sim t_\nu$.
- F compares two independent variance quantities: if $U \sim \chi_r^2$ and $V \sim \chi_\nu^2$ are independent, then $F = (U/r)/(V/\nu) \sim F_{r,\nu}$.

For the general linear hypothesis, r is the number of independent restrictions and $\nu = n - p$ is the residual degrees of freedom. The normal model gives the required independence between the numerator quadratic form and the residual sum of squares.

The full explanations, density derivations, independence proof, regression applications, and reference books are in the Basic Mathematical Statistics appendix.

Section 4

Model Comparison, Partial F Tests, and ANOVA

In this chapter, we study the ANOVA decomposition for linear regression, the overall significance test, and the comparison of nested models. These ideas connect the geometry of least squares with the inferential tools developed in Chapter 2 — Least Squares Estimation and Chapter 3 — Distribution Theory of OLS and Inference.

Learning Objectives

By the end of this chapter, students should be able to:

- define the total, regression, and error sums of squares;
- explain the ANOVA decomposition in regression with an intercept;
- interpret the degrees of freedom associated with SST, SSR, and SSE;
- perform the overall F test for regression;
- compare nested linear models using extra sums of squares;
- interpret ANOVA tables produced by statistical software.

Reading

Recommended reading for this chapter:

- Seber and Lee:
  - sections on analysis of variance in regression
  - sums of squares and decomposition of variability
  - tests for nested models
- Montgomery, Peck, and Vining:
  - sections on the ANOVA table
  - overall model significance
  - partial and sequential sums of squares

i Motivation: does measuring tree height add useful information?

Suppose a forestry team wants to estimate timber volume before felling a tree. Diameter is already available; measuring height requires additional effort. Does height contribute information beyond diameter?
The real `trees` dataset in R lets us explore this question with measurements on 31 felled black cherry trees. Consider three candidate mean models:

Model	Information used	Mean volume
M_0	No tree measurements	β_0
M_1	Diameter	$\beta_0 + \beta_1$ Diameter
M_2	Diameter and height	$\beta_0 + \beta_1$ Diameter $+ \beta_2$ Height

Discuss before calculating: Which comparison answers whether height adds information? Why is choosing the smallest training SSE insufficient? Keep these questions in mind; the optional data example below returns to them with fitted models.

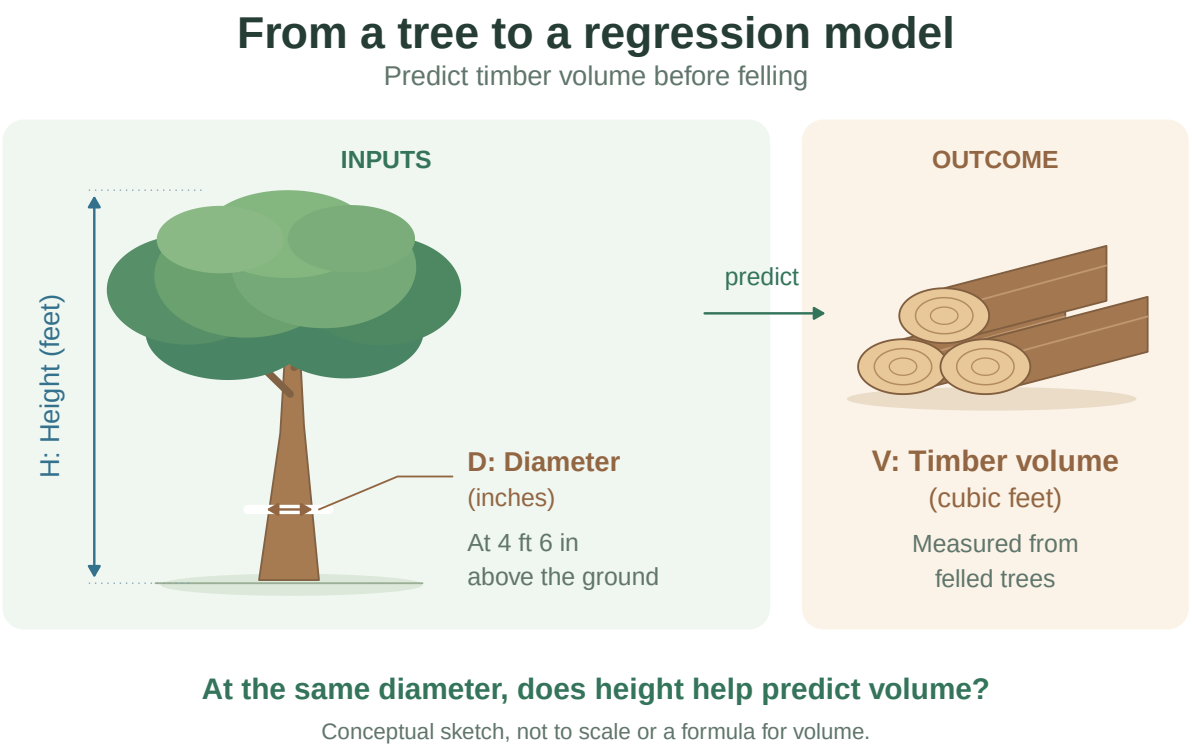


Figure 4.1: From measurements to a prediction question. Diameter and height are predictors; measured timber volume is the response. The tree and timber drawings are conceptual, not to scale.

Recap of the Linear Model

Recall the normal linear model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n).$$

When $\mathbf{X}$ has full column rank, the ordinary least squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y},$$

and the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}.$$

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

From Chapter 2 — Least Squares Estimation, we know that $\hat{\mathbf{Y}}$ and $\mathbf{e}$ are orthogonal. From Chapter 3 — Distribution Theory of OLS and Inference, we know that this leads to useful distributional results for inference. In this chapter, we organize these ideas into the ANOVA framework.

Vector notation. Following the convention used in this course, $\mathbf{a} = (a_1, a_2, a_3)$ denotes a column vector, while $\mathbf{a}^\top = (a_1\ a_2\ a_3)$ denotes a row vector. Commas distinguish the column-vector tuple; spaces separate entries of the row vector.

Section 4.1 Total, Explained, and Unexplained Variation

A central question in regression is: How much of the variation in the response can be explained by the model? To answer this, we decompose the total variation in $\mathbf{Y}$ into:

- variation explained by regression;
- variation left unexplained by the model.

When the model includes an intercept, this decomposition takes a particularly simple and important form.

Definition 4.1 (Regression Sums of Squares)
For an OLS fit with an intercept, let $\bar{Y} = n^{-1}\sum_{i=1}^n Y_i$, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$, and $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$. Define the **total**, **error**, and **regression sums of squares** by

$$\begin{aligned}
\mathrm{SST} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 = \|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2, \\
\mathrm{SSE} &= \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \mathbf{e}^\top \mathbf{e}, \\
\mathrm{SSR} &= \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2.
\end{aligned}$$

SST measures variation around the sample mean; SSE measures deviations from the fitted model; SSR measures how much the fitted values vary around the sample mean. These are definitions. Their additive relationship below follows from OLS orthogonality and the inclusion of an intercept.

! A deviation may be negative; a sum of squares cannot

The subtraction inside each definition produces a **signed deviation**, which can be negative, zero, or positive. We then **square that deviation** and add the squared values. Thus

$$\text{SST} \geq 0, \qquad \text{SSR} \geq 0, \qquad \text{SSE} \geq 0.$$

For example, $y_i - \hat{y}_i = -0.6$ is a residual, while its contribution to SSE is $(-0.6)^2 = 0.36$. A negative residual is not a negative SSE. Also, $(a-b)^2$ is not $a^2 - b^2$. Reversing the direction of a deviation changes its sign but leaves its square unchanged.

Theorem 4.1 (ANOVA Decomposition with an Intercept)
For an ordinary least squares fit whose design matrix contains an intercept,

$$\text{SST} = \text{SSR} + \text{SSE}.$$

The centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to the residual vector $\mathbf{e}$. Thus total variation around the sample mean splits into explained variation and residual variation. This geometric identity does not require normal errors.

Why the Decomposition Holds

The key reason is orthogonality. We can write

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + (\mathbf{Y} - \hat{\mathbf{Y}}).$$

That is,

$$\mathbf{Y} - \bar{Y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) + \mathbf{e}.$$

Because the model contains an intercept, the vector $\bar{Y}\mathbf{1}$ lies in the column space of $\mathbf{X}$. Hence both $\hat{\mathbf{Y}}$ and $\bar{Y}\mathbf{1}$ lie in the model space, so their difference also lies in the model space. Since the residual vector $\mathbf{e}$ is orthogonal to the model space, we have

$$(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1})^\top \mathbf{e} = 0.$$

Therefore, by the Pythagorean theorem,

$$\|\mathbf{Y} - \bar{Y}\mathbf{1}\|^2 = \|\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}\|^2 + \|\mathbf{e}\|^2,$$

which is exactly

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Exercise 4.1 (Why Does the Intercept Matter?)
Why does the usual centered decomposition $\text{SST} = \text{SSR} + \text{SSE}$ require the model space to contain the constant vector? Which step could fail without an intercept? Does this identity require normal errors?

Solution 4.1
With an intercept, $\mathbf{1}$ and therefore $\bar{Y}\mathbf{1}$ belong to the model space. Thus the centered fitted vector $\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}$ is orthogonal to $\mathbf{e}$, and the cross term vanishes when its squared norm is expanded. Without a constant vector in the model space, $\sum_i e_i$ need not be zero, so the cross term $\mathbf{e}^\top(\hat{\mathbf{Y}} - \bar{Y}\mathbf{1}) = -\bar{Y}\sum_i e_i$ need not vanish. The identity follows from OLS geometry, not normality.

Exercise 4.2 (One Observation, Three Deviations)
For one observation, $y_i = 1$, $\hat{y}_i = 1.2$, and $\bar{y} = 3$. In one minute, identify the total, explained, and residual deviations. Does the squared total deviation equal the sum of the other two squared deviations for this observation?

Solution 4.2
The deviations are $y_i - \bar{y} = -2$, $\hat{y}_i - \bar{y} = -1.8$, and $y_i - \hat{y}_i = -0.2$. They add as signed deviations: $-2 = -1.8 - 0.2$. Their squares do not add: $4 \neq 3.24 + 0.04$. The ANOVA identity holds after summing over all observations because the fitted and residual vectors are orthogonal; it is not a point-by-point identity for squared deviations.

Degrees of Freedom

The ANOVA decomposition is accompanied by a decomposition of degrees of freedom. When the model includes an intercept and $\mathbf{X}$ has rank p, we have:

- total degrees of freedom: $n - 1$;
- regression degrees of freedom: $p - 1$;
- error degrees of freedom: $n - p$.

Thus,

$$n - 1 = (p - 1) + (n - p).$$

These match the sum of squares decomposition:

$$\text{SST} = \text{SSR} + \text{SSE}.$$

Definition 4.2 (Regression and Error Mean Squares)
A **mean square** is a sum of squares divided by its associated degrees of freedom. For a full-rank regression design with an intercept and $1 < p < n$, define

$$\text{MSR} = \frac{\text{SSR}}{p - 1}, \qquad \text{MSE} = \frac{\text{SSE}}{n - p}.$$

Here p includes the intercept coefficient.

Dividing by degrees of freedom accounts for the different dimensions contributing to each sum of squares. Under the linear-model moment assumptions from Chapter 3 — Distribution Theory of

OLS and Inference, MSE is an unbiased estimator of σ^2. MSR estimates the same noise level under the overall null hypothesis, which motivates their ratio.

Section 4.2 The Overall F Test

A major inferential question is whether the regression model provides any explanatory power beyond the intercept-only model. Suppose the model includes an intercept and $p-1$ additional predictors. The null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{p-1} = 0,$$

where β_0 is the intercept and p counts all regression coefficients. Equivalently, under H_0, the mean response does not depend on the predictors. The alternative is that at least one non-intercept coefficient is nonzero.

Theorem 4.2 (Overall F Test)
Under the normal linear model with a fixed, full-rank design containing an intercept and $1 < p < n$, the null hypothesis that all non-intercept coefficients are zero gives

$$F = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} = \frac{\text{MSR}}{\text{MSE}} \sim F_{p-1,n-p}.$$

Large values provide evidence against the intercept-only mean model. Exact finite-sample inference requires the normal, common-variance error assumptions; the sum-of-squares identity alone does not supply this distribution.

Interpretation of the Overall F Test

The numerator measures explained variation per regression degree of freedom. The denominator measures unexplained variation per residual degree of freedom. So the F statistic compares:

- how much signal the model explains;
- how much noise remains in the residuals.

If the predictors have no effect, then both quantities should be of similar size, and the ratio should not be unusually large. If the predictors explain substantial variation, then the numerator should be much larger than the denominator.

Exercise 4.3 (What Does an Overall Rejection Mean?)
An intercept model with two predictors is fitted to $n = 25$ observations using a full-rank design. Write the overall null hypothesis and the F reference degrees of freedom under the normal linear model. How does this joint test differ from the individual t tests? If it rejects, must both individual slope tests also reject?

Solution 4.3
Here $p = 3$, so $H_0 : \beta_1 = \beta_2 = 0$ and the reference distribution is $F_{2,22}$. Rejection gives

evidence that the two slopes are not jointly zero. It does not show that both are nonzero, and it does not require either individual test to reject at the same level. Each individual t test asks whether one slope is zero while retaining the other predictor; the overall F test asks whether both slopes are jointly zero. These are different questions, especially when predictors are correlated.

Relationship to the Intercept-Only Model

The overall F test compares two models:

- the reduced model: intercept only;
- the full model: intercept plus predictors.

Thus the ANOVA decomposition provides the basis for formal model comparison. This leads naturally to the idea of nested models.

Section 4.3 Nested Models

Before comparing models, we need to say what their regression parameters mean.

Definition 4.3 (Regression Coefficients and the Mean Response)
For fixed predictor values, consider the additive model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i, \qquad \mathbb{E}(\varepsilon_i) = 0.$$

The **mean response function** is $\mu(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$. If predictors are random, read this as a conditional mean, with $\mathbb{E}(\varepsilon \mid x_1, x_2) = 0$.

Parameter	Interpretation in this additive model
β_0	The intercept: mean response when both $x_1 = 0$ and $x_2 = 0$.
β_1	Difference in mean response for a one-unit increase in x_1, holding x_2 fixed.
β_2	Difference in mean response for a one-unit increase in x_2, holding x_1 fixed.

For example,

$$\mu(x_1 + h, x_2) - \mu(x_1, x_2) = h\beta_1.$$

Thus a positive β_1 means a higher mean response at larger x_1 when x_2 is fixed; a negative value means a lower mean. Its units are **response units per unit of** x_1. A zero coefficient means no change in this model's mean as x_1 varies with x_2 fixed; it does not rule out an unadjusted association between x_1 and Y.

Parameter, estimate, and observation. The coefficients $\beta_0, \beta_1, \beta_2$ are unknown model parameters. The fitted coefficients $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2$ are estimates calculated from a sample and vary across samples. Substituting them gives a fitted mean $\hat{\mu}$. An individual observed response also contains its error term, so a slope does not prescribe the exact difference between two observations. These mean comparisons do not by themselves establish a causal effect.

Example 4.1 (Read a Fitted Equation in Its Units)
For the trees introduced earlier in this chapter, a fitted equation is

$$\widehat{\text{Volume}} = -57.988 + 0.339\,\text{Height} + 4.708\,\text{Diameter}.$$

Volume is in cubic feet, height in feet, and diameter in inches.

- $\hat{\beta}_{\text{Height}} = 0.339$: at the **same diameter**, trees one foot apart in height differ by an estimated 0.339 cubic feet in mean volume. A five-foot difference corresponds to $5(0.339) = 1.695$ cubic feet, using the rounded slope.
- $\hat{\beta}_{\text{Diameter}} = 4.708$: at the **same height**, a one-inch diameter difference corresponds to an estimated 4.708-cubic-foot difference in mean volume.
- $\hat{\beta}_0 = -57.988$: the fitted mean at height zero and diameter zero. Those values are outside the observed data, so this intercept is an extrapolated baseline, not a useful prediction for an actual tree.

The larger numerical diameter coefficient does not by itself make diameter the more important predictor: the predictors have different units and ranges, and the two coefficients describe different comparisons. The fitted coefficients also do not tell us their own uncertainty; that requires the standard errors and inference from Chapter 3.

We can now ask how these interpretations change when a predictor is added.

Definition 4.4 (Nested Linear Models)
For the same response vector and observations, let the reduced and full models have mean spaces $\mathcal{C}(\mathbf{X}_R)$ and $\mathcal{C}(\mathbf{X}_F)$. The models are **nested** when

$$\mathcal{C}(\mathbf{X}_R) \subseteq \mathcal{C}(\mathbf{X}_F).$$

Thus every mean vector allowed by the reduced model is also allowed by the full model. A strict comparison requires ranks $p_R < p_F$; adding columns that are linear combinations of existing columns does not enlarge the mean space.

Example 4.2 (Adding a Predictor)
Use different parameter symbols to keep the two candidate mean models distinct:

$$\mu_R(x_1) = \alpha_0 + \alpha_1 x_1,$$
$$\mu_F(x_1, x_2) = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

Coefficient	Mean comparison described by its model
α_1	A one-unit difference in x_1, without adjusting for x_2.
β_1	A one-unit difference in x_1, holding x_2 fixed.
β_2	A one-unit difference in x_2, holding x_1 fixed.

The reduced intercept is the mean at $x_1 = 0$ under the reduced model; the full intercept is the mean at $x_1 = x_2 = 0$ under the full model. Each interpretation depends on the specified mean function and on whether its baseline is supported by the data.
Setting $\beta_2 = 0$ recovers the reduced mean function, with $\alpha_0 = \beta_0$ and $\alpha_1 = \beta_1$ **under that restriction**. If the x_2 column is not in the span of the intercept and x_1, the full model adds one degree of freedom.
When we fit the two models separately, OLS re-estimates every free coefficient. Nestedness therefore does **not** require $\hat{\alpha}_1 = \hat{\beta}_1$. Even if the reduced model is true, fitting the unrestricted full model can give different estimates in a finite sample. Chapter 5 illustrates this with the same height measurements before and after adjustment for diameter.

i Interpret the restriction in the full model

For the models above, the partial F comparison tests $H_0 : \beta_2 = 0$: does x_2 contribute to the mean after accounting for x_1? It does not directly test equality of the two fitted x_1 coefficients. A change in the size, sign, or p-value of the x_1 coefficient can accompany the change in adjustment; compare the scientific questions before comparing printed numbers.

Exercise 4.4 (Which Parameter Is Being Tested?)
Compare `Volume ~ Height` with `Volume ~ Height + Diameter` on the same trees. Which null hypothesis does their partial F test address?
A. The two fitted height coefficients are equal.
B. The diameter coefficient in the full model is zero, after accounting for height.
C. The height coefficient in the full model is zero, after accounting for diameter.

Solution 4.4
B. Diameter is the added predictor. Statement C would instead compare `Volume ~ Diameter` with `Volume ~ Diameter + Height`. Neither comparison is a direct test that the height estimates from two different models are equal.

Definition 4.5 (Extra Sum of Squares)
For nested models fitted by OLS to the same response and observations, define the **extra sum of squares** for the full model relative to the reduced model as

$$\mathrm{SS}_{\mathrm{extra}} = \mathrm{SSE}_R - \mathrm{SSE}_F.$$

Its associated degrees of freedom are $p_F - p_R$, where p_R and p_F are the design-matrix ranks.

Exercise 4.5 (Can the Larger Model Fit Worse?)
Why must $\mathrm{SSE}_F \leq \mathrm{SSE}_R$ when nested models are fitted by OLS to the same observations? Must the inequality be strict whenever a genuinely new predictor direction is added?

Solution 4.5
Every fitted mean vector allowed by the reduced model is also available to the full model. Minimizing the same residual sum of squares over this larger set cannot give a larger minimum. Equality is possible even if the model space grows: for the observed response, the added directions may provide no further reduction in residual variation. Thus greater flexibility guarantees a weak improvement in training fit, not a strictly positive improvement or a useful predictor.

A positive extra sum of squares alone is not evidence against the reduced model: the F test scales that improvement by its degrees of freedom and the estimated noise variance.

Theorem 4.3 (Partial F Test for Nested Normal Linear Models)
Fit both models to the **same response and the same observations**. Suppose their fixed design matrices have nested column spaces, with ranks $p_R < p_F < n$, and the full model has independent $N(0, \sigma^2)$ errors. Under the null hypothesis that the mean belongs to the reduced model,

$$F = \frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(p_F - p_R)}{\mathrm{SSE}_F/(n - p_F)} \sim F_{p_F - p_R, n - p_F}.$$

Large values give evidence against the reduced mean model. The numerator measures improvement per added degree of freedom; the denominator estimates the unexplained variance under the full model.

The appendix at the end of this chapter derives $\mathrm{SSE}_R - \mathrm{SSE}_F$ geometrically and shows how Theorem 4.2 is the special case of Theorem 4.3 in which the reduced model contains only an intercept.

! Compare the models, not just the printed p-values

Different missing-value patterns can cause two R fits to use different observations. Construct a common analysis dataset before comparing them. For one added coefficient, the partial F statistic equals the square of its two-sided t-test statistic from the full model.

Exercise 4.6 (A Better Fit, but Enough Improvement?)
Two nested models use the same $n = 30$ observations. The reduced model has rank $p_R = 2$ and $\mathrm{SSE}_R = 90$; the full model has rank $p_F = 4$ and $\mathrm{SSE}_F = 78$. Compute the extra sum of squares, its degrees of freedom, the full-model MSE, and the partial F statistic. Does a reduction in SSE by itself justify rejecting the reduced model?

Solution 4.6
The improvement is $90 - 78 = 12$ on $4 - 2 = 2$ degrees of freedom. The full-model MSE is $78/(30 - 4) = 3$, so $F = (12/2)/3 = 2$. Under the normal-model null, compare this with $F_{2,26}$; the upper-tail p-value is about 0.156. The improvement is insufficient for rejection at the 5% level. More flexibility always weakly reduces training SSE, including when the added coefficients are zero.

Connection with General Linear Hypotheses

Nested column-space models correspond to linear restrictions on the full-model coefficients. More generally, Chapter 3 — Distribution Theory of OLS and Inference allows a specified affine restriction of the form

$$H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}.$$

So the ANOVA comparison of nested models is another way of expressing the general F test from Chapter 3 — Distribution Theory of OLS and Inference. In practice, this is one of the most common uses of regression ANOVA tables.

Definition 4.6 (Sequential and Partial Sums of Squares)
For a term in a specified regression model, both quantities are extra sums of squares from a nested comparison on the same observations:

- **Sequential sum of squares:** the reduction in SSE when the term is added after the preceding terms in a specified order.
- **Partial sum of squares:** the increase in SSE when the specified term is removed from a full model while the other specified terms are retained.

When predictors are correlated, sequential sums of squares can depend on their order. Partial sums of squares answer a comparison conditional on the terms retained in the reduced model. With interactions or rank deficiency, state the exact restriction and respect model hierarchy rather than relying only on a Type I, II, or III label.

Definition 4.7 (Coefficient of Determination)
For an OLS fit with an intercept and $\mathrm{SST} > 0$, the **coefficient of determination** is

$$R^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}.$$

By the ANOVA decomposition, R^2 lies between 0 and 1 and measures the proportion of centered variation explained by the fitted model. If every observed response is identical, $\mathrm{SST} = 0$ and this ratio is undefined.

Interpretation of R Squared

- If R^2 is close to 1, the model explains a large proportion of the variation in the response.
- If R^2 is close to 0, the model explains little of the variation.

However, R^2 alone does not guarantee that the model is appropriate. A high R^2 does not ensure that assumptions are satisfied, and a low R^2 does not necessarily imply the model is useless.

Definition 4.8 (Adjusted Coefficient of Determination)
For an OLS fit with an intercept, $\mathrm{SST} > 0$, and design rank $p < n$, define the **adjusted coefficient of determination** as

$$R^2_{\text{adj}} = 1 - \frac{\text{SSE}/(n-p)}{\text{SST}/(n-1)}.$$

Ordinary R^2 cannot decrease when the model space is enlarged on the same data. Adjusted R^2 accounts for the loss of residual degrees of freedom and can decrease or become negative. It compares residual mean square with the sample variance of the response; it is not a literal proportion of explained variation or an estimate of out-of-sample predictive performance.

Example 4.3 (A Small ANOVA Calculation)
For the column vectors $\mathbf{x} = (0, 1, 2, 3)$ and $\mathbf{y} = (1, 3, 3, 5)$, the fitted line is $\hat{y} = 1.2 + 1.2x$. Therefore $\hat{\mathbf{y}} = (1.2, 2.4, 3.6, 4.8)$, $\mathbf{e} = (-0.2, 0.6, -0.6, 0.2)$, and $\bar{y} = 3$.

$$\begin{aligned}
\text{SST} &= (1-3)^2 + (3-3)^2 + (3-3)^2 + (5-3)^2 = 8,\\
\text{SSE} &= (-0.2)^2 + 0.6^2 + (-0.6)^2 + 0.2^2 = 0.8,\\
\text{SSR} &= \text{SST} - \text{SSE} = 8 - 0.8 = 7.2.
\end{aligned}$$

Here $n = 4$ and $p = 2$, so the regression, residual, and total degrees of freedom are 1, 2, and 3, respectively.

Source	Sum of squares	df	Mean square	F
Regression	7.2	1	7.2	18
Error	0.8	2	0.4	
Total	8.0	3		

The model has $R^2 = 7.2/8 = 0.9$, yet the F-test p-value is $P(F_{1,2} \geq 18) \approx 0.0513$. With only two residual degrees of freedom, a high training R^2 does not automatically give evidence at the 5% significance level.

Exercise 4.7 (Can Two Fit Summaries Disagree?)
For the same $n = 10$ observations with SST $= 100$, an intercept model has rank 2 and SSE 20. Adding a predictor whose column is linearly independent of the existing columns increases the rank to 3 and reduces SSE to 19. Compute R^2 and adjusted R^2 for both fits. Did the one-unit reduction in SSE improve both summaries? Why can relying on R^2 alone be misleading?

Solution 4.7
Ordinary R^2 increases from $1 - 20/100 = 0.80$ to $1 - 19/100 = 0.81$. Adjusted R^2 decreases from $1 - (20/8)/(100/9) = 0.775$ to $1 - (19/7)/(100/9) \approx 0.756$. The extra predictor reduces SSE but increases residual mean square, from 2.5 to about 2.714. Training R^2 cannot decrease when the model space grows, even when the new term only fits noise. A high R^2 also does not establish a correct mean function, appropriate error assumptions, or causality. Neither summary alone establishes better prediction on new observations.

ANOVA Table Structure

A standard regression ANOVA table has the following structure:

Source	Sum of Squares	Degrees of Freedom	Mean Square	F
Regression	SSR	$p-1$	MSR	MSR/MSE
Error	SSE	$n-p$	MSE	
Total	SST	$n-1$		

Students should learn to move fluently between:

- formulas;
- geometric interpretation;
- software output.

Section 4.4 R Demonstration

Fit a simple regression model

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

x <- c(0, 1, 2, 3)
y <- c(1, 3, 3, 5)

fit <- lm(y ~ x)
summary(fit)
```

```
Call:
lm(formula = y ~ x)

Residuals:
   1    2    3    4
-0.2  0.6 -0.6  0.2

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.2000     0.5292   2.268   0.1515
x             1.2000     0.2828   4.243   0.0513 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6325 on 2 degrees of freedom
Multiple R-squared:    0.9, Adjusted R-squared:   0.85
F-statistic:    18 on 1 and 2 DF,  p-value: 0.05132
```

Obtain the ANOVA table

```
anova(fit)
```

```
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value  Pr(>F)
x          1    7.2     7.2      18 0.05132 .
Residuals  2    0.8     0.4
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Verify sums of squares manually

```
ybar <- mean(y)
yhat <- fitted(fit)
e <- resid(fit)

SST <- sum((y - ybar)^2)
SSE <- sum(e^2)
SSR <- sum((yhat - ybar)^2)

c(SST = SST, SSR = SSR, SSE = SSE)
```

```
SST SSR SSE
8.0 7.2 0.8
```

```
SST - SSR - SSE
```

```
[1] 3.330669e-16
```

From deviations to sums of squares. Figure 4.2 separates three operations: **subtract, square, then add**. Each column is one of the four observations. Read downward from a signed difference to its nonnegative squared contribution, then add across the row to obtain SS. The minus signs in the first row describe deviations; the boxed totals are the sums of squares.

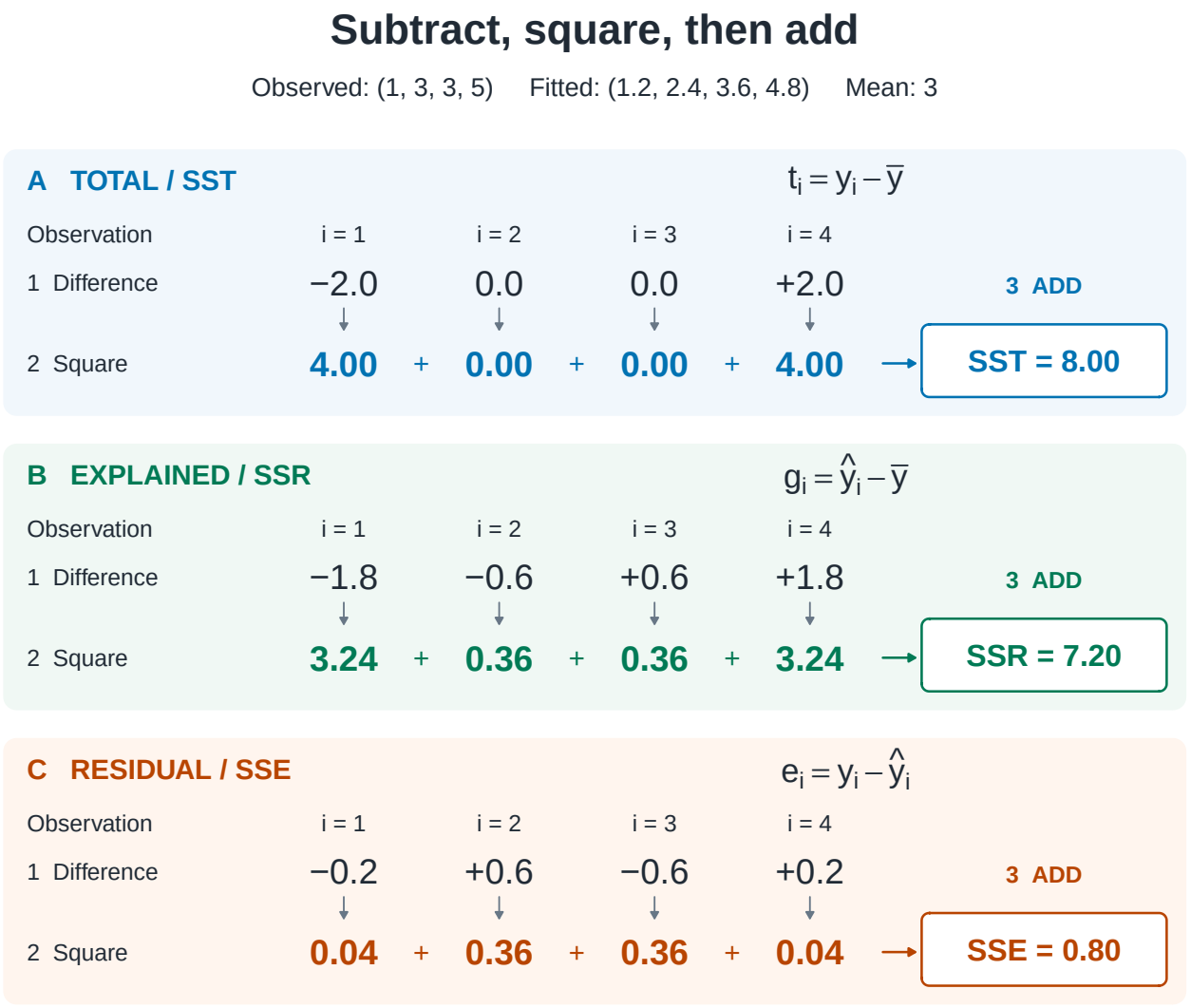


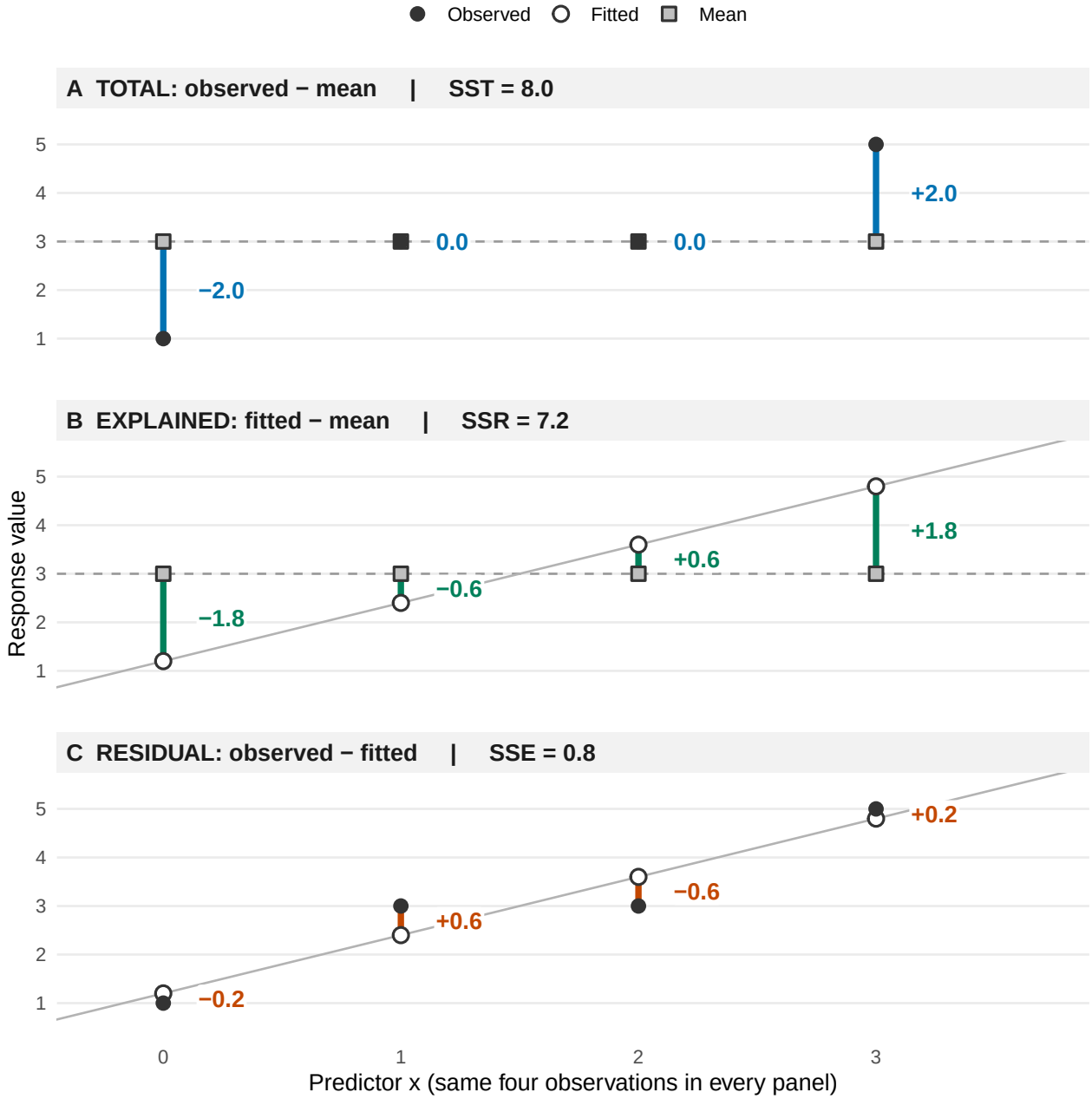
Figure 4.2: How to compute SST, SSR, and SSE from the same four observations. The first row in each panel contains signed deviations, the second contains their squares, and the box at the right contains the sum of those squares. Negative deviations become positive contributions when squared. The resulting sums are SST = 8.00, SSR = 7.20, and SSE = 0.80; none is negative.

Adding deviations is different from adding their squares. For these residuals,

$$\sum_i e_i = -0.2 + 0.6 - 0.6 + 0.2 = 0,$$
$$\text{SSE} = \sum_i e_i^2 = 0.04 + 0.36 + 0.36 + 0.04 = 0.80.$$

The signed residuals sum to zero because the OLS model includes an intercept. Their squared values do not cancel. Here all three SS values are strictly positive; in general an SS can equal zero when every corresponding deviation is zero.

The same deviations in the x, y plane. Each point represents one of the four observations. The three panels use identical axes to show total, explained, and residual deviations. Squaring and adding the colored vertical differences gives the SS values in Figure 4.2.



Companion x, y view for Figure 4.2. Filled circles are observed responses, open circles are fitted values, and squares mark the sample mean. The dashed line is the mean; the thin gray line is the fitted regression. Colored vertical segments show total (blue), explained (green), or residual (orange) deviations; their labels are signed differences, not sums of squares. Squaring and adding within each panel gives SST = 8.0, SSR = 7.2, and SSE = 0.8.

For example, at $x = 1$, the observed value equals the mean, so the **total deviation is zero**. Nevertheless, the explained deviation is $2.4 - 3 = -0.6$ and the residual is $3 - 2.4 = +0.6$. They

cancel as signed differences: $0 = -0.6 + 0.6$. Their squares do not cancel. Only after summing across all four observations do we obtain $8.0 = 7.2 + 0.8$.

Compute R squared manually

```
R2 <- SSR / SST
R2
```

```
[1] 0.9
```

```
summary(fit)$r.squared
```

```
[1] 0.9
```

```
summary(fit)$adj.r.squared
```

```
[1] 0.85
```

```
pf(18, df1 = 1, df2 = 2, lower.tail = FALSE)
```

```
[1] 0.0513167
```

Compare nested models

```
dat <- data.frame(
  y = c(4, 5, 7, 10, 8, 12, 13, 14),
  x1 = c(1, 2, 3, 4, 5, 6, 7, 8),
  x2 = c(2, 1, 3, 2, 5, 4, 6, 5)
)

fit_reduced <- lm(y ~ x1, data = dat)
fit_full <- lm(y ~ x1 + x2, data = dat)

anova(fit_reduced, fit_full)
```

```
Analysis of Variance Table

Model 1: y ~ x1
Model 2: y ~ x1 + x2
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1      6 6.8214
2      5 4.6613  1    2.1601 2.3171 0.1884
```

Inspect the two fitted models

```r
summary(fit_reduced)
```

```
Call:
lm(formula = y ~ x1, data = dat)

Residuals:
     Min       1Q   Median       3Q      Max
-1.85714 -0.30357  0.03571  0.33036  1.60714

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.5357     0.8308   3.052 0.022453 *
x1            1.4643     0.1645   8.900 0.000112 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.066 on 6 degrees of freedom
Multiple R-squared:  0.9296,    Adjusted R-squared:  0.9178
F-statistic: 79.21 on 1 and 6 DF,  p-value: 0.0001121
```

```r
summary(fit_full)
```

```
Call:
lm(formula = y ~ x1 + x2, data = dat)

Residuals:
      1       2       3       4       5       6       7       8
 0.4032 -1.0403  0.3306  0.8871 -1.1371  0.4194  0.7903 -0.6532

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.9677     0.8041   3.691  0.01413 *
x1            1.8387     0.2876   6.394  0.00139 **
x2           -0.6048     0.3973  -1.522  0.18845
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9655 on 5 degrees of freedom
Multiple R-squared:  0.9519,    Adjusted R-squared:  0.9326
F-statistic: 49.46 on 2 and 5 DF,  p-value: 0.0005079
```

Interpretation of Software Output

For a fitted model in R:

- `anova(fit)` gives sequential sums of squares in the order terms enter a single model;
- `anova(fit_reduced, fit_full)` compares nested models;
- `summary(fit)` reports the overall F statistic, R^2, and adjusted R^2.

Students should understand that these outputs are not separate topics. They are all built from the same least squares geometry and distribution theory.

Example 4.4 (Optional Real-Data Example: Is Tree Height Useful Beyond Diameter?)
Return to the forestry question. R's `trees` data contain timber volume in cubic feet, height in feet, and a variable named `Girth` that is actually **diameter in inches**, measured 4 feet 6 inches above the ground. We rename it `Diameter` to avoid that labeling trap. Each row represents one of 31 felled black cherry trees. See the official dataset documentation for the source and measurement definitions.
State the question first. Comparing M_0 with M_2 asks whether diameter and height together improve on a constant mean. Comparing M_1 with M_2 asks whether height contributes after adjusting for diameter. For the second question, $H_0 : \beta_{\text{Height}} = 0$. These are distinct questions even though both use an F test.

```
tree_dat <- datasets::trees
tree_dat$Diameter <- tree_dat$Girth
tree_dat <- tree_dat[, c("Volume", "Diameter", "Height")]
stopifnot(!anyNA(tree_dat))
fit_tree0 <- lm(Volume ~ 1, data = tree_dat)
fit_tree_d <- lm(Volume ~ Diameter, data = tree_dat)
fit_tree_dh <- lm(Volume ~ Diameter + Height, data = tree_dat)
knitr::kable(data.frame(
  Model = c("Constant mean", "Diameter", "Diameter + height"),
  Rank = c(fit_tree0$rank, fit_tree_d$rank, fit_tree_dh$rank),
  SSE = c(deviance(fit_tree0), deviance(fit_tree_d),
          deviance(fit_tree_dh))), digits = 2)
```

Model	Rank	SSE
Constant mean	1	8106.08
Diameter	2	524.30
Diameter + height	3	421.92

```
anova(fit_tree_d, fit_tree_dh)
```

```
Analysis of Variance Table

Model 1: Volume ~ Diameter
Model 2: Volume ~ Diameter + Height
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     29 524.30
2     28 421.92  1    102.38 6.7943 0.01449 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Read the comparison. The SSE drops from about 524.30 to 421.92. The added term uses one degree of freedom, and the full model has $31 - 3 = 28$ residual degrees of freedom. Thus

$$F = \frac{(524.30 - 421.92)/1}{421.92/28} \approx 6.79, \qquad p \approx 0.0145.$$

Under the stated normal, common-variance error model, this provides evidence against a zero height coefficient after accounting for diameter. It is not the overall test of both slopes.

```r
library(ggplot2)
tree_comparison <- rbind(
  transform(tree_dat, prediction = fitted(fit_tree_d),
    model = "Diameter only"),
  transform(tree_dat, prediction = fitted(fit_tree_dh),
    model = "Diameter + height"))
tree_comparison$model <- factor(tree_comparison$model,
                                levels = c("Diameter only",
                                  "Diameter + height"))
ggplot(tree_comparison, aes(Diameter, Volume)) +
  geom_segment(aes(xend = Diameter, yend = prediction, colour = "Residual"),
    linewidth = 0.65, show.legend = TRUE) +
  geom_point(aes(colour = "Observed"), shape = 16, size = 1.8,
    show.legend = TRUE) +
  geom_point(aes(y = prediction, colour = "Fitted"), shape = 1, size = 2,
    show.legend = TRUE) +
  scale_colour_manual(
    values = c(Observed = "black", Fitted = "#0072B2", Residual = "#B84400"),
    breaks = c("Observed", "Fitted", "Residual"),
    labels = c("Observed volume", "Fitted volume",
               "Residual (observed minus fitted)")) +
  guides(colour = guide_legend(nrow = 3, byrow = TRUE,
    override.aes = list(shape = c(16, 1, NA), linetype = c(0, 0, 1),
                        size = 2.8, linewidth = 0.8))) +
  facet_wrap(~ model, nrow = 1) +
  labs(x = "Diameter (inches)", y = "Timber volume (cubic feet)",
       colour = NULL) +
  theme_minimal(base_size = 11) +
  theme(legend.position = "bottom", legend.text = element_text(size = 11),
        panel.grid.minor = element_blank(),
        legend.key.width = grid::unit(1.3, "lines"),
        legend.key.height = grid::unit(1, "lines"))
```

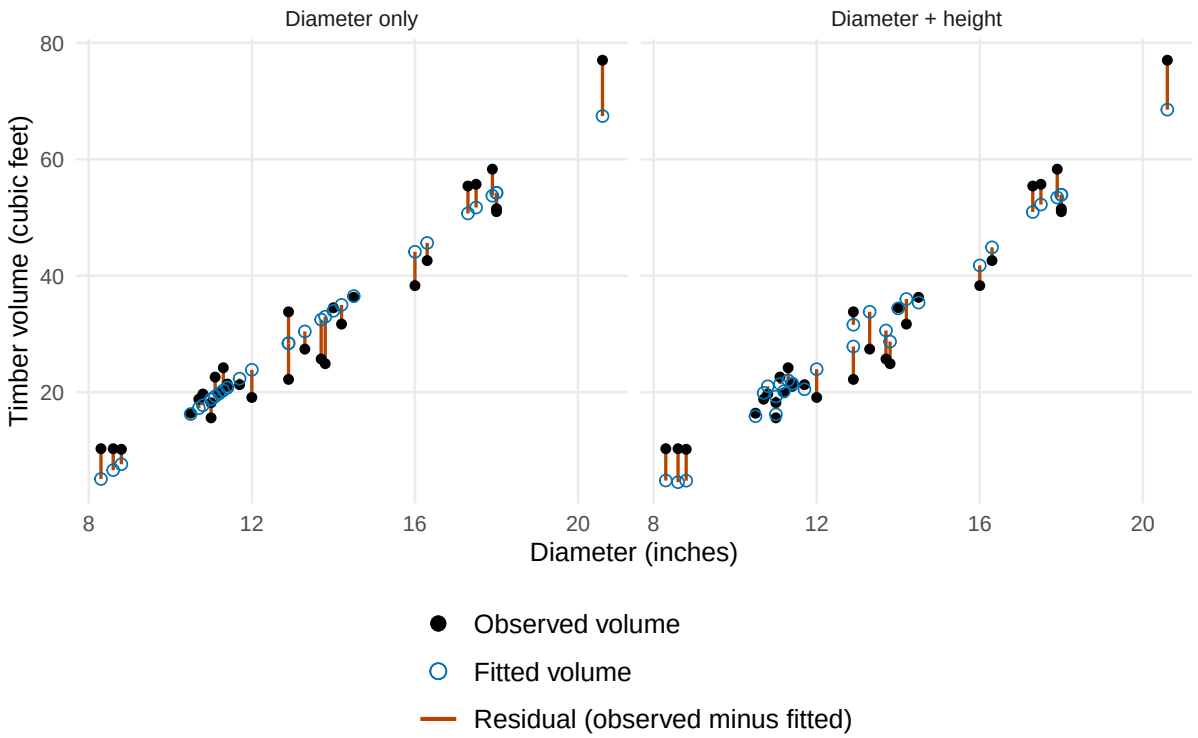


Figure 4.3: The same 31 trees under two mean models. The legend identifies black filled dots as observed volumes, blue open circles as fitted volumes, and orange vertical segments as residuals (observed minus fitted). Each segment connects the observation and prediction for one tree. Adding height reduces the sum of squared residual lengths, but the F test asks whether that improvement is large relative to the estimated noise.

Check your understanding. Why use 28 rather than 29 in the variance estimate? What would change if some trees had missing heights? The denominator comes from the full model. With missing heights, first form one common dataset and refit both models on it; comparing fits on different observations confounds the added predictor with a changed sample.

What this example does not settle. These are observational measurements, so the comparison is not a causal effect of making a tree taller. A statistically significant height term also does not decide whether measuring height is worth its cost or how much it helps predict new trees. Those decisions require practical error tolerances and validation. Inspect residuals and the mean function before using the raw-scale linear model operationally; transformations and model assessment appear later in the course.

Section 4.5 Practice Problems

Conceptual

1. Explain the meaning of SST, SSR, and SSE in words.
2. Explain why the regression degrees of freedom are $p - 1$ when the model includes an intercept.
3. Explain the difference between the overall F test and a test for a single coefficient.

Computational

Suppose a regression model with intercept has:

- $n = 20$,
- $p = 4$,
- $\text{SST} = 100$,
- $\text{SSE} = 40$.

Compute:

1. SSR,
2. the degrees of freedom for regression and error,
3. MSR,
4. MSE,
5. the overall F statistic,
6. R^2.

Nested Model Problem

A reduced model has $\text{SSE}_R = 120$ with $p_R = 3$, and a full model has $\text{SSE}_F = 90$ with $p_F = 5$. If $n = 30$, compute the nested-model F statistic.

Suggested Take home exercise

Complete the following tasks:

- prove the decomposition $\text{SST} = \text{SSR} + \text{SSE}$ when the model includes an intercept;
- derive the overall F statistic from the ANOVA decomposition;
- fit a regression model in R and reproduce the ANOVA table by hand;
- compare two nested models using an extra sum of squares test;
- interpret both R^2 and adjusted R^2 for a chosen dataset.

Summary / Recap

- Sums of squares are nonnegative squared lengths. For an OLS model with an intercept, $\text{SST} = \text{SSR} + \text{SSE}$.
- For nested models fitted to the same observations, $\text{SSE}_R - \text{SSE}_F \geq 0$ measures the extra fitted component. It equals SSR when the reduced model contains only an intercept.
- The partial F statistic divides the extra sum of squares per added parameter by the full-model MSE. Its exact null distribution requires the normal linear-model assumptions.
- For one fixed restriction under the same fitted model, $F = t^2$. An overall rejection does not identify which individual coefficients differ from zero.
- Adding predictors cannot decrease training R^2; adjusted R^2 can decrease. Neither alone measures prediction performance on new observations.

Next, Chapter 5 turns scientific comparisons into model columns, coefficients and restrictions.

Appendix: From the SSE Difference to the Overall F Test

Theorem 4.2 is the intercept-only special case of Theorem 4.3: the extra sum of squares $\mathrm{SSE}_R - \mathrm{SSE}_F$ then equals SSR.

A geometric picture of the extra sum of squares

Each vertex in Figure 4.4 represents a vector of n values. The green edge joins the reduced and full fitted vectors; its **squared length** is $\mathrm{SSE}_R - \mathrm{SSE}_F$.

For an intercept-only reduced model, $\hat{\mathbf{Y}}_R = (\bar{Y}, \ldots, \bar{Y})^\top = \bar{Y}\,\mathbf{1}_n$, where $\mathbf{1}_n$ is the all-ones vector. Panel B uses the same four observations as Figure 4.2, with $\bar{Y} = 3$.

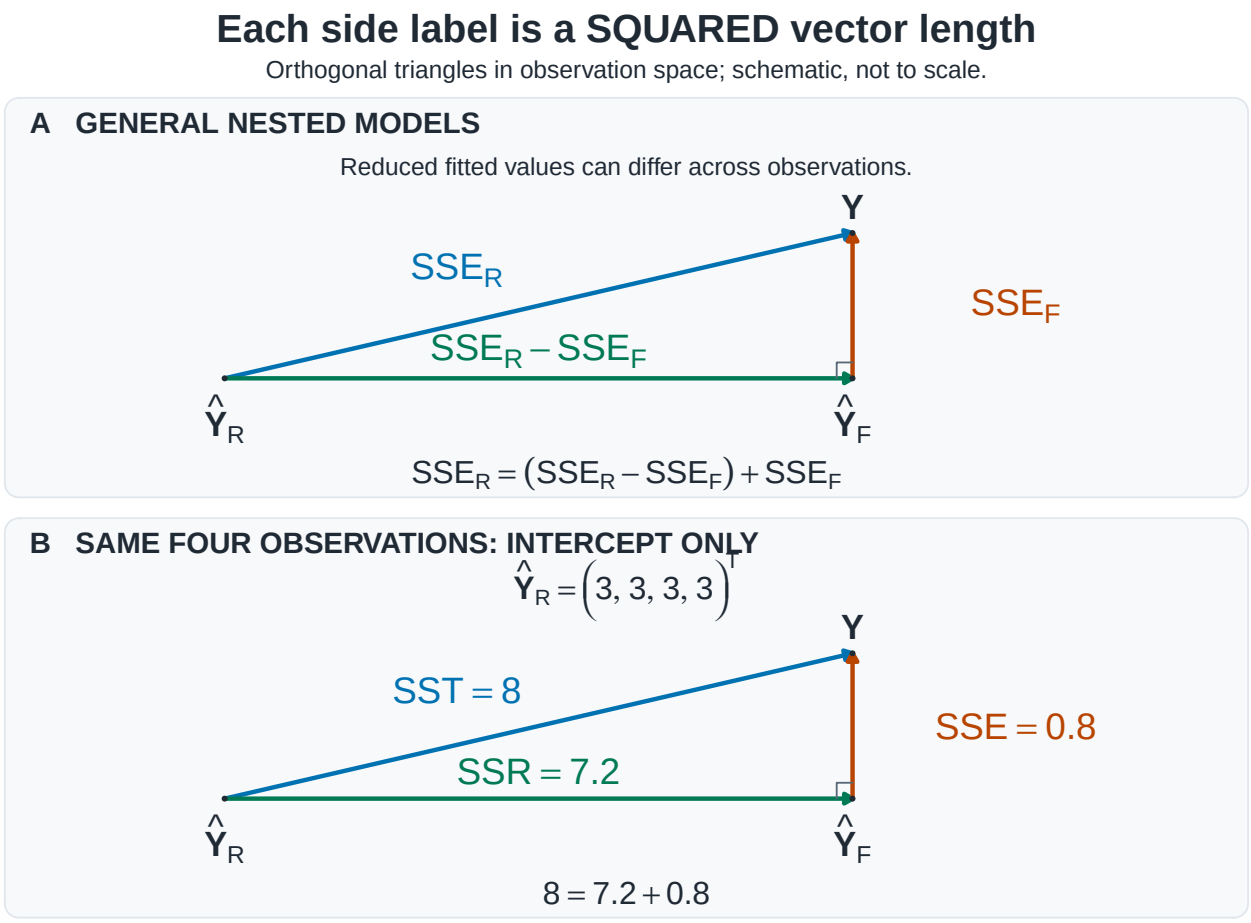


Figure 4.4: Squared lengths for nested OLS fits. Panel B uses the data in Figure 4.2: $\mathrm{SSE}_R = \mathrm{SST} = 8$, $\mathrm{SSE}_F = \mathrm{SSE} = 0.8$, and the green squared length is $\mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SSR} = 7.2$. The diagrams are schematic, not to scale.

Why subtract the two residual sums of squares?

Proof. Fit nested OLS models to the same response vector and observations. Write $\mathbf{P}_R$ and $\mathbf{P}_F$ for the orthogonal projections onto their mean spaces, so

$$\hat{\mathbf{Y}}_R = \mathbf{P}_R\mathbf{Y}, \qquad \hat{\mathbf{Y}}_F = \mathbf{P}_F\mathbf{Y}, \qquad \mathbf{e}_R = \mathbf{Y} - \hat{\mathbf{Y}}_R, \qquad \mathbf{e}_F = \mathbf{Y} - \hat{\mathbf{Y}}_F.$$

The reduced-model residual splits into two parts:

$$\underbrace{\mathbf{Y} - \hat{\mathbf{Y}}_R}_{\mathbf{e}_R} = \underbrace{\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R}_{\text{change in fitted values}} + \underbrace{\mathbf{Y} - \hat{\mathbf{Y}}_F}_{\mathbf{e}_F}.$$

Both fitted vectors belong to the full-model space. Their difference therefore lies in that space and is orthogonal to $\mathbf{e}_F$. Taking squared lengths gives

$$\mathrm{SSE}_R = \|\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R\|^2 + \mathrm{SSE}_F,$$

and hence

$$\mathrm{SSE}_R - \mathrm{SSE}_F = \|\hat{\mathbf{Y}}_F - \hat{\mathbf{Y}}_R\|^2 = \mathbf{Y}^\top(\mathbf{P}_F - \mathbf{P}_R)\mathbf{Y}.$$

□

The difference is nonnegative and measures the part of the reduced-model residual that the full model can fit. This identity requires nested OLS fits on the same observations; normal errors and an intercept are not needed.

Recovering Theorem 4.2 from Theorem 4.3

Proof. Now choose the reduced model to be **intercept only**, with rank $p_R = 1$. Its fitted value for every observation is $\bar{Y}$, so

$$\hat{\mathbf{Y}}_R = \bar{Y}\,\mathbf{1}_n, \qquad \mathrm{SSE}_R = \sum_{i=1}^{n}(Y_i - \bar{Y})^2 = \mathrm{SST}.$$

Let the full model contain an intercept and have rank $p_F = p$. Then

$$\mathrm{SSE}_F = \mathrm{SSE}, \qquad \mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SST} - \mathrm{SSE} = \mathrm{SSR}, \qquad p_F - p_R = p - 1.$$

Substituting these quantities into Theorem 4.3 gives exactly Theorem 4.2:

$$\underbrace{\frac{(\mathrm{SSE}_R - \mathrm{SSE}_F)/(p_F - p_R)}{\mathrm{SSE}_F/(n - p_F)}}_{\text{partial F statistic}} = \underbrace{\frac{\mathrm{SSR}/(p-1)}{\mathrm{SSE}/(n-p)}}_{\text{overall F statistic}}.$$

The null hypothesis also becomes the overall null: the full-model mean is constant, or equivalently, all its non-intercept coefficients are zero. □

For the four observations in Figure 4.2, the intercept-only fit has $\mathrm{SSE}_R = 8.0$, and the fitted line has $\mathrm{SSE}_F = 0.8$. Thus

$$F = \frac{(8.0 - 0.8)/(2-1)}{0.8/(4-2)} = \frac{7.2}{0.4} = 18,$$

which is the same calculation as MSR/MSE in Example 4.3.

i When the reduced model already has predictors

If both models contain an intercept, they share the same centered SST. Therefore

$$\text{SSE}_R - \text{SSE}_F = (\text{SST} - \text{SSR}_R) - (\text{SST} - \text{SSR}_F) = \text{SSR}_F - \text{SSR}_R.$$

The numerator uses the **additional** explained variation beyond the reduced model. It is generally not the full model's entire SSR_F. For example, adding height to diameter explains about $524.30 - 421.92 = 102.38$ additional squared-volume units; the variation already explained by diameter is excluded.

Why does the ratio have an F distribution?

Proof. Let $q = p_F - p_R$ and assume the normal, common-variance model in Theorem 4.3. Under H_0, the mean vector $\boldsymbol{\mu}$ lies in the reduced-model space. Nestedness implies

$$\mathbf{P}_F\mathbf{P}_R = \mathbf{P}_R\mathbf{P}_F = \mathbf{P}_R.$$

Consequently, $\mathbf{A} = \mathbf{P}_F - \mathbf{P}_R$ and $\mathbf{B} = \mathbf{I} - \mathbf{P}_F$ are symmetric idempotent matrices of ranks q and $n - p_F$. Also, $\mathbf{AB} = \mathbf{0}$, $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$, and $\mathbf{B}\boldsymbol{\mu} = \mathbf{0}$. The normal projection results from Chapter 3 therefore give

$$\frac{\text{SSE}_R - \text{SSE}_F}{\sigma^2} = \frac{\mathbf{Y}^\top\mathbf{AY}}{\sigma^2} \sim \chi^2_q, \qquad \frac{\text{SSE}_F}{\sigma^2} = \frac{\mathbf{Y}^\top\mathbf{BY}}{\sigma^2} \sim \chi^2_{n-p_F},$$

independently. Their ratio, after dividing each by its degrees of freedom, has distribution $F_{q,n-p_F}$. □

Under H_0, both mean squares estimate σ^2. The full-model residual provides an independent noise estimate; SSE_R includes the improvement being tested. The geometric identity alone does not establish the F distribution: normal, common-variance errors are also required.

Section 5

Multiple Regression and Categorical Predictors

Chapter 4 compared models. Here we learn to **build and interpret** them when predictors include groups and interactions.

Learning Objectives

By the end of this chapter, students should be able to:

- compare unadjusted and adjusted associations;
- code a factor and count its design-matrix columns;
- interpret additive and interaction models;
- explain how centering and reference groups affect coefficient meanings;
- choose a model comparison for a scientific question and check it in R.

Reading. Multiple regression, qualitative predictors, and interactions in Montgomery, Peck, and Vining (Montgomery et al. 2021) and Seber and Lee (Seber and Lee 2003).

i Recap of the Linear Model

Chapter 1, Chapter 2, and Chapter 3 gave us the design-matrix notation and least-squares estimator. For a full-column-rank design,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \qquad \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}.$$

The fitted means are $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$. Under the normal-model assumptions, Chapter 4's partial F test compares nested mean models. We now use these tools to interpret the columns and coefficients: **which mean responses are being compared, and what is held fixed?** An adjusted association alone does not establish causation.

i Motivation: what comparison does a coefficient describe?

Trees. Chapter 4 asked whether height adds information about timber volume beyond diameter. Now compare two trees of the same diameter: what mean-volume difference corresponds to one extra foot of height? Why might that answer differ from a regression using height alone?
Vitamin C. At the same dose, how does mean odontoblast length differ between orange juice and ascorbic acid? Could that difference change with dose?
These questions lead from adjusted slopes to group indicators and interactions. We will write the mean model first, then read the corresponding coefficient or combination of coefficients.

Section 5.1 Multiple regression and adjusted interpretation

Chapter 4 used an additive model with two predictors. We now make the word **additive** explicit and connect the mean function to the matrix notation from Chapter 1 and Chapter 2.

Definition 5.1 (Additive Linear Mean Model)
At predictor values $(x_1, \dots, x_k)$, let $\mu(x_1, \dots, x_k)$ denote the **mean response** at that predictor profile. An additive linear mean model specifies

$$\mu(x_1, \dots, x_k) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k.$$

Additive means that the intercept and the separate predictor contributions are added together. There are no interaction terms, such as $\beta_{12} x_1 x_2$: the contribution associated with changing one predictor does not depend on the values of the others. **Linear** means that each contribution here has the form $\beta_j x_j$.

i Additivity concerns the mean function

Predictors may be correlated; additivity does **not** require their independence. More generally, a mean of the form $\beta_0 + \sum_j f_j(x_j)$ is additive even when the separate functions are curved. Here we use $f_j(x_j) = \beta_j x_j$. An interaction term couples predictors and changes this additive structure; Section 5.4 develops that extension.

Connection with the earlier notation. As in Chapter 1 and Chapter 2, first treat predictor values as fixed. For observation i, write $\mu_i = \mu(x_{i1}, \dots, x_{ik})$ and

$$Y_i = \mu_i + \varepsilon_i, \qquad \mu_i = \beta_0 + \sum_{j=1}^{k} \beta_j x_{ij} = \mathbf{x}_i^\top \boldsymbol{\beta}, \qquad \mathbb{E}(\varepsilon_i) = 0.$$

Following Chapter 2, $\mathbf{x}_i = (1, x_{i1}, \dots, x_{ik})$ is the **column vector including the intercept entry**. Its transpose $\mathbf{x}_i^\top = (1 \; x_{i1} \; \cdots \; x_{ik})$ is row i of the design matrix. Thus μ_i is the systematic mean part, while ε_i is the individual deviation from it.

Design matrix and parameter count. Stacking these rows gives the familiar matrix form:

$$\mathbf{X} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1k} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nk} \end{pmatrix} \in \mathbb{R}^{n \times (k+1)}, \qquad \boldsymbol{\beta} = (\beta_0, \beta_1, \dots, \beta_k), \qquad p = k + 1.$$

The mean vector is $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n) = \mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, so $\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. This is the same model as before, now written to show what each mean contains. If predictors are random, use the conditional statements $\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ and $\mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \mathbf{0}$.

We assume full column rank, so $\operatorname{rank}(\mathbf{X}) = p$. Here k counts **predictor columns**, not necessarily named variables: a factor can contribute several indicator columns. The tree model below has height and diameter, giving $k = 2$ and $p = 3$. Later, adding interaction columns preserves the matrix form but changes the additive mean structure.

Definition 5.2 (Adjusted Regression Coefficient)
In the additive linear mean model of Definition 5.1, choose $j \in \{1, \dots, k\}$. Then β_j is the mean-response difference for a one-unit increase in x_j, **holding the other predictors fixed**:

$$\mu(x_1, \dots, x_j + 1, \dots, x_k) - \mu(x_1, \dots, x_j, \dots, x_k) = \beta_j.$$

Why does the difference equal the coefficient? With the other predictors fixed, $C_j = \beta_0 + \sum_{\ell \neq j} \beta_\ell x_\ell$ is the same in both means. Thus the one-unit mean difference is

$$\Delta_j = [C_j + \beta_j(x_j + 1)] - [C_j + \beta_j x_j] = \beta_j.$$

The intercept and all unchanged predictor contributions cancel, leaving $\beta_j[(x_j + 1) - x_j] = \beta_j$. A change of h units gives the mean difference $h\beta_j$ in this additive model.
The intercept is the mean when $x_1 = \cdots = x_k = 0$; see Chapter 4's coefficient interpretation.

Example 5.1 (The Same Height Measurement, Two Comparisons)
Return to the 31 black cherry trees. Compare the height coefficient with and without adjustment for diameter. As before, `Girth` records diameter in inches; `Height` is in feet and `Volume` in cubic feet.

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))
ch5_trees <- transform(datasets::trees, Diameter = Girth)
ch5_tree_simple <- lm(Volume ~ Height, data = ch5_trees)
ch5_tree_adjusted <- lm(Volume ~ Height + Diameter, data = ch5_trees)
knitr::kable(data.frame(
  Model = c("Height alone", "Height adjusted for diameter"),
  Height_coefficient = c(coef(ch5_tree_simple)["Height"],
                         coef(ch5_tree_adjusted)["Height"])),
  digits = 3, col.names = c("Model", "Height coefficient"),
  row.names = FALSE, align = "c")
```

Model	Height coefficient
Height alone	1.543
Height adjusted for diameter	0.339

The height-only slope is **1.543** cubic feet per foot. After adjustment, it is **0.339**: among trees of the same diameter, a one-foot height difference corresponds to 0.339 more cubic feet of fitted mean volume.

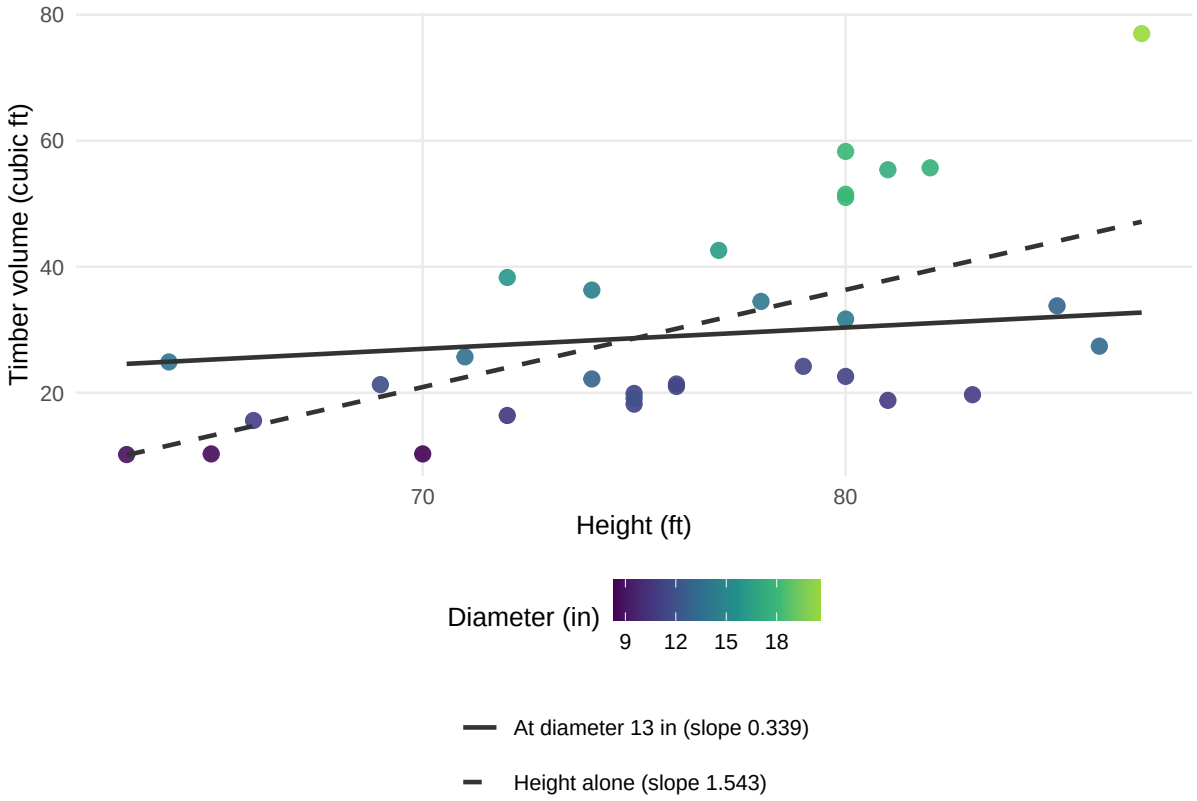


Figure 5.1: Dashed: height-only fit. Solid: adjusted mean at diameter 13 inches. Color shows observed diameter; distances to the solid line are not residuals because the trees have different diameters.

Taller trees in this sample also tend to have larger diameters. The height-only slope includes that accompanying variation; the adjusted slope compares heights at fixed diameter. The two slopes answer different questions.

Exercise 5.1 (Interpret 0.339)
Which interpretation matches the adjusted height coefficient?
A. Increasing any tree's height by one foot causes its volume to increase by 0.339 cubic feet.
B. At the same diameter, a one-foot height difference corresponds to a 0.339-cubic-foot difference in fitted mean volume.
C. Every tree one foot taller has exactly 0.339 more cubic feet of observed volume.

Solution 5.1
B. The coefficient describes an adjusted mean association. It is neither an automatic causal effect nor an exact difference between individual responses.

Section 5.2 Categorical predictors are columns of a design matrix

Start with three treatment methods named A, B, and C. These are names for different methods, not amounts of a treatment. Calling a method B does not mean that it lies halfway between methods A and C.

Why not enter 1, 2, and 3 as one numeric predictor? Suppose we store A as $s = 1$, B as $s = 2$, and C as $s = 3$, then fit $\mathbb{E}(Y \mid s) = \alpha + \gamma s$. Writing μ_A, μ_B, μ_C for the group means, the model requires

$$\mu_A = \alpha + \gamma, \qquad \mu_B = \alpha + 2\gamma, \qquad \mu_C = \alpha + 3\gamma.$$

Both A-to-B and B-to-C increase the score by one, so **both changes in the mean must equal the same slope**:

$$\mu_B - \mu_A = \gamma = \mu_C - \mu_B.$$

For example, means of 18, 22, and 19 give successive differences of $+4$ and -3. The one-slope model cannot represent that mean pattern exactly. The equal-gap restriction comes from treating the arbitrary codes as a numeric predictor; it does not follow from having three groups.

Use a factor when each group should have its own mean. In R, `y ~ factor(group)` treats group values as categories, even if they are stored as the numbers 1, 2, and 3. With an intercept and two indicators, the model can represent any three group means, including 18, 22, and 19. The indicator coefficients below show how.

If the groups instead correspond to measured doses, such as 0.5, 1, and 2 mg/day, the numbers have a quantitative meaning. A numeric-dose model still assumes a particular response shape, such as a straight line; a factor-dose model allows separate means at those observed doses. See Section 5.3 and Section 5.5.

Definition 5.3 (Reference-Group Indicator Coding)
Let G_i be the group of observation i. Suppose A, B, and C are all possible groups and every observation belongs to exactly one of them. With A as the **reference group**, define

$$D_{Bi} = \mathbb{1}\{G_i = \mathrm{B}\}, \qquad D_{Ci} = \mathbb{1}\{G_i = \mathrm{C}\},$$

where an indicator is 1 when its stated condition holds and 0 otherwise. The mean model is

$$\mathbb{E}(Y_i \mid G_i) = \beta_0 + \beta_B D_{Bi} + \beta_C D_{Ci}.$$

The row of the design matrix is $(1\ D_{Bi}\ D_{Ci})$:

Group	Intercept column	D_{Bi}	D_{Ci}	Mean
A	1	0	0	β_0
B	1	1	0	$\beta_0 + \beta_B$
C	1	0	1	$\beta_0 + \beta_C$

Writing μ_A, μ_B, μ_C for the three group means gives

$$\beta_0 = \mu_A, \qquad \beta_B = \mu_B - \mu_A, \qquad \beta_C = \mu_C - \mu_A.$$

Thus the B-minus-C difference is $\beta_B - \beta_C$. More generally, for g groups, reference coding uses an intercept and $g-1$ indicators, giving g identifiable parameters in this group-only model when every group is observed.

A is represented, not omitted. An A observation has $D_{Bi} = D_{Ci} = 0$, so its mean is β_0. Omitting a separate A indicator does not remove any A observations. The intercept column is 1 for **every** observation: its contribution is $1 \times \beta_0$. This does not mean that the intercept coefficient equals 1.

Use `model.matrix()` to check the coding; `(Intercept)`, `groupB`, and `groupC` correspond to $\mathbf{1}$, $\mathbf{D}_B$, and $\mathbf{D}_C$. The **Group** column in the displayed tables labels the rows; it is not an additional column of the design matrix:

```
ch5_groups <- data.frame(group = factor(c("A", "B", "C"),
                                        levels = c("A", "B", "C")))
contrasts(ch5_groups$group) <- contr.treatment(levels(ch5_groups$group), base = 1)
ch5_X_reference <- model.matrix(~ group, data = ch5_groups)
knitr::kable(
  data.frame(Group = ch5_groups$group, ch5_X_reference,
             check.names = FALSE), row.names = FALSE, align = "c")
```

Group	(Intercept)	groupB	groupC
A	1	0	0
B	1	1	0
C	1	0	1

! Why adding an A indicator would be redundant

To examine what would happen if we also included an A indicator, **first define**

$$D_{Ai} = \mathbb{1}\{G_i = \text{A}\} = \begin{cases} 1, & G_i = \text{A}, \\ 0, & G_i \neq \text{A}. \end{cases}$$

Since each observation belongs to exactly one of the three groups,

$$D_{Ai} + D_{Bi} + D_{Ci} = 1, \qquad D_{Ai} = 1 - D_{Bi} - D_{Ci}.$$

Stacking the values over observations gives $\mathbf{D}_A = \mathbf{1} - \mathbf{D}_B - \mathbf{D}_C$. Therefore $[\mathbf{1}\ \mathbf{D}_A\ \mathbf{D}_B\ \mathbf{D}_C]$ has four columns but rank only 3 when all three groups occur. Its four coefficients cannot be uniquely identified: adding a constant to the intercept and subtracting it from each group coefficient leaves every fitted mean unchanged.
The original matrix $[\mathbf{1}\ \mathbf{D}_B\ \mathbf{D}_C]$ has no such redundancy. Also, $\mathbf{D}_A$ is **not** the intercept column: $\mathbf{D}_A$ is 1 only for A observations, whereas $\mathbf{1}$ is 1 for every group.

An equivalent model without a separate intercept. We can instead use all three indicators and write

$$\mathbb{E}(Y_i \mid G_i) = \mu_A D_{Ai} + \mu_B D_{Bi} + \mu_C D_{Ci}.$$

Each coefficient now represents its group's mean directly. In R, `y ~ 0 + group` removes the intercept term. Its design matrix makes the distinction clear:

```
ch5_X_means <- model.matrix(~ 0 + group, data = ch5_groups)
knitr::kable(
  data.frame(Group = ch5_groups$group, ch5_X_means,
             check.names = FALSE), row.names = FALSE, align = "c")
```

Group	groupA	groupB	groupC
A	1	0	0
B	0	1	0
C	0	0	1

Here `groupA`, `groupB`, and `groupC` are the three indicator columns. Both valid codings span the same mean vectors and give the same fitted values, residuals, and SSE. Removing the separate intercept here does **not** force the A mean to zero; it is estimated by the `groupA` coefficient.

Check the column counts and ranks in R:

```
ch5_X_redundant <- cbind(`(Intercept)` = 1, ch5_X_means)
ch5_codings <- list(ch5_X_reference, ch5_X_redundant, ch5_X_means)
knitr::kable(data.frame(
  Coding = c("Intercept + B + C", "Intercept + A + B + C", "A + B + C"),
  Columns = vapply(ch5_codings, ncol, integer(1)),
  Rank = vapply(ch5_codings, function(X) qr(X)$rank, integer(1))),
  row.names = FALSE, align = "c")
```

Coding	Columns	Rank
Intercept + B + C	3	3
Intercept + A + B + C	4	3
A + B + C	3	3

The two full-rank codings each estimate three group means; the four-column coding adds a redundant coefficient, not another identifiable mean.

Example 5.2 (Different Codings, the Same Group Means in R)
Suppose we have nine observations, divided into three groups of three. The sample means for A, B, and C are 18, 22, and 19. To interpret the printed t tests, assume independent normal

errors with a common variance, as in Chapter 3.

```
ch5_coding_data <- data.frame(
  group = factor(rep(c("A", "B", "C"), each = 3),
                 levels = c("A", "B", "C")),
  y = c(16, 18, 20, 20, 22, 24, 17, 19, 21))
contrasts(ch5_coding_data$group) <-
  contr.treatment(levels(ch5_coding_data$group), base = 1)
ch5_fit_reference <- lm(y ~ group, data = ch5_coding_data)
ch5_fit_means <- lm(y ~ 0 + group, data = ch5_coding_data)
```

With an intercept: A is the reference. Display the coefficient table from R's `summary()` output:

```
coef(summary(ch5_fit_reference))
```

```
            Estimate Std. Error    t value     Pr(>|t|)
(Intercept)       18   1.154701 15.5884573 4.412347e-06
groupB             4   1.632993  2.4494897 4.982526e-02
groupC             1   1.632993  0.6123724 5.627645e-01
```

The estimates are $\hat{\beta}_0 = 18$, $\hat{\beta}_B = 4$, and $\hat{\beta}_C = 1$. The intercept estimates the A mean; `groupB` estimates the B-minus-A difference $22 - 18$; `groupC` estimates the C-minus-A difference $19 - 18$.
Without an intercept: each coefficient is a group mean.

```
coef(summary(ch5_fit_means))
```

```
       Estimate Std. Error  t value     Pr(>|t|)
groupA       18   1.154701 15.58846 4.412347e-06
groupB       22   1.154701 19.05256 1.351732e-06
groupC       19   1.154701 16.45448 3.210729e-06
```

Now `groupA`, `groupB`, and `groupC` estimate μ_A, μ_B, and μ_C directly: 18, 22, and 19. The `groupB` standard error is about 1.155 here, compared with 1.633 for the B-minus-A coefficient above, because the two standard errors refer to different quantities.
Read the null hypothesis before comparing p-values. In the intercept model, the `groupB` row tests $H_0 : \mu_B - \mu_A = 0$ (p-value about 0.0498). In the no-intercept model, its row tests $H_0 : \mu_B = 0$ (p-value about 1.35×10^{-6}). These are different tests. To test B versus A with the no-intercept model, use the contrast $\mu_B - \mu_A$, whose estimate is still $22 - 18 = 4$; see the contrasts appendix.
Change the reference group to B. We keep the same observations and make B the first factor level:

```
ch5_coding_B <- ch5_coding_data
ch5_coding_B$group <- relevel(ch5_coding_B$group, ref = "B")
ch5_fit_reference_B <- lm(y ~ group, data = ch5_coding_B)
```

Reference	Intercept	Other coefficients
A	18	B minus A: 4; C minus A: 1
B	22	A minus B: -4; C minus B: -3

With B as the reference, the intercept is the B mean, 22. The other coefficients compare A and C with B. Changing the reference changes those comparisons, while the fitted group means remain 18, 22, and 19.

Check that all three codings give the same fit.

```
all.equal(fitted(ch5_fit_reference), fitted(ch5_fit_means))
```

```
[1] TRUE
```

```
all.equal(residuals(ch5_fit_reference), residuals(ch5_fit_means))
```

```
[1] TRUE
```

```
all.equal(fitted(ch5_fit_reference), fitted(ch5_fit_reference_B))
```

```
[1] TRUE
```

```
all.equal(residuals(ch5_fit_reference), residuals(ch5_fit_reference_B))
```

```
[1] TRUE
```

```
c(SSE_reference_A = deviance(ch5_fit_reference),
  SSE_no_intercept = deviance(ch5_fit_means),
  SSE_reference_B = deviance(ch5_fit_reference_B))
```

```
 SSE_reference_A SSE_no_intercept  SSE_reference_B
              24               24               24
```

All four equality checks return `TRUE`, and all three SSEs are 24. The coefficient meanings and their individual tests change, while the fitted group means, residuals, and overall fit stay the same.

Exercise 5.2 (Count Columns, Not Variable Names)
A model contains an intercept, two continuous predictors, and one four-level factor using reference coding. Assume all columns are independent and $n = 40$.

1. How many regression parameters are there?
2. What are the residual degrees of freedom?
3. Would replacing the three factor indicators by all four indicators while retaining the

intercept add an identifiable parameter?

Solution 5.2
There are $p = 1 + 2 + 3 = 6$ parameters and $40 - 6 = 34$ residual degrees of freedom. **No:** all four indicators together with the intercept introduce a redundant column.

Section 5.3 A continuous predictor and a group indicator

R's `ToothGrowth` data record odontoblast length in 60 guinea pigs. Each animal received vitamin C at 0.5, 1, or 2 mg/day, delivered as orange juice (`OJ`) or ascorbic acid (`VC`).

Question. How does the mean response vary with dose, and is the OJ-minus-VC mean difference the same at every dose?

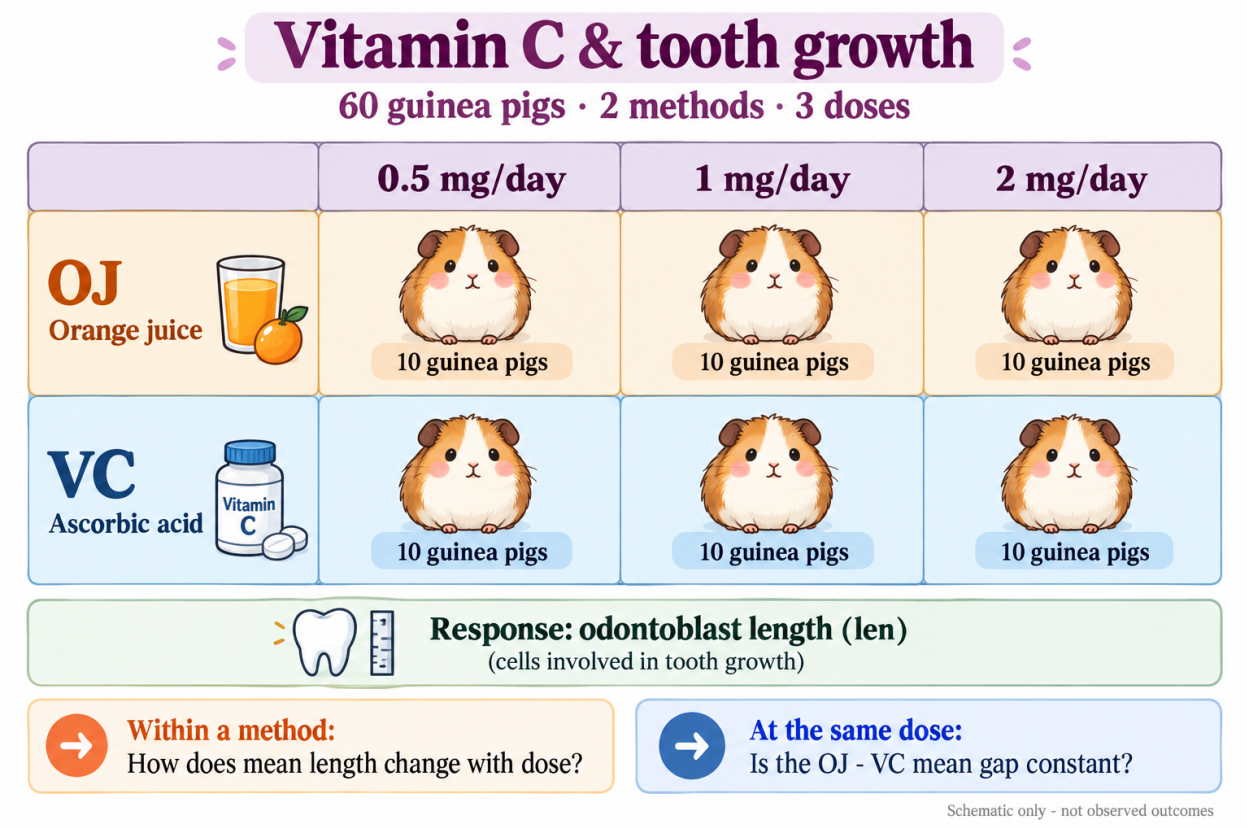


Figure 5.2: Each cell represents 10 animals. Compare doses across a row and OJ versus VC within each column. Does the mean gap change across columns? This is a schematic, not observed outcomes.

```
ch5_tooth <- datasets::ToothGrowth
ch5_tooth$supp <- factor(ch5_tooth$supp, levels = c("VC", "OJ"))
ch5_tooth$dose_f <- factor(ch5_tooth$dose, levels = c(0.5, 1, 2))
```

```
contrasts(ch5_tooth$supp) <-
  contr.treatment(levels(ch5_tooth$supp), base = 1)
contrasts(ch5_tooth$dose_f) <-
  contr.treatment(levels(ch5_tooth$dose_f), base = 1)
stopifnot(!anyNA(ch5_tooth))
ch5_cell_summary <- aggregate(
  len ~ supp + dose, data = ch5_tooth,
  FUN = function(y) c(n = length(y), mean = mean(y)))
ch5_summary_table <- data.frame(ch5_cell_summary[1:2],
                                ch5_cell_summary$len)
```

```
knitr::kable(ch5_summary_table, digits = 2, align = "c",
  col.names = c("Method", "Dose (mg/day)", "n", "Mean length"))
```

Method	Dose (mg/day)	n	Mean length
VC	0.5	10	7.98
OJ	0.5	10	13.23
VC	1.0	10	16.77
OJ	1.0	10	22.70
VC	2.0	10	26.14
OJ	2.0	10	26.06

Both methods have 10 animals at each dose, allowing comparisons at the same observed dose.

i Define centered dose as $x_c = x - 1$

We choose **1 mg/day as a reference dose** and let $x_c = x - 1$ measure the difference from it, in mg/day:

Actual dose x	Centered dose x_c
0.5	-0.5
1	0
2	1

Thus $x_c = 0$ means **1 mg/day**, and $x_c = -0.5$ means 0.5 mg/day, not a negative administered dose. A one-unit change in x_c is still a one-mg/day change in x; we shift the origin without changing the units.

Why choose 1? Both methods were observed at 1 mg/day, whereas zero dose was not observed. Using this reference lets the intercept describe a mean within the observed dose range instead of a mean extrapolated to zero dose. Another scientifically useful reference, such as 0.5 or 2 mg/day, could also be used; centering is optional, and the reference need not be the sample mean.

With an intercept and the corresponding model terms retained, centering changes coefficient meanings but leaves fitted values, residuals, and SSE unchanged. In the additive model below only the intercept changes; Section 5.4 explains why a group coefficient also changes when there is an interaction.

```
ch5_tooth$dose_c <- ch5_tooth$dose - 1
```

Definition 5.4 (Additive Continuous-by-Group Model)
Let x be dose, $x_c = x - 1$, and $z = \mathbb{1}\{\text{OJ}\}$. The additive model is

$$\mu(x, z) = \beta_0 + \beta_x x_c + \beta_g z.$$

Method	Mean function	Slope in dose
VC ($z = 0$)	$\beta_0 + \beta_x(x - 1)$	β_x
OJ ($z = 1$)	$\beta_0 + \beta_g + \beta_x(x - 1)$	β_x

The lines are parallel, with constant OJ-minus-VC difference β_g. Centering dose at 1 makes β_0 the VC mean at **1 mg/day**. The design row is $(1\ x_c\ z)$, so $p = 3$ when full rank.

```
ch5_fit_add <- lm(len ~ dose_c + supp, data = ch5_tooth)
```

The fitted mean is

$$\hat{\mu}(x, z) = 15.336 + 9.764(x - 1) + 3.700z.$$

At a fixed method, a one-mg/day increase corresponds to **9.764** more units of fitted mean length. At a fixed dose, OJ exceeds VC by **3.700** units. This model imposes a straight-line dose response and the same slope for both methods.

Fitted values at the observed doses. For animal i, the fitted value is $\hat{y}_i = \hat{\mu}(x_i, z_i)$. Animals with the same method and dose receive the same fitted value, although their observed lengths can differ. Compare the mean length in each method-dose group with predictions from the fitted parallel lines:

```
ch5_additive_fitted <- ch5_summary_table[c("supp", "dose", "mean")]
ch5_additive_fitted$supp <- factor(ch5_additive_fitted$supp,
                                   levels = c("VC", "OJ"))
ch5_additive_fitted$dose_c <- ch5_additive_fitted$dose - 1
ch5_additive_fitted$fitted <- predict(
  ch5_fit_add, newdata = ch5_additive_fitted)
knitr::kable(
  ch5_additive_fitted[c("supp", "dose", "mean", "fitted")],
  digits = 2, row.names = FALSE, align = "c",
  col.names = c("Method", "Dose (mg/day)", "Mean length", "Fitted value"))
```

Method	Dose (mg/day)	Mean length	Fitted value
VC	0.5	7.98	10.45
OJ	0.5	13.23	14.15
VC	1.0	16.77	15.34
OJ	1.0	22.70	19.04
VC	2.0	26.14	25.10
OJ	2.0	26.06	28.80

The fitted OJ-minus-VC difference is 3.70 at every dose. Fitted values need not equal the group mean lengths: the additive model fits all six cells using two parallel straight lines.

Read the new zero point in the additive model. At $x = 1$, we have $x_c = 0$, so the VC mean is β_0 and the OJ mean is $\beta_0 + \beta_g$. The constant OJ-minus-VC difference is still β_g at every dose. If we instead use uncentered x, the intercept is the VC mean at $x = 0$, namely $\beta_0 - \beta_x$.

Compare the same fit in R using the two dose variables:

```
ch5_fit_add_raw <- lm(len ~ dose + supp, data = ch5_tooth)
knitr::kable(data.frame(
  Term = c("Intercept", "VC dose slope", "OJ coefficient"),
  Using_dose = unname(coef(ch5_fit_add_raw)),
  Using_centered_dose = unname(coef(ch5_fit_add))),
  digits = 4, row.names = FALSE, align = "c",
  col.names = c("Term", "Using dose", "Using dose - 1"))
```

Term	Using dose	Using dose - 1
Intercept	5.5725	15.3361
VC dose slope	9.7636	9.7636
OJ coefficient	3.7000	3.7000

```
all.equal(fitted(ch5_fit_add_raw), fitted(ch5_fit_add))
```

```
[1] TRUE
```

The intercept moves from about 5.57 (the fitted VC mean at zero dose) to 15.34 (the fitted VC mean at 1 mg/day). The common dose slope and the constant group difference are unchanged. `TRUE` confirms that both parameterizations give the same fitted values.

Section 5.4 Interactions and conditional comparisons

Why consider an interaction? The additive model in Section 5.3 assumes that increasing dose has the same effect for VC and OJ. Its fitted lines are parallel, so its OJ-minus-VC difference must stay at 3.70 for every dose. That is a restriction imposed by the model.

The observed mean differences are 5.25 at 0.5 mg/day, 5.93 at 1 mg/day, and -0.08 at 2 mg/day. These sample differences can vary because of random error, but they motivate a question: **does the dose slope depend on the delivery method?** Equivalently, does the OJ-minus-VC mean difference change with dose?

Adding $x_c z$ lets the two groups have different slopes. For VC ($z = 0$), the term is zero; for OJ ($z = 1$), it adds β_{xg} to the dose slope. Thus $\beta_{xg} = 0$ recovers the additive model, while a nonzero interaction allows the group difference to vary linearly with dose. Comparing the models evaluates whether the data support this extra flexibility; unequal sample differences alone do not establish an interaction. Both models still assume straight lines within each group.

Definition 5.5 (Continuous-by-Group Interaction)
Add the product column $x_c z$:

$$\mu(x, z) = \beta_0 + \beta_x x_c + \beta_g z + \beta_{xg} x_c z.$$

Method	Mean function	Slope in dose
VC	$\beta_0 + \beta_x(x - 1)$	β_x
OJ	$(\beta_0 + \beta_g) + (\beta_x + \beta_{xg})(x - 1)$	$\beta_x + \beta_{xg}$

The OJ-minus-VC mean difference now depends on dose:

$$\Delta(x) = \beta_g + \beta_{xg}(x - 1).$$

Here β_g is the difference at 1 mg/day, and β_{xg} is the difference in slopes. The design row is $(1\ x_c\ z\ x_c z)$, giving $p = 4$ when full rank.

With an interaction, the reference dose also defines the group comparison

At $x = 1$, both x_c and $x_c z$ are zero. Consequently,

$$\mu(1,0) = \beta_0, \qquad \mu(1,1) = \beta_0 + \beta_g, \qquad \Delta(1) = \beta_g.$$

The intercept is the VC mean at **1 mg/day**, and the group coefficient is the OJ-minus-VC difference **at that same dose**. At another dose the difference is $\Delta(x) = \beta_g + \beta_{xg}(x-1)$, so it need not equal β_g.
Without centering, the group coefficient would describe the difference at **zero dose**, $\Delta(0) = \beta_g - \beta_{xg}$, outside the observed dose range. This is the key difference from Section 5.3: in an additive model the group difference is constant, but with an interaction changing the reference dose changes which group difference the coefficient represents. The VC slope β_x and slope difference β_{xg} stay the same.

In R, `dose_c * supp` expands to `dose_c + supp + dose_c:supp`. The last design column is zero for VC and equals centered dose for OJ:

```
ch5_profiles <- expand.grid(dose_c = c(-0.5, 0, 1),
                            supp = c("VC", "OJ"))
ch5_profiles$supp <- factor(ch5_profiles$supp, levels = c("VC", "OJ"))
contrasts(ch5_profiles$supp) <- contr.treatment(levels(ch5_profiles$supp), base = 1)
```

```
knitr::kable(model.matrix(~ dose_c * supp, data = ch5_profiles),
             row.names = FALSE, align = "c")
```

(Intercept)	dose_c	suppOJ	dose_c:suppOJ
1	-0.5	0	0.0
1	0.0	0	0.0
1	1.0	0	0.0
1	-0.5	1	-0.5
1	0.0	1	0.0
1	1.0	1	1.0

```
ch5_fit_int <- lm(len ~ dose_c * supp, data = ch5_tooth)
```

The fitted mean is

$$\hat{\mu}(x,z) = 15.011 + 11.716(x-1) + 4.351z - 3.904(x-1)z.$$

The VC slope is **11.716**; the OJ slope is **7.811**, using unrounded estimates. Their fitted mean difference is about **4.351** at 1 mg/day and **0.446** at 2 mg/day. The negative interaction means the gap decreases with dose, while both slopes remain positive.

Verify what centering changes in the interaction fit.

```
ch5_fit_int_raw <- lm(len ~ dose * supp, data = ch5_tooth)
knitr::kable(data.frame(
  Term = c("Intercept", "VC dose slope", "OJ coefficient",
           "Slope difference"),
  Using_dose = unname(coef(ch5_fit_int_raw)),
  Using_centered_dose = unname(coef(ch5_fit_int))),
  digits = 4, row.names = FALSE, align = "c",
  col.names = c("Term", "Using dose", "Using dose - 1"))
```

Term	Using dose	Using dose - 1
Intercept	3.2950	15.0107
VC dose slope	11.7157	11.7157
OJ coefficient	8.2550	4.3507
Slope difference	-3.9043	-3.9043

```
all.equal(fitted(ch5_fit_int_raw), fitted(ch5_fit_int))
```

```
[1] TRUE
```

```
all.equal(deviance(ch5_fit_int_raw), deviance(ch5_fit_int))
```

```
[1] TRUE
```

The OJ coefficient changes from **8.255**, the model's extrapolated difference at zero dose, to **4.351**, its difference at 1 mg/day. Using the uncentered coefficients gives the same difference at 1 mg/day: approximately $8.255 - 3.904(1) = 4.351$. The VC intercept similarly changes from 3.295 at zero dose to 15.011 at 1 mg/day. Both equality checks return `TRUE`: centering moves the reference point, not the fitted lines or SSE.

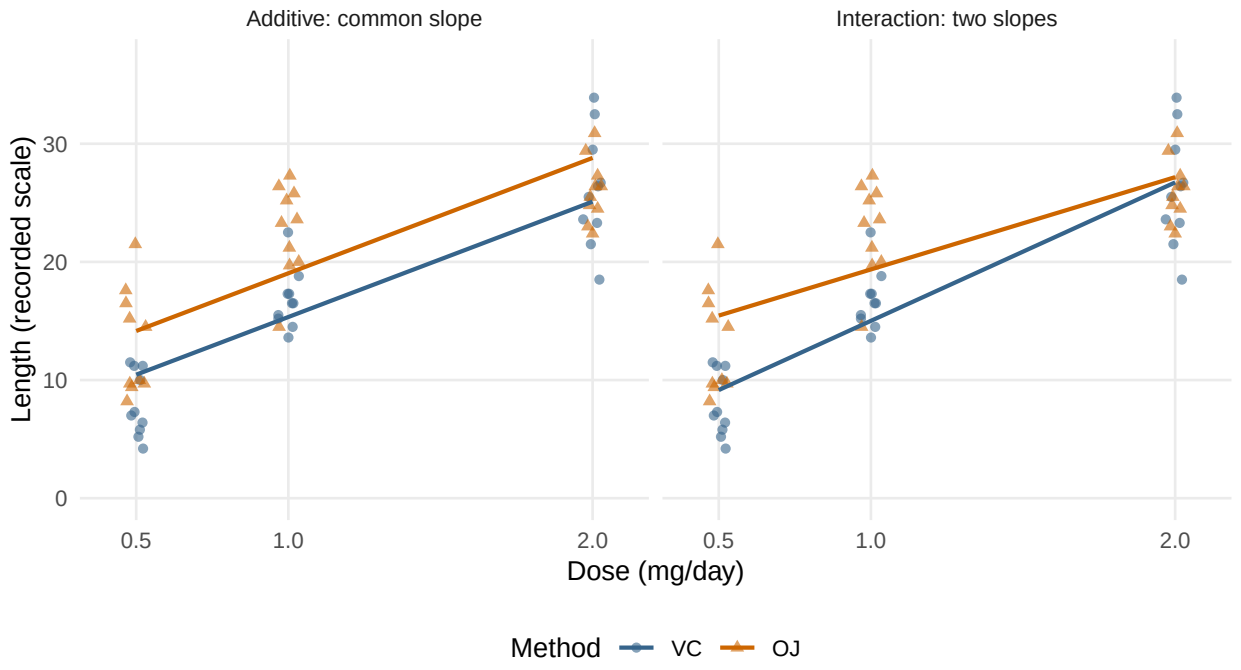


Figure 5.3: The same observations fitted with parallel lines (additive) or separate slopes (interaction). Horizontal jitter is for display only; both models use the original doses.

Two choices of reference. Centering at dose c makes the interaction model's group coefficient the difference at c. Separately, choosing OJ instead of VC as the reference group makes the main dose coefficient the OJ slope. With all model terms retained, both recodings preserve fitted values and SSE.

! Keep the lower-order terms

Retain `dose_c` and `supp` when interpreting `dose_c:supp`. Dropping a main effect imposes a baseline-dependent restriction; its individual p-value is not a reason to remove it automatically.

Exercise 5.3 (Which Difference Is Six?)
Suppose $x_c = x - 1$ and $\hat{\mu} = 10 + 2x_c + 6z - x_c z$.

1. What is the slope for $z = 1$?
2. At which original x value is the group difference 6?
3. What is the group difference at $x = 3$?

Solution 5.3
The slope for $z = 1$ is $2 - 1 = 1$. The group difference is 6 at $x = 1$, where $x_c = 0$. At $x = 3$, $x_c = 2$ and the difference is $6 - 2 = 4$.

Section 5.5 Match the hypothesis to the model comparison

Apply the partial F test from Chapter 4, using the same observations and the normal-model assumptions from the recap. The new task is to choose the restriction:

Question	Reduced model	Full model
Equal slopes: $\beta_{xg} = 0$?	`dose_c + supp`	`dose_c * supp`
Identical lines: $\beta_g = \beta_{xg} = 0$?	`dose_c`	`dose_c * supp`
Straight-line means within each method?	`dose_c * supp`	`dose_f * supp`

The first test removes only the slope difference; the second removes both the slope and intercept differences. For the third, treating dose as a factor permits **six separate method-dose means** ($p = 6$), rather than two straight lines ($p = 4$).

```
ch5_fit_dose <- lm(len ~ dose_c, data = ch5_tooth)
ch5_fit_cells <- lm(len ~ dose_f * supp, data = ch5_tooth)
ch5_tests <- list(
  "Equal slopes" = anova(ch5_fit_add, ch5_fit_int),
  "Identical lines" = anova(ch5_fit_dose, ch5_fit_int),
  "Straight-line means" = anova(ch5_fit_int, ch5_fit_cells))
ch5_test_results <- do.call(rbind, lapply(names(ch5_tests), function(name) {
  a <- ch5_tests[[name]]
  data.frame(Comparison = name,
    df = sprintf("(%d, %d)", a$Df[2], a$Res.Df[2]),
    F = round(a$F[2], 3),
    p_value = format.pval(a$`Pr(>F)`[2], digits = 3, eps = 0.0001))
}))
knitr::kable(ch5_test_results, row.names = FALSE, align = "c",
             col.names = c("Comparison", "F degrees of freedom", "F", "p-value"))
```

Comparison	F degrees of freedom	F	p-value
Equal slopes	(1, 56)	5.333	0.0246
Identical lines	(2, 56)	8.825	0.000466
Straight-line means	(2, 54)	8.399	0.000667

At the **5% significance level**, the three tests lead to these conclusions:

- **Equal slopes:** $H_0 : \beta_{xg} = 0$. With $F_{1,56} = 5.333$ and $p = 0.0246 < 0.05$, reject equal slopes. Within the straight-line model, allowing different slopes improves the fit; the additive model's parallel-line restriction is not supported.
- **Identical lines:** $H_0 : \beta_g = \beta_{xg} = 0$. With $F_{2,56} = 8.825$ and $p = 0.000466 < 0.05$, reject a single common line. Within the two-line model, the methods differ in intercept, slope, or both. This joint test alone does not establish that both coefficients are nonzero.

- **Straight-line means:** H_0 says that the three population means for each method lie on a straight line. With $F_{2,54} = 8.399$ and $p = 0.000667 < 0.05$, reject this restriction. Even allowing different slopes does not adequately describe the six method-dose means. Use the **six-cell factor-dose model** for dose-specific comparisons. The first two tests compare line models; this third test shows why separate lines are still insufficient. In the plot below, the mean lengths at 1 mg/day lie above the fitted lines.

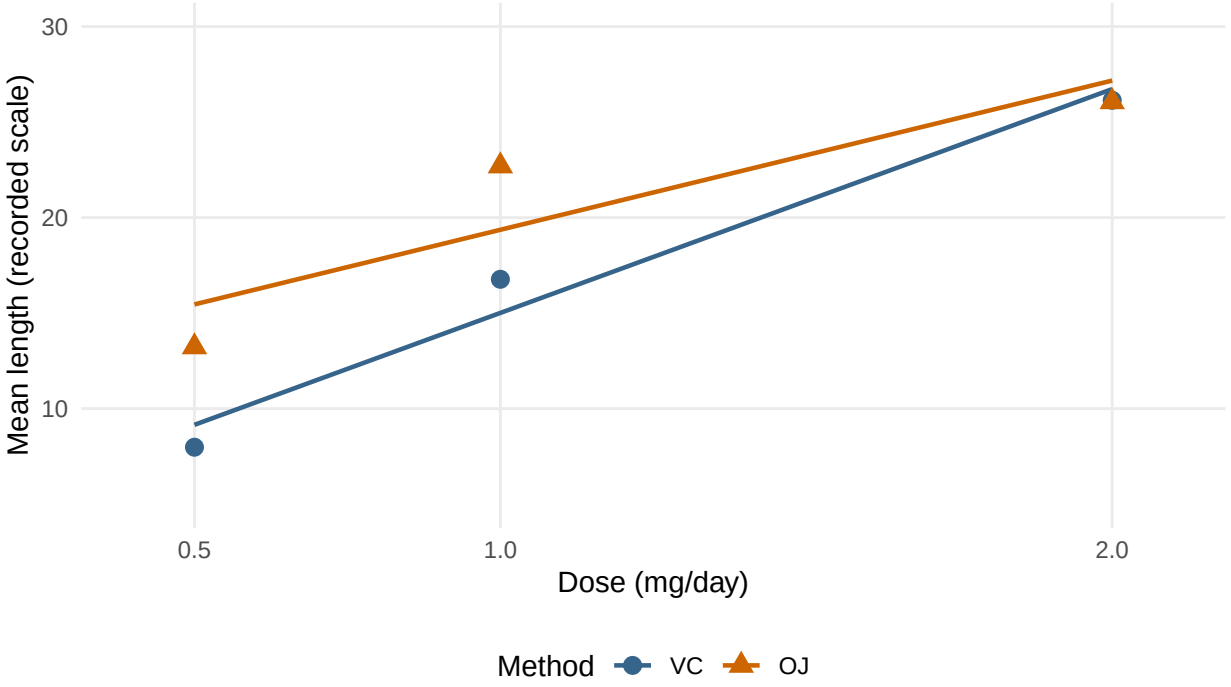


Figure 5.4: Points show the six observed method-dose means; lines show the interaction fit. A factor-dose model reproduces each point, while straight lines cannot.

Section 5.6 Summary

Choose the model by the question you want to answer.

Question	R model	Interpretation
Describe the overall association with `x`.	`y ~ x`	Unadjusted slope; see Section 5.1.
Compare `x` at fixed values of other predictors.	`y ~ x + w`	Adjusted slope, holding `w` fixed; see Section 5.1.
Estimate the means of several groups.	`y ~ group`	Reference mean plus group differences; see Section 5.2.
Assume a common dose slope for both methods.	`len ~ dose_c + supp`	Parallel lines and a constant group gap; see Section 5.3.
Allow dose slopes to differ by method.	`len ~ dose_c * supp`	A group gap that changes linearly with dose; see Section 5.4.
Compare methods at each observed dose without assuming straight lines.	`len ~ dose_f * supp`	Six separate method-dose means; see Section 5.5.

Choose coding for the quantities you want to read. For a group-only model, an intercept makes the coefficients a reference mean and differences from it; `y ~ 0 + group` makes them the group means themselves. Both give the same fitted values, but their individual coefficient tests ask different questions. Use one intercept with $g-1$ indicators, or no intercept with all g indicators.

Choose a meaningful reference dose. With $x_c = x - 1$, the intercept is the VC mean at 1 mg/day. In the interaction model, the group coefficient is the OJ-minus-VC difference at that dose; elsewhere use $\Delta(x) = \beta_g + \beta_{xg}(x-1)$. Centering preserves fitted values.

For these data, use the six-cell model for dose-specific comparisons. Section 5.5 finds evidence against both equal slopes and straight-line shape. Its comparison table distinguishes the tests for equal slopes, identical lines, and straight-line means; choose the test that matches the question.

Section 5.7 Practice Problems

Exercise 5.4 (Concept Check: True or False?)

1. With an intercept, a three-level factor contributes two independent indicator columns when all levels are observed.
2. Changing the reference group changes the fitted means and SSE.
3. A negative interaction coefficient implies a negative slope in the non-reference group.
4. Replacing numeric dose by a three-level factor must preserve the fitted means.

Solution 5.4

1. **True.** One level is the reference.
2. **False.** The same mean model is represented with different coefficients.
3. **False.** The non-reference slope is the main slope plus the interaction coefficient.
4. **False.** Factor coding can relax the straight-line restriction.

Exercise 5.5 (Practice: State the Restrictions and Degrees of Freedom)
Consider $n = 48$ observations, a binary group z, and a continuous predictor x. Assume the relevant design matrices have full rank and the normal-model assumptions hold. The full model is $\mu = \beta_0 + \beta_x x + \beta_g z + \beta_{xg} xz$.

1. Write the null hypothesis for parallel lines and give both F degrees of freedom.
2. Write the null hypothesis for identical lines and give both F degrees of freedom.
3. Express the group difference at $x = 2$ as a linear combination of coefficients.
4. If the full model has SSE 440 and the additive model has SSE 500, compute the partial F statistic for equal slopes.

Solution 5.5

1. $H_0 : \beta_{xg} = 0$; degrees of freedom $(1, 44)$.
2. $H_0 : \beta_g = \beta_{xg} = 0$; degrees of freedom $(2, 44)$.
3. $\Delta(2) = \beta_g + 2\beta_{xg}$, with column vector $\mathbf{l} = (0, 0, 1, 2)$ in the displayed coefficient order.
4. $F = \{(500 - 440)/1\}/(440/44) = 6$.

Section 6

Outliers, Leverage, and Influence

A fitted regression line summarizes the observations, but some observations may matter much more than others. We ask three different questions: **is the response unusual given the predictors, are the predictor values unusual, and does the observation substantially change the fit?** These lead to outliers, leverage, and influence. We use the same small dataset throughout so that every diagnostic can be connected to a visible change in the estimates.

Learning Objectives

By the end of this chapter, students should be able to:

- distinguish a response outlier, a high-leverage observation, and an influential observation;
- compare coefficient estimates and predictions with and without one observation;
- interpret leverage, standardized residuals, and externally studentized residuals;
- explain how Cook's distance combines response discrepancy and predictor position;
- investigate a flagged observation and report a sensitivity analysis.

Quick Recap: What We Will Use

- **Mean model and design — Chapter 1.** Write $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, with $\mathbb{E}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \mathbf{0}$. A row describes an observation; a column multiplies a coefficient. Let p count the columns, including the intercept. We assume full column rank and $n > p$.
- **Fitted values and residuals — Chapter 2.** OLS gives $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ and $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$. A residual is an observed deviation from the fitted mean, not the unobserved error.
- **Noise and uncertainty — Chapter 3.** Under $\mathrm{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2\mathbf{I}$, estimate σ^2 by $s^2 = \mathrm{SSE}/(n-p)$. Here $\mathrm{SSE} = \sum_i e_i^2$ is the unexplained variation from Chapter 4. Estimated coefficient covariance is $s^2(\mathbf{X}^\top\mathbf{X})^{-1}$.
- **Interpret the quantity that changes — Chapter 5.** An additive slope describes a mean-response difference holding other predictors fixed. To assess influence, compare the coefficients and predictions relevant to that question, using the same model specification.

Section 6.1 See how one observation changes the fit

First look at the data and the fitted lines. We will name the three concepts after seeing how they differ, then introduce their numerical measures.

Example 6.1 (Three Points, Three Different Stories)

Look at the lines first. Every panel starts with the same six green observations, whose fitted line is $\hat{y} = 2 + x$. We add **one orange point at a time**. The dashed gray line omits that point; the solid purple line includes it. Compare the point's vertical position, its horizontal position, and how much the line moves.

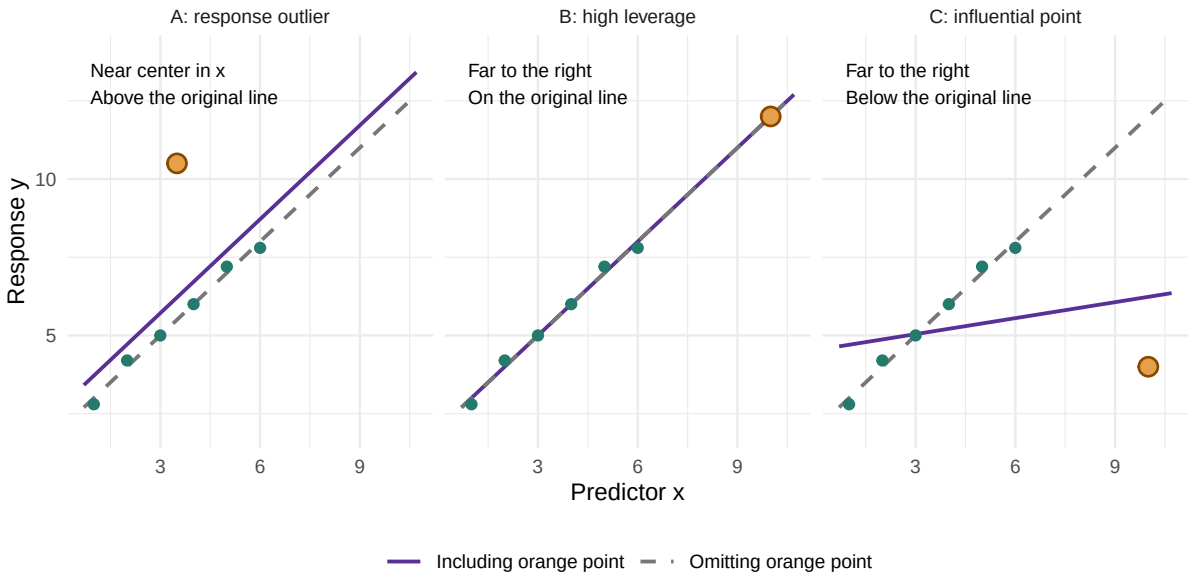


Figure 6.1: The same six baseline observations appear in every panel. A shifts the line upward without changing its slope. B lies far to the right but leaves the line unchanged. C is equally far to the right and pulls the line downward, sharply reducing its slope.

- **A: unusual response, ordinary predictor value.** At $x = 3.5$, the original line predicts 5.5, but the orange response is 10.5. This point is near the center of the predictor values and far above the line. It mainly shifts the line upward.
- **B: high leverage without a change in the fitted line.** The new $x = 10$ lies beyond the original range of 1 to 6. Its response, 12, lies exactly on the extension of the original line. The point has the potential to pull the fit, but here the two lines coincide.
- **C: high leverage with a large change in the fitted line.** The predictor is again $x = 10$, but the response is 4 instead of 12. The point pulls down the right side of the line, substantially reducing the estimated slope.

These descriptions can overlap. C is also a high-leverage point; its **actual effect on the fit** makes it influential. A also affects the fit, even though its effect on the slope differs from C's.
Now put numbers on the changes. The six baseline observations are:

Variable	Obs. 1	Obs. 2	Obs. 3	Obs. 4	Obs. 5	Obs. 6
x	1.0	2.0	3.0	4.0	5.0	6.0
y	2.8	4.2	5.0	6.0	7.2	7.8

Fit `lm(y ~ x)` to the baseline data, then add A, B, or C separately. The table reports the intercept, slope, fitted mean at $x = 6$, and residual standard deviation s. Removing the orange point from any of the three augmented datasets returns the baseline fit.

Fit	Intercept	Slope	Fitted mean at 6	Residual SD
Baseline	2.000	1.000	8.000	0.200
Baseline + A	2.714	1.000	8.714	2.078
Baseline + B	2.000	1.000	8.000	0.179
Baseline + C	4.532	0.170	5.553	1.899

- **A changes the intercept and the noise estimate.** The intercept rises from 2.000 to 2.714, while the slope stays at 1.000. Here $x = 3.5$ is the baseline mean of x, so this added point does not change the slope. The residual SD rises from 0.200 to 2.078: an outlying response can inflate estimated noise even when the slope is unchanged.
- **B leaves both coefficient estimates unchanged.** The fitted mean at $x = 6$ remains 8.000. High leverage alone does not imply that a point changes the estimated line; it can still affect the precision of the estimates.
- **C changes the estimated relationship.** The slope falls from 1.000 to 0.170 and the intercept rises to 4.532. The fitted mean at $x = 6$ falls from 8.000 to 5.553. This is a substantial change in a prediction within the original predictor range, not just at the distant added point.

These pictures motivate three distinct definitions.

Definition 6.1 (Outlier, Leverage, and Influence)

- A **response outlier** has an unusual response *given its predictor values and the specified model*. An extreme response value alone does not make an observation a response outlier.
- A **high-leverage observation** has an unusual configuration of predictor values. Leverage concerns the predictors jointly and does not depend on the observed responses. It measures the potential to pull the fit.
- An **influential observation** changes the fitted coefficients or predictions substantially when omitted. Influence concerns the sensitivity of the analysis, rather than unusualness alone.

Section 6.2 High leverage: unusual predictor values

Leverage asks **where the observation sits among the predictor rows**. It does not use its response. From the hat matrix, define h_{ii} as diagonal entry i. With an intercept and one predictor,

$$h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

The first term is shared by all observations; the second grows as x_i moves away from the predictor center. The mean and sum in this formula use **all observations in the fit being diagnosed**, including the point of interest. With several predictors, leverage measures unusualness of the whole predictor combination, not just one variable at a time.

Point	Added x	Leverage
A	3.500	0.143
B	10.000	0.721
C	10.000	0.721

For A, $x = 3.5$ remains the predictor mean, so its leverage is $1/7 = 0.143$. B and C have the **same predictor rows**, so both have leverage 0.721 even though their responses are different. B leaves the estimated line unchanged; C changes it substantially. Thus high leverage indicates **potential influence**, not a conclusion that the fit has changed.

Because $\sum_i h_{ii} = p$, average leverage is p/n. In this example, $p = 2$ and $n = 7$, so the average is $2/7 = 0.286$. Comparing a value with $2p/n$ is a rough screening rule, not a test or a reason to delete an observation. R returns leverages with `hatvalues(fit)`.

Derivation: leverage and its average

1. Center the predictor without changing the fitted model. Set $z_i = x_i - \bar{x}$, $\mathbf{z} = (z_1, \ldots, z_n)^\top$, and $S_{xx} = \sum_i z_i^2 > 0$. The columns of $\mathbf{Z} = (\mathbf{1}, \mathbf{z})$ span the same space as $(\mathbf{1}, \mathbf{x})$, so they give the same hat matrix. Since $\mathbf{1}^\top \mathbf{z} = 0$,

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} n & 0 \\ 0 & S_{xx} \end{pmatrix}, \qquad \mathbf{H} = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top = \frac{\mathbf{1}\mathbf{1}^\top}{n} + \frac{\mathbf{z}\mathbf{z}^\top}{S_{xx}}.$$

2. Read diagonal entry i. It is

$$h_{ii} = \frac{1}{n} + \frac{z_i^2}{S_{xx}} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{\sum_j (x_j - \bar{x})^2}.$$

3. Sum the leverages in a general full-rank model. By the cyclic property of the trace,

$$\sum_i h_{ii} = \operatorname{tr}(\mathbf{H}) = \operatorname{tr}\left[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\right] = \operatorname{tr}(\mathbf{I}_p) = p.$$

Thus average leverage is p/n. This uses the projection matrix from Chapter 2, not the responses.

A well-measured high-leverage observation can be useful: under an appropriate model it can provide information about the slope. In B, the slope estimate stays at 1.000 while its estimated standard error falls from 0.048 to 0.024. Whether the model is credible at that distant predictor value still needs scientific judgment.

Section 6.3 Outliers: unusual responses given the predictors

An outlier is unusual **relative to the mean predicted by the specified model**. B has the largest response in the example, 12, yet it lies on the fitted line and has zero residual. A's response is smaller, 10.5, but is far above the expected response near $x = 3.5$. Looking only at the distribution of Y would miss this distinction.

First fix the notation. Let $e_i = Y_i - \hat{Y}_i$ be the raw residual from the full fit, h_{ii} its leverage, and $\nu = n - p$ its residual degrees of freedom. Under the mean model and constant error variance,

$$\mathrm{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}).$$

Different observations therefore have different residual variances. Both residuals below account for this using $\sqrt{1 - h_{ii}}$; they differ in **how the unknown error standard deviation σ is estimated**. Assume $h_{ii} < 1$ and positive estimated error variance.

Definition 6.2 (Standardized Residual (Internally Studentized))
Estimate the error variance using **all n observations**:

$$s^2 = \frac{\mathrm{SSE}}{n - p} = \frac{\sum_{j=1}^{n} e_j^2}{n - p}.$$

The standardized residual is

$$r_i = \frac{e_i}{s\sqrt{1 - h_{ii}}}.$$

Observation i helps determine both the fitted line and s. An unusual response can inflate s and make its own scaled residual appear less extreme. This residual is also called **internally studentized**. In R, use `rstandard(fit)` with its default settings.

Definition 6.3 (Externally Studentized Residual (Studentized Deleted Residual))
Refit the **same model after removing observation** i, giving $\hat{\boldsymbol{\beta}}_{(-i)}$. Estimate the error variance from the remaining $n - 1$ observations:

$$s^2_{(-i)} = \frac{\mathrm{SSE}_{(-i)}}{n - p - 1}, \qquad \mathrm{SSE}_{(-i)} = \sum_{j \neq i} \left(Y_j - \mathbf{x}_j^\top \hat{\boldsymbol{\beta}}_{(-i)}\right)^2.$$

The externally studentized residual is

$$t_i = \frac{e_i}{s_{(-i)}\sqrt{1 - h_{ii}}}.$$

Here e_i **and** h_{ii} **still come from the full fit**; only the error-scale estimate is replaced. The deleted fit must retain rank p, with $n - p - 1 > 0$ and $s_{(-i)} > 0$. In R, use `rstudent(fit)`.

In this chapter, **standardized** means internally studentized, and **studentized** by itself means externally studentized, matching these two R diagnostic functions. Other texts may use *studentized* for either type; check which variance estimate appears in the denominator.

Feature	Standardized r_i	Externally studentized t_i
Numerator and leverage	Full-fit e_i, h_{ii}	Same full-fit e_i, h_{ii}
Error-scale estimate	s, using all observations	$s_{(-i)}$, refitting without i
Error degrees of freedom	$n-p$	$n-p-1$
Main use	Compare residual sizes on a common scale	Assess a response without letting it inflate its own noise estimate

Neither scaling makes residuals independent. With independent normal errors and a correctly specified, fixed model, each pre-specified t_i has a t_{n-p-1} distribution; r_i does **not** have that exact t distribution. Checking many points also requires accounting for multiple comparisons.

Derivation: why divide by the square root of one minus leverage?

1. Express the residual vector using the errors. From the residual maker in Chapter 2, $\mathbf{M} = \mathbf{I} - \mathbf{H}$ is symmetric and idempotent, and $\mathbf{MX} = \mathbf{0}$. Therefore

$$\mathbf{e} = \mathbf{MY} = \mathbf{M}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \mathbf{M}\boldsymbol{\varepsilon}.$$

2. Compute its covariance. Under $\mathrm{Var}(\boldsymbol{\varepsilon} \mid \mathbf{X}) = \sigma^2\mathbf{I}$,

$$\mathrm{Var}(\mathbf{e} \mid \mathbf{X}) = \mathbf{M}(\sigma^2\mathbf{I})\mathbf{M}^\top = \sigma^2\mathbf{M}^2 = \sigma^2(\mathbf{I} - \mathbf{H}).$$

The ith diagonal entry is $\sigma^2(1 - h_{ii})$, so the true standard deviation of e_i is $\sigma\sqrt{1 - h_{ii}}$. Replacing σ with s gives r_i; replacing it with $s_{(-i)}$ gives t_i.

3. Keep variance scaling separate from independence. For $i \neq j$, $\mathrm{Cov}(e_i, e_j \mid \mathbf{X}) = -\sigma^2 h_{ij}$, generally nonzero. Dividing residuals by estimated scales does not remove their shared dependence on the fitted model. A high-leverage point has a smaller raw-residual variance, which is why raw residuals alone are not comparable.

Derivation: deleting one observation and relating the two residuals

1. Find the coefficient change. Write $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$ and let $\mathbf{x}_i$ be the column vector whose transpose is row i. The full and deleted normal equations are

$$\mathbf{A}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top\mathbf{Y}, \qquad (\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^\top)\hat{\boldsymbol{\beta}}_{(-i)} = \mathbf{X}^\top\mathbf{Y} - \mathbf{x}_i Y_i.$$

For $\boldsymbol{\delta}_i = \hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(-i)}$, subtracting gives

$$(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^\top)\boldsymbol{\delta}_i = \mathbf{x}_i e_i.$$

Since $h_{ii} = \mathbf{x}_i^\top\mathbf{A}^{-1}\mathbf{x}_i$, $(\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^\top)\mathbf{A}^{-1}\mathbf{x}_i = (1 - h_{ii})\mathbf{x}_i$. When the deleted fit is estimable, this identifies the solution:

$$\boldsymbol{\delta}_i = \frac{\mathbf{A}^{-1}\mathbf{x}_i e_i}{1 - h_{ii}}.$$

2. Distinguish a deleted residual from a full-fit residual. Predict the omitted response from the deleted fit:

$$d_i = Y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}_{(-i)} = e_i + \mathbf{x}_i^\top \boldsymbol{\delta}_i = \frac{e_i}{1-h_{ii}}.$$

Thus d_i is not the numerator e_i in the displayed definition of t_i.

3. Recalculate the residual sum of squares. At all original rows, the deleted-fit residual vector is $\mathbf{e} + \mathbf{X}\boldsymbol{\delta}_i$. Remove its ith squared component d_i^2 and use $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$:

$$\begin{aligned}
\mathrm{SSE}_{(-i)} &= \|\mathbf{e} + \mathbf{X}\boldsymbol{\delta}_i\|^2 - d_i^2 \\
&= \mathrm{SSE} + \boldsymbol{\delta}_i^\top \mathbf{A}\boldsymbol{\delta}_i - d_i^2 \\
&= \mathrm{SSE} + \frac{e_i^2 h_{ii}}{(1-h_{ii})^2} - \frac{e_i^2}{(1-h_{ii})^2} \\
&= \mathrm{SSE} - \frac{e_i^2}{1-h_{ii}}.
\end{aligned}$$

It is **not** simply $\mathrm{SSE} - e_i^2$: refitting changes the other residuals too.

4. Relate the two scales and residuals. With $\nu = n-p$, $\mathrm{SSE} = \nu s^2$, and $r_i^2 = e_i^2/[s^2(1-h_{ii})]$,

$$s_{(-i)}^2 = s^2 \frac{\nu - r_i^2}{\nu - 1}, \qquad t_i = r_i \frac{s}{s_{(-i)}} = r_i \sqrt{\frac{\nu - 1}{\nu - r_i^2}}.$$

For positive deleted-fit variance, $r_i^2 < \nu$. A very unusual point can make r_i approach its finite bound while t_i becomes much larger. The two values have the same sign, but need not have similar magnitudes; $|t_i|$ is not always larger than $|r_i|$.

Derivation: why the externally studentized residual has a t distribution

Assume $\boldsymbol{\varepsilon} \mid \mathbf{X} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$ and fix the model and index i in advance. Use the normal-theory OLS results in Chapter 3 on the deleted dataset.

1. Standardize the deleted prediction error. Because Y_i is independent of the observations used to fit $\hat{\boldsymbol{\beta}}_{(-i)}$,

$$\begin{aligned}
\mathbb{E}(d_i \mid \mathbf{X}) &= 0, \\
\mathrm{Var}(d_i \mid \mathbf{X}) &= \sigma^2[1 + \mathbf{x}_i^\top (\mathbf{A} - \mathbf{x}_i\mathbf{x}_i^\top)^{-1}\mathbf{x}_i] \\
&= \sigma^2\left(1 + \frac{h_{ii}}{1-h_{ii}}\right) = \frac{\sigma^2}{1-h_{ii}}.
\end{aligned}$$

The inverse identity follows from step 1 of the deletion derivation. As a linear combination of normal observations, $Z_i = d_i\sqrt{1-h_{ii}}/\sigma \sim N(0,1)$.

2. Use an independent variance estimate. The deleted fit has $n-1-p = \nu - 1$ residual degrees of freedom, so

$$U_i = \frac{(\nu-1)s^2_{(-i)}}{\sigma^2} \sim \chi^2_{\nu-1}.$$

Under normality, the deleted-fit coefficient estimate is independent of its SSE. Also, the held-out Y_i is independent of that entire deleted dataset. Hence d_i, and therefore Z_i, is independent of U_i.

3. Form the t ratio. By the definition of the t distribution,

$$\frac{Z_i}{\sqrt{U_i/(\nu-1)}} = \frac{d_i\sqrt{1-h_{ii}}}{s_{(-i)}} = \frac{e_i}{s_{(-i)}\sqrt{1-h_{ii}}} = t_i \sim t_{\nu-1}.$$

The internal residual r_i uses the full-fit s, which shares the contribution from observation i with its numerator. This independence argument does not apply to r_i; its bound $|r_i| \le \sqrt{\nu}$ also rules out an exact t distribution, whose support is unbounded.

Compare the same three added points. Each row uses its own augmented fit. The s column uses all seven observations; $s_{(-i)}$ comes from refitting after removing that row's added point. The last column is the externally studentized residual.

Point	Residual e_i	s	$s_{(-i)}$	Standardized r_i	Studentized t_i
A	4.286	2.078	0.200	2.228	23.146
B	0.000	0.179	0.200	0.000	0.000
C	-2.234	1.899	0.200	-2.226	-21.138

For A, the full-fit residual SD is $s = 2.078$, but deleting A gives $s_{(-A)} = 0.200$. The same full-fit residual and leverage therefore produce $r_A = 2.228$ and $t_A = 23.146$. A was inflating the noise estimate used to assess itself. For C the scales are 1.899 and 0.200, with the same effect. Deleting either point returns the six baseline observations from Section 6.1. B has zero residual, so both scaled residuals are zero regardless of the different scales. These diagnostics identify a response discrepancy under the model; they do not establish a recording error.

A and C have similar absolute standardized residuals, yet C changes the slope much more. **Outlyingness and influence answer different questions.**

Section 6.4 Influence: changes in estimates and Cook's distance

Influence asks **how much the fitted result changes when an observation is omitted**. Compare the same model on the remaining observations: changes in the slope, intercept, or scientifically relevant prediction describe what that observation was doing. Cook's distance summarizes movement of the fitted mean vector using both the residual and leverage.

Definition 6.4 (Cook's Distance)
Let $\hat{\boldsymbol{\beta}}_{(-i)}$ be the OLS estimate after omitting observation i, and let $\hat{Y}_{j(-i)} = \mathbf{x}_j^\top \hat{\boldsymbol{\beta}}_{(-i)}$ be its prediction at **each of the original n predictor rows**, including row i. Cook's distance is

$$D_i = \frac{\sum_{j=1}^{n}(\hat{Y}_j - \hat{Y}_{j(-i)})^2}{ps^2} = \frac{e_i^2}{ps^2}\frac{h_{ii}}{(1-h_{ii})^2} = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}.$$

Here s^2 comes from the full fit. These formulas apply to ordinary unweighted least squares with full column rank and positive residual degrees of freedom; the deleted fit must also be estimable ($h_{ii} < 1$).

Read the two parts. The first expression asks how much the *whole fitted mean vector* moves after deletion, measured relative to the noise scale and model size. The last combines response discrepancy (r_i^2) with leverage ($h_{ii}/(1-h_{ii})$). Thus the largest residual need not produce the largest Cook's distance. D_i is dimensionless; it is not a p-value. Cook's original article.

Derivation: the equivalent formulas for Cook's distance

1. Express movement of the fitted means through the coefficient change. Use $\boldsymbol{\delta}_i = \hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(-i)}$ from the deletion derivation in Section 6.3. Predictions are evaluated at **all n original rows**, so

$$\sum_{j=1}^{n}(\hat{Y}_j - \hat{Y}_{j(-i)})^2 = \|\mathbf{X}\boldsymbol{\delta}_i\|^2 = \boldsymbol{\delta}_i^\top \mathbf{A}\boldsymbol{\delta}_i.$$

2. Substitute the deleted-coefficient identity. With $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$,

$$\begin{aligned}\boldsymbol{\delta}_i^\top \mathbf{A}\boldsymbol{\delta}_i &= \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i^\top \mathbf{A}^{-1}\mathbf{A}\mathbf{A}^{-1}\mathbf{x}_i \\ &= \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}.\end{aligned}$$

3. Scale the change and rewrite using r_i. Dividing by ps^2 gives

$$D_i = \frac{e_i^2}{ps^2}\frac{h_{ii}}{(1-h_{ii})^2} = \frac{r_i^2}{p}\frac{h_{ii}}{1-h_{ii}}, \qquad r_i^2 = \frac{e_i^2}{s^2(1-h_{ii})}.$$

Cook's distance uses the **full-fit** noise scale s, so this expression uses the internally studentized r_i. Substituting the externally studentized t_i into the same expression would not give Cook's distance. The algebra does not require normal errors; the variance model supplies the interpretation of the scaling.

Apply Cook's distance to the same three points. In each augmented fit, $n = 7$ and $p = 2$. The diagnostics for its orange point are:

Point	Leverage	Standardized residual	Cook's distance
A	0.143	2.228	0.414
B	0.721	0.000	0.000
C	0.721	-2.226	6.395

A and C have almost the same absolute standardized residual, but C has much higher leverage and a much larger Cook's distance: 6.395 versus 0.414. B has the same leverage as C, yet its zero residual gives zero Cook's distance. These numbers quantify what the plots and coefficient table already showed: **influence depends on both the response discrepancy and the predictor position**.

Exercise 6.1 (Which Point Moves the Fit More?)
In a model with $p = 2$, point A has $r_A = 3$ and $h_{AA} = 0.05$; point B has $r_B = 1$ and $h_{BB} = 0.50$. Compute their Cook's distances. Which point is more influential by this measure? Does either number establish a data error?

Solution 6.1
$D_A = (9/2)(0.05/0.95) \approx 0.237$, while $D_B = (1/2)(0.50/0.50) = 0.50$. B has greater influence despite its smaller standardized residual. Neither value establishes an error or justifies deletion; inspect the observations and the scientific consequences of the changed fit.

Section 6.5 Decide what is unusual, influential, and worth investigating

Make two separate judgments: **is the response unusual under the model, and does the observation materially affect the fitted result?** A point can satisfy both. A diagnostic flag tells us what to investigate; it does not by itself tell us to remove the observation.

1. Use a criterion that matches the question.

- **Response outlier:** inspect the externally studentized residual t_i from Section 6.3. A large $|t_i|$ indicates an unusual response given the predictors and model. The familiar $|t_i| > 2$ rule is only a rough screen; use the t-based rule below when making a formal outlier assessment.
- **High leverage:** compare h_{ii} with the average p/n. The rule $h_{ii} > 2p/n$ flags unusual predictor positions for inspection. It does not establish a response outlier or actual influence.
- **Possible influence:** inspect Cook's distances and flag values that stand out, with $D_i > 4/n$ as one rough screening guide. **Confirm the practical importance by comparing fits with and without the point.** There is no universal significance cutoff for Cook's distance, and a value below $4/n$ does not establish that a point is harmless.

A formal outlier check under the normal linear model. Let $d = n - p - 1$ and let $t_{d,q}$ denote the q quantile of the t distribution with d degrees of freedom. Under the fixed, correctly specified normal model in Section 6.3, a two-sided level-α check for **one pre-specified observation** uses $|t_i| > t_{d,1-\alpha/2}$. If all n observations are inspected, use a multiplicity adjustment. A Bonferroni rule controlling the probability of at least one false flag at at most α under the model is

$$|t_i| > t_{d,\,1-\alpha/(2n)}.$$

Equivalently, compute the two-sided t p-value and multiply it by n, capped at 1. This is the approach used by `car::outlierTest()`. Rejection is evidence of a response outlier **relative to the specified model**, not proof of a data error. Failure to reject does not prove that the observation is ordinary; poor model fit and multiple unusual points can hide one another.

Apply the rules to A, B, and C. Within each separate augmented dataset, $n = 7$, $p = 2$, and $d = 4$. At $\alpha = 0.05$, the pre-specified-point cutoff is 2.776, whereas inspecting all seven points gives a Bonferroni cutoff of **5.068**. The leverage and Cook screening values happen to be the same here: $2p/n = 4/n = 0.571$.

- **A is a response outlier with ordinary leverage.** Its $|t_A| = 23.146$ exceeds 5.068, while $h_{AA} = 0.143$ is not high. Its $D_A = 0.414$ does not cross the Cook screen, but Section 6.1 shows changes in the intercept and noise estimate. Report those effects; do not conclude that A has no impact merely because it does not cross that screen.
- **B has high leverage but is not a response outlier and does not change the fitted mean.** Its $h_{BB} = 0.721$ is high, but $t_B = 0$ and $D_B = 0$. It still affects precision. If its measurements and membership in the target population are valid, its high leverage alone is no reason to exclude it.
- **C is both a response outlier and influential, with high leverage.** Its $|t_C| = 21.138$ exceeds 5.068, $h_{CC} = 0.721$, and $D_C = 6.395$. Omitting C changes the slope from 0.170 back to 1.000 and the fitted mean at $x = 6$ from 5.553 back to 8.000. The observed change in the fit confirms the influence suggested by Cook's distance.

2. Quantify influence for the question being answered. Refit the same model with and without each flagged point. Compare the relevant coefficients, standard errors or confidence intervals, and predictions at the same predictor values. State the size of the change in the original units and whether it alters a scientific conclusion or decision. A small effect on one slope does not imply a small effect on every coefficient, prediction, or uncertainty estimate. Single-case deletion can also miss the joint effect of several unusual observations.

3. Decide what to do using evidence about the data and model.

- **Confirmed recording or measurement error:** correct it from the original record, then refit and repeat the diagnostics. If it cannot be recovered and is unusable, document the data-quality reason for excluding it.
- **Observation outside the intended population or study protocol:** apply the same justified eligibility rule to every observation, and report the exclusion. Do not invent a rule because a result is inconvenient.
- **Valid but unusual observation:** retain it in the main analysis. If it materially changes the result, also report the analysis without it as a sensitivity check and explain the difference. A small p-value or large Cook's distance alone is not a reason to delete valid data.
- **Valid observations exposing an inadequate model:** investigate the mean function, error pattern, and omitted structure. Revise the model when scientifically justified and repeat the diagnostics; Chapter 7 develops transformations and model terms. Do not try to repair a systematic model problem by removing individual rows.

4. Report the decision and the sensitivity. For example: "C has externally studentized residual -21.138 and Cook's distance 6.395. With C, the slope is 0.170 and the fitted mean at $x = 6$ is 5.553; without C, they are 1.000 and 8.000. If the record is verified as valid and eligible, we retain C and report both fits because the conclusion is sensitive to it." State what was checked, what was retained, corrected, or excluded, and why.

Section 6.6 Summary

Concept	Question	Useful measure	What it does not establish
Response outlier	Is the response unusual given the predictors and model?	Standardized or externally studentized residual	That the observation has high leverage or must be deleted.
High leverage	Is the predictor configuration unusual?	Hat-matrix diagonal h_{ii}	That the observation actually changes the estimated line.
Influence	Does omitting the observation change the fitted result?	Coefficient/prediction comparison and Cook's distance	That the observation is a data error.

Follow Section 6.5: **screen, verify the data, compare fits, act on evidence, and report sensitivity**. Outlier evidence concerns the response under the model; influence concerns changes in the fitted result. Chapter 7 considers how to revise the functional form and select model terms.

Section 6.7 Practice Problems

Exercise 6.2 (Practice: Diagnose and Explain)

1. B has a larger response than A but a zero residual. Why is the response value alone insufficient to identify an outlier?
2. Why do B and C have the same leverage but different Cook's distances?
3. In A, the slope is unchanged but the intercept and residual SD increase. Is it correct to say the point has no effect on estimation?
4. A high-leverage point has a small raw residual. Explain why this alone does not establish that it is harmless.
5. A student deletes every point with $D_i > 4/n$. State what information and analysis are missing.

Solution 6.2

1. Outlyingness concerns the response relative to the fitted conditional mean. B lies on the original line; A lies well above it at an ordinary predictor value.
2. Their predictor rows are identical, so leverage is identical. Their responses differ: B leaves the line unchanged, while C moves it substantially.
3. No. A affects the intercept, predictions, and estimated noise even though its position at the predictor mean leaves the slope unchanged.
4. A high-leverage observation can pull the fit toward itself. Examine a scaled residual and compare the fitted result with and without that observation.

5. The cutoff is only a screening guide. Verify the observation, examine the model, compare scientifically relevant estimates and predictions with and without the point, and justify any exclusion using evidence.

Section 7

Transformations and Variable Selection

Chapter 6 examined how individual observations affect a fitted model. We now ask **which functions of the predictors to use, and which terms to retain**. First choose and check a plausible functional form, then compare variable-selection and shrinkage procedures, and finally assess predictions on observations kept out of model development.

Learning Objectives

By the end of this chapter, students should be able to:

- distinguish a curved relationship from a model nonlinear in its parameters;
- choose, fit, and check a suitable transformation;
- interpret predictions after transforming the response;
- carry out forward selection, backward elimination, and stepwise selection;
- explain ridge and lasso, and choose a penalty using cross-validation;
- separate model development from final prediction assessment.

i Quick Recap: Tools for Model Development

- Chapter 2: OLS fits the mean patterns represented by the design columns.
- Chapter 3: coefficient tests and intervals rely on assumptions for a specified model.
- Chapter 4: compare nested models on the same observations and response; a lower training SSE alone does not establish better prediction.
- Chapter 5: interpret coefficients conditionally, code factors as whole terms, and retain the lower-order terms needed for interactions.
- Chapter 6: distinguish response outliers, leverage, and influence; investigate flags and recheck sensitivity when the model changes.

Section 7.1 A curved relationship can still be a linear model

Definition 7.1 (Linear in the Parameters)
A regression model is linear in its parameters when it can be written as $Y_i = \beta_0 + \sum_{j=1}^{d} \beta_j h_j(\mathbf{x}_i) + \varepsilon_i$, where the functions h_j are specified independently of the unknown coefficients. The design row is $(1 \; h_1(\mathbf{x}_i) \; \cdots \; h_d(\mathbf{x}_i))$; the response need not be a straight-line function of the original predictors.

Relationship	Model fitted by least squares	What changes?
Curvature in x	$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$	Add a polynomial column; retain the response scale.
Diminishing change, $x > 0$	$Y = \beta_0 + \beta_1 \log x + \varepsilon$	Transform a predictor.
Exponential growth, $Y > 0$	$\log Y = \beta_0 + \beta_1 x + \varepsilon$	Transform the response; specify errors on the log scale.
Power relationship, $Y, x > 0$	$\log Y = \beta_0 + \beta_1 \log x + \varepsilon$	Transform both variables.

In R, write a quadratic as `lm(y ~ x + I(x^2), data = dat)`; `I()` makes the square arithmetic. For a log-log model, use `lm(log(y) ~ log(x), data = dat)`. An interaction from Chapter 5 is another constructed column, $x_1 x_2$.

! Transform the statistical model, including its errors

$Y = a\exp(bx)\exp(\varepsilon)$ becomes $\log Y = \log a + bx + \varepsilon$. In contrast, taking logs of $Y = a\exp(bx) + \varepsilon$ does **not** produce that same additive-error model. Likewise, a saturation model such as $Y = ax/(b + x) + \varepsilon$ is nonlinear in a, b; reciprocating it changes the errors and may be inappropriate. Some problems need nonlinear least squares or another response model, not forced linearization.

Exercise 7.1 (Linear in Which Sense?)
Which can be fitted by `lm()` using known predictor columns?

1. $Y = \beta_0 + \beta_1\sqrt{x} + \varepsilon$, with $x \geq 0$.
2. $Y = \beta_0 + \beta_1 x^{\beta_2} + \varepsilon$, with unknown β_2.
3. $\log Y = \beta_0 + \beta_1 x + \varepsilon$, with $Y > 0$.

Solution 7.1
Models 1 and 3 are linear in their coefficients. Model 2 has an unknown exponent inside the predictor function; ordinary `lm()` cannot estimate all three coefficients as written. Model 3 models the conditional mean of $\log Y$, not directly the conditional mean of Y.

Section 7.2 Choose a transformation, fit, and check again

Motivation: measuring tree volume. Measuring a standing tree's diameter and height is easier than measuring its timber volume. The trees from Chapter 5 suggest a physical starting point: volume is roughly proportional to cross-sectional area times height, so a power relationship such as $V \approx aD^bH^c$ may be more natural than an additive straight-line model. The exponents are to be estimated, not fixed by the cylinder analogy. Data documentation.

A practical procedure.

1. **State the goal and inspect the data.** Identify the response units, positive/zero values, measurement process, and a scientifically plausible shape. For prediction, reserve test observations before exploring relationships.

2. **Fit a baseline and inspect residuals.** Curvature suggests a mean-shape problem; a fan-shaped spread suggests nonconstant variance. A skewed marginal response alone is not a reason to transform it: regression assumptions concern conditional errors.
3. **Choose a small candidate set.** A predictor transformation can address shape while keeping response units. A response transformation changes the target and error model; log responses are natural for positive multiplicative variation.
4. **Refit and recheck.** Examine residuals versus fitted values and predictors, a normal Q-Q plot when using normal-model inference, and influential observations. A transformation does not fix dependence or a poor study design.
5. **Interpret and validate.** Explain the new coefficients, back-transform appropriately, and compare predictions on a common scale. If choices are learned from data, repeat that learning within training folds.

Example 7.1 (From Tree Geometry to a Log-Linear Model)
For the 31 trees, consider

$$V_i = aD_i^b H_i^c \exp(\varepsilon_i) \quad \Longrightarrow \quad \log V_i = \beta_0 + b \log D_i + c \log H_i + \varepsilon_i, \qquad \beta_0 = \log a.$$

Here D is diameter in inches, H is height in feet, and V is volume in cubic feet. We use all 31 trees to illustrate specification and diagnostics; this is not a held-out performance claim.

```
library(ggplot2)
ch7_trees <- transform(datasets::trees, Diameter = Girth)
ch7_raw <- lm(Volume ~ Diameter + Height, data = ch7_trees)
ch7_log <- lm(log(Volume) ~ log(Diameter) + log(Height), data = ch7_trees)
knitr::kable(data.frame(Term = names(coef(ch7_log)),
  Estimate = unname(coef(ch7_log))), digits = 3, row.names = FALSE)
```

Term	Estimate
(Intercept)	-6.632
log(Diameter)	1.983
log(Height)	1.117

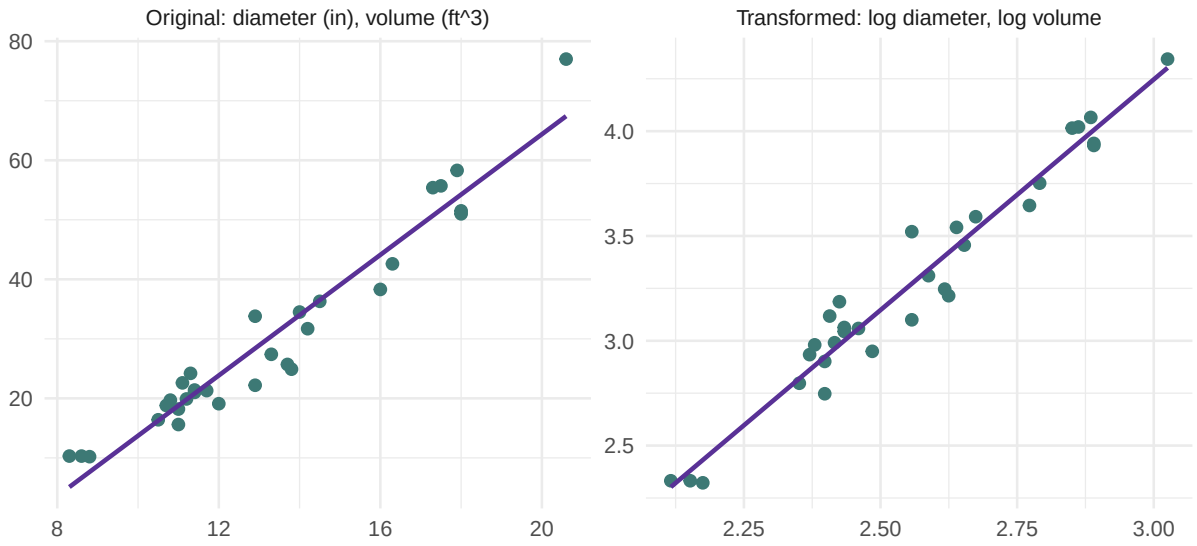


Figure 7.1: The same 31 trees on two scales, each with a simple least-squares line using diameter alone. The log-log view motivates a power relationship; the fitted multiple regression above also adjusts for height. This plot does not establish that the transformed model is adequate.

The estimated diameter and height exponents are about **1.983** and **1.117**, close to the geometric motivation but not constrained to it. Holding height fixed, multiplying diameter by r changes the fitted log-volume by $\hat{b} \log r$. Under the lognormal model below, the fitted median volume is multiplied by $r^{\hat{b}}$. Thus a 1% diameter increase corresponds approximately to a $\hat{b}$% change in that median; the exact percentage is $100(1.01^{\hat{b}} - 1)$.

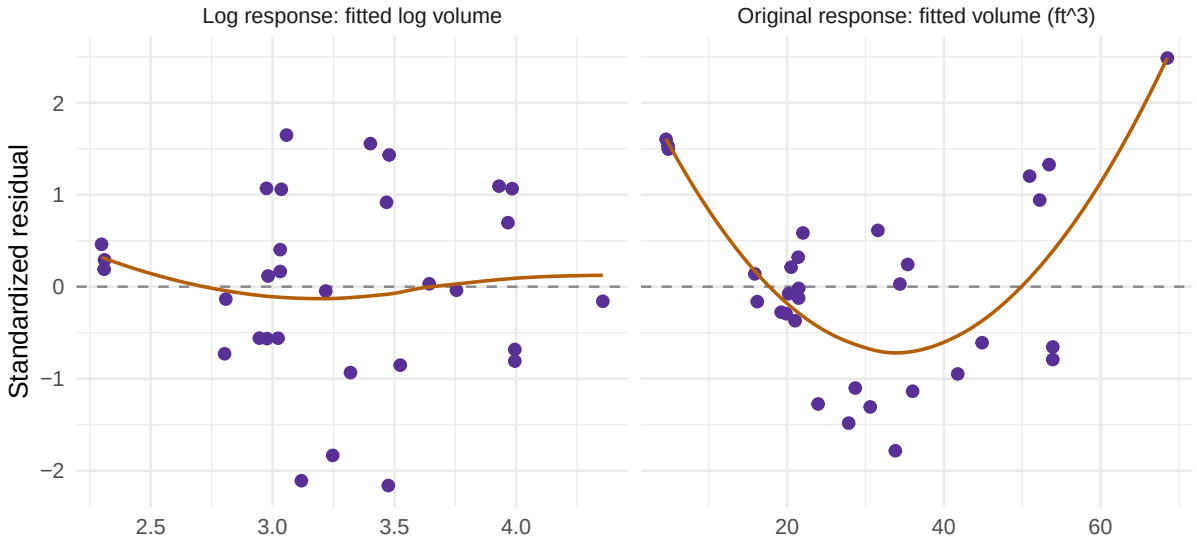


Figure 7.2: Standardized residuals from the additive original-scale model and the log-log model, both using diameter and height. The smooth orange line helps reveal remaining mean structure. Inspect each pattern; their fitted-value scales differ, so raw SSE or R-squared values across response scales would not be a fair comparison.

The log model is a candidate to examine, not a guaranteed cure. Both panels still deserve scrutiny. With only 31 observations, a large search over powers and interactions would be hard to justify.

Definition 7.2 (Back-Transformation Under Lognormal Errors)
If $\log Y = \eta(\mathbf{x}) + \varepsilon$ with $\varepsilon \mid \mathbf{x} \sim N(0, \sigma^2)$, then

$$\text{Median}(Y \mid \mathbf{x}) = \exp\{\eta(\mathbf{x})\}, \qquad \mathbb{E}(Y \mid \mathbf{x}) = \exp\{\eta(\mathbf{x}) + \sigma^2/2\}.$$

Exponentiating a fitted log mean therefore estimates a conditional median, not the original-scale mean. A plug-in mean prediction uses an estimate of σ^2; it is not an exactly unbiased estimator.

```
ch7_newtree <- data.frame(Diameter = 15, Height = 80)
ch7_eta <- as.numeric(predict(ch7_log, newdata = ch7_newtree))
knitr::kable(data.frame(
  `Median volume` = exp(ch7_eta),
  `Mean volume, lognormal correction` = exp(ch7_eta + sigma(ch7_log)^2 / 2),
  check.names = FALSE), digits = 2,
  caption = "Predicted cubic feet for diameter 15 inches and height 80 feet.")
```

Table 7.1: Predicted cubic feet for diameter 15 inches and height 80 feet.

Median volume	Mean volume, lognormal correction
37.82	37.94

With homoscedastic errors independent of the predictors, a less distribution-specific alternative is the smearing estimate $\exp(\hat{\eta})\, n^{-1} \sum_i \exp(e_i)$, where e_i are training log residuals. A single correction can fail if the error distribution changes with the predictors. Estimate any correction within each training fold when validating.

Other powers. For positive responses, Box-Cox considers $g_\kappa(y) = (y^\kappa - 1)/\kappa$ for $\kappa \neq 0$ and $g_0(y) = \log y$. Thus $\kappa = 1$ leaves the response shape unchanged, $\kappa = 1/2$ gives a square-root scale, and $\kappa = 0$ gives logs. `MASS::boxcox(ch7_raw)` profiles the transformation parameter for the specified mean model. Treat it as guidance, then refit and inspect; choosing the power is itself model selection. Box-Cox documentation.

⚠ Keep scales and data comparable

Do not rank an `lm(y ~ ...)` fit against an `lm(log(y) ~ ...)` fit using their raw SSE, R^2, or unadjusted software AIC. Their responses differ. Compare held-out predictions of the **same target in the same units**, or use likelihoods expressed on a common response scale with the transformation Jacobian. Logs require positive values; adding an arbitrary constant changes the model. Binary or count responses may instead call for a GLM.

Exercise 7.2 (What Does Exponentiating Predict?)
Suppose $\log Y = 2 + 0.4x + \varepsilon$, with $\varepsilon \sim N(0, 0.25)$. At $x = 0$, give the conditional median and mean of Y. Does increasing x by one add 0.4 units to the original-scale mean?

Solution 7.2
The median is $e^2 \approx 7.39$ and the mean is $e^{2+0.25/2} \approx 8.37$. A one-unit increase in x multiplies both by $e^{0.4} \approx 1.49$ because the log-error variance is constant. It does not add 0.4 response units.

Recheck influence after transforming. A new response scale changes residuals, and new predictor columns can change leverage. Apply the diagnostics from Chapter 6 to the fitted log model as well.

Example 7.2 (Does One Tree Drive the Power Relationship?)
Return to the log-volume model. A wrongly recorded diameter could distort a timber prediction rule, but an accurately measured unusual tree may reveal where the rule needs improvement. First identify observations worth inspecting:

```
ch7_diag <- data.frame(Tree = seq_len(nrow(ch7_trees)),
  Studentized = rstudent(ch7_log), Leverage = hatvalues(ch7_log),
  Cook = cooks.distance(ch7_log))
ch7_top <- head(order(ch7_diag$Cook, decreasing = TRUE), 3)
```

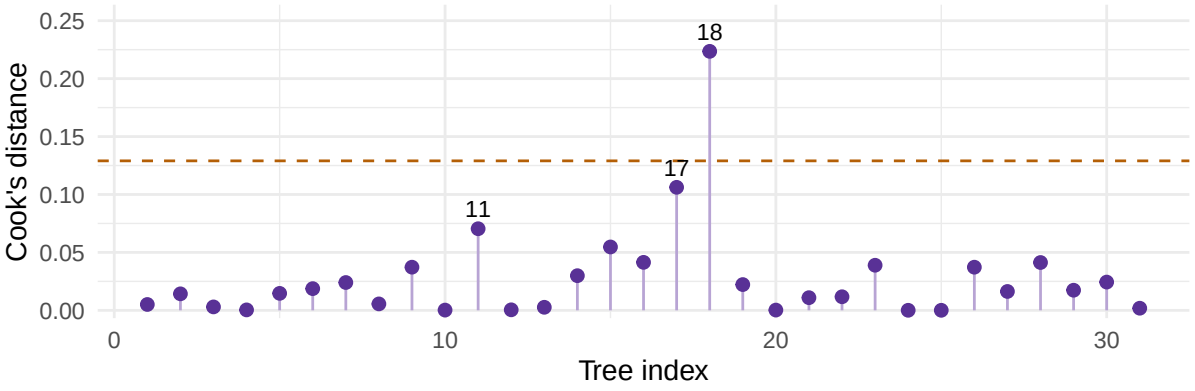


Figure 7.3: Cook's distances for the log-volume model. Labels mark the three largest values; the dashed 4/n line is only a screening guide. Tree 18 deserves a sensitivity check, not automatic deletion.

Tree	Studentized	Leverage	Cook
18	-2.326	0.126	0.224
17	1.598	0.116	0.106
11	1.703	0.072	0.070

Refit to understand sensitivity. Temporarily omit the largest-D_i tree and compare the same coefficients; keep the original fit as the reference.

```
ch7_flag <- which.max(ch7_diag$Cook)
ch7_log_without <- update(ch7_log, data = ch7_trees[-ch7_flag, ])
knitr::kable(data.frame(Term = names(coef(ch7_log)),
  All_trees = unname(coef(ch7_log)),
  Omit_flagged_tree = unname(coef(ch7_log_without))),
  digits = 3, row.names = FALSE,
  col.names = c("Term", "All trees", "Omit flagged tree"))
```

Term	All trees	Omit flagged tree
(Intercept)	-6.632	-7.185
log(Diameter)	1.983	1.958
log(Height)	1.117	1.261

Tree **18** has $D_i \approx 0.224$. Omitting it changes the diameter exponent from **1.983 to 1.958**, and the height exponent from **1.117 to 1.261**. The estimated height contribution is more sensitive here; influence can matter differently for different scientific questions. This calculation supplies no evidence of a recording error. Report the sensitivity and investigate the measurement before deciding how to handle it.

Why recheck before selection? One influential observation can make a predictor appear useful or change which term a search retains. Resolve data-quality questions and inspect model form first, then assess whether the selection is stable. Recheck the final selected model as well. Ridge and lasso still use squared-error loss here; shrinkage is not an automatic remedy for outliers. For prediction assessment, all outcome-driven diagnostic and model-development decisions belong within training data, not the held-out test set.

Section 7.3 From candidate terms to variable selection

After checking functional form and influential observations, return to the candidate terms. Transformations, indicators, polynomials, and interactions all create candidate columns. Adding all of them improves training fit but can make a model unstable and hard to interpret. Selection asks which **terms** to retain within a scientifically defensible candidate set.

Definition 7.3 (Variable Selection)
Variable selection chooses a subset of candidate predictors or model terms using a specified rule. A term may contain several columns, such as a factor. The candidate set, terms that must remain, comparison criterion, and stopping rule are part of the procedure.

Choose the rule before the search. For Gaussian models fitted to the same observations and response, $\text{AIC} = -2\widehat{\ell} + 2k$ and $\text{BIC} = -2\widehat{\ell} + k \log n$, where $k = p + 1$ includes p regression coefficients and the variance parameter. Smaller values are preferred. Ordinary R^2 alone cannot stop a search

because it never decreases when columns are added. These criteria are scores, not significance levels. R documentation.

Method	Start	One step	Stop
Forward selection	Required terms, often just the intercept	Try each admissible addition; take the greatest score improvement.	No addition improves the score.
Backward elimination	Full candidate model	Try each admissible removal; take the greatest improvement.	No removal improves the score.
Stepwise selection	A specified starting model	Reconsider additions and removals as the fit changes.	Neither move improves the score.

These are greedy searches; different starting points or directions can yield different models. **Best subset selection** evaluates every allowed subset and finds the best score within that finite set, but d optional terms can require 2^d fits. Backward elimination requires an estimable starting model with positive residual degrees of freedom for the usual Gaussian criterion.

! Protect model structure

Keep treatment assignments, blocking variables, and scientifically required adjustment terms when the question requires them. Retain parent terms with interactions and lower powers with polynomial terms. Search a multi-level factor as a whole term. R's formula-based `step()` respects interaction hierarchy, but it will not infer every scientific constraint, such as retaining `x` whenever `I(x^2)` is present.

Example 7.3 (Which Sensor Measurements Help Predict an Outcome?)
Imagine a manufacturing outcome and eight candidate sensor readings. We simulate data so the useful predictors are known: $Y = 5 + 3x_1 - 2x_2 + 1.5x_3 + \varepsilon$, with independent $N(0, 4)$ errors. Predictors x_1, x_2 are correlated; $x_4, \ldots, x_8$ have no additional mean contribution. All methods receive the same eight candidates. This is a teaching simulation, not an industrial dataset.

```r
set.seed(8561)
ch7_n <- 160
ch7_X <- matrix(rnorm(ch7_n * 8), nrow = ch7_n)
ch7_X[, 2] <- 0.7 * ch7_X[, 1] + sqrt(1 - 0.7^2) * ch7_X[, 2]
colnames(ch7_X) <- paste0("x", 1:8)
ch7_dat <- data.frame(y = 5 + 3 * ch7_X[, 1] - 2 * ch7_X[, 2] +
  1.5 * ch7_X[, 3] + rnorm(ch7_n, sd = 2), ch7_X)
ch7_id <- sample(seq_len(ch7_n), 120)
ch7_train <- ch7_dat[ch7_id, ]
ch7_test <- ch7_dat[-ch7_id, ]
ch7_folds <- sample(rep(1:5, length.out = nrow(ch7_train)))
```

We reserve 40 observations now and fit all searches on the other 120. The candidate set contains only main effects, so no polynomial or interaction hierarchy needs to be imposed in this example.

```r
ch7_null <- lm(y ~ 1, data = ch7_train, na.action = na.fail)
ch7_full <- lm(y ~ ., data = ch7_train, na.action = na.fail)
ch7_scope <- list(lower = formula(ch7_null), upper = formula(ch7_full))
ch7_forward <- step(ch7_null, scope = ch7_scope,
  direction = "forward", trace = 0)
ch7_backward <- step(ch7_full, scope = ch7_scope,
  direction = "backward", trace = 0)
ch7_both <- step(ch7_null, scope = ch7_scope,
  direction = "both", trace = 0)
ch7_searches <- list(Forward = ch7_forward, Backward = ch7_backward,
  Stepwise = ch7_both)
knitr::kable(data.frame(
  Search = names(ch7_searches),
  Selected = vapply(ch7_searches, function(m)
    paste(attr(terms(m), "term.labels"), collapse = ", "), character(1)),
  AIC = vapply(ch7_searches, AIC, numeric(1))), digits = 2, row.names = FALSE)
```

Search	Selected	AIC
Forward	x1, x2, x3, x4	494.56
Backward	x1, x2, x3, x4	494.56
Stepwise	x1, x2, x3, x4	494.56

Here all three searches retain x_1, x_2, x_3, x_4. The first three generate the mean, but x_4 is noise: selection can retain a chance association.
`step()` uses AIC by default; `k = log(nrow(ch7_train))` changes the penalty to the BIC-style rule. Fix the analysis rows across all candidates, especially when missing values occur. The trace uses `extractAIC()`, whose additive constant differs from `AIC()` for `lm`; compare scores computed with the same function. Selection documentation.

Exercise 7.3 (Take One Forward Step)
The current AIC is 210. Adding x_2, x_3, or x_4 gives 207, 209, or 212. Which move is made? At the next step, the best available AIC is 207.4. Continue or stop? Would a different answer prove that one variable has no scientific role?

Solution 7.3
Add x_2, producing the largest AIC decrease. Stop at the next step because 207.4 is worse than 207. This is a conditional search result under the chosen candidates and score, not proof that an excluded variable has no scientific role.

Section 7.4 Shrinkage: ridge, lasso, and elastic net

Subset selection makes discrete include-or-exclude decisions. **Regularization** instead penalizes coefficient size: it accepts some bias to reduce sensitivity to the sample. Lasso can also produce exact zeros.

Definition 7.4 (Penalized Least Squares)
Let $\mathbf{x}_i$ contain d candidate predictor columns, excluding the intercept, and let $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_d)$. On n_{tr} training observations, estimate the intercept and slopes by minimizing

$$\frac{1}{2n_{\text{tr}}} \sum_i (y_i - \beta_0 - \mathbf{x}_i^\top \boldsymbol{\beta})^2 + \lambda \left\{ \frac{1-\alpha}{2} \sum_j \beta_j^2 + \alpha \sum_j |\beta_j| \right\}.$$

The intercept is not penalized. Ridge uses $\alpha = 0$, lasso uses $\alpha = 1$, and elastic net uses $0 < \alpha < 1$. Larger λ puts more weight on shrinkage. At $\lambda = 0$ the objective is OLS; uniqueness still depends on the design rank.

Why can lasso give zeros? Near zero, a squared penalty becomes nearly flat, while an absolute-value penalty keeps a nonzero slope on each side of its corner. A small improvement in fit may therefore fail to move a lasso coefficient away from zero. Ridge usually reduces its magnitude continuously instead.

Method	Main behavior	Interpretation
Ridge	Shrinks slopes, usually without exact zeros.	Useful when many correlated columns carry information.
Lasso	Can set slopes exactly to zero.	A sparse prediction rule; correlated predictors can compete for selection.
Elastic net	Combines both penalties.	Can retain correlated predictors together; both α and λ need a selection rule.

Why scaling matters. Converting a predictor from meters to millimeters changes its coefficient by a factor of 1,000. Standardize predictor columns using training data so the penalty does not favor

a unit convention. In `glmnet`, `standardize = TRUE` performs this within each fit, including CV training folds, while predictions use the original input units. An ordinary lasso selects columns; it does not automatically preserve an entire factor or interaction hierarchy. Package documentation.

Choose the penalty on training data. Divide the 120 training observations into five folds. Fit on four and score predictions on the fifth, repeating for all folds. `lambda.min` minimizes average CV error; `lambda.1se` is the largest penalty within one estimated standard error of the minimum. The latter is a simplicity rule, not a hypothesis test. A single `cv.glmnet()` call tunes λ at a fixed α. CV documentation.

Install `glmnet` once with `install.packages("glmnet")` if needed. Continue the same sensor example:

```
library(glmnet)
```

```
Loading required package: Matrix
```

```
Loaded glmnet 5.1
```

```
ch7_xtrain <- as.matrix(ch7_train[paste0("x", 1:8)])
ch7_xtest <- as.matrix(ch7_test[paste0("x", 1:8)])
ch7_ridge <- cv.glmnet(ch7_xtrain, ch7_train$y,
  alpha = 0, foldid = ch7_folds, standardize = TRUE, type.measure = "mse")
ch7_lasso <- cv.glmnet(ch7_xtrain, ch7_train$y,
  alpha = 1, foldid = ch7_folds, standardize = TRUE, type.measure = "mse")
```

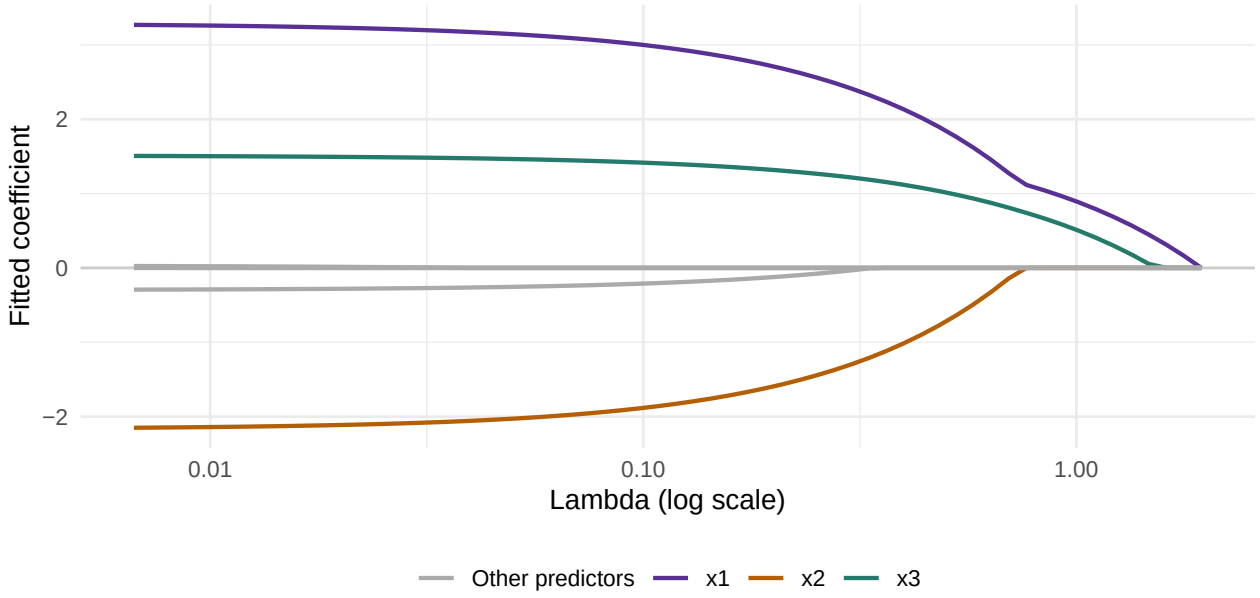


Figure 7.4: Lasso coefficient paths from the sensor training data. Larger penalties move slopes toward zero. The three true signal predictors are colored; the other five are gray. A path displays fitted coefficients, not confidence intervals.

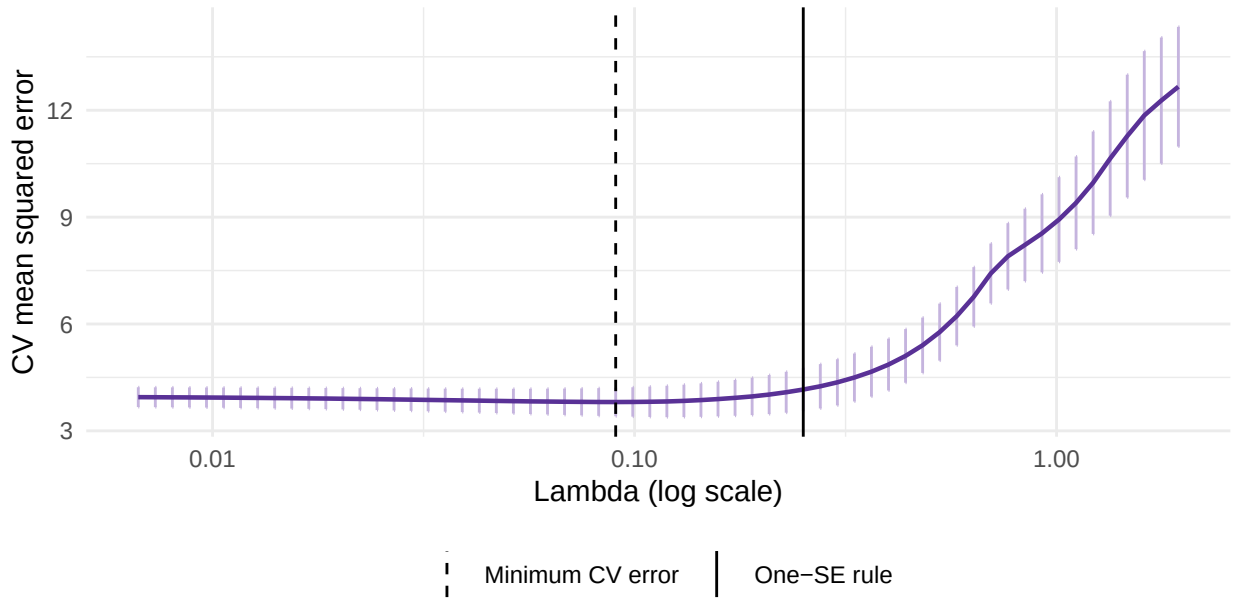


Figure 7.5: Five-fold CV within the sensor training set. Bars show plus or minus one estimated CV standard error, not confidence intervals for individual responses. Dashed: minimum-error lambda. Solid: the more regularized one-SE choice used below.

From one column to many. Ridge remains computable with more columns than observations; ordinary full-model OLS then cannot have full column rank. Lasso can give sparse fits in that setting, but it does not guarantee recovery of the true variables. Here we focus on a modest candidate set so the classical searches and penalized fits are directly comparable.

Section 7.5 Evaluate the complete procedure

Our choices are now fixed: AIC for the three subset searches and the one-SE rule for ridge and lasso. Evaluate all five on the **same untouched 40 observations**. Add full-model OLS and a training-mean baseline for context. For m test observations, RMSE $= \{m^{-1}\sum_i (y_i - \hat{y}_i)^2\}^{1/2}$, in response units.

```
ch7_predictions <- cbind(
  Baseline = rep(mean(ch7_train$y), nrow(ch7_test)),
  Full_OLS = predict(ch7_full, newdata = ch7_test),
  Forward = predict(ch7_forward, newdata = ch7_test),
  Backward = predict(ch7_backward, newdata = ch7_test),
  Stepwise = predict(ch7_both, newdata = ch7_test),
  Ridge = as.numeric(predict(ch7_ridge, newx = ch7_xtest, s = "lambda.1se")),
  Lasso = as.numeric(predict(ch7_lasso, newx = ch7_xtest, s = "lambda.1se")))
knitr::kable(data.frame(Method = colnames(ch7_predictions),
  Test_RMSE = sqrt(colMeans((ch7_predictions - ch7_test$y)^2))),
  digits = 3, row.names = FALSE)
```

Method	Test_RMSE
Baseline	2.972
Full_OLS	2.182
Forward	2.185
Backward	2.185
Stepwise	2.185
Ridge	2.095
Lasso	2.131

```
ch7_lasso_coef <- as.matrix(coef(ch7_lasso, s = "lambda.1se"))[-1, 1]
knitr::kable(data.frame(Predictor = names(ch7_lasso_coef),
  Lasso_coefficient = unname(ch7_lasso_coef)), digits = 3, row.names = FALSE)
```

Predictor	Lasso_coefficient
x1	2.562
x2	-1.445
x3	1.267
x4	-0.078
x5	0.000
x6	0.000
x7	0.000
x8	0.000

Ridge and lasso have slightly smaller test RMSE than full OLS on this split. Lasso keeps a small coefficient for the noise predictor x_4, while setting $x_5, \ldots, x_8$ to zero. Equal search results are possible, and no method must win on every split. Report the comparison without revising the methods after looking at test outcomes. If choosing a winning method from this table, its same test error is no longer an independent final assessment of that choice.

! Validate the search, not just its final formula

If CV is used to evaluate a selection procedure, rerun the entire variable search inside each training fold. Selecting on all observations first and then cross-validating only the selected formula leaks information. Learned transformations, imputation, scaling, and tuning follow the same rule. Final performance claims require an untouched test set or appropriately nested resampling.

Prediction and inference answer different questions. A selected coefficient is not automatically a scientific discovery; ordinary fixed-model t tests and confidence intervals do not automatically remain valid after selection or lasso followed by OLS refitting. Likewise, a good predictive score does not establish a causal effect.

> **Exercise 7.4** (Find the Leakage)
> A student selects predictors using all 160 outcomes, then splits the observations and reports

test RMSE. Another student reserves 40 observations first and runs selection only on the other 120. Which analysis gives an honest held-out assessment of its fitted procedure? Does that make its selected-coefficient p-values valid?

Solution 7.4
Only the second keeps test outcomes out of model development, provided the test results are not used to revise the procedure. That separation assesses prediction; it does not repair ordinary post-selection coefficient inference.

Section 7.6 Summary

- A linear model is linear in its coefficients; transformed predictors can describe curved relationships.
- Choose a transformation for a stated reason, refit, inspect residuals, and interpret the new response scale.
- Forward, backward, and stepwise searches use a candidate set and a stopping rule; they need not find the globally best subset.
- Ridge shrinks; lasso can also select. Scale predictors and tune penalties within training data.
- Compare predictions on the same observations and response scale. Keep the entire development procedure out of final test outcomes.

The next design unit asks which terms must be retained because of randomization, blocking, and the scientific question. That provides a useful boundary for automatic selection. Dose-specific comparisons from Chapter 5 are retained in Contrasts and Estimable Comparisons (Optional).

Section 7.7 Practice Problems

Exercise 7.5 (Practice: Specify, Select, and Assess)

1. A positive outcome grows approximately exponentially with time. Write a multiplicative-error model and its log-linear form. Explain why an additive original-scale error would change the argument.
2. A quadratic mean fits better than a straight line. Does it cease to be a linear regression model? State the columns of its design matrix.
3. A backward search considers removing a three-level factor. How many coefficient restrictions are imposed when there is an intercept and no interaction? Why should its indicators be treated together?
4. Explain why ridge is not generally a variable-selection method and why lasso selections can vary among correlated predictors.
5. Compare `lambda.min` with `lambda.1se`. Where should the folds, transformation choices, and final test observations enter the procedure?
6. A tree exceeds the $4/n$ Cook's-distance guide. A student deletes it and runs forward

selection. Explain what is missing from this procedure and propose a sensitivity analysis.

Solution 7.5

1. $Y = a\exp(bt)\exp(\varepsilon)$ gives $\log Y = \log a + bt + \varepsilon$. Logging $a\exp(bt) + \varepsilon$ does not yield this additive log-error structure.
2. It remains linear in the coefficients; the columns are $\mathbf{1}$, $\mathbf{x}$, and the vector of squared predictor values.
3. Two restrictions. The two columns represent one factor; selecting them separately makes the search depend on the chosen reference coding.
4. Ridge usually leaves all slopes nonzero. Correlated predictors offer similar information, so small sample changes can alter which one lasso retains.
5. `lambda.min` minimizes CV error; `lambda.1se` prefers the largest penalty within one estimated SE of that minimum. Choose transformations and tune models within training data/folds, then evaluate the fixed procedure on an untouched test set.
6. A screening flag does not establish an error. Check the measurement and model specification, compare scientifically relevant estimates and predictions with and without the tree, and examine whether the selected terms change. Explain any exclusion and retain an honest test set.

Part I

Optional topic

Quantile Regression

> **Self-study topic**
>
> This reading extends the regression framework to conditional quantiles. It is supplementary to the main lecture sequence. Read Chapter 2 — Least Squares Estimation, Chapter 5 — Multiple Regression and Categorical Predictors, and Chapter 6 — Outliers, Leverage, and Influence first: least squares, coefficient interpretation, and diagnostics are the useful prerequisites. The R examples use the `quantreg` package; install it once with `install.packages("quantreg")` if needed.

When the mean is not the whole question

OLS describes a conditional mean. In some studies, the scientific question concerns a typical response, a low outcome, or an unusually high outcome. For example, average delivery time and the 90th percentile of delivery time answer different service-planning questions. A covariate may change the spread of the response even when its relationship with the mean is simple.

Definition 7.5 (Conditional Quantile)
For $0 < \tau < 1$, define

$$Q_\tau(Y \mid \mathbf{X} = \mathbf{x}) = \inf\{q : P(Y \le q \mid \mathbf{X} = \mathbf{x}) \ge \tau\}.$$

Here Y is a scalar response, $\mathbf{X}$ is a random predictor vector, and $\mathbf{x}$ is its observed value. A linear quantile model specifies $Q_\tau(Y \mid \mathbf{X} = \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}_\tau$. The coefficients may vary with τ. At $\tau = 0.5$, the target is a conditional median.

Interpretation. Without an interaction involving x_j, $\beta_{\tau,j}$ is the change in the modeled conditional τ-quantile for a one-unit change in x_j, holding the other predictors fixed. This compares conditional distributions. It does not track a particular person who remains at the same percentile, and it is not automatically a causal effect.

> **Use all observations to estimate a conditional quantile**
>
> Fitting the 90th percentile does not mean selecting the largest 10% of responses and fitting OLS to them. Quantile regression uses the full dataset with a loss function that gives different penalties to positive and negative residuals. A marginal percentile of all responses is also different from a percentile conditional on the predictors.

Why a different loss? Least squares penalizes a residual $u = y - \mathbf{x}^\top \mathbf{b}$ by u^2. Quantile regression instead uses the check loss, also called pinball loss:

$$\rho_\tau(u) = u\{\tau - \mathbb{1}\{u < 0\}\} = \begin{cases} \tau u, & u \geq 0, \\ (1-\tau)(-u), & u < 0, \end{cases}$$

$$\hat{\boldsymbol{\beta}}_\tau \in \arg\min_{\mathbf{b}} \sum_{i=1}^{n} \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b}).$$

At $\tau = 0.5$, minimizing check loss is equivalent to minimizing the sum of absolute residuals. At $\tau = 0.9$, underpredicting the response costs nine times as much per unit as overpredicting it. The fitted target consequently moves toward the upper tail.

Proposition 7.1 (Why Check Loss Targets a Quantile)
If $\mathbb{E}(|Y|) < \infty$, the minimizers of $\mathbb{E}\{\rho_\tau(Y - a)\}$ are characterized by $F(a-) \leq \tau \leq F(a)$; the conditional-quantile definition above selects the lower endpoint when the minimizer is not unique. Where F is continuous, the derivative with respect to a is $F(a) - \tau$. It is nonpositive below a minimizing quantile and nonnegative above it. The same argument applies to the conditional distribution at a fixed $\mathbf{x}$.

Scope of the linear model. The population argument allows a different quantile at every $\mathbf{x}$. A linear quantile regression restricts these values to one linear function. If the true conditional quantile is nonlinear, the fitted line is a best approximation under the chosen loss, rather than an exact description at every predictor value.

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

check_loss <- function(u, tau) u * (tau - (u < 0))
taus <- c(0.1, 0.5, 0.9)
loss <- expand.grid(u = seq(-3, 3, length.out = 301), tau = taus)
loss$value <- check_loss(loss$u, loss$tau)
ggplot(loss, aes(u, value, colour = factor(tau))) +
  geom_vline(xintercept = 0, linetype = "dotted", colour = "grey70") +
  geom_line(linewidth = 0.9) +
  scale_colour_manual(values = c("steelblue4", "grey35", "darkorange3")) +
  labs(x = "Residual: observed - predicted", y = "Check loss",
    colour = "Quantile")
```

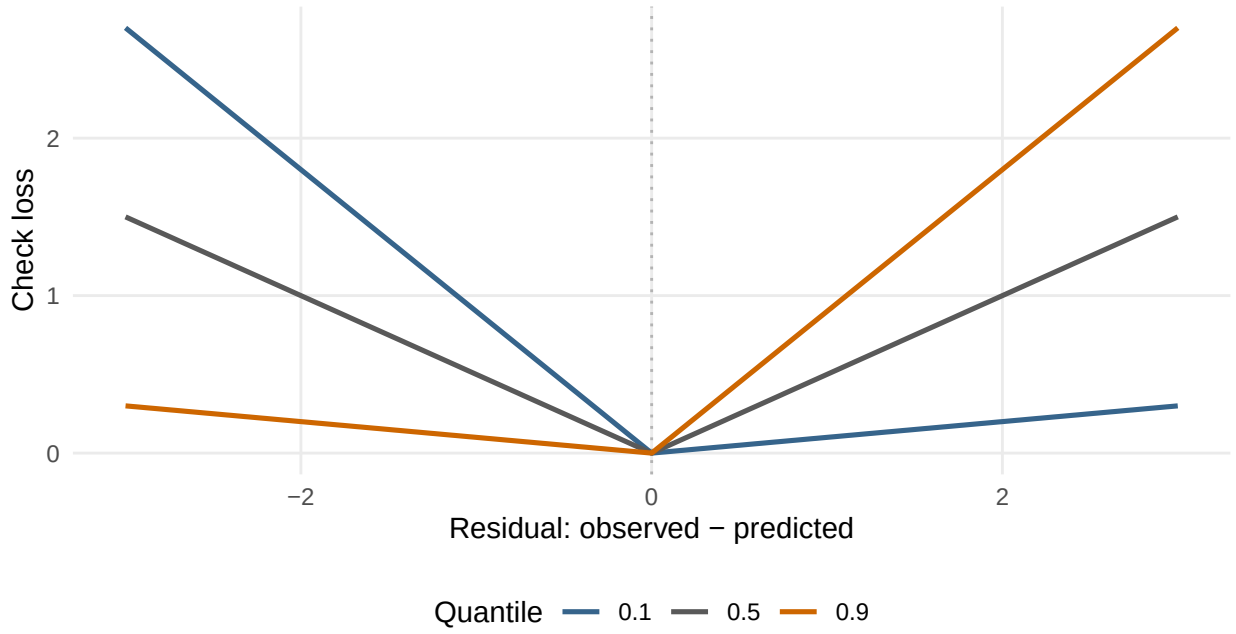


Figure 7.6: Check loss for three quantiles. A positive residual means the prediction is too low. The upper-quantile loss penalizes that side more strongly.

A mean model can miss changing spread

Consider the simulated model $Y = 5 + 2x + (0.5 + 0.8x)Z$, with $0 \le x \le 4$ and $Z \sim N(0,1)$ independent of x. Its conditional mean and median are both $5 + 2x$, while

$$Q_\tau(Y \mid x) = 5 + 2x + (0.5 + 0.8x)\Phi^{-1}(\tau).$$

The conditional-quantile slopes are $2 + 0.8\Phi^{-1}(\tau)$. Different slopes here reflect changing spread, even though every observation follows the same data-generating model. Quantile regression need not be reserved for skewed distributions.

```
set.seed(8561)
n <- 350
x <- runif(n, 0, 4)
dat_qr <- data.frame(x, y = 5 + 2 * x +
                       (0.5 + 0.8 * x) * rnorm(n))
fit_mean <- lm(y ~ x, data = dat_qr)
fit_quantiles <- quantreg::rq(y ~ x, tau = taus, data = dat_qr)
round(coef(fit_quantiles), 3)
```

```
            tau= 0.1 tau= 0.5 tau= 0.9
(Intercept)    4.509    5.078    5.850
x              0.768    1.816    2.965
```

```
data.frame(tau = taus, true_slope = 2 + 0.8 * qnorm(taus),
           estimated_slope = unname(coef(fit_quantiles)["x", ]))
```

```
  tau true_slope estimated_slope
1 0.1  0.9747587       0.7676385
2 0.5  2.0000000       1.8156789
3 0.9  3.0252413       2.9651924
```

The matrix returned by `coef()` contains one column per requested quantile. R's `rq()` documentation describes its formula interface and the `tau` argument.

```
new_x <- data.frame(x = seq(0, 4, length.out = 200))
qhat <- predict(fit_quantiles, newdata = new_x)
quantile_lines <- data.frame(x = rep(new_x$x, times = length(taus)),
                             prediction = as.vector(qhat),
                             target = rep(paste("Quantile", taus),
                               each = nrow(new_x)))
prediction_lines <- rbind(quantile_lines,
  data.frame(x = new_x$x, prediction = predict(fit_mean, newdata = new_x),
             target = "OLS mean"))
prediction_lines$target <- factor(prediction_lines$target,
                                  levels = c(paste("Quantile", taus),
                                    "OLS mean"))
ggplot(dat_qr, aes(x, y)) +
  geom_point(alpha = 0.35, colour = "grey40", size = 1.5) +
  geom_line(data = prediction_lines,
            aes(y = prediction, colour = target, linetype = target),
              linewidth = 0.8) +
  scale_colour_manual(values = c("steelblue4", "grey35", "darkorange3",
    "black")) +
  scale_linetype_manual(values = c("solid", "solid", "solid", "dashed")) +
  labs(x = "x", y = "Response", colour = NULL, linetype = NULL)
```

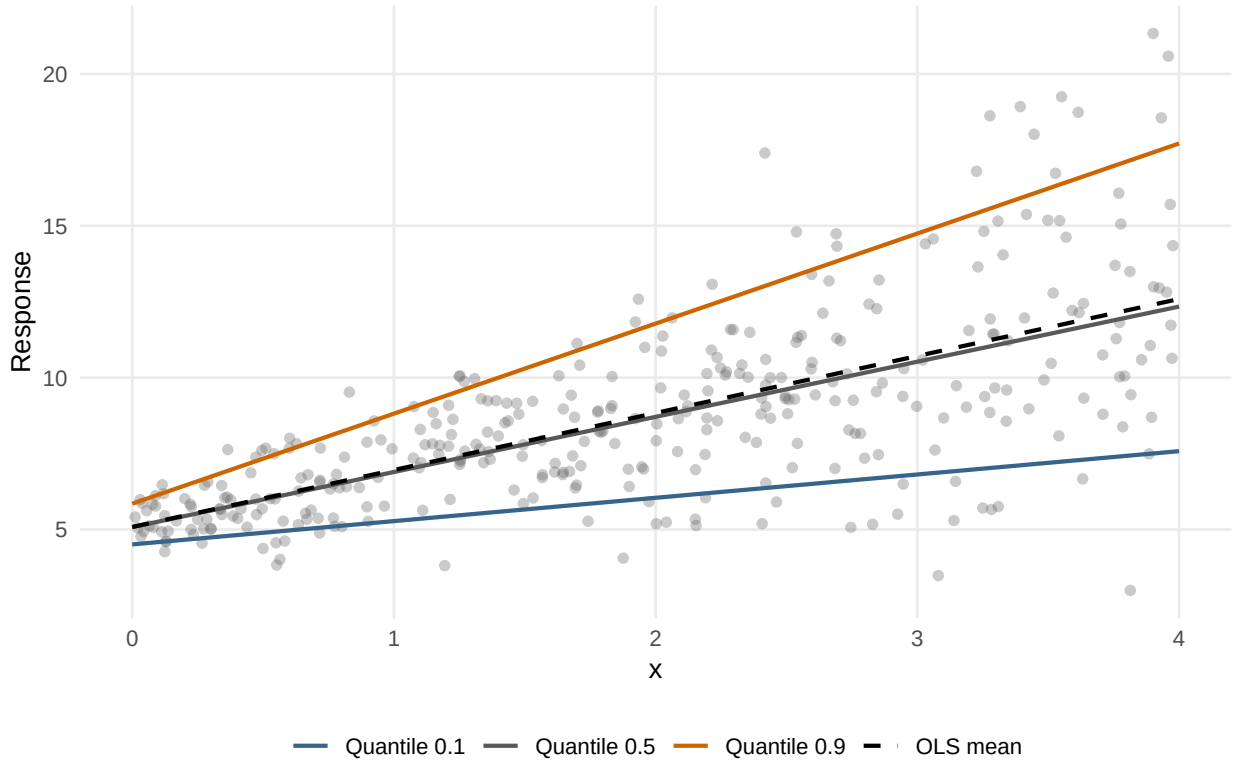


Figure 7.7: Fitted conditional quantiles and the OLS mean in simulated data with increasing spread. The growing distance between the lower and upper quantiles reveals variation that a mean line alone does not describe.

Read the graph. Compare the predicted 10th, 50th, and 90th percentiles at a fixed x. The gap between the 10th and 90th percentiles describes an interval containing the central 80% of a continuous conditional distribution when the quantile model is correct. The fitted gap is not a confidence interval for the mean or for a regression coefficient, and its predictive coverage is not guaranteed in a finite sample.

Uncertainty and practical checks

Quantile regression avoids a normal-error or constant-variance requirement for defining its target, but it is not assumption-free. Reliable estimation and inference still depend on the quantile specification, the design, the sampling structure, and sufficient information near the quantile of interest. Very extreme quantiles are difficult to estimate from small datasets.

A pairs-bootstrap illustration. These simulated observations are independent draws, so resampling whole (x_i, y_i) pairs is appropriate. The code requests a bootstrap standard error for the median-regression slope and forms an approximate normal interval. The replicate count is kept moderate for the notes; increase it and check Monte Carlo stability in a substantive analysis.

```r
set.seed(8561)
fit_median <- quantreg::rq(y ~ x, tau = 0.5, data = dat_qr)
median_summary <- summary(fit_median, se = "boot", R = 499,
                          bsmethod = "xy")
b <- coef(fit_median)["x"]
se_b <- median_summary$coefficients["x", "Std. Error"]
c(slope = unname(b), bootstrap_SE = se_b,
  lower = unname(b - qnorm(0.975) * se_b),
  upper = unname(b + qnorm(0.975) * se_b))
```

```
      slope bootstrap_SE       lower       upper
   1.815679     0.136845    1.547468    2.083890
```

This is not the exact finite-sample OLS t interval from Chapter 3 — Distribution Theory of OLS and Inference. For dependent or clustered observations, an independent-pairs bootstrap generally fails to preserve the sampling structure. The package's `summary.rq()` documentation describes alternative inference methods.

> **!** Two fitted quantiles need a joint comparison
>
> If one slope is significant at $\tau = 0.1$ and another is not significant at $\tau = 0.9$, that does not establish that the slopes differ. They are estimated from the same observations and are dependent. A difference such as $\beta_{0.9,j} - \beta_{0.1,j}$ requires joint inference that retains this dependence, for example refitting both quantiles in each common bootstrap sample.

Before interpreting a fit: check the functional form, influential predictor values, the amount of information in the tails, and whether separately fitted quantiles cross. Linear loss is less sensitive to very large response residuals than squared loss, but it does not eliminate leverage problems. Crossing fitted quantiles cannot describe a valid ordered distribution at that x; investigate misspecification or use methods that enforce ordering rather than silently relabeling the curves.

Evaluate the stated target. For held-out data, compare candidate predictions at the same τ using mean check loss. A coverage diagnostic compares the fraction of responses below the predicted quantile with τ, including checks across meaningful predictor ranges. Good overall coverage can hide poor conditional coverage. Use splits that match future prediction, as in Section 7.5 — Evaluate the Complete Procedure.

Application and self-study

> **i** Application: household expenditure
>
> The `engel` dataset bundled with `quantreg` contains income and food expenditure. Use `data("engel", package = "quantreg")`, plot the variables, and fit the 0.1, 0.5, and 0.9 conditional quantiles of `foodexp` given `income`. Explain how a household-expenditure quantile differs from mean expenditure. Treat the fitted relationships as descriptive associations, and

avoid extrapolating beyond the observed incomes.

1. Use the loss definition to explain why the 90th percentile is penalized more for underprediction.
2. Derive the true intercept and slope at each quantile in the simulated example. Explain why the median and mean agree here.
3. Predict the 10th and 90th percentiles at $x = 1$ and $x = 3$. Explain what their widening gap means.
4. Design a bootstrap for comparing two quantile slopes. State which observations must stay together if the data are clustered.

Suggested reading. Start with the `quantreg` package documentation, then the vignette available through `vignette("rq", package = "quantreg")`. Roger Koenker's *Quantile Regression* develops the estimation and inference theory in greater depth.

Mixed-Effects Models

> **i** Self-study topic
>
> This reading extends linear models to grouped and repeated observations. It is supplementary to the main lecture sequence. Chapter 1 — Matrix Representation and Notation and Chapter 3 — Distribution Theory of OLS and Inference supply the covariance and inference prerequisites. Also distinguish the experimental unit from repeated measurements: several measurements on one unit do not create several independently randomized units. The R examples use `lme4` and `patchwork`; install them once with `install.packages(c("lme4", "patchwork"))` if needed. The covariance connection to GLS is developed here, so the Generalized Least Squares and Correlated Errors notes is not a prerequisite.

Why group membership changes the analysis

Measurements from the same person, classroom, clinic, or production batch often share influences. Treating all rows as independent can give misleading standard errors. Repeated measurements provide information about within-unit change, but they do not create additional independently sampled or randomized units.

A linear mixed-effects model combines shared regression coefficients with group-specific deviations. **Fixed effects** describe common coefficients, possibly including measured group characteristics. **Random effects** represent latent group deviations through a distribution. These terms describe components of the model; they do not mean that a predictor must be nonrandom or that treatment assignment was randomized.

Definition 7.6 (A Random-Intercept Model)
For observation i in group j, let

$$Y_{ij} = \beta_0 + \beta_1 x_{ij} + b_j + \varepsilon_{ij},$$
$$b_j \overset{\text{iid}}{\sim} N(0, \tau^2), \qquad \varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2).$$

Condition on the observed predictors. Assume that the group effects and residual errors are mutually independent, with the displayed distributions unchanged after conditioning on those predictors. Conditional on b_j, the group has intercept $\beta_0 + b_j$. After integrating over b_j, its mean is $\beta_0 + \beta_1 x_{ij}$.

For two distinct observations in the same group, the shared b_j gives covariance τ^2; their marginal variances are $\tau^2 + \sigma^2$. Therefore the within-group correlation in this simple model is

$$\text{ICC} = \frac{\tau^2}{\tau^2 + \sigma^2}.$$

This intraclass correlation describes the fraction of conditional-on-predictors variation attributable to a shared group intercept. It is not a universal correlation formula: random slopes or additional residual correlation change it.

Connection to the course. In matrix notation,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{b} + \boldsymbol{\varepsilon}, \qquad \mathbf{b} \sim N(\mathbf{0}, \mathbf{G}), \qquad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \mathbf{R}),$$
$$\text{Var}(\mathbf{Y} \mid \mathbf{X}, \mathbf{Z}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R},$$

with $\mathbf{b}$ and $\boldsymbol{\varepsilon}$ independent and the design conditioned on. For a known covariance $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$, estimation of $\boldsymbol{\beta}$ has the GLS form. In practice, the covariance parameters are estimated too, so the exact OLS t/F theory does not transfer unchanged.

! A random intercept does not repair every dependence problem

The basic model assumes independent groups and independent residual errors conditional on the random effects. Serial correlation, household relationships across groups, or an omitted random slope may require a different covariance structure. A random-effects model also does not remove confounding: correlation between group effects and predictors can invalidate its interpretation unless modeled appropriately.

Partial pooling and shrinkage

Three possible analyses of group means illustrate what the random-effects assumption adds:

Approach	What is shared?	Main tradeoff
Complete pooling	One mean for every group	Ignores genuine group differences.
Separate group means	Each group has its own unrestricted mean	Small groups can have very noisy estimates.
Partial pooling	Groups share a distribution of deviations	Stabilizes estimates using an explicit exchangeability assumption.

Proposition 7.2 (Shrinkage in the Intercept-Only Model)
For $Y_{ij} = \mu + b_j + \varepsilon_{ij}$ under the Gaussian assumptions above, suppose μ, $\tau^2 > 0$, and $\sigma^2 > 0$ are known. If group j has n_j observations, then

$$\mathbb{E}(\mu + b_j \mid \bar{Y}_j) = \mu + w_j(\bar{Y}_j - \mu), \qquad w_j = \frac{\tau^2}{\tau^2 + \sigma^2/n_j}.$$

The conditional group mean is a weighted average of the observed group mean and the common

mean. Small groups have less precise sample means and receive more shrinkage toward the common mean. Larger between-group variation reduces shrinkage.

Why the weight appears. The sample mean contains a group signal b_j with variance τ^2 and averaged residual noise with variance σ^2/n_j. Conditioning a jointly normal pair yields the signal-to-total-variance weight. Plugging estimated parameters into this expression gives an empirical best linear unbiased prediction (empirical BLUP) in this simple model. More general random-slope models have a matrix version rather than this scalar weight.

```
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

set.seed(8561)
n_group <- c(3, 4, 5, 6, 8, 10, 12, 16, 20, 24, 30, 40)
group <- factor(rep(LETTERS[seq_along(n_group)], n_group))
b_true <- rnorm(length(n_group), sd = 2)
dat_group <- data.frame(group,
  y = 12 + b_true[as.integer(group)] + rnorm(length(group), sd = 3))
fit_pool <- lme4::lmer(y ~ 1 + (1 | group), data = dat_group,
                       REML = TRUE)
mu_hat <- unname(lme4::fixef(fit_pool)[1])
tau2_hat <- as.numeric(lme4::VarCorr(fit_pool)$group[1, 1])
sigma2_hat <- sigma(fit_pool)^2
raw_mean <- tapply(dat_group$y, dat_group$group, mean)
shrunk_mean <- mu_hat + lme4::ranef(fit_pool)$group[, "(Intercept)"]
w_hat <- tau2_hat / (tau2_hat + sigma2_hat / n_group)
round(data.frame(n = n_group, raw_mean, mixed_mean = shrunk_mean,
                 weight_on_group_mean = w_hat), 3)
```

```
   n raw_mean mixed_mean weight_on_group_mean
A  3   16.277     14.780                0.647
B  4    9.330     10.117                0.709
C  5   15.146     14.380                0.753
D  6   12.371     12.300                0.786
E  8   12.053     12.051                0.830
F 10   11.601     11.663                0.859
G 12    7.901      8.399                0.880
H 16   11.816     11.837                0.907
I 20   12.033     12.033                0.924
J 24   16.297     16.025                0.936
K 30    8.738      8.909                0.948
L 40   11.989     11.991                0.961
```

```
c(ICC = tau2_hat / (tau2_hat + sigma2_hat))
```

```
      ICC
0.3790268
```

The syntax `(1 | group)` requests a random intercept for each group. The shared intercept is returned by `fixef()`, while `ranef()` returns estimated group deviations. These are different quantities; adding them gives the fitted group mean in this intercept-only example. See the `lmer()` reference for model syntax.

```
pooling <- data.frame(label = paste0(levels(group), " (", n_group, ")"),
                      raw = as.numeric(raw_mean),
                        pooled = as.numeric(shrunk_mean))
pooling$label <- factor(pooling$label, levels = pooling$label)
ggplot(pooling, aes(y = label)) +
  geom_vline(xintercept = mu_hat, linetype = "dashed",
    colour = "darkorange3") +
  geom_segment(aes(x = raw, xend = pooled, yend = label), colour = "grey60") +
  geom_point(aes(x = raw, colour = "Separate means",
    shape = "Separate means"), size = 2.3) +
  geom_point(aes(x = pooled, colour = "Partial pooling",
    shape = "Partial pooling"), size = 2.3) +
  scale_colour_manual(values = c("Separate means" = "black",
    "Partial pooling" = "steelblue4")) +
  scale_shape_manual(values = c("Separate means" = 1,
    "Partial pooling" = 16)) +
  labs(x = "Group mean", y = "Group (sample size)", colour = NULL,
    shape = NULL)
```

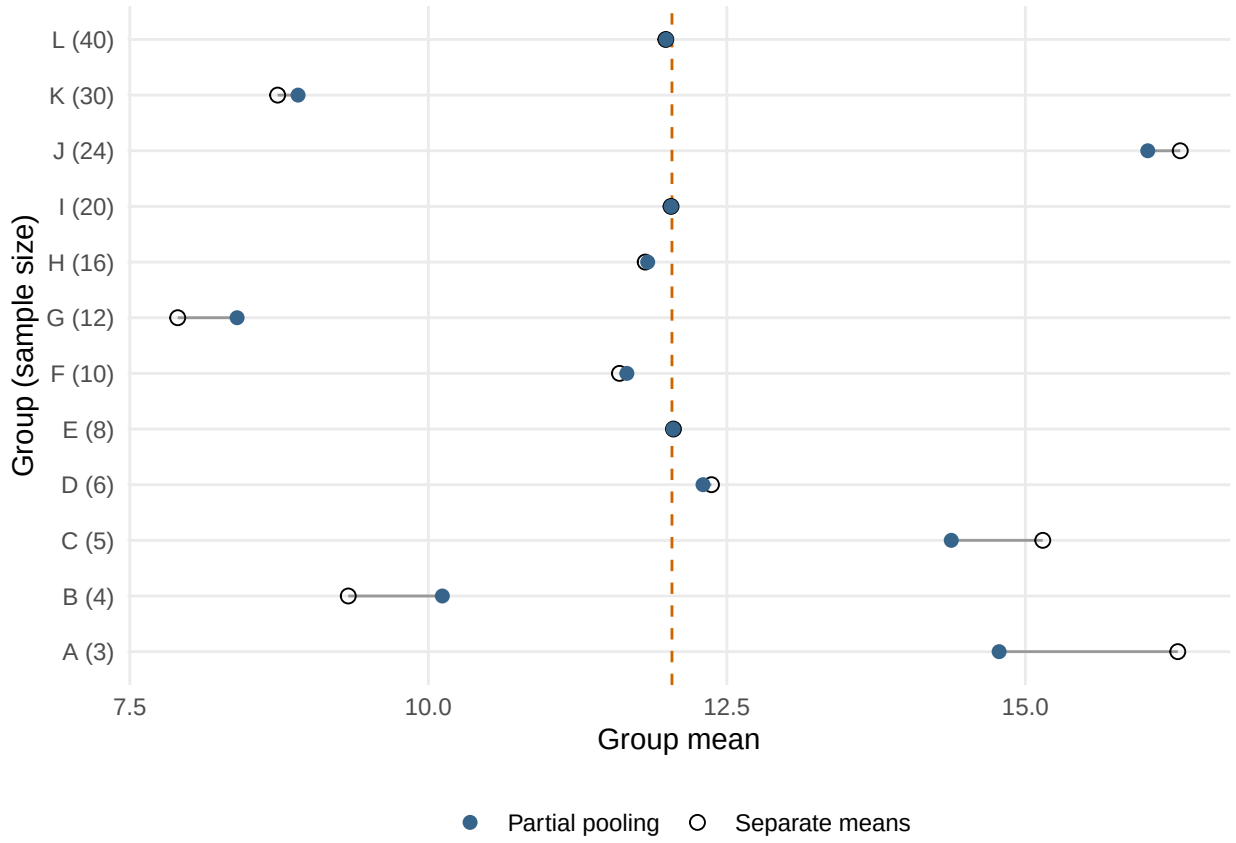


Figure 7.8: Unpooled sample means and partially pooled group means in simulated data. The dashed line is the fitted common mean. The amount of shrinkage reflects group sample size and the estimated variance components.

What shrinkage does not establish. It is not evidence that a small group's true mean equals the common mean. It reflects uncertainty and the assumed distribution of group effects. Point predictions of group deviations are also not confidence intervals or a justified ranking of all groups.

Application: repeated measurements and random slopes

> **i** Application: reaction times during sleep restriction
>
> The public `sleepstudy` data bundled with `lme4` contain repeated reaction-time measurements for 18 subjects. `Reaction` is the average reaction time for a subject on a recorded day, measured in milliseconds. Days 0–1 are adaptation/training and day 2 is baseline. We use baseline and later measurements, with `day = Days - 2`. The independent grouping unit is the subject. This linear-trend example is not an estimate of a randomized treatment contrast. See the dataset documentation for the study and variables.

A random intercept shifts a subject's line vertically. A **random slope** also allows the subject's rate of change to vary:

$$Y_{ij} = \beta_0 + \beta_1 t_{ij} + b_{0j} + b_{1j} t_{ij} + \varepsilon_{ij}, \qquad \begin{pmatrix} b_{0j} \\ b_{1j} \end{pmatrix} \sim N_2(\mathbf{0}, \mathbf{G}).$$

Here β_1 describes the average linear trend across the modeled subject population, while $\beta_1 + b_{1j}$ is the subject-specific slope. Allowing covariance between b_{0j} and b_{1j} recognizes that baseline level and rate of change may be related.

A covariance model as well as a mean model. For two measurements on the same subject, at distinct occasions t and s, the covariance contributed by the random effects is $g_{00} + (t+s)g_{01} + tsg_{11}$. Thus correlations can depend on the observation times. One constant random-intercept ICC no longer summarizes the whole dependence structure.

```
data("sleepstudy", package = "lme4")
study <- subset(sleepstudy, Days >= 2)
study$day <- study$Days - 2
fit_sleep <- lme4::lmer(Reaction ~ day + (day | Subject),
                        data = study, REML = TRUE)
lme4::fixef(fit_sleep)
```

```
(Intercept)         day
  267.96742    11.43543
```

```
lme4::VarCorr(fit_sleep)
```

```
 Groups   Name        Std.Dev. Corr
 Subject  (Intercept) 30.9573
          day          6.7659  0.178
 Residual             25.5264
```

```
lme4::isSingular(fit_sleep)
```

```
[1] FALSE
```

In `(day | Subject)`, the random intercept is included along with the random slope, and their covariance is estimated. This is a model for continuous Gaussian responses. Binary or count outcomes with random effects lead to **generalized linear mixed models**, which add a link and response family; they require further study beyond this topic.

```r
subjects <- levels(study$Subject)[1:4]
plot_grid <- expand.grid(day = seq(0, 7, length.out = 40), Subject = subjects)
plot_grid$conditional <- predict(fit_sleep, newdata = plot_grid)
plot_grid$population <- predict(fit_sleep, newdata = plot_grid, re.form = NA)
subject_lines <- rbind(
  transform(plot_grid, prediction = conditional, fit = "Subject-specific"),
  transform(plot_grid, prediction = population, fit = "Population"))
ggplot(subset(study, Subject %in% subjects), aes(day, Reaction)) +
  geom_point(size = 1.7) +
  geom_line(data = subject_lines,
            aes(y = prediction, colour = fit, linetype = fit),
              linewidth = 0.8) +
  scale_colour_manual(values = c("Subject-specific" = "steelblue4",
    Population = "black")) +
  scale_linetype_manual(values = c("Subject-specific" = "solid",
    Population = "dashed")) +
  facet_wrap(~ Subject, ncol = 2, labeller = label_both) +
  coord_cartesian(ylim = range(study$Reaction)) +
  labs(x = "Days since baseline", y = "Reaction time (ms)", colour = NULL,
    linetype = NULL)
```

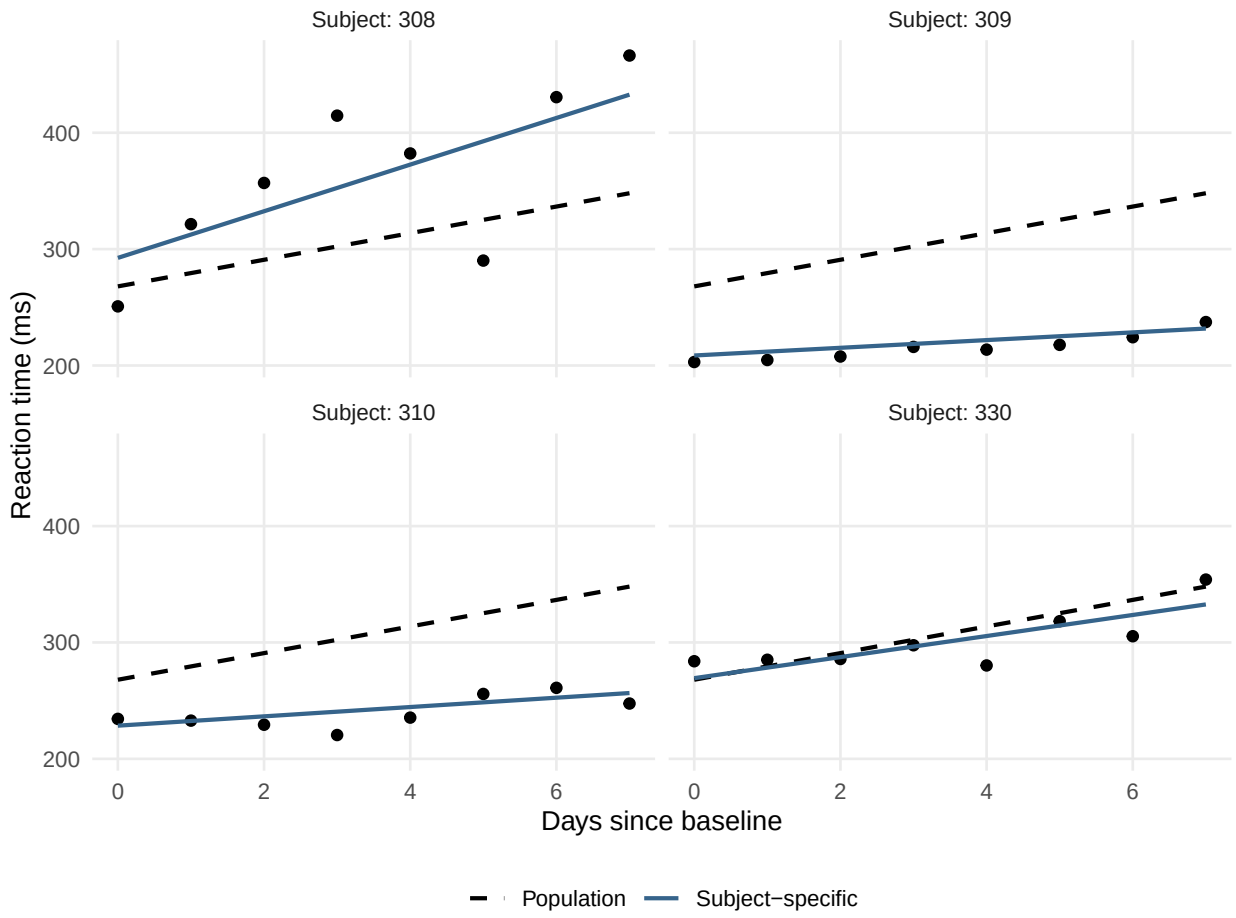


Figure 7.9: Four subjects from the sleepstudy data. Blue lines include each subject's fitted random effects; dashed lines use only the common fixed effects. A common linear trend does not require identical subject trajectories.

Prediction target. For a previously observed subject, predictions can use that subject's estimated random effects. For a new subject with no response data yet, those deviations are unknown; setting them to zero gives the population mean in this Gaussian identity-link model. A prediction interval for the new subject must still include between-subject and residual variation, together with parameter uncertainty. A fixed-effect line alone is not such an interval. With a nonlinear GLMM link, setting random effects to zero generally does not equal averaging the response over their distribution.

Inference, model checks, and further reading

ML and REML. Maximum likelihood estimates regression and covariance parameters through the marginal model. Restricted maximum likelihood (REML) accounts for the estimation of fixed effects when fitting covariance parameters. When comparing different fixed-effect structures by likelihood, use ML fits on the same observations and keep the specified random-effects structure comparable; the REML criterion is not directly comparable across different fixed-effect designs.

The following comparison asks whether the average linear trend is zero while allowing the same subject-specific random intercept and slope structure in both models. Its likelihood-ratio reference is asymptotic, not the exact F distribution from Chapter 4 — Model Comparison, Partial F Tests, and ANOVA.

```
fit_sleep_ml <- update(fit_sleep, REML = FALSE)
fit_no_trend_ml <- lme4::lmer(Reaction ~ 1 + (day | Subject),
                              data = study, REML = FALSE)
anova(fit_no_trend_ml, fit_sleep_ml, refit = FALSE)
```

```
Data: study
Models:
fit_no_trend_ml: Reaction ~ 1 + (day | Subject)
fit_sleep_ml: Reaction ~ day + (day | Subject)
                npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
fit_no_trend_ml    5 1444.4 1459.3 -717.21    1434.4
fit_sleep_ml       6 1425.2 1443.0 -706.58    1413.2 21.266  1  3.997e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> **!** Variance components require special care
>
> Testing whether a random-effect variance is zero places the null parameter on a boundary. The usual chi-square rule based only on the parameter-count difference need not apply. Small numbers of independent groups also limit large-sample approximations. Methods such as parametric bootstrap or appropriate small-sample corrections need additional justification. Do not use the ordinary OLS F test simply because the model contains a linear predictor.

Check what the design can support. Examine conditional residuals versus fitted values, residual patterns across time and groups, convergence messages, and near-singularity. A singular fit means the fitted random-effects covariance is on or near a lower-dimensional boundary, such as an estimated variance near zero or a redundant intercept/slope combination. It is not synonymous with an optimizer failing to converge. Review the design and model complexity before changing optimizer settings. See `isSingular()`.

```
conditional_diagnostics <- data.frame(fitted = fitted(fit_sleep),
                                      residual = resid(fit_sleep))
residual_plot <- ggplot(conditional_diagnostics, aes(fitted, residual)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_point(alpha = 0.5, colour = "steelblue4", size = 1.5) +
  labs(x = "Conditional fitted value", y = "Conditional residual")
qq_plot <- ggplot(conditional_diagnostics, aes(sample = residual)) +
  stat_qq_line(colour = "darkorange3") +
  stat_qq(size = 1.5, colour = "steelblue4") +
  labs(x = "Theoretical normal quantile", y = "Conditional residual")
patchwork::wrap_plots(residual_plot, qq_plot, nrow = 1)
```

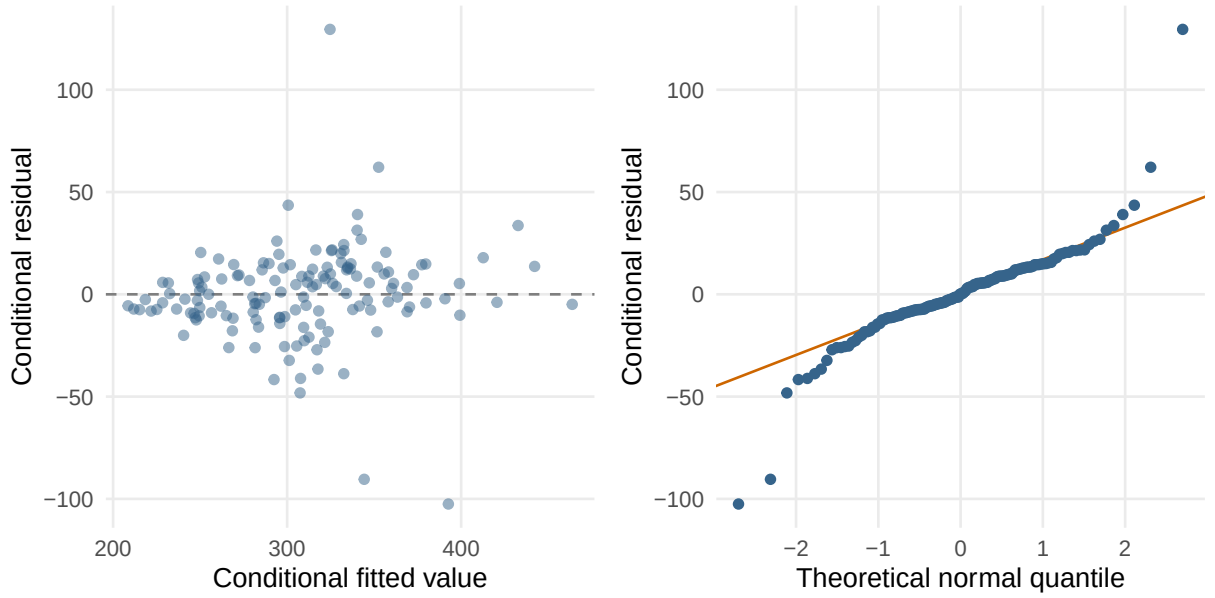


Figure 7.10: Basic conditional-residual checks for the fitted sleepstudy model. These plots can reveal mean or residual-distribution problems, but they do not prove that the full dependence structure is correct.

Plots of estimated random effects can be informative, but they display shrunken estimates with unequal uncertainty, not directly observed independent normal draws. With few groups, distributional checks are particularly limited. For prediction to new subjects, evaluate with subject-level holdouts; random row-wise splitting can let the same subject appear in both training and test data.

Self-study questions.

1. Derive the covariance and ICC for the random-intercept model. Explain why increasing observations within a group is different from adding new independent groups.
2. In the simulation, verify that `mu_hat + w_hat * (raw_mean - mu_hat)` matches the mixed-model group predictions. Explain the role of n_j.
3. Write down the fixed and random portions of the `sleepstudy` model. Explain what changes if `(day | Subject)` is replaced with `(1 | Subject)`.
4. Explain why a new subject's prediction interval is wider than an interval that conditions on known subject effects, and why a row-wise validation split may answer the wrong question.
5. State the null hypothesis in the ML comparison and explain why testing a random-effect variance of zero is a different inferential problem.

Suggested reading. The authors' Fitting Linear Mixed-Effects Models Using lme4 develops the model and computation. The package's inference guide discusses likelihood, bootstrap, and small-sample approaches. Continue to generalized mixed models only after distinguishing conditional and marginal prediction targets in the Gaussian case.

Contrasts and Estimable Comparisons

A scientific question often concerns a difference between means rather than one coefficient printed by software. This optional reading uses the linear-combination inference from Chapter 3 — Distribution Theory of OLS and Inference to express those questions. The experimental-design unit connects these comparisons to the design that generated the observations.

Learning objectives. Express a comparison as a linear function of regression coefficients, obtain its estimate and interval, interpret a conditional comparison, and distinguish one comparison from a family of comparisons.

> **i** Optional reading after Chapter 5
>
> The main sequence now develops transformations and variable selection in Chapter 7. This reading preserves the dose-specific ToothGrowth comparisons, individual and simultaneous intervals, and additional contrast practice. Use it when a scientific question requires a specified difference between means.

Application: does the method difference depend on dose? The plot continues Chapter 5. Compare OJ with VC separately at each observed dose before deciding which quantity to estimate. The ToothGrowth documentation describes 60 guinea pigs, two delivery methods and three doses in mg/day; the response is recorded odontoblast length.

Comparisons of group means

Suppose the conditional means of three groups are μ_A, μ_B, and μ_C. A question such as "Does A differ from the average of B and C?" concerns $\theta = \mu_A - (\mu_B + \mu_C)/2$.

Definition 7.7 (Contrast of Group Means)
A contrast is a linear combination $\theta = \sum_{j=1}^{g} c_j \mu_j$ whose weights satisfy $\sum_{j=1}^{g} c_j = 0$. The zero-sum condition makes the comparison unchanged when the same constant is added to every group mean.

From means to coefficients. With A as the reference group, write $\boldsymbol{\beta} = (\beta_0, \beta_B, \beta_C)$. Then $\mu_A = \beta_0$, $\mu_B = \beta_0 + \beta_B$, and $\mu_C = \beta_0 + \beta_C$, so

$$\theta = -\tfrac{1}{2}\beta_B - \tfrac{1}{2}\beta_C = \mathbf{a}^\top \boldsymbol{\beta}, \qquad \mathbf{a} = (0, -\tfrac{1}{2}, -\tfrac{1}{2}).$$



Figure 7.11: Individual responses and cell means in ToothGrowth. Circles show animals; diamonds show means. Panel headings give dose in mg/day. The OJ-minus-VC difference appears different across doses. Horizontal jitter separates points for display only.

> **!** The zero-sum condition applies to mean weights
>
> The weights on (μ_A, μ_B, μ_C) sum to zero. The equivalent weights on treatment-coded regression coefficients need not sum to zero. Always translate the scientific comparison through the chosen parameterization.

Under the full-rank normal linear model, use

$$\hat{\theta} = \mathbf{a}^\top \hat{\boldsymbol{\beta}}, \qquad \mathrm{SE}(\hat{\theta}) = \hat{\sigma}\sqrt{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}}.$$

A confidence interval is $\hat{\theta} \pm t_{1-\alpha/2, n-p}\, \mathrm{SE}(\hat{\theta})$. This is the same procedure as in Chapter 3 — Distribution Theory of OLS and Inference, applied to a different target. For one fixed restriction under the same fitted normal model, the corresponding F statistic equals the squared t statistic. Several linearly independent restrictions can be collected into $\mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, with $\mathrm{rank}(\mathbf{C}) = r$, and tested jointly using the general F test.

Interpretation. Report the contrast estimate in the units of the response, its interval, and which population means it compares. Equality of two slopes is only a meaningful scientific question when their units and scales make that comparison defensible.

Conditional comparisons in the ToothGrowth data

Chapter 5 compared two straight lines with a model for all six method-dose means. The straight-line restriction did not fit these data well. We therefore use the six-cell model for the comparisons below. The target at an observed dose d is

$$\Delta_d = \mu_{\mathrm{OJ},d} - \mu_{\mathrm{VC},d}, \qquad d \in \{0.5, 1, 2\}.$$

Each dose defines a different comparison. Combining all doses into one average would require specifying the averaging weights and would answer another question.

```
ch6_dat <- datasets::ToothGrowth
ch6_dat$supp <- factor(ch6_dat$supp, levels = c("VC", "OJ"))
ch6_dat$dose_f <- factor(ch6_dat$dose, levels = c(0.5, 1, 2))
ch6_fit <- lm(len ~ dose_f * supp, data = ch6_dat)
ch6_profiles <- expand.grid(dose_f = levels(ch6_dat$dose_f),
                            supp = levels(ch6_dat$supp))
ch6_X <- model.matrix(delete.response(terms(ch6_fit)),
                      data = ch6_profiles, contrasts.arg = ch6_fit$contrasts,
                      xlev = ch6_fit$xlevels)
# First three rows are VC; last three are OJ, in the same dose order.
ch6_L <- ch6_X[4:6, , drop = FALSE] - ch6_X[1:3, , drop = FALSE]
ch6_est <- drop(ch6_L %*% coef(ch6_fit))
ch6_V <- ch6_L %*% vcov(ch6_fit) %*% t(ch6_L)
ch6_se <- sqrt(diag(ch6_V))
ch6_df <- df.residual(ch6_fit)
ch6_crit <- qt(0.975, ch6_df)
ch6_results <- data.frame(
  Dose = levels(ch6_dat$dose_f), Estimate = ch6_est, SE = ch6_se,
  Lower = ch6_est - ch6_crit * ch6_se,
  Upper = ch6_est + ch6_crit * ch6_se,
  p_value = 2 * pt(-abs(ch6_est / ch6_se), df = ch6_df))
knitr::kable(ch6_results, digits = 3, row.names = FALSE,
             caption = "OJ minus VC at each dose; individual 95% intervals.")
```

Table 7.2: OJ minus VC at each dose; individual 95% intervals.

Dose	Estimate	SE	Lower	Upper	p_value
0.5	5.25	1.624	1.994	8.506	0.002
1	5.93	1.624	2.674	9.186	0.001
2	-0.08	1.624	-3.336	3.176	0.961

```
stopifnot(ch6_df == 54, max(abs(ch6_est - c(5.25, 5.93, -0.08))) < 1e-10)
```

Where do the weights come from? A fitted mean at a predictor profile is a design row times $\hat{\boldsymbol{\beta}}$. Subtract the VC row from the OJ row to obtain the weights for their difference:

$$\hat{\Delta}_d = (\mathbf{x}_{\mathrm{OJ},d} - \mathbf{x}_{\mathrm{VC},d})^\top \hat{\boldsymbol{\beta}}, \qquad \widehat{\mathrm{Var}}(\hat{\Delta}_d) = \mathbf{a}_d^\top \widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) \mathbf{a}_d.$$

The estimate is 5.25 at dose 0.5, 5.93 at dose 1, and −0.08 at dose 2. The small last estimate does not establish equivalence; its interval describes uncertainty under the independent, common-variance normal model.

Do the two differences differ? Compare dose 1 with dose 2 using $\theta = \Delta_1 - \Delta_2$, rather than comparing their separate significance labels. Its estimate is $5.93 - (-0.08) = 6.01$.

```
ch6_a <- ch6_L[2, ] - ch6_L[3, ]
ch6_dd <- sum(ch6_a * coef(ch6_fit))
ch6_dd_se <- sqrt(drop(t(ch6_a) %*% vcov(ch6_fit) %*% ch6_a))
knitr::kable(data.frame(
  Comparison = "Difference at dose 1 minus difference at dose 2",
  Estimate = ch6_dd, SE = ch6_dd_se,
  Lower = ch6_dd - ch6_crit * ch6_dd_se,
  Upper = ch6_dd + ch6_crit * ch6_dd_se),
  digits = 3, row.names = FALSE)
```

Comparison	Estimate	SE	Lower	Upper
Difference at dose 1 minus difference at dose 2	6.01	2.297	1.405	10.615

The same calculation works for a continuous-by-group model. For $\Delta(x_0) = \beta_g + (x_0 - 1)\beta_{xg}$, the estimated variance is

$$\widehat{\mathrm{Var}}(\hat{\beta}_g) + (x_0 - 1)^2 \widehat{\mathrm{Var}}(\hat{\beta}_{xg}) + 2(x_0 - 1)\widehat{\mathrm{Cov}}(\hat{\beta}_g, \hat{\beta}_{xg}).$$

Do not add coefficient standard errors, or omit their covariance. A separate significant/non-significant label for each estimate is not a test of their difference.

One comparison and a family of comparisons

For one prespecified comparison, the individual t interval answers the stated question. When reporting all three dose-specific OJ-minus-VC differences as a family, account for the three opportunities to make a false rejection. Prespecifying several tests does not remove this family-wise issue.

For this **family of three tests**, Holm adjusts the p-values. Bonferroni simultaneous intervals use critical value $t_{1-\alpha/(2m),n-p}$ with $m = 3$; their joint coverage is at least $1 - \alpha$ under the model. These are two different adjustment procedures, so an adjusted p-value and an interval should be labeled with the procedure that produced it.

```
ch6_m <- nrow(ch6_results)
ch6_bonf <- qt(1 - 0.05 / (2 * ch6_m), ch6_df)
knitr::kable(data.frame(
  Dose = ch6_results$Dose,
  Holm_p = p.adjust(ch6_results$p_value, method = "holm"),
  Bonferroni_lower = ch6_est - ch6_bonf * ch6_se,
```

```
  Bonferroni_upper = ch6_est + ch6_bonf * ch6_se),
  digits = 3, row.names = FALSE)
```

Dose	Holm_p	Bonferroni_lower	Bonferroni_upper
0.5	0.004	1.237	9.263
1	0.002	1.917	9.943
2	0.961	-4.093	3.933

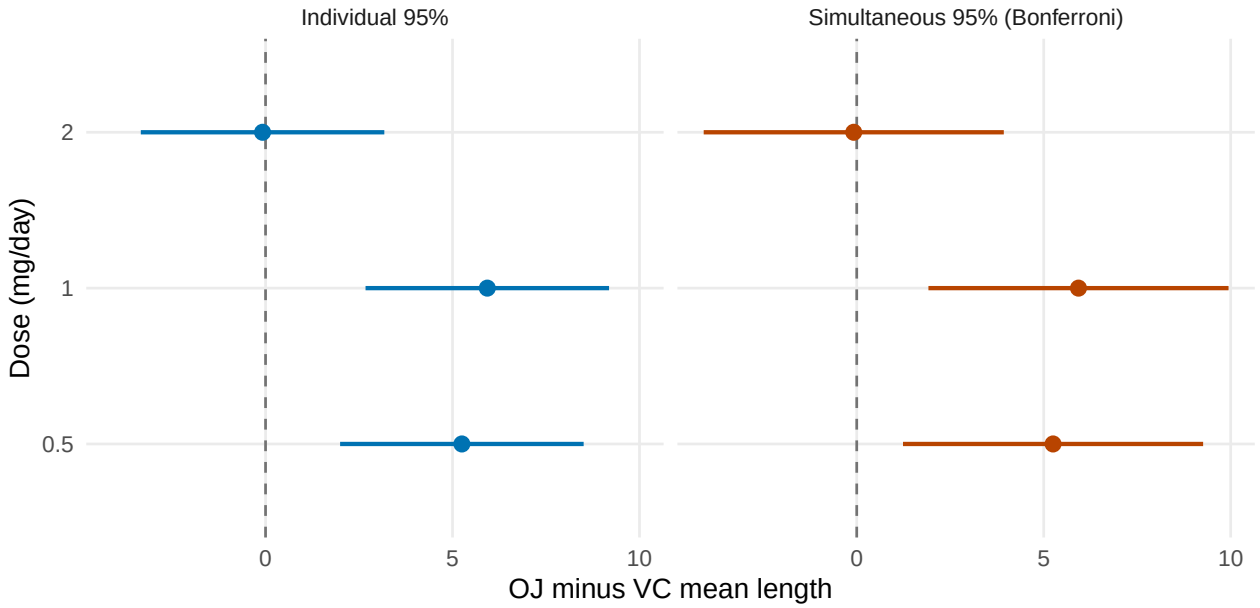


Figure 7.12: The same three OJ-minus-VC estimates with individual 95% intervals (blue) and Bonferroni simultaneous 95% intervals for the three-comparison family (orange). The estimates do not change; joint coverage requires wider intervals. The vertical dashed line is zero.

The difference-of-differences above is a separate teaching example. If it is also part of the reported inferential family, include it when defining that family and its adjustment. For all pairwise differences among levels of a simple one-way model, Tukey simultaneous intervals are another useful procedure. R documentation: Holm adjustment and Tukey intervals.

Exercise 7.6 (Check the Comparison)

1. Write the mean weights for $\Delta_1 - \Delta_2$ in the order $(\mu_{\mathrm{VC},1}, \mu_{\mathrm{OJ},1}, \mu_{\mathrm{VC},2}, \mu_{\mathrm{OJ},2})$.
2. Does changing the reference method change the OJ-minus-VC target?
3. Why are simultaneous intervals wider than individual intervals in the plot?

Solution 7.6

1. The weights are $(-1, 1, 1, -1)$; they sum to zero.
2. No. Re-express the same target using the new coefficient coding; its estimate and SE remain the same.
3. The simultaneous procedure protects the coverage of the whole stated family, rather than each interval separately.

Coding and estimability

Changing the reference group changes the coefficient labels and values, but not the fitted group means or a fixed comparison of those means. The vector $\mathbf{a}$ must change with the coding so that it represents the same target.

Example 7.4 (An Intercept and All Group Indicators)
In $Y_{ij} = \mu + \tau_i + \varepsilon_{ij}$, adding a constant to every τ_i and subtracting it from μ leaves every group mean unchanged. The separate parameters are not identified without a constraint, but $\tau_A - \tau_B$ is unchanged and is estimable.

The practical lesson is to state the target in terms of observable mean responses before reading coefficients. The row-space criterion and generalized-inverse formulation are beyond this reading.

Summary / Recap

- Start with the comparison of interest, then translate its mean weights into coefficient weights. A contrast of group means has weights summing to zero.
- Estimate $\mathbf{c}^\top \boldsymbol{\beta}$ by $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$, with squared SE $\mathbf{c}^\top \widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}})\mathbf{c}$. Include covariances.
- With an interaction, compare groups at specified predictor values. In ToothGrowth, the OJ-minus-VC comparison is estimated separately at each dose.
- A prespecified comparison and a family of comparisons have different error guarantees. State the family and distinguish individual from simultaneous intervals.
- Coding can change coefficients without changing an estimable comparison. One fixed restriction gives $t^2 = F$ under the same normal model.

The experimental-design unit asks which observations supply independent information for these comparisons.

Further practice: a three-group comparison

This simulated example separates the known data-generating means from their noisy estimates. Changing the seed changes the sample, not the target comparison.

```r
library(ggplot2)
theme_set(theme_minimal(base_size = 11) +
            theme(legend.position = "bottom",
              panel.grid.minor = element_blank()))

set.seed(8561)
group <- factor(rep(c("A", "B", "C"), each = 12))
mu <- c(A = 10, B = 13, C = 15)
dat <- data.frame(group,
                  y = unname(mu[group]) + rnorm(length(group), sd = 2))
fit <- lm(y ~ group, data = dat)

# A minus the average of B and C, under treatment coding.
a <- c(0, -1/2, -1/2)
estimate <- sum(a * coef(fit))
se <- sqrt(drop(t(a) %*% vcov(fit) %*% a))
df <- df.residual(fit)
ci <- estimate + c(-1, 1) * qt(0.975, df) * se
c(estimate = estimate, SE = se, lower = ci[1], upper = ci[2],
  p_value = 2 * pt(-abs(estimate / se), df))
```

```
     estimate            SE         lower         upper       p_value
-3.3388312952  0.9030487544 -5.1760978004 -1.5015647901  0.0007874472
```

Verify the same question under another coding. A cell-means model directly estimates (μ_A, μ_B, μ_C), so its contrast weights are $(1, -1/2, -1/2)$.

```r
fit_means <- lm(y ~ 0 + group, data = dat)
c_mean <- c(1, -1/2, -1/2)
estimate_means <- sum(c_mean * coef(fit_means))
se_means <- sqrt(drop(t(c_mean) %*% vcov(fit_means) %*% c_mean))
rbind(treatment_coding = c(estimate, se),
      cell_means_coding = c(estimate_means, se_means))
```

```
                      [,1]      [,2]
treatment_coding  -3.338831 0.9030488
cell_means_coding -3.338831 0.9030488
```

```r
contrast_result <- data.frame(estimate, lower = ci[1], upper = ci[2],
                              comparison = "A - (B + C) / 2")
ggplot(contrast_result, aes(estimate, comparison)) +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey50") +
  geom_segment(aes(x = lower, xend = upper, yend = comparison),
               linewidth = 1, colour = "steelblue4") +
  geom_point(size = 3, colour = "steelblue4") +
```

```
  expand_limits(x = 0) +
  labs(x = "Estimated contrast and 95% confidence interval", y = NULL)
```

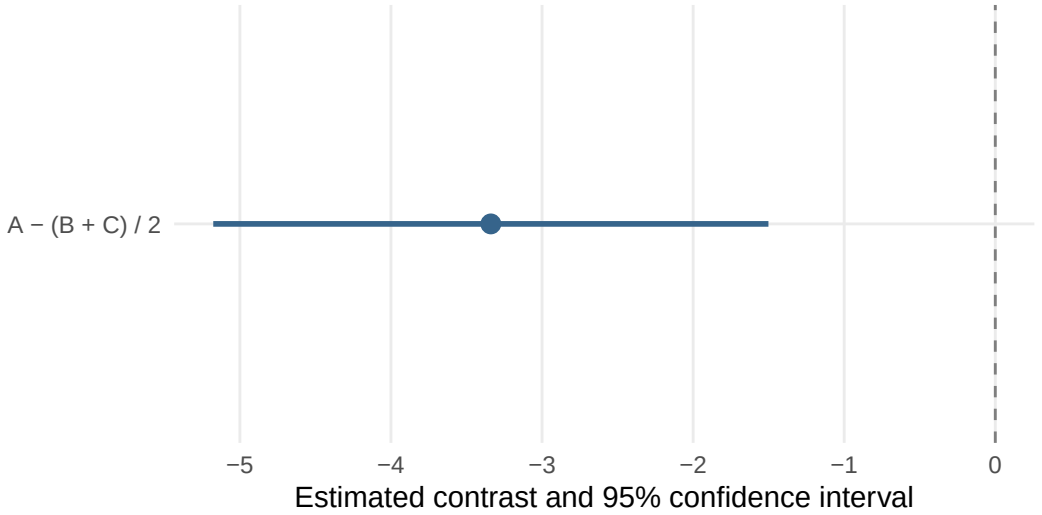


Figure 7.13: The estimated A-minus-average-of-B-and-C contrast and its 95% confidence interval. Zero represents no difference for this particular comparison.

A family of comparisons. For all three pairwise group differences, fit the one-way ANOVA and request simultaneous Tukey intervals:

```
TukeyHSD(aov(y ~ group, data = dat))
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = y ~ group, data = dat)

$group
         diff        lwr      upr     p adj
B-A 2.8862540  0.3275576 5.444950 0.0242654
C-A 3.7914086  1.2327122 6.350105 0.0026175
C-B 0.9051546 -1.6535418 3.463851 0.6639004
```

These intervals answer a different collection of questions from the single contrast above. Do not infer a difference between group means by inspecting whether two separate mean-response intervals overlap.

Practice

1. Express $\mu_B - \mu_C$ under treatment coding with A as reference, then under cell-means coding.

2. Verify that the estimate and standard error agree under the two codings.
3. Explain why the coefficients can change after changing the reference group while the scientific comparison remains unchanged.
4. Specify the comparison family before reporting all pairwise differences, and explain the role of multiplicity adjustment.

References

Montgomery, Douglas C. 2017. *Design and Analysis of Experiments*. John wiley & Sons.

Montgomery, Douglas C, Elizabeth A Peck, and G Geoffrey Vining. 2021. *Introduction to Linear Regression Analysis*. John Wiley & Sons.

Petersen, Kaare Brandt, and Michael Syskind Pedersen. 2008. *The Matrix Cookbook*. Technical University of Denmark.

Seber, George AF, and Alan J Lee. 2003. *Linear Regression Analysis*. John Wiley & Sons.

Part II

Appendix

Matrix Algebra Review

Matrix dimensions

If $\mathbf{A}$ is $m \times n$ and $\mathbf{B}$ is $n \times p$, then $\mathbf{AB}$ is defined and is $m \times p$.

Transpose rules

For conformable matrices,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top.$$

Inverse rule

If $\mathbf{A}$ and $\mathbf{B}$ are invertible, then

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1} \mathbf{A}^{-1}.$$

Symmetric matrices

A matrix $\mathbf{A}$ is symmetric if

$$\mathbf{A}^\top = \mathbf{A}.$$

Covariance matrices are always symmetric.

A good reference of the linear algebra can be found from Petersen and Pedersen (2008).

- Petersen, Kaare Brandt, and Michael Syskind Pedersen. (2008). *The Matrix Cookbook*. Technical University of Denmark.

Basic Mathematical Statistics

The chi-square, Student's t, and F distributions all arise from normal random variables. This appendix explains the constructions, proves the distributional results, and connects them to the general linear hypothesis in Chapter 3 — Distribution Theory of OLS and Inference, Section 3.8: General Linear Hypotheses.

The logical order is **normal variables** $\longrightarrow$ **sums of squares** $\longrightarrow$ **standardized ratios**. Degrees of freedom count independent normal components. Independence is essential when forming the ratios.

Chi-square: a sum of squared standard normals

Definition 7.8 (Chi-square distribution)
If $Z_1, \dots, Z_r$ are independent $N(0,1)$ variables, then

$$U = \sum_{j=1}^{r} Z_j^2 \sim \chi_r^2.$$

Each squared standardized component contributes one degree of freedom. Thus U measures a total squared distance from zero. Its density is

$$f_U(u) = \frac{u^{r/2-1} e^{-u/2}}{2^{r/2}\Gamma(r/2)}, \qquad u > 0.$$

This is a gamma density with shape $r/2$ and **scale** 2. Here $\Gamma(a) = \int_0^\infty x^{a-1}e^{-x}\,dx$ for $a > 0$. For the sum-of-squares construction, r is a positive integer; the density also defines χ_r^2 for any real $r > 0$.

Proof: chi-square density and moments

For $s < 1/2$,

$$\begin{aligned}\mathbb{E}(e^{sZ_j^2}) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1-2s}{2} z^2\right\} dz \\ &= (1-2s)^{-1/2}.\end{aligned}$$

The substitution $w = \sqrt{1-2s}\, z$ gives the second equality. Independence therefore gives

$$M_U(s) = \prod_{j=1}^{r} \mathbb{E}(e^{sZ_j^2}) = (1-2s)^{-r/2}.$$

To identify the law, integrate the displayed gamma density against e^{su}:

$$\int_0^\infty e^{su}\frac{u^{r/2-1}e^{-u/2}}{2^{r/2}\Gamma(r/2)}\,du = \frac{\Gamma(r/2)(1/2-s)^{-r/2}}{2^{r/2}\Gamma(r/2)} = (1-2s)^{-r/2}.$$

Moment-generating functions finite in a neighborhood of zero uniquely determine a probability law, so U has the claimed density. Differentiating at zero gives

$$\mathbb{E}(U) = r, \qquad \mathrm{Var}(U) = 2r.$$

In particular, U/r has mean 1 and variance $2/r$: it estimates a unit variance more stably as the degrees of freedom increase.

Why normal quadratic forms are chi-square

Theorem 7.1 (Normal quadratic forms)
Let $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I}_n)$, and let $\mathbf{A}$ be symmetric and idempotent, with rank $r \geq 1$. Then

$$\mathbf{Z}^\top \mathbf{A}\mathbf{Z} \sim \chi^2_r.$$

Proof: normal quadratic forms

The spectral theorem gives an orthogonal matrix $\mathbf{Q}$ such that

$$\mathbf{A} = \mathbf{Q}\,\mathrm{diag}(\mathbf{I}_r, \mathbf{0})\mathbf{Q}^\top.$$

Indeed, idempotence implies $\lambda^2 = \lambda$, so every eigenvalue is 0 or 1; exactly r eigenvalues are 1. Set $\mathbf{W} = \mathbf{Q}^\top\mathbf{Z}$. An orthogonal transformation preserves the standard multivariate normal law, because its mean is zero and its covariance is $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}_n$. Hence the W_j are independent $N(0,1)$ variables, and

$$\mathbf{Z}^\top\mathbf{A}\mathbf{Z} = \mathbf{W}^\top\,\mathrm{diag}(\mathbf{I}_r, \mathbf{0})\mathbf{W} = \sum_{j=1}^{r} W_j^2 \sim \chi^2_r.$$

Geometrically, $\mathbf{A}$ projects onto an r-dimensional subspace. The quadratic form is the squared length of the projected normal vector. If $r = 0$, it is identically zero rather than having a positive-degree chi-square density.

Corollary 7.1 (Residual variation in the normal linear model)
Throughout the regression applications, assume

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n), \quad \sigma^2 > 0, \quad \mathrm{rank}(\mathbf{X}) = p < n,$$

with fixed $\mathbf{X}$. Write

$$\mathbf{V} = (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \mathbf{H} = \mathbf{X}\mathbf{V}\mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad \nu = n - p.$$

Since $\mathbf{M}$ is symmetric and idempotent with rank ν, and $\mathbf{M}\mathbf{X} = \mathbf{0}$,

$$\frac{\text{SSE}}{\sigma^2} = \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right)^\top \mathbf{M} \left(\frac{\boldsymbol{\varepsilon}}{\sigma}\right) \sim \chi^2_\nu.$$

Consequently, $\hat{\sigma}^2 = \text{SSE}/\nu$ is unbiased for σ^2.

Proposition 7.3 (Independence of OLS estimates and residual variation)
Both $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{V}\mathbf{X}^\top \boldsymbol{\varepsilon}$ and $\mathbf{e} = \mathbf{M}\boldsymbol{\varepsilon}$ are jointly normal, and

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \sigma^2 \mathbf{V}\mathbf{X}^\top \mathbf{M} = \mathbf{0}.$$

A jointly normal vector with zero cross-covariance has a characteristic function that factors into the two marginal characteristic functions. Thus these vectors are independent, even though the residual covariance is singular. Since $\text{SSE} = \mathbf{e}^\top \mathbf{e}$, $\hat{\boldsymbol{\beta}}$ and SSE are independent. Zero covariance alone would not suffice without joint normality.

Student's t: estimating an unknown scale

Definition 7.9 (Student's t distribution)
Let $Z \sim N(0,1)$ and $U \sim \chi^2_\nu$ be independent, with $\nu > 0$. Then

$$T = \frac{Z}{\sqrt{U/\nu}} \sim t_\nu.$$

The denominator is a random estimate of a standard deviation. It can be small, producing larger standardized values than a normal denominator fixed at 1 would produce. The result is a symmetric density with heavier tails than the standard normal density.

Proof: Student t density

Use $t = z/\sqrt{u/\nu}$ and retain u. The inverse map and absolute Jacobian determinant are

$$z = t\sqrt{u/\nu}, \qquad \left|\frac{\partial(z,u)}{\partial(t,u)}\right| = \sqrt{u/\nu}.$$

Independence gives a product joint density. Integrating out u yields

$$\begin{aligned} f_T(t) &= \int_0^\infty \frac{e^{-t^2 u/(2\nu)}}{\sqrt{2\pi}} \frac{u^{\nu/2-1} e^{-u/2}}{2^{\nu/2}\Gamma(\nu/2)} \sqrt{\frac{u}{\nu}}\, du \\ &= \frac{1}{\sqrt{2\pi\nu}\, 2^{\nu/2}\Gamma(\nu/2)} \int_0^\infty u^{(\nu+1)/2-1} e^{-(1+t^2/\nu)u/2}\, du. \end{aligned}$$

Using $\int_0^\infty u^{a-1} e^{-bu}\, du = \Gamma(a) b^{-a}$ for $a, b > 0$, we obtain

$$f_T(t) = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi}\,\Gamma(\nu/2)} \left(1 + \frac{t^2}{\nu}\right)^{-(\nu+1)/2}, \qquad t \in \mathbb{R}.$$

This is the Student t density. Its dependence on t^2 proves symmetry. Its tails decay polynomially, as $|t|^{-(\nu+1)}$, rather than exponentially as for a normal density. Also, $\mathrm{Var}(U/\nu) = 2/\nu \to 0$, so $U/\nu \to 1$ in probability and t_ν approaches $N(0,1)$ as $\nu \to \infty$.

Corollary 7.2 (Student t inference for a regression contrast)
For any fixed nonzero $\mathbf{a} \in \mathbb{R}^p$, let $v_a = \mathbf{a}^\top \mathbf{V} \mathbf{a} > 0$. Then

$$Z = \frac{\mathbf{a}^\top \hat{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta}}{\sigma\sqrt{v_a}} \sim N(0,1), \qquad U = \frac{\mathrm{SSE}}{\sigma^2} \sim \chi^2_\nu.$$

The normal quadratic-form section proves their independence. Therefore

$$\frac{\mathbf{a}^\top \hat{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta}}{\hat{\sigma}\sqrt{v_a}} \sim t_\nu.$$

Under $H_0 : \mathbf{a}^\top \boldsymbol{\beta} = d$, replace the true contrast in the numerator by d. Choosing a coordinate vector for $\mathbf{a}$ gives the usual coefficient t test.

F: comparing two independent sums of squares

Definition 7.10 (F distribution)
If $U_1 \sim \chi^2_r$ and $U_2 \sim \chi^2_\nu$ are independent, with $r, \nu > 0$, then

$$F = \frac{U_1/r}{U_2/\nu} \sim F_{r,\nu}.$$

The two degrees of freedom have different roles: r belongs to the numerator and ν to the denominator. Both normalized sums of squares have expectation 1. Their ratio is positive and can be large when the numerator is unusually large or the denominator is small. The ratio itself does not have mean 1: its mean is $\nu/(\nu-2)$ when $\nu > 2$.

Proof: F density

Put $a = r/2$, $b = \nu/2$, and $c = r/\nu$. Transform (U_1, U_2) to (F, V), where $V = U_2$. The inverse map is

$$u_1 = cfv, \qquad u_2 = v, \qquad f > 0,\ v > 0,$$

and its absolute Jacobian determinant is

$$\left|\det \begin{pmatrix} cv & cf \\ 0 & 1 \end{pmatrix}\right| = cv.$$

Using independence and the two chi-square densities,

$$
\begin{aligned}
f_F(f) &= \int_0^\infty \frac{(cfv)^{a-1}e^{-cfv/2}}{2^a\Gamma(a)} \frac{v^{b-1}e^{-v/2}}{2^b\Gamma(b)}\, cv\, dv \\
&= \frac{c^a f^{a-1}}{2^{a+b}\Gamma(a)\Gamma(b)} \int_0^\infty v^{a+b-1}e^{-(1+cf)v/2}\, dv \\
&= \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} c^a f^{a-1}(1+cf)^{-(a+b)}.
\end{aligned}
$$

Substituting the definitions of a, b, c gives the F density:

$$
f_F(f) = \frac{\Gamma((r+\nu)/2)}{\Gamma(r/2)\Gamma(\nu/2)} \left(\frac{r}{\nu}\right)^{r/2} f^{r/2-1} \left(1+\frac{r}{\nu}f\right)^{-(r+\nu)/2}, \quad f > 0.
$$

Remark: normalization and order of the degrees of freedom

The raw ratio U_1/U_2 equals $(r/\nu)F$, and hence is not generally $F_{r,\nu}$. Reversing the numerator and denominator gives $1/F \sim F_{\nu,r}$.

Proof of the general linear hypothesis F test

Theorem 7.2 (F test for the general linear hypothesis)
Consider $H_0 : \mathbf{C}\boldsymbol{\beta} = \mathbf{d}$, where $\mathbf{C}$ is $r \times p$ with row rank r, $1 \le r \le p$, and $\mathbf{d} \in \mathbb{R}^r$. Keep the normal-model assumptions from the normal quadratic-form section, and write

$$
\mathbf{D} = \mathbf{C}\mathbf{V}\mathbf{C}^\top, \qquad \mathbf{L} = \mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}.
$$

Under H_0, the statistic

$$
F = \frac{\mathbf{L}^\top \mathbf{D}^{-1}\mathbf{L}}{r\hat{\sigma}^2} \sim F_{r,n-p}.
$$

Proof: general linear hypothesis F test

Step 1: standardize the restrictions. The matrix $\mathbf{D}$ is positive definite: for nonzero $\mathbf{b}$, full row rank implies $\mathbf{C}^\top\mathbf{b} \neq \mathbf{0}$, so $\mathbf{b}^\top\mathbf{D}\mathbf{b} = (\mathbf{C}^\top\mathbf{b})^\top\mathbf{V}(\mathbf{C}^\top\mathbf{b}) > 0$. Under H_0, $\mathbf{L} \sim N_r(\mathbf{0}, \sigma^2\mathbf{D})$. Consequently,

$$
\mathbf{W} = \sigma^{-1}\mathbf{D}^{-1/2}\mathbf{L} \sim N_r(\mathbf{0}, \mathbf{I}_r),
$$

where $\mathbf{D}^{-1/2}$ is the symmetric positive definite inverse square root. The numerator quadratic form therefore satisfies

$$
Q = \frac{\mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L}}{\sigma^2} = \mathbf{W}^\top\mathbf{W} \sim \chi^2_r.
$$

Step 2: identify the independent denominator. We already proved $U = \mathrm{SSE}/\sigma^2 \sim \chi^2_\nu$, with $\nu = n - p$. The quantity Q is a function of $\hat{\boldsymbol{\beta}}$, whereas U is a function of SSE. Their

independence follows from the normal-model independence proved in the normal quadratic-form section.

Step 3: form the ratio. Apply the F ratio result and cancel σ^2:

$$F = \frac{Q/r}{U/\nu} = \frac{\mathbf{L}^\top \mathbf{D}^{-1}\mathbf{L}}{r\hat{\sigma}^2} \sim F_{r,n-p} \quad \text{under } H_0.$$

Substituting $\mathbf{L} = \mathbf{C}\hat{\boldsymbol{\beta}} - \mathbf{d}$ and $\mathbf{D} = \mathbf{C}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{C}^\top$ recovers the statistic in Chapter 3 — Distribution Theory of OLS and Inference, Section 3.8: General Linear Hypotheses. Thus r counts the independent restrictions and $n-p$ counts residual degrees of freedom. Large values indicate a departure from H_0; a level-α test rejects when $F > F_{1-\alpha;r,n-p}$, where the right-hand side is an upper quantile. The central F law above is a null distribution; under an alternative the numerator is generally noncentral chi-square.

Corollary 7.3 (The identity $T^2 = F$ for one restriction)
When $r = 1$ and $\mathbf{C} = \mathbf{a}^\top$,

$$F = \frac{(\mathbf{a}^\top\hat{\boldsymbol{\beta}} - d)^2}{\hat{\sigma}^2\mathbf{a}^\top\mathbf{V}\mathbf{a}} = T^2.$$

Equivalently, squaring the defining ratio for T gives $T^2 = (Z^2/1)/(U/\nu)$, and $Z^2 \sim \chi^2_1$ independently of U. Hence $T^2 \sim F_{1,\nu}$. The upper-tail F test matches the **two-sided** t test, not a one-sided test.

A compact regression example and reading guide

Example 7.5 (Regression with 20 observations and 3 coefficients)
Suppose the normal linear model has $n = 20$ observations and $p = 3$ coefficients, including an intercept. Then $\nu = 17$. If SSE $= 34$, the observed variance estimate is $\hat{\sigma}^2 = 34/17 = 2$.

- The sampling law of SSE/σ^2 is χ^2_{17}; the observed value 34 does not identify the unknown σ^2.
- For $H_0 : \beta_1 = 0$, suppose $\hat{\beta}_1 = 1.5$ and $V_{22} = 0.05$ (the intercept is the first coordinate). Then

$$T = \frac{1.5}{\sqrt{2(0.05)}} \approx 4.743, \qquad F = T^2 = 22.5.$$

The null reference distributions are t_{17} and $F_{1,17}$, respectively.

- For a joint test of two independent restrictions, $r = 2$. If the observed unscaled numerator quadratic form $\mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L}$ is 12, then $F = 12/(2 \times 2) = 3$, compared with $F_{2,17}$.

What to check before using these results

Confirm the normal-model assumptions, the rank of the projection or restriction matrix, the residual degrees of freedom, and the required independence. A symmetric matrix alone is not enough for the quadratic-form chi-square result: idempotence is also needed. Likewise, correct marginal distributions alone do not establish a t or F ratio law.

Reference books.

1. **Seber, G. A. F., and Lee, A. J. (2003). *Linear Regression Analysis*, 2nd ed. Wiley.** Read Chapter 2, *Multivariate Normal Distribution*, for the normal-vector background; Chapter 3, *Linear Regression: Estimation and Distribution Theory*, for regression sampling distributions; and Chapter 4, *Hypothesis Testing*, for linear restrictions. This is the main reference for the regression applications above. Publisher and contents.
2. **Casella, G., and Berger, R. (2024 reprint). *Statistical Inference*, 2nd ed. Chapman & Hall/CRC.** Chapters 3, *Common Families of Distributions*, and 5, *Properties of a Random Sample*, provide the probability background for chi-square, t, and F distributions. The publisher identifies this as a reprint of the earlier second edition. Publisher and contents.

Part III

Assignment

Assignment 1

STAT 8561 | Fall 2026
Total: 40 points
Coverage: Chapter 1 — Matrix Representation and Notation and Chapter 2 — Least Squares Estimation.

This assignment focuses on matrix representation, least squares estimation, projection geometry, and linear regression in R.

Question	Topic	Points
1	Matrix representation	8
2	Least squares and projection	12
3	Consequences and geometry of least squares	12
4	Regression in R	8
Total		**40**

Instructions

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 1** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A1_LastName_FirstName.pdf`.
2. Its matching source file, named `A1_LastName_FirstName.qmd` or `A1_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- For each multiple-choice item, select **exactly one** answer and report its letter. Include an explanation only when requested.
- For True/False items, enter only `T` or `F`.
- For short-answer questions, one or two clear sentences are sufficient unless a derivation is requested.
- Round numerical answers to **three decimal places**, retaining full precision during calculations.
- Include all R code used to obtain your answers in executable chunks.
- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

Question 1: Matrix Representation of the Linear Model [8]

Consider the simple linear regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, 5,$$

with the following observed data:

i	x_i	Y_i
1	0	2
2	1	4
3	2	5
4	3	8
5	4	9

(a) [2]

Which of the following is the correct design matrix $\mathbf{X}$?

A.
$$\begin{pmatrix} 0 & 2 \\ 1 & 4 \\ 2 & 5 \\ 3 & 8 \\ 4 & 9 \end{pmatrix}$$

B.
$$\begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{pmatrix}$$

C.
$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \\ 3 & 1 \\ 4 & 1 \end{pmatrix}$$

D.
$$\begin{pmatrix} 1 & 2 \\ 1 & 4 \\ 1 & 5 \\ 1 & 8 \\ 1 & 9 \end{pmatrix}$$

Solution 1(a)

B. Each row is $(1, x_i)$: the first column represents the intercept, and the second contains the predictor values.

(b) [2]

Fill in the following table.

Quantity	Dimension / Rank
$\mathbf{Y}$	___
$\mathbf{X}$	___
$\boldsymbol{\beta}$	___
$\text{rank}(\mathbf{X})$	___

Solution 1(b)

Quantity	Dimension / Rank
$\mathbf{Y}$	5×1
$\mathbf{X}$	5×2
$\boldsymbol{\beta}$	2×1
$\text{rank}(\mathbf{X})$	2

The two columns of $\mathbf{X}$ are linearly independent because the predictor is not constant.

(c) [2]

Which pair is correct?

A.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

B.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 5 \\ 5 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}$$

C.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 30 & 10 \\ 10 & 5 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 77 \\ 28 \end{pmatrix}$$

D.
$$\mathbf{X}^\top\mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 20 \end{pmatrix}, \qquad \mathbf{X}^\top\mathbf{Y} = \begin{pmatrix} 28 \\ 77 \end{pmatrix}$$

Solution 1(c)

A. Here $\sum_i x_i = 10$, $\sum_i x_i^2 = 30$, $\sum_i Y_i = 28$, and $\sum_i x_i Y_i = 74$. Thus

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 5 & 10 \\ 10 & 30 \end{pmatrix}, \qquad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} 28 \\ 74 \end{pmatrix}.$$

(d) [2]

Assume

$$\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}, \qquad \mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_5.$$

For each statement, enter `T` or `F`.

Statement	T/F
$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$	---
$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$	---
$\mathbb{E}[\mathbf{Y}] = \mathbf{0}$	---
The five errors all have variance σ^2	---

Solution 1(d)

Statement	T/F
$\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$	T
$\mathrm{Var}(\mathbf{Y}) = \sigma^2 \mathbf{I}_5$	T
$\mathbb{E}[\mathbf{Y}] = \mathbf{0}$	F
The five errors all have variance σ^2	T

Question 2: Least Squares and Projection [12]

Continue using the data and design matrix from Question 1.

(a) [4]

Starting from

$$S(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}),$$

derive the normal equations

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y},$$

and then state the OLS estimator when $\mathbf{X}$ has full column rank.

Keep your derivation concise.

Solution 2(a)

Expanding,

$$S(\boldsymbol{\beta}) = \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Differentiating,

$$\frac{\partial S(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}.$$

Setting the derivative equal to zero gives

$$\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}.$$

Hence,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{Y}.$$

(b) [3]

Using the quantities from Question 1, compute

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \end{pmatrix}.$$

Fill in the blanks:

$$\hat{\beta}_0 = _____, \qquad \hat{\beta}_1 = _____.$$

Hence,

$$\hat{Y} = _____ + _____ x.$$

Solution 2(b)

Using the data in Question 1,

$$\hat{\boldsymbol{\beta}} = \frac{1}{50}\begin{pmatrix} 30 & -10 \\ -10 & 5 \end{pmatrix}\begin{pmatrix} 28 \\ 74 \end{pmatrix} = \begin{pmatrix} 2.0 \\ 1.8 \end{pmatrix}.$$

Therefore $\hat{\beta}_0 = 2.000$, $\hat{\beta}_1 = 1.800$, and

$$\hat{Y} = 2.0 + 1.8x.$$

These are the coefficients used in Question 3.

(c) [2]

For the least squares residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}},$$

which statement must be true?

A. $\mathbf{Xe} = \mathbf{0}$

B. $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$

C. $\mathbf{e}^\top \mathbf{e} = 0$

D. $\mathbf{He} = \mathbf{e}$

In one sentence, explain the geometric meaning of your choice.

Your explanation:

Solution 2(c)

B. The residual vector is orthogonal to the column space of $\mathbf{X}$.

(d) [3]

For the hat matrix

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top,$$

enter `T` or `F`.

Statement	T/F
$\mathbf{H}^\top = \mathbf{H}$	___
$\mathbf{H}^2 = \mathbf{H}$	___
$\hat{\mathbf{Y}} = \mathbf{HY}$	___
$\mathbf{H}$ is an orthogonal projection matrix	___

Solution 2(d)

Statement	T/F
$\mathbf{H}^\top = \mathbf{H}$	T
$\mathbf{H}^2 = \mathbf{H}$	T
$\hat{\mathbf{Y}} = \mathbf{HY}$	T
$\mathbf{H}$ is an orthogonal projection matrix	T

Question 3: Consequences and Geometry of Least Squares [12]

Continue using the data and fitted regression model from Questions 1 and 2.

Recall that

$$\hat{\beta}_0 = 2.0, \qquad \hat{\beta}_1 = 1.8.$$

(a) Fitted Values and Residuals [3]

Compute the fitted values

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$$

and the residual vector

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}.$$

Fill in the blanks:

$$\hat{\mathbf{Y}} = \begin{pmatrix} _____ \\ _____ \\ _____ \\ _____ \\ _____ \end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix} _____ \\ _____ \\ _____ \\ _____ \\ _____ \end{pmatrix}.$$

Solution 3(a)

$$\hat{\mathbf{Y}} = \begin{pmatrix} 2.0 \\ 3.8 \\ 5.6 \\ 7.4 \\ 9.2 \end{pmatrix}, \qquad \mathbf{e} = \begin{pmatrix} 0 \\ 0.2 \\ -0.6 \\ 0.6 \\ -0.2 \end{pmatrix}.$$

(b) Consequences of Including an Intercept [3]

For each statement, enter `T` or `F`.

Statement	T/F
The residuals satisfy $\sum_{i=1}^{n} e_i = 0$.	___
The fitted values satisfy $\hat{Y} = \bar{Y}$.	___
The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$.	___
These properties necessarily hold for every regression model without an intercept.	___

Solution 3(b)

Statement	T/F
The residuals satisfy $\sum_{i=1}^{n} e_i = 0$.	T
The fitted values satisfy $\hat{Y} = \bar{Y}$.	T
The vector $\mathbf{1}_n$ belongs to $\mathcal{C}(\mathbf{X})$.	T
These properties necessarily hold for every regression model without an intercept.	F

Because the model contains an intercept, $\mathbf{1}_n$ is a column of $\mathbf{X}$. Since $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$,

$$\mathbf{1}_n^\top \mathbf{e} = 0,$$

which implies

$$\sum_{i=1}^{n} e_i = 0$$

and therefore

$$\bar{\hat{Y}} = \bar{Y}.$$

(c) Residual-Maker Matrix [3]

Define

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H}.$$

For each statement, enter `T` or `F`.

Statement	T/F
$\mathbf{e} = \mathbf{M}\mathbf{Y}$	___
$\mathbf{M}^\top = \mathbf{M}$	___

Statement	T/F
$\mathbf{M}^2 = \mathbf{M}$	---
$\mathbf{HM} = \mathbf{0}$	---

Solution 3(c)

Statement	T/F
$\mathbf{e} = \mathbf{MY}$	T
$\mathbf{M}^\top = \mathbf{M}$	T
$\mathbf{M}^2 = \mathbf{M}$	T
$\mathbf{HM} = \mathbf{0}$	T

The matrix $\mathbf{H}$ projects onto $\mathcal{C}(\mathbf{X})$, whereas $\mathbf{M}$ projects onto $\mathcal{C}(\mathbf{X})^\perp$.

(d) Rank and Uniqueness [3]

Suppose now that a design matrix $\mathbf{X}$ does **not** have full column rank.

For each statement, enter `T` or `F`.

Statement	T/F
$(\mathbf{X}^\top\mathbf{X})^{-1}$ necessarily exists.	---
The least squares coefficient vector $\hat{\boldsymbol{\beta}}$ may not be unique.	---
The fitted vector $\hat{\mathbf{Y}}$ is still unique.	---
The residual vector $\mathbf{e}$ is still unique.	---

Solution 3(d)

Statement	T/F
$(\mathbf{X}^\top\mathbf{X})^{-1}$ necessarily exists.	F
The least squares coefficient vector $\hat{\boldsymbol{\beta}}$ may not be unique.	T
The fitted vector $\hat{\mathbf{Y}}$ is still unique.	T
The residual vector $\mathbf{e}$ is still unique.	T

Even when several coefficient vectors produce the same least squares solution, the orthogonal projection of $\mathbf{Y}$ onto $\mathcal{C}(\mathbf{X})$ is unique.

Question 4: Linear Regression in R [8]

For this question, use the built-in R dataset `cars`.

The variables are:

- `speed`: speed of the car in miles per hour;
- `dist`: stopping distance in feet.

Consider the model

$$\text{dist}_i = \beta_0 + \beta_1 \text{speed}_i + \varepsilon_i.$$

Load the data using

```
data(cars)
```

(a) [4]

Fit the model using `lm()` and complete the table.

```
fit <- lm(dist ~ speed, data = cars)
summary(fit)
```

Quantity	Your answer
$\hat{\beta}_0$	---
$\hat{\beta}_1$	---

Which interpretation of $\hat{\beta}_1$ is correct?

A. For each 1 mph increase in speed, the stopping distance increases by exactly 3.9324 feet for every car.

B. For each 1 mph increase in speed, the fitted mean stopping distance increases by approximately 3.9324 feet.

C. The average speed increases by 3.9324 mph for every extra foot of stopping distance.

D. A car travelling at 0 mph has a stopping distance of 3.9324 feet.

Solution 4(a)

```
data(cars)
fit <- lm(dist ~ speed, data = cars)
coef(fit)
```

```
(Intercept)       speed
```

```
-17.579095     3.932409
```

$$\hat{\beta}_0 = -17.5791, \qquad \hat{\beta}_1 = 3.9324.$$

The correct interpretation is **B**.

(b) [4]

Construct $\mathbf{Y}$ and $\mathbf{X}$ manually and reproduce the OLS coefficients using matrix operations.

Complete the following code:

```r
Y <- matrix(cars$dist, ncol = 1)

X <- cbind(
  1,
  cars$speed
)

# Replace NULL with your matrix expression
beta_hat <- NULL

beta_hat
```

Then report

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} _____ \\ _____ \end{pmatrix}.$$

Solution 4(b)

```r
Y <- matrix(cars$dist, ncol = 1)
X <- cbind(1, cars$speed)
beta_hat <- solve(t(X) %*% X) %*% t(X) %*% Y
beta_hat
```

```
           [,1]
[1,] -17.579095
[2,]   3.932409
```

The result agrees with `lm()`:

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} -17.579095 \\ 3.932409 \end{pmatrix} \approx \begin{pmatrix} -17.579 \\ 3.932 \end{pmatrix}.$$

Assignment 2

STAT 8561 | Fall 2026
Due date: September 30, 2026
Total: 40 points
Coverage: Chapter 2 — Least Squares Estimation and Chapter 3 — Distribution Theory of OLS and Inference.

This assignment emphasizes conceptual understanding of least squares, projection, and inference. Read the assumptions in each question carefully.

Question	Topic	Points
1	Least squares: assumptions and consequences	6
2	Projection geometry and rank	8
3	Sampling properties and the role of normality	8
4	Interpreting confidence and prediction intervals	6
5	Distribution and inference for OLS	12
Total		**40**

i Instructions

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 2** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A2_LastName_FirstName.pdf`.
2. Its matching source file, named `A2_LastName_FirstName.qmd` or `A2_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- For each multiple-choice item, select **exactly one** answer. Enter only the letter.
- For True/False items, enter only `T` or `F`. For matching items, enter one letter per row; letters may be reused.
- Questions 1-4 require **no written explanations, derivations, or R output**. Work through the reasoning before selecting your answers.
- Question 5 requires only the requested choices and numerical answers. Round numerical answers to **three decimal places**, retaining full precision during calculations.
- R is optional for checking calculations. If you use R, include your code after your answers in executable chunks.

- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

All necessary information is supplied in the questions. No external data or additional R packages are required.

Question 1: Least Squares: Assumptions and Consequences [6]

Consider $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is fixed, has full column rank, and $n > p$. Select **one answer per part**.

(a) Changing the distributional assumption [2]

Two students use the same observed $\mathbf{X}$ and $\mathbf{Y}$. One assumes normal errors; the other does not specify an error distribution. Both define their estimate by minimizing

$$S(\boldsymbol{\beta}) = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2.$$

Which statement is correct?

A. The second student cannot compute a least squares estimate.

B. The estimates must differ because the distributional assumptions differ.

C. The estimates are the same because they minimize the same criterion for the same data.

D. The estimates are the same only if the observed residuals are all zero.

Solution 1(a)

C. Both students minimize the same function of the same observed data. Normality is not needed to compute the least squares estimate; it is used for exact distributional inference.

(b) What makes the minimizer unique? [2]

Which statement explains why solving the normal equations gives a **unique** least squares minimizer under the stated assumptions?

A. Full column rank makes $\mathbf{X}^\top\mathbf{X}$ positive definite.

B. Normal errors make every system of normal equations invertible.

C. Having more observations than parameters guarantees full column rank for any design matrix.

D. Minimizing the sum of squared residuals forces every residual to be zero.

Solution 1(b)

A. Full column rank makes $\mathbf{X}^\top\mathbf{X}$ positive definite because

$$\mathbf{a}^\top\mathbf{X}^\top\mathbf{X}\mathbf{a} = \|\mathbf{X}\mathbf{a}\|^2 > 0 \quad \text{for every } \mathbf{a} \neq \mathbf{0}.$$

The least squares criterion therefore has a unique minimizer. The condition $n > p$ alone does not rule out linearly dependent columns.

(c) Shifting every response [2]

Now suppose the model includes an intercept. Add 5 to **every observed response**, keep all predictors unchanged, and refit the same model by least squares. What happens?

A. Every slope increases by 5, while the intercept is unchanged.

B. The coefficients are unchanged, and every residual increases by 5.

C. The intercept increases by 5, and every residual increases by 5.

D. The intercept increases by 5, the slopes are unchanged, and the residuals are unchanged.

Solution 1(c)

D. Adding 5 to the intercept adds 5 to every fitted response. The slopes are unchanged, and

$$(\mathbf{Y} + 5\mathbf{1}_n) - (\hat{\mathbf{Y}} + 5\mathbf{1}_n) = \mathbf{e},$$

so the residuals are unchanged.

Question 2: Projection Geometry and Rank [8]

For parts (a) and (c), let $\mathbf{X}$ have full column rank. Define

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top, \quad \mathbf{M} = \mathbf{I}_n - \mathbf{H}, \quad \hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}, \quad \mathbf{e} = \mathbf{M}\mathbf{Y}.$$

(a) Projecting a second time [2]

A student applies $\mathbf{H}$ to the fitted vector and to the residual vector. Which pair of results must hold?

A. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{0}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

B. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{0}$.

C. $\mathbf{H}\hat{\mathbf{Y}} = \mathbf{Y}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

D. $\mathbf{H}\hat{\mathbf{Y}} = \hat{\mathbf{Y}}$ and $\mathbf{H}\mathbf{e} = \mathbf{e}$.

Solution 2(a)

B. Since $\mathbf{H}^2 = \mathbf{H}$,

$$\mathbf{H}\hat{\mathbf{Y}} = \mathbf{H}^2\mathbf{Y} = \hat{\mathbf{Y}}, \qquad \mathbf{He} = \mathbf{H}(\mathbf{I}_n - \mathbf{H})\mathbf{Y} = \mathbf{0}.$$

The fitted vector lies in the model space, and the residual vector is orthogonal to it.

(b) Adding a redundant predictor [2]

Suppose $\mathbf{x}$ is not constant. Compare the least squares fits using

$$\mathbf{X} = [\mathbf{1}_n, \mathbf{x}] \quad \text{and} \quad \mathbf{W} = [\mathbf{1}_n, \mathbf{x}, 2\mathbf{x}]$$

for the **same response vector**. Which statement is correct?

A. The extra column must strictly reduce SSE.

B. The fit using $\mathbf{W}$ does not exist because $\mathbf{W}^\top\mathbf{W}$ is singular.

C. The fit using $\mathbf{W}$ has nonunique coefficients, but its fitted values and SSE equal those from $\mathbf{X}$.

D. The fit using $\mathbf{W}$ has nonunique coefficients and therefore nonunique fitted values.

Solution 2(b)

C. Adding $2\mathbf{x}$ does not change the column space. Both models project the response onto the same space, giving identical fitted values, residuals, and SSE. The coefficients for $\mathbf{W}$ are nonunique because its columns are linearly dependent.

(c) Geometric consequences [4]

Enter `T` or `F`. Each statement is worth **1 point**. Here $\bar{Y} = n^{-1}\sum_i Y_i$.

Statement	T/F
(i) The uncentered identity $\mathbf{Y}^\top\mathbf{Y} = \hat{\mathbf{Y}}^\top\hat{\mathbf{Y}} + \mathbf{e}^\top\mathbf{e}$ holds for the least squares fit.	___
(ii) The centered identity $\sum_i (Y_i - \bar{Y})^2 = \sum_i (\hat{Y}_i - \bar{Y})^2 + \text{SSE}$ is guaranteed for every least squares model, even one without an intercept.	___
(iii) If the model includes an intercept, the average fitted value equals the average observed response.	___
(iv) The residual vector must be orthogonal to the observed response vector $\mathbf{Y}$ itself.	___

Solution 2(c)

(i) T; (ii) F; (iii) T; (iv) F.

- **(i)** $\hat{\mathbf{Y}}^\top \mathbf{e} = 0$ gives the uncentered sum-of-squares decomposition.
- **(ii)** The centered decomposition holds when the model includes an intercept, but is not guaranteed without one.
- **(iii)** An intercept gives $\mathbf{1}_n^\top \mathbf{e} = 0$, so the observed and fitted means agree.
- **(iv)** In fact, $\mathbf{e}^\top \mathbf{Y} = \mathbf{e}^\top (\hat{\mathbf{Y}} + \mathbf{e}) = \text{SSE}$, which need not be zero.

Question 3: Sampling Properties and the Role of Normality [8]

Throughout this question, $\mathbf{X}$ is fixed with full column rank, $n > p$, and $\sigma^2 > 0$.

(a) Match the result to its assumptions [4]

Choose the **weakest sufficient set of assumptions among A, B, and C** for each result. Each row is worth **1 point**. Letters may be reused.

A. The least squares criterion and full column rank alone; no error moment or distribution assumptions.

B. $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$, without assuming normality.

C. The normal linear model, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Result	Letter
(i) The least squares coefficient vector is the unique minimizer of $S(\boldsymbol{\beta})$.	___
(ii) $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.	___
(iii) $\text{SSE}/\sigma^2 \sim \chi^2_{n-p}$.	___
(iv) $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$.	___

Solution 3(a)

(i) A; (ii) B; (iii) C; (iv) B.

- **(i)** Uniqueness of the minimizer is an algebraic consequence of full column rank.
- **(ii)** Zero-mean errors give $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$.
- **(iii)** The exact chi-square distribution requires the normal linear model among the listed assumptions.
- **(iv)** The covariance formula follows from $\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ without normality.

For (ii), zero-mean errors alone suffice; **B** is the weakest sufficient set *among the choices provided*.

(b) Why does independence follow? [2]

Under the normal linear model, which argument correctly explains why $\hat{\boldsymbol{\beta}}$ and SSE are independent?

A. $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are jointly normal with zero covariance, and SSE is a function of $\mathbf{e}$.

B. Any two uncorrelated random vectors are independent, regardless of their distribution.

C. They estimate different parameters, so they must be independent.

D. Each observed residual is independent of every other observed residual in every normal linear model.

Solution 3(b)

A. Both $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$ are linear transformations of the normal response vector, so they are jointly normal. Also,

$$\mathrm{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{e}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{M} = \mathbf{0}.$$

Joint normality turns zero covariance into independence. Since $\mathrm{SSE} = \mathbf{e}^\top\mathbf{e}$ is a function of $\mathbf{e}$, it is independent of $\hat{\boldsymbol{\beta}}$. Uncorrelatedness alone would not suffice without joint normality.

(c) Increasing the error variance [2]

Keep $\mathbf{X}$ and $\boldsymbol{\beta}$ fixed. In a second model, the errors still have mean zero, but their covariance is $4\sigma^2\mathbf{I}_n$ instead of $\sigma^2\mathbf{I}_n$. Which statement describes the **sampling distribution's mean and covariance** for the OLS estimator?

A. Its mean and covariance are both multiplied by 4.

B. Its mean is unchanged, and its covariance is multiplied by 16.

C. Its mean and covariance are both unchanged.

D. Its mean is unchanged, and its covariance is multiplied by 4; each coefficient's standard deviation doubles.

Solution 3(c)

D. The estimator remains unbiased, while its covariance becomes

$$4\sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}.$$

Each coefficient's variance is multiplied by 4, so its standard deviation is multiplied by $\sqrt{4} = 2$.

Question 4: Interpreting Confidence and Prediction Intervals [6]

Assume a correctly specified normal linear model. For prediction, a future observation has an error independent of the original data and with the same error variance.

(a) Mean response versus one future response [2]

At the **same predictor value**, software reports two intervals at the 95% level:

- Interval I: $(2.493, 3.507)$;
- Interval II: $(1.659, 4.341)$.

One is a confidence interval for the mean response; the other is a prediction interval for one future response. Which interpretation is correct?

A. I is the prediction interval because predicting one observation involves less uncertainty than estimating a mean.

B. Both intervals describe the same target; their different widths are only rounding error.

C. I is the mean-response interval; II is the prediction interval because it also accounts for the new observation's error.

D. II is the mean-response interval because averaging adds the new observation's error variance.

Solution 4(a)

C. At the same predictor value, the mean-response interval uses standard error $\hat{\sigma}\sqrt{h_0}$, whereas the prediction interval uses $\hat{\sigma}\sqrt{1+h_0}$, where $h_0 = \mathbf{x}_0^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_0$. The extra 1 accounts for the new observation's error variance. Thus the narrower interval I concerns the mean and the wider interval II concerns one future response.

(b) Connecting a confidence interval and a test [2]

A 95% confidence interval for β_1 is $(0.4, 1.6)$. Consider the corresponding two-sided t test of

$$H_0 : \beta_1 = 1 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 1$$

at $\alpha = 0.05$. Which conclusion is correct?

A. Reject H_0 because zero is outside the interval.

B. Fail to reject H_0 because 1 is inside the interval; this does not prove that $\beta_1 = 1$.

C. Accept H_0 as proven true because 1 is inside the interval.

D. No conclusion is possible because a t test can only test a coefficient against zero.

Solution 4(b)

B. The hypothesized value 1 lies inside $(0.4, 1.6)$, so the corresponding two-sided test fails to reject $H_0 : \beta_1 = 1$ at the 5% level. This does not prove the null hypothesis. Whether zero lies in the interval is irrelevant to this stated null value.

(c) What an interval means [2]

Enter `T` or `F`. Each statement is worth **1 point**. Assume a positive estimated error variance and $\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 > 0$.

Statement	T/F
(i) Holding the data and predictor value fixed, changing a mean-response confidence level from 95% to 99% makes the interval wider.	___
(ii) A 95% confidence interval for the mean response is designed to contain 95% of individual future responses at that predictor value.	___

Solution 4(c)

(i) T; (ii) F.

- **(i)** Holding the data fixed, a higher confidence level uses a larger critical value and produces a wider interval.
- **(ii)** A confidence interval for the mean concerns the unknown mean response. It does not describe a range containing 95% of individual future responses; those require a prediction interval.

Question 5: Distribution and Inference [12]

Suppose a multiple linear regression model has $n = 20$ observations and $p = 3$ parameters: an intercept and two slopes. The first column of $\mathbf{X}$ is $\mathbf{1}_{20}$. Assume

$$\mathbf{Y} \sim N_{20}(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_{20}) .$$

Suppose

$$\hat{\boldsymbol{\beta}} = \begin{pmatrix} 2.0 \\ 1.5 \\ -0.5 \end{pmatrix}, \qquad (\mathbf{X}^\top \mathbf{X})^{-1} = \begin{pmatrix} 0.05 & 0 & 0 \\ 0 & 0.05 & 0 \\ 0 & 0 & 0.08 \end{pmatrix},$$

and $\text{SSE} = 34$. Use $t_{0.975,17} = 2.110$.

(a) [3]

Choose the correct distributions.

For $\hat{\boldsymbol{\beta}}$:

A. $N_3(\mathbf{0}, \sigma^2 \mathbf{I}_3)$

B. $N_3(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$

C. t_{17}

D. χ^2_{17}

For SSE/σ^2:

A. $N(0, 1)$

B. t_{17}

C. χ^2_{17}

D. $F_{3,17}$

Solution 5(a)

B for $\hat{\boldsymbol{\beta}}$ and **C** for SSE/σ^2:

$$\hat{\boldsymbol{\beta}} \sim N_3(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}), \qquad \frac{\text{SSE}}{\sigma^2} \sim \chi^2_{17}.$$

The residual degrees of freedom are $n - p = 20 - 3 = 17$.

(b) [2]

Compute

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - p} = _____.$$

Solution 5(b)

$$\hat{\sigma}^2 = \frac{34}{20 - 3} = 2.000.$$

(c) [4]

For testing

$$H_0 : \beta_1 = 0 \qquad \text{versus} \qquad H_1 : \beta_1 \neq 0,$$

where β_1 is the second entry of $\boldsymbol{\beta}$, fill in the following. Use significance level $\alpha = 0.05$.

$$\mathrm{SE}(\hat{\beta}_1) = _____, \qquad t = _____,$$

and choose the decision:

A. Reject H_0

B. Fail to reject H_0

Solution 5(c)

Use the second diagonal entry of $(\mathbf{X}^\top\mathbf{X})^{-1}$, which is 0.05:

$$\mathrm{SE}(\hat{\beta}_1) = \sqrt{2(0.05)} = 0.3162278 \approx 0.316,$$

$$t = \frac{1.5 - 0}{\sqrt{0.1}} = 4.7434165 \approx 4.743.$$

Since $|t| > 2.110$, choose **A: Reject** H_0 at the 5% significance level.

(d) [3]

Which of the following is the 95% confidence interval for β_1?

A. $(0.8329,\ 2.1671)$

B. $(1.1838,\ 1.8162)$

C. $(-0.6671,\ 0.6671)$

D. $(0,\ 3.0000)$

Solution 5(d)

A. Using the supplied critical value,

$$1.5 \pm 2.110\sqrt{0.1} = (0.832759,\ 2.167241) \approx (0.833,\ 2.167).$$

These endpoints agree with option A at the requested three decimal places.

Assignment 3

STAT 8561 | Fall 2026
Due date: Friday, October 30, 2026
Total: 40 points
Coverage: Chapter 4 and the introductory portions of Chapter 5.

Questions 1-3 review Chapter 4. Question 4 introduces indicator coding and additive models from Chapter 5; the model and a reading guide are supplied. Complete Question 4 after those topics have been covered in class.

Question	Topic	Points
1	Model-comparison concepts	10
2	ANOVA and extra sums of squares	10
3	Reading a nested-model comparison	8
4	Guided introduction: two groups and one predictor	12
Total		**40**

i Instructions

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 3** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A3_LastName_FirstName.pdf`.
2. Its matching source, named `A3_LastName_FirstName.qmd` or `A3_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning; label each question and subpart.
- For multiple-choice items, enter **exactly one letter**. For True/False items, enter only `T` or `F`.
- For numerical items, report only the requested value. Round to **three decimal places**, retaining full precision during calculations. Degrees of freedom should be integers.
- **No written explanations, derivations, or R output are required.** All information needed to answer the questions is supplied.
- R is optional for checking. If you use R, include your code after your answers in executable chunks.
- Your source must render without errors. Use LaTeX for mathematics; do not submit screenshots of answers or code.

Notation and assumptions. p counts all independent design columns, including the intercept. Use $\alpha = 0.05$. Assume fixed full-rank designs and independent $N(0, \sigma^2)$ errors under the full model. Nested comparisons use the same response and observations, except the invalid comparison described in Question 1(e).

Question 1: What Model Comparisons Establish [10]

Select **exactly one** answer for each part. No written explanation is required.

(a) Adding a predictor [2]

A full OLS model adds an independent predictor column to a reduced model. Which statement must hold for the observed data?

A. $\mathrm{SSE}_F < \mathrm{SSE}_R$ strictly.
B. $\mathrm{SSE}_F \leq \mathrm{SSE}_R$, with equality possible.
C. $\mathrm{SSE}_F > \mathrm{SSE}_R$ because one degree of freedom is lost.
D. The added predictor has a statistically significant coefficient.

(b) The overall regression sum of squares [2]

The full model includes an intercept. Which reduced model gives $\mathrm{SSE}_R - \mathrm{SSE}_F = \mathrm{SSR}_F$?

A. Any model obtained by deleting one predictor.
B. The same full model written with different coefficient names.
C. The intercept-only model.
D. The model that fixes every fitted mean at zero.

(c) An overall rejection [2]

An overall F test rejects the null that all slopes are zero. What is the appropriate conclusion under the stated model assumptions?

A. There is evidence that at least one slope is nonzero.
B. Every slope is individually significant at the 5% level.
C. Every predictor has a causal effect on the response.
D. The fitted model has the lowest possible future prediction error.

(d) Training fit [2]

Both nested models include an intercept and $\mathrm{SST} > 0$. What can happen when a predictor is added?

A. Training R^2 must increase strictly.
B. Both R^2 and adjusted R^2 must increase.
C. Training R^2 can decrease because the residual degrees of freedom decrease.
D. Training R^2 cannot decrease, but adjusted R^2 may decrease.

(e) A missing-data trap [2]

`lm(y ~ x)` uses 30 observations; `lm(y ~ x + z)` uses 26 because `z` is missing for four observations. What should be done before using the nested-model partial F formula?

A. Use both SSE values as printed; nested formulas are sufficient.
B. Refit both models to the same complete-case observations for `y`, `x`, and `z`.
C. Replace the four missing `z` values by zero without further justification.
D. Divide each SSE by its own sample size before subtracting them.

Question 2: ANOVA and Extra Sums of Squares [10]

Two models use the same $n = 24$ observations:

$$M_R: \quad \mu_R(x) = \alpha_0 + \alpha_1 x,$$
$$M_F: \quad \mu_F(x, z) = \beta_0 + \beta_1 x + \beta_2 z.$$

The design ranks are $p_R = 2$ and $p_F = 3$. You are given

$$\text{SST} = 120, \qquad \text{SSE}_R = 54, \qquad \text{SSE}_F = 42.$$

(a)-(h) Numerical answers [8]

Give **one number per row**. Each row is worth **1 point**. Here SSR_F uses the intercept-only model as its baseline.

Item	Quantity	Your answer
(a)	Full-model SSR_F	_____
(b)	Full-model error degrees of freedom	_____
(c)	Full-model MSE_F	_____
(d)	Overall F statistic for $H_0 : \beta_1 = \beta_2 = 0$	_____
(e)	Full-model R^2	_____
(f)	Full-model adjusted R^2	_____
(g)	Extra sum of squares for adding z after x	_____
(h)	Partial F statistic for $H_0 : \beta_2 = 0$	_____

Formula reminders:

$$\bar{R}^2_F = 1 - \frac{\text{SSE}_F/(n-p_F)}{\text{SST}/(n-1)}, \qquad F_{\text{partial}} = \frac{(\text{SSE}_R - \text{SSE}_F)/(p_F - p_R)}{\text{SSE}_F/(n-p_F)}.$$

(i) Decision [2]

For part (h), the null reference distribution is $F_{1,21}$, with 5% upper-tail critical value **4.325**. Select one conclusion.

A. Fail to reject: the partial F statistic does not exceed 4.325.
B. Reject: there is evidence that z contributes to the mean after accounting for x.
C. Reject: this establishes that every slope is individually significant.
D. The test cannot be performed without fitting the models in R.

Question 3: Does Horsepower Add Information Beyond Weight? [8]

R's `mtcars` data contain 32 automobile models. The response is fuel economy (`mpg`, miles per US gallon); predictors are weight (`wt`, in 1,000 lb) and gross horsepower (`hp`). The data are observational.

Figure 7.14 motivates the question: does horsepower contribute to the mean **after accounting for weight**?

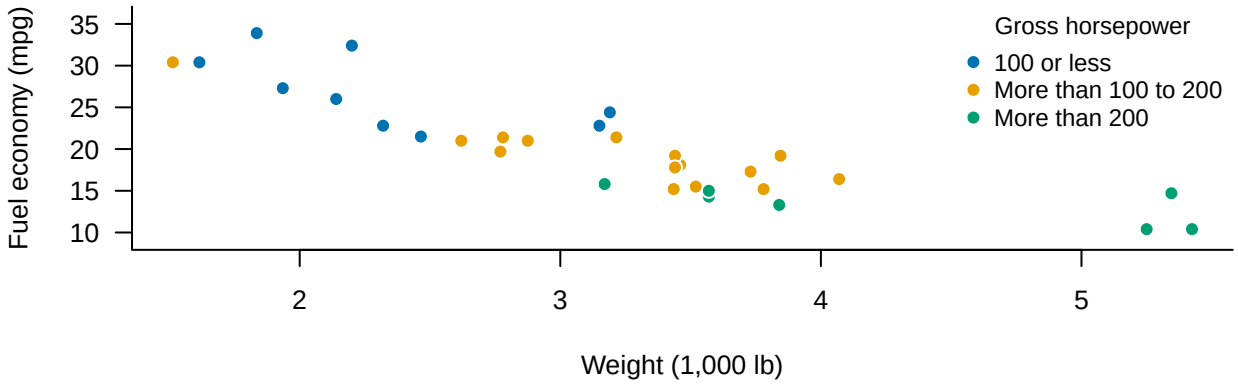


Figure 7.14: Fuel economy and weight for the 32 automobile models in R's mtcars data. Point colors identify horsepower bands (legend); horsepower is used as a continuous predictor in the models, not as these bands. The plot shows raw observations.

Model summaries (no R required).

$$M_R: \quad \mu_R(w) = \alpha_0 + \alpha_w w,$$
$$M_F: \quad \mu_F(w,h) = \beta_0 + \beta_w w + \beta_h h.$$

Both fits use the same $n = 32$ observations, with $p_R = 2$ and $p_F = 3$. In the full model, the test of $\beta_h = 0$ gives $t_h = -3.518712$ and a two-sided p-value of **0.001451**.

(a) The null hypothesis [2]

Which null tests whether horsepower adds information after accounting for weight?
A. $\beta_w = 0$; **B.** $\beta_0 = 0$; **C.** $\beta_h = 0$; **D.** $\beta_w = \beta_h = 0$.

(b) Reference distribution [2]

Which is the null reference distribution for the partial F test comparing these models?
A. $F_{2,29}$; **B.** $F_{1,30}$; **C.** $F_{2,30}$; **D.** $F_{1,29}$.

(c) Connect the tests [2]

For this single restriction, $F = t_h^2$. Report the partial F statistic to **three decimal places**.

(d) What does the result establish? [2]

Enter `T` or `F` for each statement (**1 point each**).

Item	Statement	T/F
(i)	At the 5% level, there is evidence that horsepower contributes to the mean after accounting for weight.	_____
(ii)	This result proves that changing horsepower causes a change in fuel economy.	_____

Question 4: Guided Introduction to Chapter 5 [12]

In a **hypothetical plant-growth study**, Y is plant height (cm), x is daily water volume (mL/day, observed from 8 to 12), and fertilizer is A or B. Define

$$x_c = x - 10, \qquad z = \begin{cases} 0, & \text{fertilizer A}, \\ 1, & \text{fertilizer B}. \end{cases}$$

The additive mean model and its fitted equation are

$$\mu(x, z) = \beta_0 + \beta_x(x - 10) + \beta_g z,$$
$$\hat{\mu}(x, z) = 18 + 1.5(x - 10) + 4z.$$

Reading guide. Substitute $z = 0$ for A and $z = 1$ for B. The additive model has a **common slope** because it contains no product term $(x - 10)z$. A design row uses column order $(1, x_c, z)$. See Chapter 5: indicator coding and additive models.

(a) Group B's population mean [2]

Which expression is $\mu_B(x)$?

A. $\beta_0 + \beta_x(x - 10)$; **B.** $\beta_0 + \beta_g + \beta_x(x - 10)$.
C. $\beta_0 + (\beta_x + \beta_g)(x - 10)$; **D.** $\beta_g + \beta_x x$.

(b) Design rows [2]

Which pair gives the rows for **A at** $x = 8$ and **B at** $x = 12$, in that order?

A. $(1, 8, 0)$ and $(1, 12, 1)$.
B. $(1, -2, 1)$ and $(1, 2, 0)$.
C. $(1, -2, 0)$ and $(1, 2, 1)$.
D. $(0, -2, 0)$ and $(1, 2, 1)$.

(c) Fitted means [2]

Give the fitted mean height at $x = 12$ for **A** and for **B**, in cm (**1 point each**).

(d) Adjusted group difference [2]

At the same water volume, what is the fitted **B-minus-A** mean difference, in cm? Give one number.

(e) Interpret the slope [2]

Which statement correctly interprets the fitted coefficient 1.5?

A. Within either fertilizer group, one additional mL/day of water is associated with a 1.5-cm increase in fitted mean height.
B. Every individual plant grows exactly 1.5 cm when water increases by one mL/day.
C. At the same water volume, B has a fitted mean height 1.5 cm higher than A.
D. The water slope is 1.5 for A and $1.5 + 4 = 5.5$ for B.

(f) Connect to Chapter 4 [2]

Which null hypothesis and reduced model test the fertilizer contribution after accounting for water, **within this additive model**?

A. $H_0 : \beta_x = 0$; reduced mean $\alpha_0 + \alpha_g z$.
B. $H_0 : \beta_0 = 0$; reduced mean $\alpha_x(x - 10) + \alpha_g z$.
C. $H_0 : \beta_x = \beta_g = 0$; reduced mean α_0.
D. $H_0 : \beta_g = 0$; reduced mean $\alpha_0 + \alpha_x(x - 10)$.

Assignment 4

Due date: TBA

Questions will be posted here when the assignment is released.

Instructions

Use **Quarto** or **R Markdown**. Upload **both** files to the **Assignment 4** submission folder on **iCollege**:

1. Your rendered **PDF**, named `A4_LastName_FirstName.pdf`.
2. Its matching source file, named `A4_LastName_FirstName.qmd` or `A4_LastName_FirstName.Rmd`.

The PDF must be generated from the submitted source file. **HTML alone does not satisfy the submission requirement.**

- Put your name and course number at the beginning of the PDF.
- Clearly label every question and subpart.
- Include any R code used to obtain your answers in executable chunks. Before generating random values, set a seed (e.g., `set.seed(8561)`) and keep that line in your code.
- Your source must render from beginning to end without errors. Type mathematical expressions using LaTeX; do not submit screenshots of answers or code.

Part IV

Exam

Exam 1

Term: 2026F (Fall 2026)
Instructor: Chi-Kuang Yeh
Date: October 14, 2026
Status: Completed
Time: 75 minutes
Total: [100] points
Coverage: Assignment 1–Assignment 2 and lecture notes through Section 4.2: The Overall F Test; Definition 4.4 and later material are excluded.
Emphasis: Model interpretation, least squares geometry, and statistical reasoning.
Notation: p is the number of coefficients, including any intercept. Vectors are boldfaced; comma-separated tuples are column vectors.
Sums of squares: SST = centered total; SSR = regression; SSE = residual.

Question	Format	Points
1	Multiple Choice (10 parts, [5] points each)	[50]
2	True or False (5 parts, [4] points each)	[20]
3	Conceptual (3 parts, [10] points each)	[30]
Total		**[100]**

Exam instructions

1. Work independently. Only a **non-programmable calculator** is permitted as an electronic device. No code execution is required.
2. Record Questions 1–2 on the separate answer sheet: one letter A–D or T/F per part. Only the grid is graded; no explanations are required.
3. For Question 3, use one or two sentences per numbered part on the separate answer sheet.
4. Submit only the answer sheet. You may keep the question paper.

Answer sheet: Write all final answers on the separate answer sheet.

Question 1: Multiple Choice [50]

Each part is worth **[5] points**. Select the **single best answer**: the statement guaranteed by the stated conditions. Record it on the separate answer sheet. No explanation is required.

1a)

For fixed predictor values, the mean model is $\mathbb{E}(Y_i) = \beta_1 x_i + \beta_2 z_i$, with no intercept term. At $x_i = z_i = 0$, which statement must hold?

A. The observed response must equal zero.
B. The mean response equals zero, although an individual response may differ from zero.
C. Both regression coefficients must equal zero.
D. The residuals must sum to zero for every data set.

Answer

Answer: B. With no intercept, substituting $x_i = z_i = 0$ gives a zero model mean. Random error still permits a nonzero observed response. A zero residual sum is not guaranteed without an intercept.

1b)

Each of two measurement errors has variance $\sigma^2 > 0$. Compare the variance of their arithmetic mean when their correlation is positive with the variance when it is zero. Keep the individual variances fixed. Which statement is correct?

A. The average of the uncorrelated errors has greater variance.
B. The two averages have the same variance.
C. The average of the positively correlated errors has zero variance.
D. The average of the positively correlated errors has greater variance.

Answer

Answer: D. For errors $\varepsilon_1, \varepsilon_2$, $\mathrm{Var}\{(\varepsilon_1 + \varepsilon_2)/2\} = \sigma^2/2 + \mathrm{Cov}(\varepsilon_1, \varepsilon_2)/2$. Positive covariance increases uncertainty in the average.

1c)

Fit a simple linear regression by OLS with an intercept and one nonconstant predictor x_i measured in centimeters. Refit the same data using $x_i^* = x_i/100$, measured in meters. Which relationship must hold?

A. The estimated slope is multiplied by 100; the fitted responses and residuals stay the same.
B. The estimated slope is divided by 100; the fitted responses stay the same.

C. The fitted responses are divided by 100 because the predictor is smaller.
D. The residual sum of squares must decrease because the predictor uses smaller numbers.

Answer

Answer: A. Since $x = 100x^*$, $\hat{\beta}_0 + \hat{\beta}_1 x = \hat{\beta}_0 + (100\hat{\beta}_1)x^*$. Changing predictor units reexpresses the same fitted line; it does not improve the fit.

1d)

Fit $\mathbf{Y}$ by OLS using a design matrix X with full column rank. Replace the response by $\mathbf{Y}^* = \mathbf{Y} + X\mathbf{a}$, where $\mathbf{a}$ is a fixed vector, and refit by OLS using the same X. Which relationship must hold?

A. The fitted vector is unchanged; the residual vector increases by $X\mathbf{a}$.
B. Both the fitted vector and residual vector increase by $X\mathbf{a}$.
C. The new residual vector must be zero.
D. The fitted vector increases by $X\mathbf{a}$; the residual vector is unchanged.

Answer

Answer: D. The added vector already lies in the model space. Thus $H(\mathbf{Y} + X\mathbf{a}) = \hat{\mathbf{Y}} + X\mathbf{a}$ and $(I - H)X\mathbf{a} = \mathbf{0}$. The fit absorbs the entire change.

1e)

Let $\mathbf{x}$ be nonconstant. Compare OLS using $X = \begin{pmatrix} \mathbf{1}_n & \mathbf{x} \end{pmatrix}$ with OLS using $W = \begin{pmatrix} \mathbf{1}_n & \mathbf{x} & \mathbf{x} + \mathbf{1}_n \end{pmatrix}$ on the same response vector. Which statement must hold?

A. The coefficients for W are nonunique, but its fitted vector and SSE equal those from X.
B. The third column adds a new direction, so SSE must strictly decrease.
C. The fit using W does not exist because $W^\top W$ is singular.
D. The coefficients and fitted vector for W are both nonunique.

Answer

Answer: A. The added column is a linear combination of the first two. The column space and its unique projection are unchanged, even though different coefficient vectors represent that projection.

1f)

For a fixed design matrix X with full column rank, which statement about computing the OLS coefficient estimate is correct?

A. Normally distributed errors are required to define the sum of squared residuals.
B. Normally distributed errors are required to make $X^\top X$ invertible.

C. The OLS estimate can be computed from X and $\mathbf{Y}$ without assuming normally distributed errors.
D. Without normally distributed errors, the OLS estimate can be computed only when all observed residuals are zero.

Answer

Answer: C. Full column rank makes $X^\top X$ invertible, so $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{Y}$ can be computed from the observed data without assuming normal errors. Normality is not needed to define or minimize the sum of squared residuals.

1g)

In $\mathbf{Y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, keep X and $\boldsymbol{\beta}$ fixed, with X of full column rank. Errors have mean zero in both models, but their covariance changes from $\sigma^2 I$ to $4\sigma^2 I$, where $\sigma^2 > 0$. How do the mean and standard deviation of each OLS coefficient estimator change?

A. The mean and standard deviation of each coefficient estimator both double.
B. Each mean is unchanged, and each standard deviation is multiplied by 4.
C. Each mean is unchanged, and each standard deviation doubles.
D. Each mean and each standard deviation are unchanged.

Answer

Answer: C. Mean-zero errors preserve $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$. The covariance is multiplied by 4, so standard deviations are multiplied by 2. More noise reduces precision without creating bias.

1h)

The usual two-sided 95% t confidence interval for a slope β_1 is $(-0.2, 0.6)$. What is the conclusion of the corresponding two-sided t test of $H_0 : \beta_1 = 0.5$ at significance level 0.05?

A. Reject H_0 because the interval is not centered at 0.5.
B. The interval proves that $\beta_1 = 0.5$.
C. No conclusion is possible because t tests can only test a slope against zero.
D. Fail to reject H_0 because 0.5 is inside the interval; this does not prove H_0.

Answer

Answer: D. A corresponding two-sided test rejects exactly the hypothesized values outside the confidence interval. Failure to reject is not proof of an exact parameter value.

1i)

For the same response data, an intercept-only OLS fit has residual sum of squares 80, while an OLS fit with an intercept and predictors has residual sum of squares 30. What does the reduction of 50 represent?

A. The error variance σ^2 has decreased by 50.
B. The squared prediction error on every future data set will decrease by 50.
C. The reduction in unexplained variation in these observations relative to the constant-mean fit.
D. Each predictor is individually significant because the fit improved.

Answer

Answer: C. The intercept-only fit uses $\bar{Y}$, so its SSE is the centered SST of 80. Thus SSR $= 80 - 30 = 50$ is the sample variation explained beyond a constant mean. This improvement does not establish a change in σ^2, future predictive performance, or individual significance.

1j)

In a correctly specified normal linear model with an intercept and three slopes, a valid overall F test rejects $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ at significance level 0.05. Which conclusion is justified?

A. Each of the three slopes must be individually significant at level 0.05.
B. There is evidence that at least one slope is nonzero, without identifying which one.
C. There is evidence that the intercept is nonzero.
D. The model must predict every future response accurately.

Answer

Answer: B. The overall test assesses the slopes jointly against an intercept-only model. Rejection provides evidence against all slopes being zero; it does not establish individual significance or accurate predictions.

Question 2: True or False [20]

Each part is worth **[4] points**. Record **T** or **F** in the answer grid on the separate answer sheet. No explanation is required.

2a)

The assumption $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ guarantees that the model errors satisfy $\sum_{i=1}^{n} \varepsilon_i = 0$ in every realized sample.

Answer

Answer: False. Mean zero is a statement about expectations across repeated samples. It does not impose a zero sum on a realized error vector. The zero sum of fitted OLS residuals with an intercept is a different, algebraic property.

2b)

If the OLS coefficient estimator and the residual vector have zero cross-covariance, they must be independent, regardless of the error distribution.

Answer

Answer: False. Uncorrelated random vectors need not be independent. In the normal linear model, joint normality supplies the extra condition that makes the implication valid.

2c)

For an OLS fit with an intercept, the centered decomposition $\text{SST} = \text{SSR} + \text{SSE}$ holds even without normally distributed errors.

Answer

Answer: True. With an intercept, the centered fitted vector is orthogonal to the residual vector, giving the sum-of-squares identity. Normality is not needed for this algebraic decomposition.

2d)

In an OLS model with an intercept, the null hypothesis that all slopes are zero implies that the observed regression sum of squares SSR must equal zero in every sample.

Answer

Answer: False. Even when the true mean is constant, random noise can align with predictor directions and produce a nonzero fitted slope and SSR. Under the normal-model null, MSR estimates σ^2; it is not forced to be zero.

2e)

For an OLS model whose design matrix has full column rank, with an intercept and p coefficients in total, the regression degrees of freedom associated with SSR are p.

Answer

Answer: False. The intercept accounts for the constant-mean direction. The centered regression degrees of freedom are $p-1$; the total and residual degrees of freedom are $n-1$ and $n-p$, respectively.

Question 3: Conceptual [30]

For each part, write one or two sentences on the separate answer sheet; brief formulas may be used. Equivalent explanations are accepted, with partial credit for correct reasoning.

3a) Matrix representation and least squares [20]

3a(i) [10]

A regression of exam score (points) on study time (hours) gives the fitted line $\hat{y} = 60 + 2x$. Interpret the slope. Does the model require a student who studies one hour more than another to score exactly 2 points higher? Explain briefly.

Answer

Answer: A student who studies one hour more has a fitted mean exam score that is 2 points higher (mean [3], response units [2], one-unit change [2]). The two observed scores need not differ by exactly 2 points because each can deviate from its fitted mean [3]. Do not require or award a causal interpretation.

3a(ii) [10]

For simple linear regression with an intercept and $\mathbf{x} = (0, 1, 2, 3)$, a student proposes residuals $\mathbf{e} = (1, -1, 1, -1)$ because $\sum_i e_i = 0$. Is zero sum sufficient? Evaluate the other normal equation and decide whether these can be OLS residuals.

Answer

Answer: OLS also requires $\mathbf{x}^\top \mathbf{e} = 0$, so a zero residual sum alone is insufficient [5]. Here $0(1) + 1(-1) + 2(1) + 3(-1) = -2$ [2], so these cannot be the residuals of the claimed OLS fit [3].
Full-credit short answer: "No: $\mathbf{x}^\top \mathbf{e} = -2 \neq 0$ violates the OLS normal equations."

3b) Interpreting uncertainty [10]

3b(i) [10]

From the same fitted normal linear model and data, compare the usual 95% confidence interval for the mean with the 95% prediction interval for one independent future response at the same predictor value. Assume positive estimated error variance. Which is wider? Why would individual prediction remain uncertain even if the true mean were known exactly?

Answer

Answer: The prediction interval is wider [3] because it includes additional random error from the independent future response [4]. That individual variation remains even if the true mean response is known exactly [3]. This explanation earns full credit; no formula or separate statement about mean-estimation uncertainty is required.

Answer Key and Grading Rubric

Question 1: Multiple Choice [50]

Part	1a	1b	1c	1d	1e	1f	1g	1h	1i	1j
Answer	B	D	A	D	A	C	C	D	C	B

Award **[5] points** for each correct answer and **[0] points** otherwise. No explanations are required.

Question 2: True or False [20]

Part	2a	2b	2c	2d	2e
Answer	False	False	True	False	False

Award **[4] points** for each correct answer and **[0] points** otherwise. No explanations are required.

Question 3: Conceptual [30]

Accept equivalent concise reasoning, including equations. Apply the allocations below for partial credit; deduct an arithmetic error once and retain credit for otherwise correct reasoning.

Part	Grading criteria	Points
3a(i)	Fitted/estimated mean [3]; response units [2]; per one-unit predictor increase [2]; individual outcomes still have error variation [3].	[10]
3a(ii)	Identify the additional OLS requirement $\mathbf{x}^\top \mathbf{e} = 0$ and show why the residual-sum check is insufficient [5]; compute $\mathbf{x}^\top \mathbf{e} = -2$ [2]; reject the proposed OLS fit [3].	[10]
3b(i)	Prediction interval is wider [3]; it includes additional random error from the independent future response [4]; that variation remains even if the true mean is known exactly [3].	[10]
Total		**[30]**

Part V

Project

Instruction

STAT 8561 | Fall 2026
Due on December 13, 2026
Total: 100 points
Course-grade weight: 20% (see Grade Distribution)
MS students: 1-4 students per group
PhD students: Complete the project individually
Project: Linear Regression Data Analysis and Machine Learning Comparison

Analyze a **real dataset with a continuous response** using multiple linear regression. Explain the findings in context, and compare predictions with **one machine learning regression algorithm of your choice**.

Group Members

Group	Members	Topic
1	Xiaorui Bao	TBD
2	Tommy Stell	TBD
3	Xinning Wang	TBD
4	Kimia Daliran Saravi Jada Grant Shravya Honne Airianna Jones	TBD
5	Shaswat Bhushan Jangam Prakash Karki	TBD
6	Mangsa Limbu Pheudin Seoyoung Kim	TBD
7	Hovor Moses Kwame Shudeshna Nandy Juliet Seadey	TBD
8	Qiaoting Gao Zidu Peng Zhe Zhong	TBD
9	Silas Bambil Yasmeen Babagana	TBD
10	Emmanuel Torzen Enock Ackaah	TBD
11	Elton Kwashie Seth Kwarteng Halimah Akinoso	TBD
12	Taiwo Ayeni	TBD

Project Overview

Criterion	Topic	Points
1	Question and data	10
2	Linear model and interpretation	25
3	Model comparison and diagnostics	20
4	Machine learning method	10
5	Fair predictive evaluation	15
6	Findings and critical discussion	10
7	Communication and reproducibility	10
Total		**100**

Instruction

Use **Quarto** or **R Markdown**. Upload **both** files to the **project submission area on iCollege**:

1. Your rendered **PDF**, named `Project_Report.pdf`.
2. Its matching source file, named `Project_Report.qmd` or `Project_Report.Rmd`.

The report requires **8-10 pages of main text**, including the abstract and main-text figures, tables, and captions. A separate title-only cover page and the end sections are excluded; see the complete page-count rule. List all group members and their contributions.
The submitted source and supporting files must reproduce the PDF and results. **HTML alone does not satisfy the requirement.**
Python or Jupyter Notebook: Discuss the format with the instructor in advance; a PDF and reproducible source are still required.
See Report requirements and submission for the full instructions. The submission folder will be announced on iCollege.

Report writing model

Preview the report template (PDF) to see the report layout and section headings. You may refer to published articles in The American Statistician for examples of how to organize and present your work.

Prepare the report in the style of **The American Statistician**: a clear research question, defensible statistical analysis, informative graphics, and a conclusion supported by the evidence. Use the journal references and report structure and the downloadable Quarto starter to plan the 8-10 page report.

Assessment criteria

The project totals **100 points** and is worth **20% of the course grade** (Grade Distribution). Each weight below is a percentage of the **project grade**: 25% equals 25 points. A score of 90/100 contributes 18 percentage points to the course grade.

Criterion	Weight	Full-credit evidence (points)
Question and data	10%	Clear question, observational unit, and outcome units (3); cited source and variable definitions (3); justified cleaning, missing-data handling, and dependence assessment (4).
Linear model and interpretation	25%	Correct mean model and predictor coding (5); correct interpretations of at least two coefficients or contrasts and the intercept, including units and adjustment (12); correct 95% confidence intervals and uncertainty discussion (8).
Model comparison and diagnostics	20%	Valid nested-model restriction and test, with assumptions addressed (6); explanation of retained coefficients across models (4); informative residual, Q-Q, and influence checks with justified responses to problems (10).
Machine learning method	10%	Appropriate regression method and accurate explanation (4); documented implementation, preprocessing, and final settings (6).
Fair predictive evaluation	15%	Appropriate common test set and no data leakage (6); documented training-only validation and tuning (4); correct baseline, test RMSE/MAE, and prediction plots for both models (5).
Findings and critical discussion	10%	Contextual conclusions about effect size and uncertainty (4); reasoned prediction comparison and model recommendation (3); meaningful limitations and next step (3).
Communication and reproducibility	10%	Organized report and readable, discussed figures and tables (3); runnable source that reproduces results with documented data, seed, and environment (5); citations and each member's contribution (2).
Total	**100%**	**100 points**

Subitem points appear in parentheses. Partial credit reflects accuracy, completeness, and justification; omissions earn zero. Full credit does not require machine learning to outperform linear regression.

How to get started

1. **Choose a question and data.** Identify a continuous outcome, its units, the observational unit, and plausible predictors. Cite the original data source.
2. **Plan the comparison.** Choose one machine learning method for regression, such as a regression tree, random forest, gradient boosting, or support vector regression. Decide how to reserve test observations before fitting or tuning models.

3. **Analyze and explain.** Fit and diagnose the linear regression, interpret its coefficients and uncertainty, and compare a scientifically motivated pair of nested linear models.
4. **Evaluate and communicate.** Compare both methods on the same held-out observations using RMSE and MAE. Explain what each model reveals, where it performs poorly, and which you would use for your stated purpose.

A few possible questions

- How is a property's sale price associated with floor area and other recorded characteristics, and does a random forest improve prediction?
- How is a building's energy use associated with its characteristics, and does gradient boosting capture patterns missed by the linear model?
- How is a measured environmental outcome associated with recorded conditions, and how does a regression tree compare with linear regression?

These are examples of scope; you may choose another application and another suitable regression algorithm.

> **i** What makes a strong project?
>
> A strong project gives a defensible linear regression analysis, clear interpretations, a fair prediction comparison, and reproducible evidence. A machine learning model does **not** have to outperform linear regression. Explain the result you obtain, including its limitations.
> Use existing software to fit the models, and explain the method, important settings, and analysis decisions. Record **8561** as the seed for randomized procedures. In R, use `set.seed(8561)` and prefer **ggplot2** for figures.

Next: Report requirements, assessment criteria, and submission.

Report

STAT 8561 | Fall 2026
Due on December 13, 2026
Course-grade weight: 20% (see Grade Distribution)
Points: 100
MS students: 1-4 students per group
PhD students: Complete the project individually

Use a real dataset to explain an outcome with **multiple linear regression**, and compare its predictive performance with **one machine learning regression algorithm of your choice**. Begin with the project overview.

Research question and data

Choose a continuous response and at least two scientifically meaningful candidate predictors. State what one row represents, the response's units, the population or setting of interest, and the question your analysis will answer. Choose a dataset large enough to support the proposed models and a meaningful evaluation on held-out observations.

- Link to and cite the original data source. Describe how the observations were collected, the sample size, and the variable definitions.
- Explain exclusions, missing values, unusual observations, and any transformations. Give the number of observations retained for analysis.
- Distinguish variables available when a prediction would be made from information recorded afterward. Use predictors that would actually be available in the intended application.
- Explain whether observations are independent. If rows share a subject, site, or time sequence, use a splitting strategy that respects that structure and state the implications for inference.

Before choosing models, write one question about **association or explanation** and one about **prediction**. For example, an adjusted association between size and price is a different question from predicting the price of a new property.

Linear regression analysis

Use the ideas from least squares estimation, OLS inference, model comparison, and multiple regression and categorical predictors.

1. **Specify the model.** Write the mean model, define its variables and units, and explain why the predictors belong in the analysis. State reference categories and justify any interactions or transformed terms.

2. **Explain the coefficients.** Interpret at least two scientifically relevant coefficients or contrasts in context, stating the predictor change, response units, and other variables held fixed. Explain the intercept's baseline and whether it is meaningful for the observed data. With an interaction, make clear which coefficient interpretation depends on the other variable's value.
3. **Report uncertainty.** Give estimates and 95% confidence intervals for the coefficients or contrasts you interpret. State the assumptions supporting the inference. Distinguish an interval for a model parameter from a prediction interval for an individual outcome. Do not equate a small p-value with a large practical effect.
4. **Compare nested models.** Compare one reduced and one full linear model on the same training observations and response scale. State the restriction, such as an added coefficient or group of coefficients being zero. Use an appropriate partial F test when its assumptions are defensible; otherwise explain the limitation of that test. Explain whether a retained coefficient's meaning or estimate changes when the adjustment set changes.
5. **Check the model.** Use residual-versus-fitted and normal Q-Q plots, and examine influential observations. Discuss nonlinearity, unequal variance, unusual observations, and the plausibility of independence. Justify any remedial action; do not remove an observation solely to improve a result. See observation diagnostics and checking the fitted functional form.

Use the training data for these modeling decisions. If you select terms or transformations after examining the data, report that process and treat ordinary confidence intervals and p-values as exploratory: standard fixed-model formulas do not account for data-driven selection. Treat findings from observational data as associations unless the study design justifies a causal interpretation.

Machine learning comparison

Choose **one** method suited to predicting a continuous response, for example a regression tree, random forest, gradient boosting, or support vector regression. Explain how it produces predictions, why it is appropriate for your data, and which important tuning parameters you choose. Cite the method and software documentation.

Existing implementations are appropriate. Identify the functions, software versions, preprocessing, and settings needed to reproduce the fit. If you report a feature-importance measure or an effect plot, explain what it measures; it is not automatically an adjusted regression coefficient or a causal effect.

A fair evaluation

1. **Reserve a test set first.** State the split and its rationale. For independent observations, an 80% training / 20% test split is a reasonable starting point. Use a group-aware or chronological split when appropriate. Conduct model-building exploration on the training data.
2. **Use the same comparison.** Both methods predict the same response for the same test observations and use the same available candidate predictors. Explain any model-specific encoding, transformations, or training-based variable selection.
3. **Keep learning within training data.** Estimate imputation, scaling, encoding rules, and other learned preprocessing using training data. If you use cross-validation to select a model

or tune parameters, repeat all learned preprocessing inside each training fold. Apply the fitted transformations to validation or test observations.

4. **Tune without the test outcomes.** Choose linear-model terms and machine learning settings using training data, with cross-validation or a separate validation subset as appropriate. Report the candidate settings, validation design, and selection metric. Finalize the models before evaluating them on the test set; do not revise them after seeing test performance.
5. **Report the comparison.** Evaluate a training-mean baseline, the final linear regression, and the machine learning model on the same test set. Report RMSE and MAE in the original response units. Explain how any response transformation is reversed and how any correction is estimated using training data.

For m test observations, calculate

$$\text{RMSE} = \sqrt{\frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}, \qquad \text{MAE} = \frac{1}{m}\sum_{i=1}^{m}|y_i - \hat{y}_i|.$$

The baseline predicts the **training-set mean** for every test observation. Smaller RMSE and MAE indicate smaller prediction errors. Training R^2 alone does not establish which model predicts new observations better. See model assessment and prediction and the tidymodels resampling guide.

Use this format for the main results table; fill it with your own results:

Model	Test RMSE	Test MAE
Training-mean baseline	Your result	Your result
Multiple linear regression	Your result	Your result
Chosen machine learning method	Your result	Your result

Include an observed-versus-predicted plot for both fitted models and discuss where errors are largest. Differences on one test set are estimates, not proof of universal superiority; discuss how the sample size, split, and application limit the comparison.

For implementation guidance, consult the scikit-learn guide to avoiding data leakage and the official ensemble-methods documentation when relevant to your chosen method.

Explain the findings

Your conclusion should answer the research question in ordinary language and address all of the following:

- **Substantive findings:** Which adjusted relationships are supported, how large are they in meaningful units, and how uncertain are they?
- **Model comparison:** Did the machine learning method improve test prediction, and is the difference useful in this application? Explain why a flexible model might or might not improve performance.
- **Interpretation:** What does the linear model make easy to explain? What additional patterns, if any, does the machine learning analysis suggest? Distinguish coefficient estimates from prediction-based importance measures.
- **Recommendation and limitations:** Which model would you use for the stated purpose, and why? Discuss assumptions, data quality, the scope of generalization, and one useful next step.

Report format and reproducibility

Write **8-10 pages of main text**, counted using the rule below. Use Quarto or R Markdown to produce a PDF with readable 11- or 12-point text, standard margins, and numbered sections. For a Python or Jupyter Notebook workflow, follow the prior-discussion requirement in the submission note.

> **i** Page count: 8-10 pages of main text
>
> The finished report must contain **at least 8 and at most 10 pages of main text**.
>
> - **Count:** the abstract, Sections 1-6 (Introduction through Discussion and conclusion), and all figures, tables, captions, equations, and code examples within those sections.
> - **Do not count:** a separate title-only cover page, References, Author contributions, and appendices or supplementary material. Put these end sections after the main text; start **References on a new page**.
> - A first page containing the title, author names, and abstract **counts as a main-text page**. Only a separate page containing the title and author information alone is excluded; a separate cover page is optional.
>
> The **9-page plan below is a suggested allocation**, not a different requirement. The short template PDF shows headings and prompts; its number of pages is **not** the required report length. The full submitted PDF may exceed 10 pages because the excluded material comes afterward.

Writing model: The American Statistician

Write the report in the style of an applied article in **The American Statistician (TAS)**: introduce a statistical question, explain the analysis to a statistically literate reader, and use evidence to support a clear conclusion. Connect estimates, uncertainty, diagnostic plots, and prediction results to the application. Organize the main text as a coherent article; put lengthy software output and implementation details in supporting material.

The structure below is a **course adaptation** of this publication style. Use it with this project's 8-10 page limit and existing rubric. A journal submission or an exact reproduction of the publisher's typesetting is not required. Keep the course's PDF, source-file, font-size, and margin requirements above.

Read these references before drafting:

- The American Statistician: journal and recent articles. Look at how articles introduce a problem and connect tables and figures to their argument.
- Publisher's instructions for authors, for the journal's own manuscript guidance.
- Denby and Pregibon (1987), "An Example of the Use of Graphics in Regression": an example of explaining regression through graphics and their statistical interpretation.
- Smith (2011), "Missteps in Multiple Regression Student Projects: Beyond Association-Not-Causation": a useful companion on pitfalls in student regression projects. Full articles may require access through the university library.

Suggested report structure

Use a specific title and list all authors and the course. A useful starting point is a **150-200 word abstract** followed by **3-5 keywords**. These are course writing suggestions. The abstract should state the question, data, methods, main quantitative finding, and conclusion; write it after completing the analysis.

The following is an illustrative **9-page plan**, including the abstract and main-text figures and tables. Adjust the allocation within the required **8-10 counted pages**; apply the page-count rule above.

Section	What the reader should learn	Pages
Abstract	The question, approach, main result, and takeaway.	0.5
1. Introduction	Why the application matters; the association and prediction questions; relevant background.	1
2. Data and study design	Source, units, cleaning, training-data exploration, and the train/test design. Include a motivating plot.	1.5
3. Linear regression analysis	Model and assumptions; interpreted estimates and confidence intervals; nested-model comparison; diagnostics and limitations.	2
4. Machine learning method	One chosen method, its rationale, preprocessing, tuning design, and final settings.	1
5. Results and predictive comparison	Baseline, regression, and ML test RMSE/MAE; observed-versus-predicted plots; practical meaning of the differences.	2
6. Discussion and conclusion	Answer the research question, recommend a model for the stated purpose, and explain uncertainty and limitations.	1

After the main text, include **References**, an **Author contributions** statement, and any **Supplementary material**. Cite data, methods, software, and adapted code. Use consistent author-year citations and complete references, with DOI or source links where available.

Number figures and tables, give informative captions with units and a legend where needed, and discuss each in the text. Present selected results as readable tables and plots. Keep full code, secondary diagnostics, and long output in the appendix or supporting files; include everything needed to reproduce the reported numbers.

Preview and download the template

Preview the STAT 8561 report template (PDF). This PDF is rendered directly from the downloadable template below and shows the layout, section headings, and writing prompts. It is a short outline, not an example of the required length; your completed report must contain **8-10 counted pages of main text** under the page-count rule.

You may refer to published articles in **The American Statistician** and the examples linked above for statistical exposition, figures, tables, and interpretation. Use them as models for organizing and presenting your own work.

Download the STAT 8561 report starter (ZIP). Unzip it to obtain `Project_Report.qmd` and `README.md`. The Quarto file contains the section headings and writing prompts above. Replace every prompt with your own analysis, add executable code and citations, and render `Project_Report.pdf`. The starter follows the course's TAS-inspired structure and submission format.

Set the seed to **8561** for randomized splitting, resampling, and model fitting; in Python, set the relevant `random_state` or generator seed to `8561`. Save the split identifiers and record the software environment. Prefer **ggplot2** for R figures. Use relative paths and provide instructions that work from a fresh session.

Assessment criteria

The project totals **100 points** and is worth **20% of the course grade** (Grade Distribution). Each weight below is a percentage of the **project grade**: 25% equals 25 points. A score of 90/100 contributes 18 percentage points to the course grade.

Criterion	Weight	Full-credit evidence (points)
Question and data	10%	Clear question, observational unit, and outcome units (3); cited source and variable definitions (3); justified cleaning, missing-data handling, and dependence assessment (4).
Linear model and interpretation	25%	Correct mean model and predictor coding (5); correct interpretations of at least two coefficients or contrasts and the intercept, including units and adjustment (12); correct 95% confidence intervals and uncertainty discussion (8).
Model comparison and diagnostics	20%	Valid nested-model restriction and test, with assumptions addressed (6); explanation of retained coefficients across models (4); informative residual, Q-Q, and influence checks with justified responses to problems (10).
Machine learning method	10%	Appropriate regression method and accurate explanation (4); documented implementation, preprocessing, and final settings (6).
Fair predictive evaluation	15%	Appropriate common test set and no data leakage (6); documented training-only validation and tuning (4); correct baseline, test RMSE/MAE, and prediction plots for both models (5).
Findings and critical discussion	10%	Contextual conclusions about effect size and uncertainty (4); reasoned prediction comparison and model recommendation (3); meaningful limitations and next step (3).
Communication and reproducibility	10%	Organized report and readable, discussed figures and tables (3); runnable source that reproduces results with documented data, seed, and environment (5); citations and each member's contribution (2).
Total	**100%**	**100 points**

Subitem points appear in parentheses. Partial credit reflects accuracy, completeness, and justification; omissions earn zero. Full credit does not require machine learning to outperform linear regression.

Submission

Report submission checklist

MS students work in groups of **1-4 students**. **PhD students** must complete the project **individually**. Upload the following to the **iCollege project submission area** by **December 13, 2026**. The submission folder will be announced on iCollege:

- `Project_Report.pdf`.
- Its matching `Project_Report.qmd` or `Project_Report.Rmd` source (the default format; see the Python/Jupyter Notebook note below).

If the source depends on supporting files, include them with the source in a ZIP. Provide any required scripts, data that can be redistributed or precise data-access instructions, split identifiers, and software versions and rendering instructions.
Put all author names and **STAT 8561** on the report. The PDF must be generated from the submitted source file; **HTML alone does not satisfy the submission requirement**. Check that the submitted files reproduce the figures, tables, and model-comparison results from beginning to end.

Python and Jupyter Notebook submissions

If you would like to use **Python** or submit a **Jupyter Notebook (`.ipynb`)**, discuss your plan with the instructor **in advance** and agree on the submission format. Python/Jupyter Notebook submissions are subject to that discussion; they are not an automatic alternative to the default `.qmd`/`.Rmd` requirement.
Any agreed alternative must still include a **PDF report**, the **reproducible source** (including the notebook if applicable), and the supporting files needed to regenerate the analysis. The same report requirements and grading criteria apply.

Related: Project Instruction.